# SnapGene User Guide

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# Licenses

## Download and Register SnapGene

After purchasing SnapGene, how can I download and register the software?

### Obtain the Download Link from Email

> Dear SnapGene Test Lab,
>
> Please follow the link below to download SnapGene. This link is unique for your group.
>
> http://www.snapgene.com/support/downloads/?group=%2Fx851EbDqgL8umKjABE7zjB64c8bszgf7Bl28FcuRs1uOYobxEZqY8GiHQrCAnJU
>
> If you are unable to follow the link, please copy it and paste it into your browser's URL bar. Although your email client may display the link on more than one line, it is a single line.
>
> Please forward this message to other authorized SnapGene users in your group so that they can download and install the software.
>
> Best regards,
> The SnapGene Team

After purchase, the account manager will receive an email (shown above) with a download link. This email can be forwarded to others in the group who will be using SnapGene.

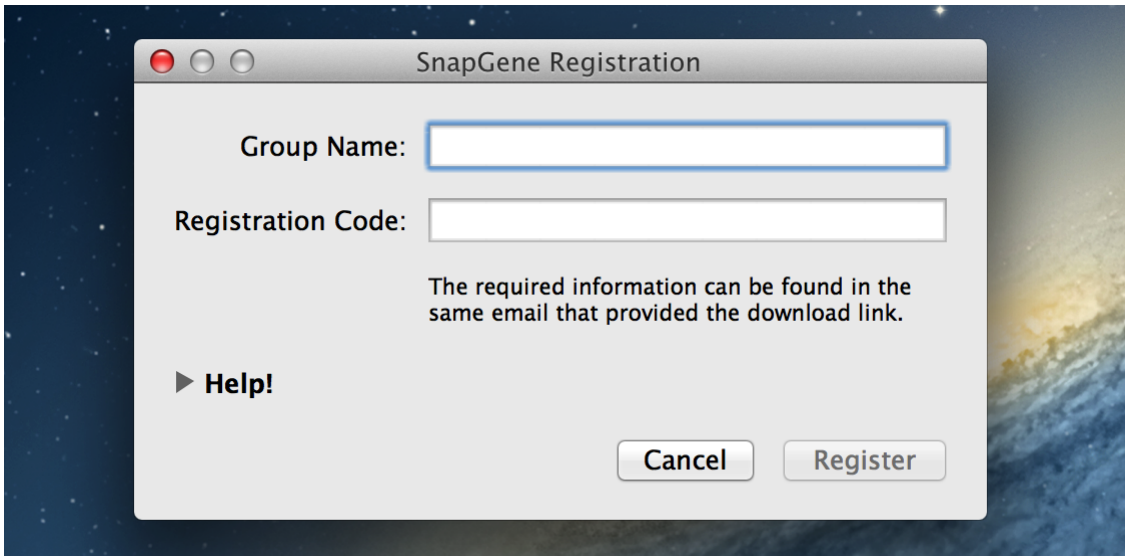
At any time, the account manager can generate a new email to be forwarded (see lesson: Generate an Email with a Download Link).

### Download the Appropriate Version of SnapGene

| Operating System | Best SnapGene Version | |
|---|---|---|
| Mac OS 10.5 (Leopard) or higher | SnapGene 1.5.3 for Mac OS X | Download |
| Windows XP or higher | SnapGene 1.5.3 for Windows | Download |

The link in the email will take you to a web page with two download buttons. Click on the button for either the Mac or the Windows version of SnapGene.

## Install and Run SnapGene




Install the software. When you run it for the first time, the SnapGene Registration dialog will appear.

## Obtain the Group Name and Registration Code

**Group Name:** SnapGene Test Lab

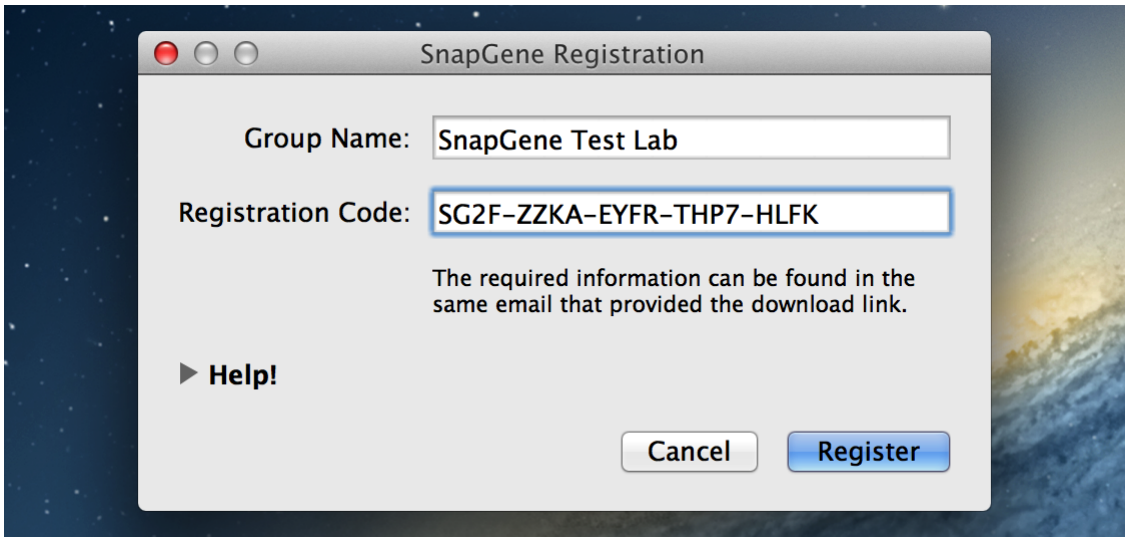
**Registration Code:** SG2F-ZZKA-EYFR-THP7-HLFK

The following SnapGene versions are available for your registration code.

| Operating System | Best SnapGene Version |
|---|---|

On the same web page as the download buttons, you will find the group name and registration code.

## Register SnapGene




Enter the Group Name and Registration Code. You can copy these items from the web page, and paste them in the SnapGene Registration dialog.

Then click **Register**.

## Begin Using SnapGene

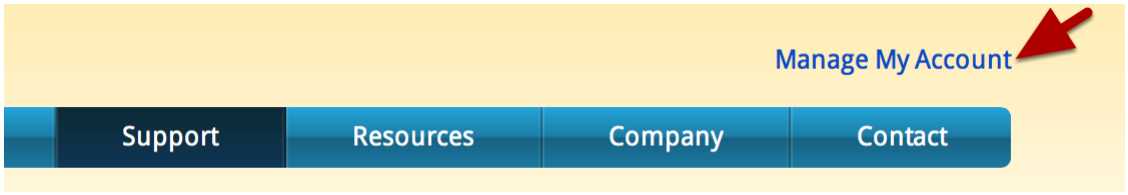
After a few moments, SnapGene will confirm the registration. Cick **OK**.

**You must be connected to the internet to register. From then on, SnapGene will run without requiring an internet connection.**

## Generate an Email with a Download Link

As the account manager, how can I send an email with a download link to the authorized SnapGene users in my group?

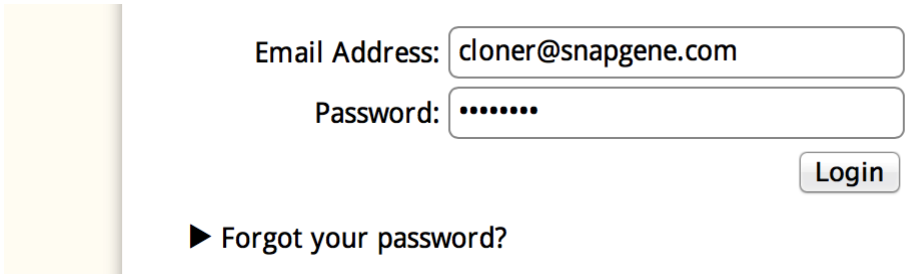
### Go to the Login Page




For this procedure, you must be the account manager for the SnapGene licenses.

Go to snapgene.com, and click on the "Manage My Account" link in the upper right corner.

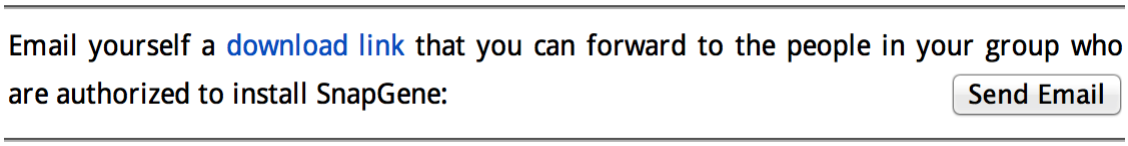
### Login




Enter the email address and password for the account manager.

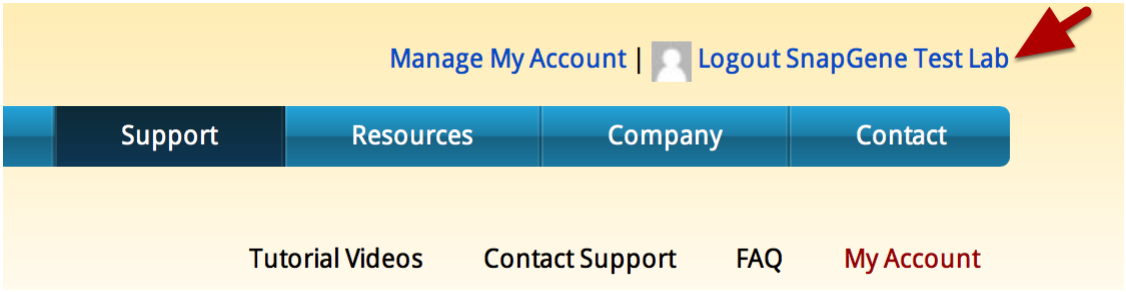
The original password was sent in an email immediately after purchase. If you have lost the password, click the triangle to show controls for obtaining a new password by email.

### Email Yourself a Download Link




On the account page, click **Send Email** to send yourself an email with a download link. Forward this email to the authorized SnapGene users in your group.

## Logout

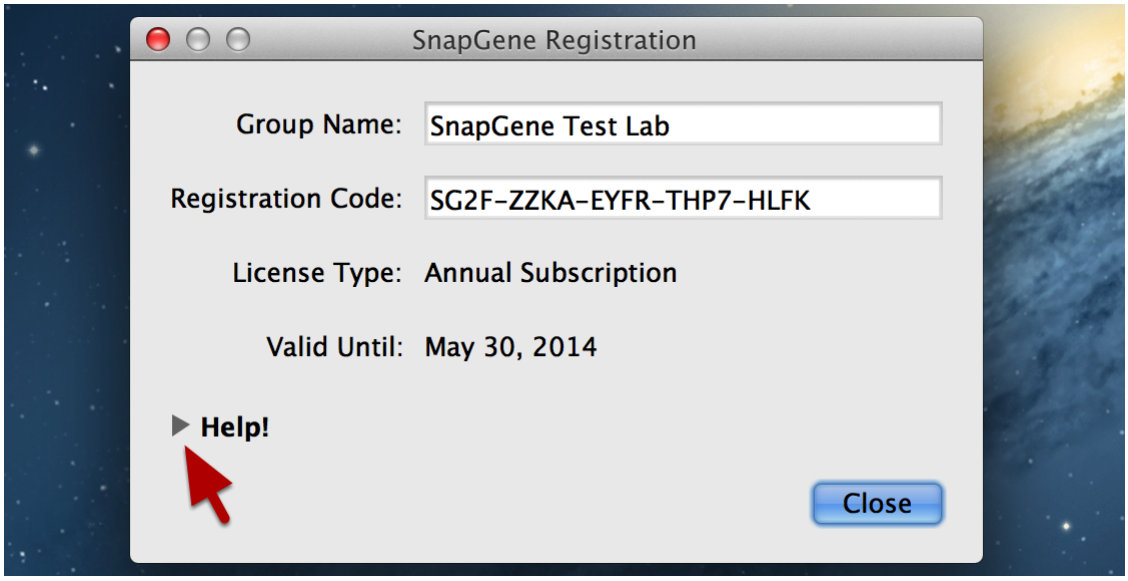



When you are finished, click on the "Logout" link in the upper right corner.

## Unregister the Computer You Are Using

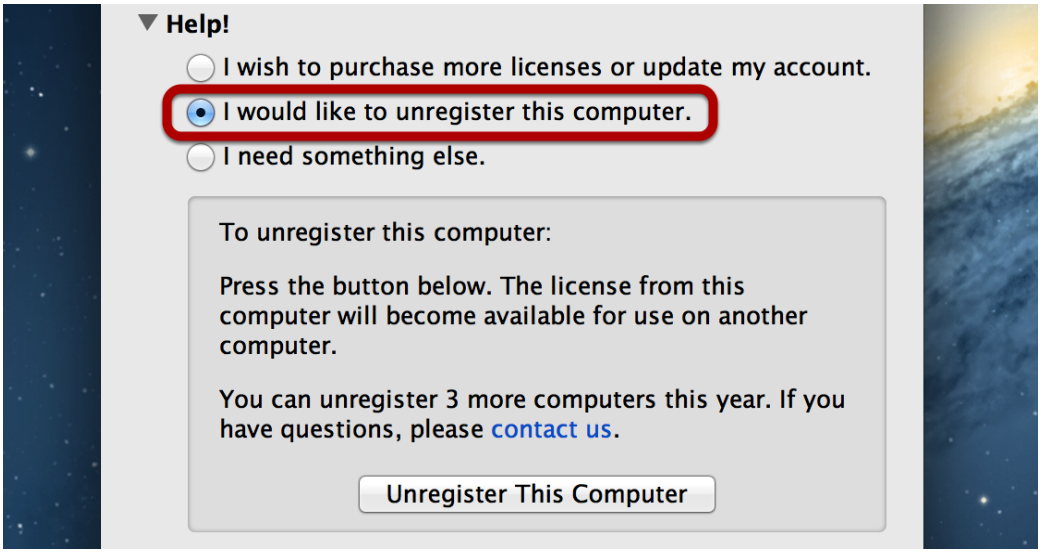
How can I unregister SnapGene on the computer I am currently using, freeing the license for use on another computer?

### Open and Expand the Registration Dialog




To open the Registration Dialog, click **SnapGene** → **Registration** (Mac) or **Help** → **Registration** (Windows). Expand the Registration dialog by clicking the "Help!" triangle.

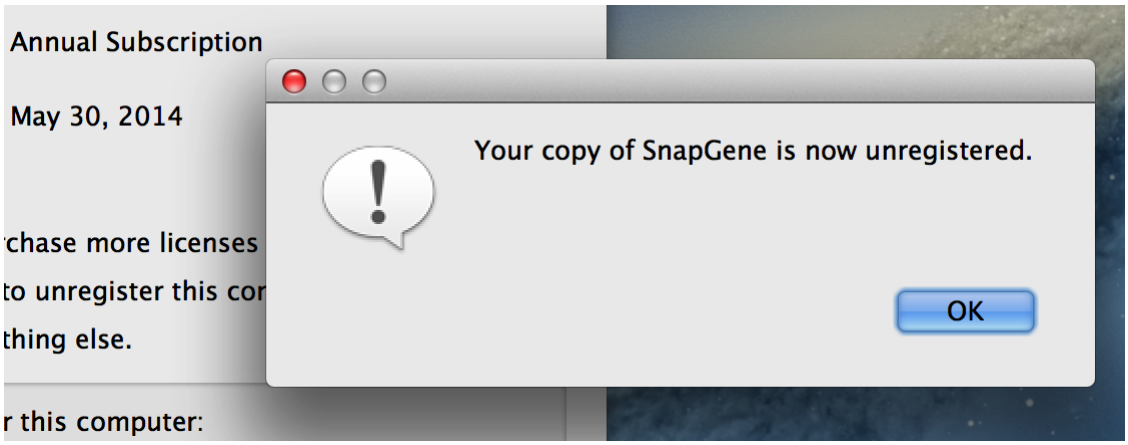
### Choose the Second Option




Choose "**I would like to unregister this computer.**"

Click **Unregister This Computer**.

## Complete the Unregistration

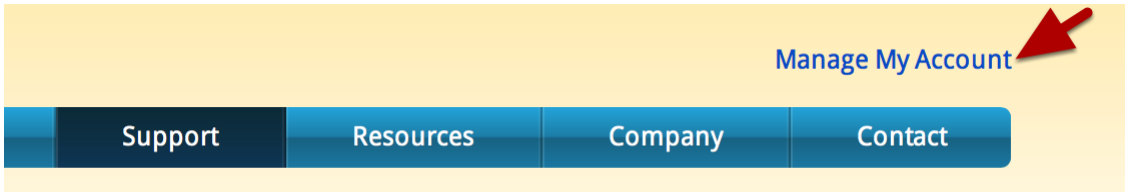



SnapGene will no longer run on the same computer unless you register again. The license is available for use on another computer.

## Unregister a Computer Remotely

As the account manager, how can I unregister a SnapGene license on a computer that is not physically accessible?

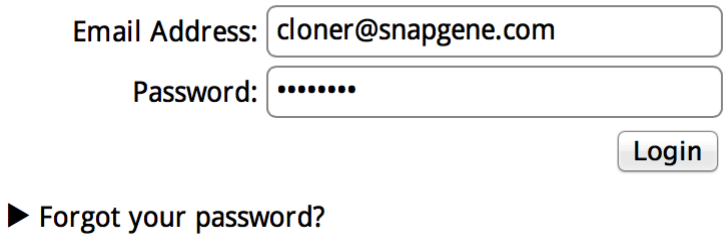
### Go to the Login Page




For this procedure, you must be the account manager for the SnapGene licenses.

Go to snapgene.com, and click on the "Manage My Account" link in the upper right corner.

### Login



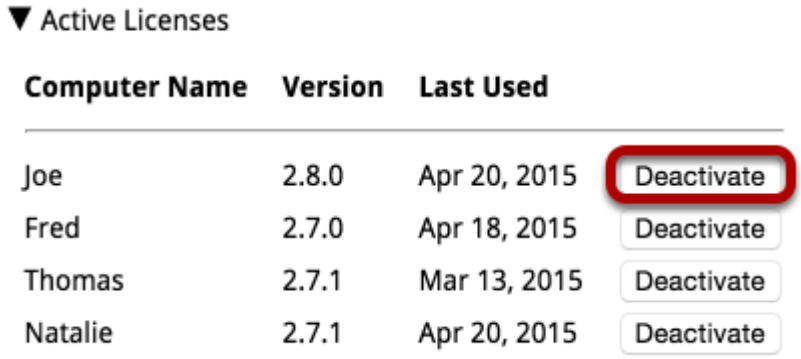

Enter the email address and password for the account manager.

The original password was sent in an email immediately after purchase. If you have lost the password, click the triangle to show controls for obtaining a new password by email.

### Display the Active Licenses

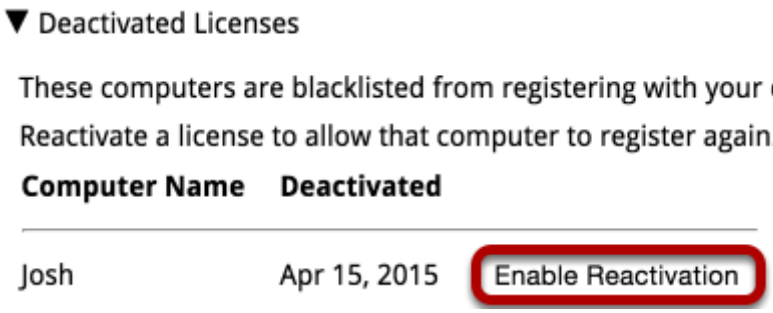
On the account page, click the "Active Licenses" triangle to show the SnapGene licenses in use by your group.

## Deactivate the License Remotely



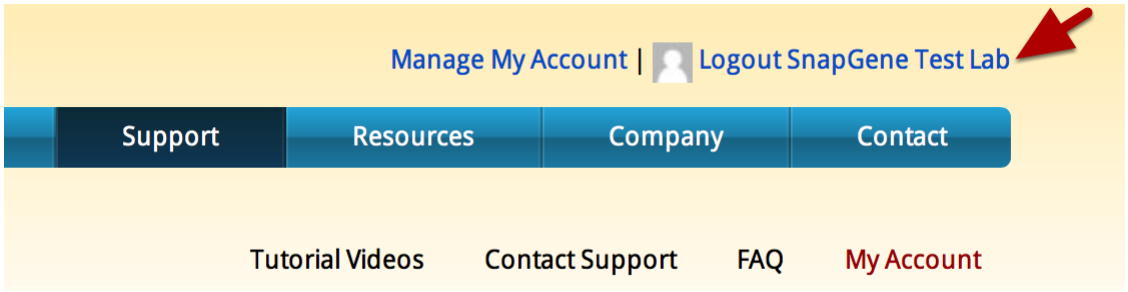

To deactivate a license, press the **Deactivate** button next to the name of the relevant computer. The computer will no longer be listed in the Active Licenses list, but instead will be listed in the Deactivated Licenses list. The released license will be immediately available for use on another computer.

## Re-enable a Computer Remotely




To allow the computer owner to use a SnapGene license from your group again, press the **Enable Reactivation** button next to the name of the relevant computer. After the computer owner registers with your code, that computer name will appear in the Active Licenses list.

## Logout

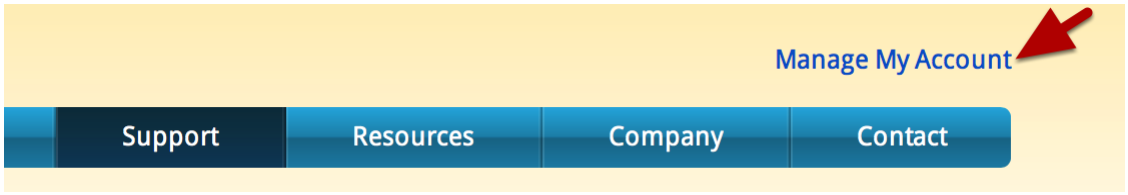



When you are finished, click on the "Logout" link in the upper right corner.

## Retrieve a License from a Computer

A computer in my lab died or was stolen. As the account manager, how can I retrieve the SnapGene license for use on another computer?

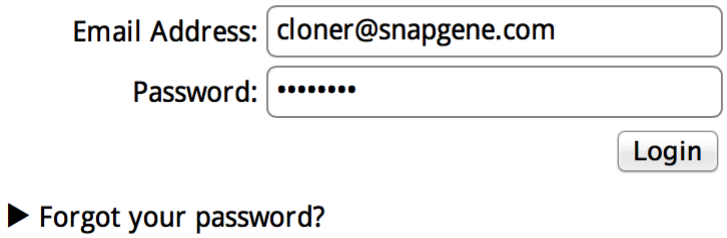
### Go to the Login Page




For this procedure, you must be the account manager for the SnapGene licenses.

Go to snapgene.com, and click on the "Manage My Account" link in the upper right corner.

### Login



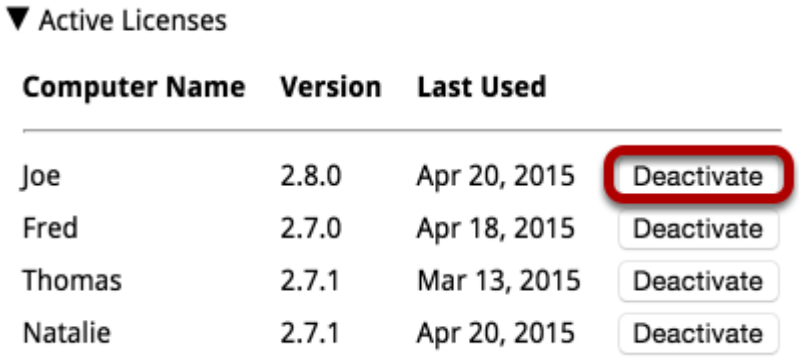

Enter the email address and password for the account manager.

The original password was sent in an email immediately after purchase. If you have lost the password, click the triangle to show controls for obtaining a new password by email.

### Display the Active Licenses

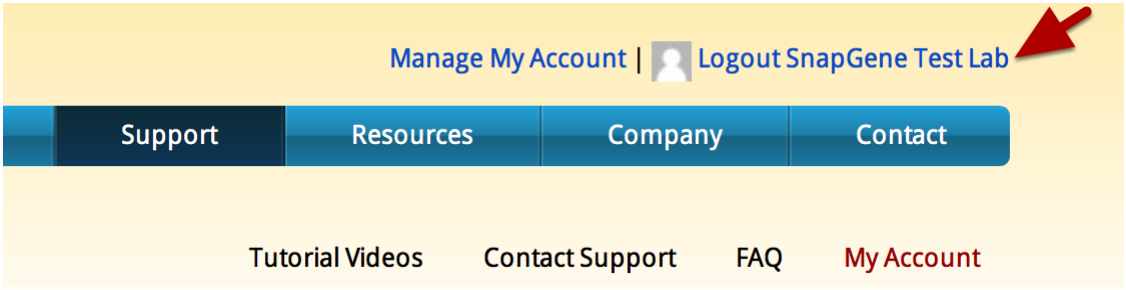
On the account page, click the "Active Licenses" triangle to show the SnapGene licenses in use by your group.

## Deactivate the License Online



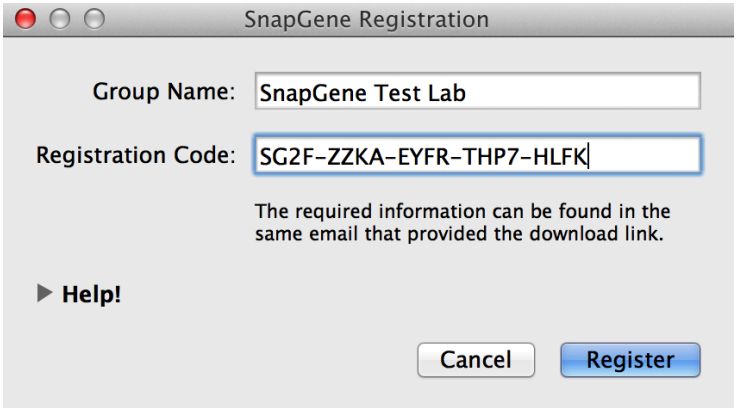

To deactivate a license, press the **Deactivate** button next to the name of the relevant computer. The computer will no longer be listed in the Active Licenses list, but instead will be listed in the Deactivated Licenses list. The released license will be immediately available for use on another computer.

## Logout




When you are finished, click on the "Logout" link in the upper right corner.

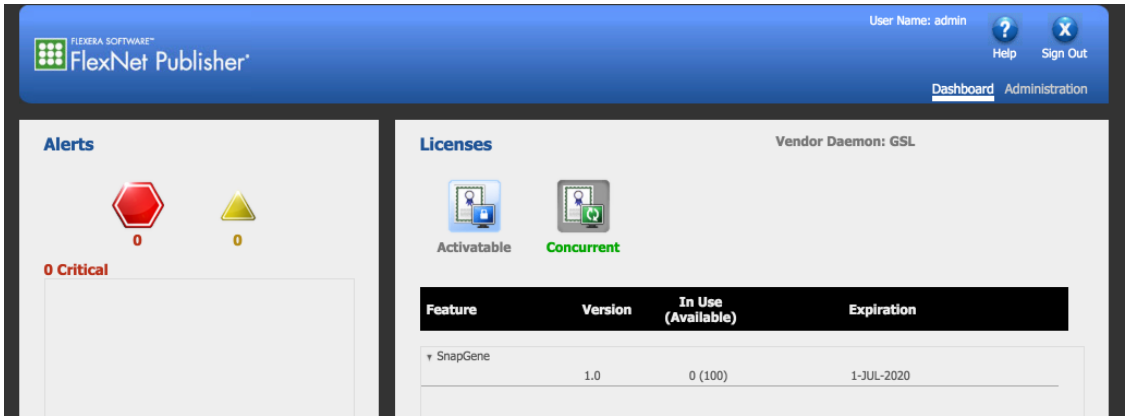
## Download and Register SnapGene




On the new computer, download and register SnapGene (see lesson: Download and Register SnapGene).

## Set Up a License Server on a Mac

As the account manager, how can I set up a server on a Mac for sharing SnapGene licenses among several computers?




To share SnapGene licenses across several computers, first install the FLEXnet License Server on a network computer that will always be available. The SnapGene team will provide a server installation file called **FLEXnet_GSL_mac.dmg**.

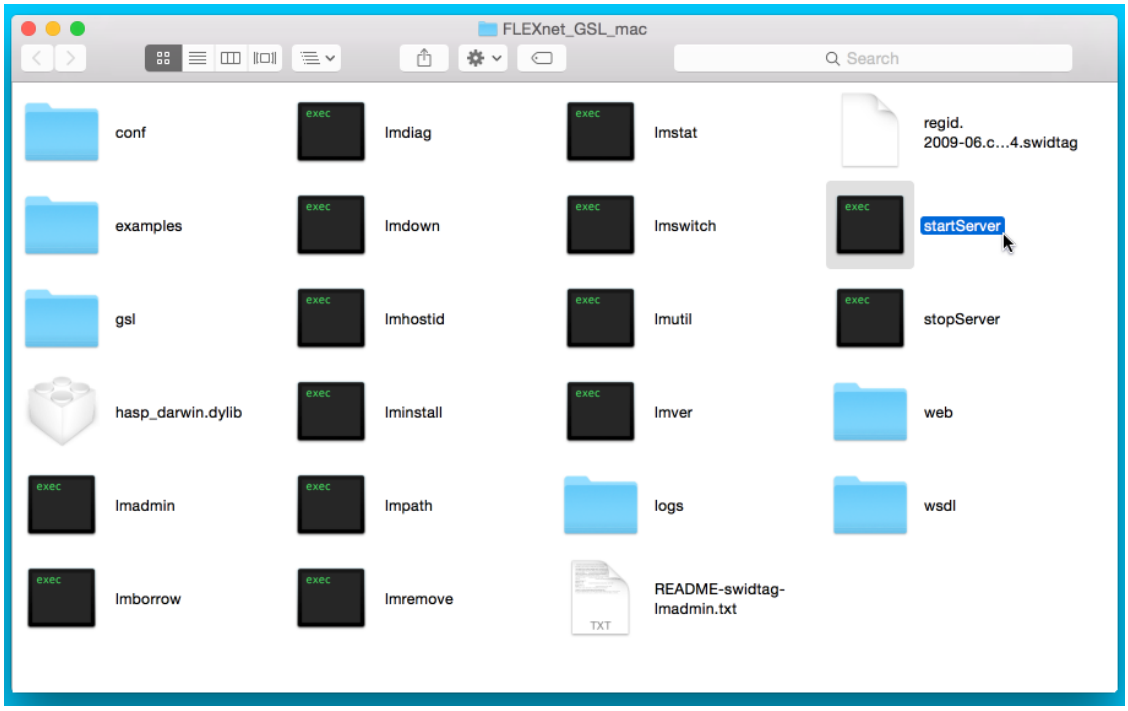
### Mount the Server Folder

To mount the server folder, double-click on **FLEXnet_GSL_mac.dmg**.

## Install the Server Folder

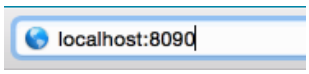
To install the server folder, drag the mounted folder to a location on your hard drive, then double-click to open the folder.

## Start the Server



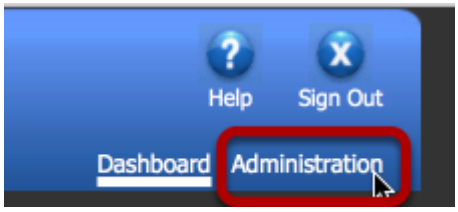

Double-click the **startServer** file in the FLEXnet_GSL_mac folder.

## Access the Server

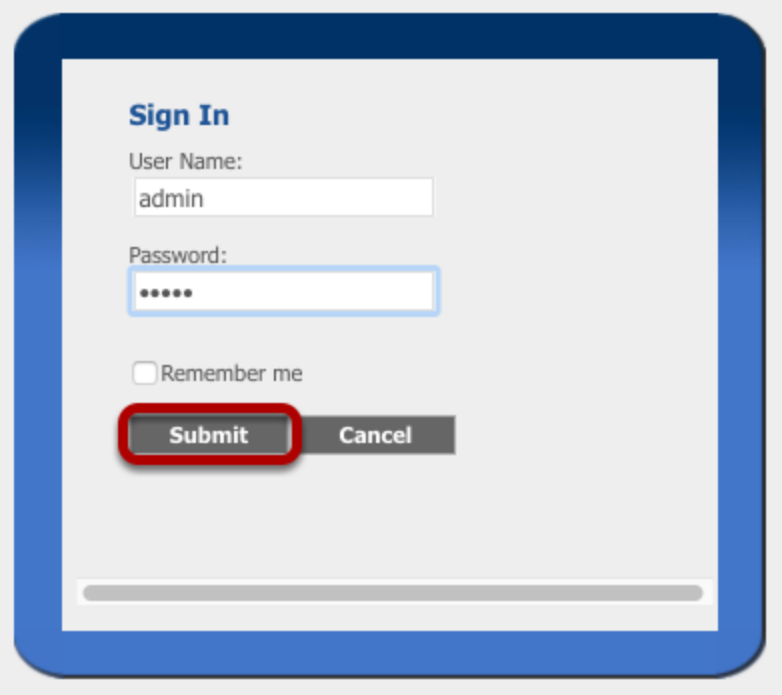



To access the server, open a browser window, then type **localhost:8090**.

## Sign In to the Server

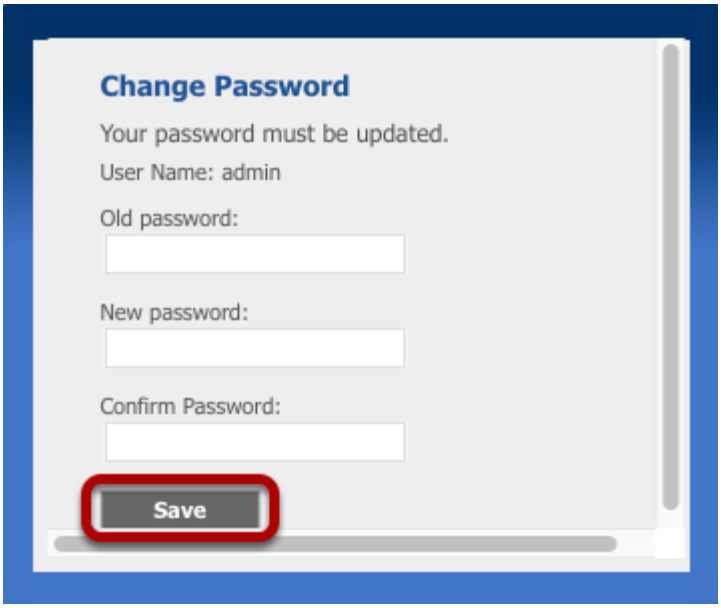



Click the **Administration** link at the top right of the window.



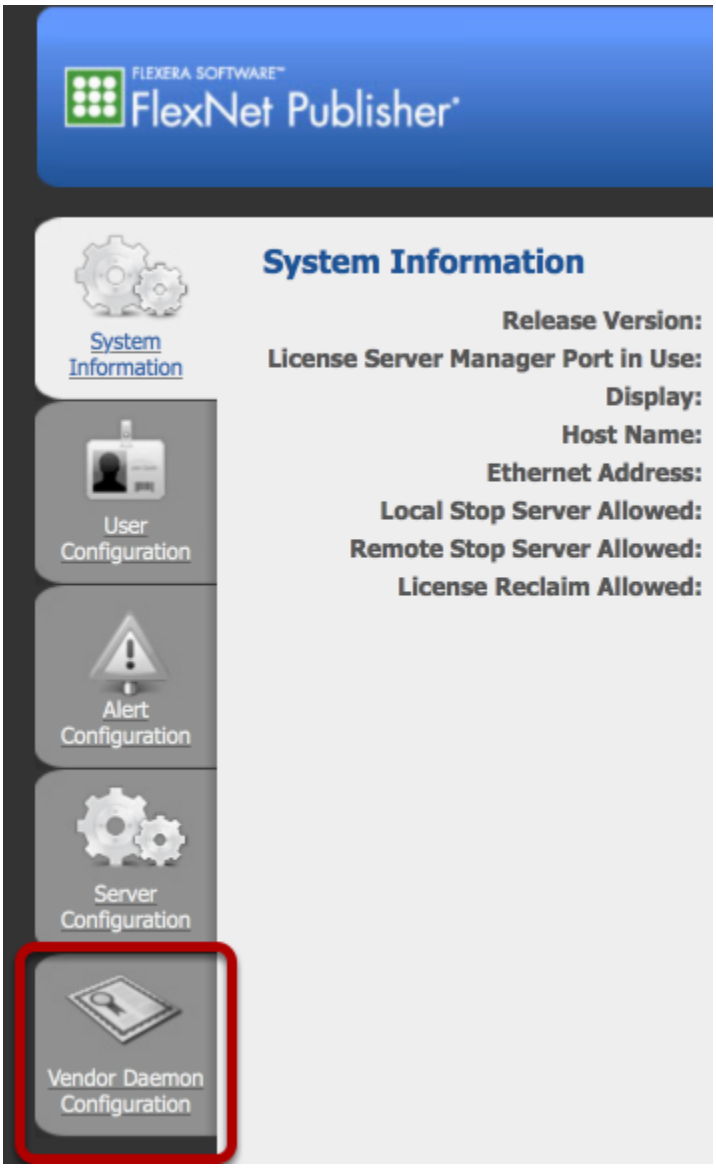

To sign in, type the default User Name: **admin** and Password: **admin**, then click **Submit.**

## Change the Password



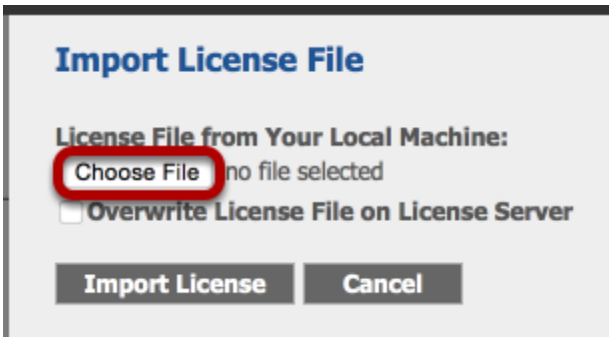

To change the password, type the old and new passwords, then click **Save**.

## Import the SnapGene Vendor Daemon License

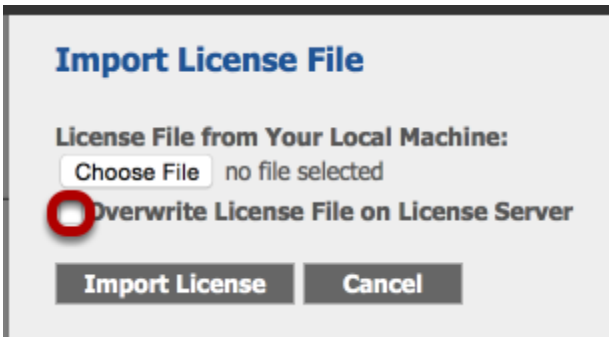



To import the SnapGene Vendor Daemon license, click the **Vendor Daemon Configuration** button.

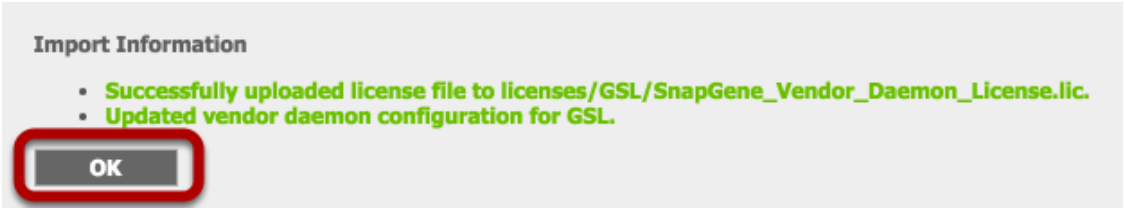
Then click the **Import License** button.




Click **Choose File**, browse to the .lic file sent to you by the SnapGene team, then click **Import License**.

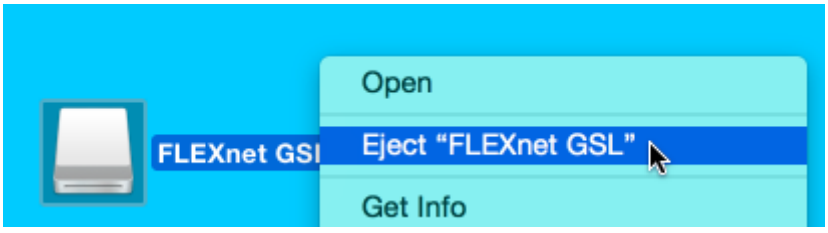



Note: If you received a new license file (e.g., for additional licenses), instead check the box for **Overwrite License File on License Server**, then click **Import License**.



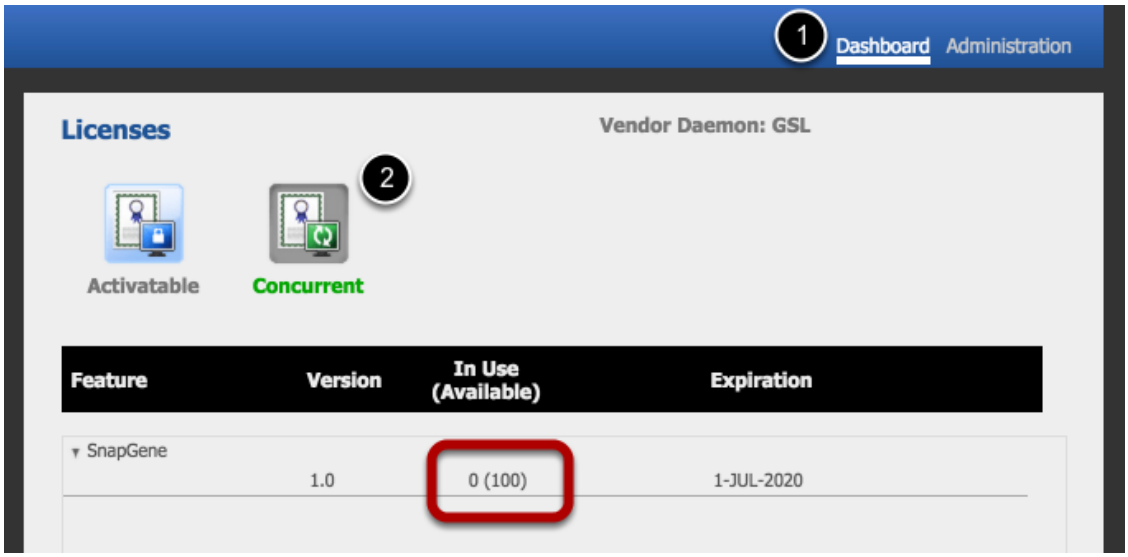

Configuration of the license server is complete. Click **OK.**

## Eject the Drive



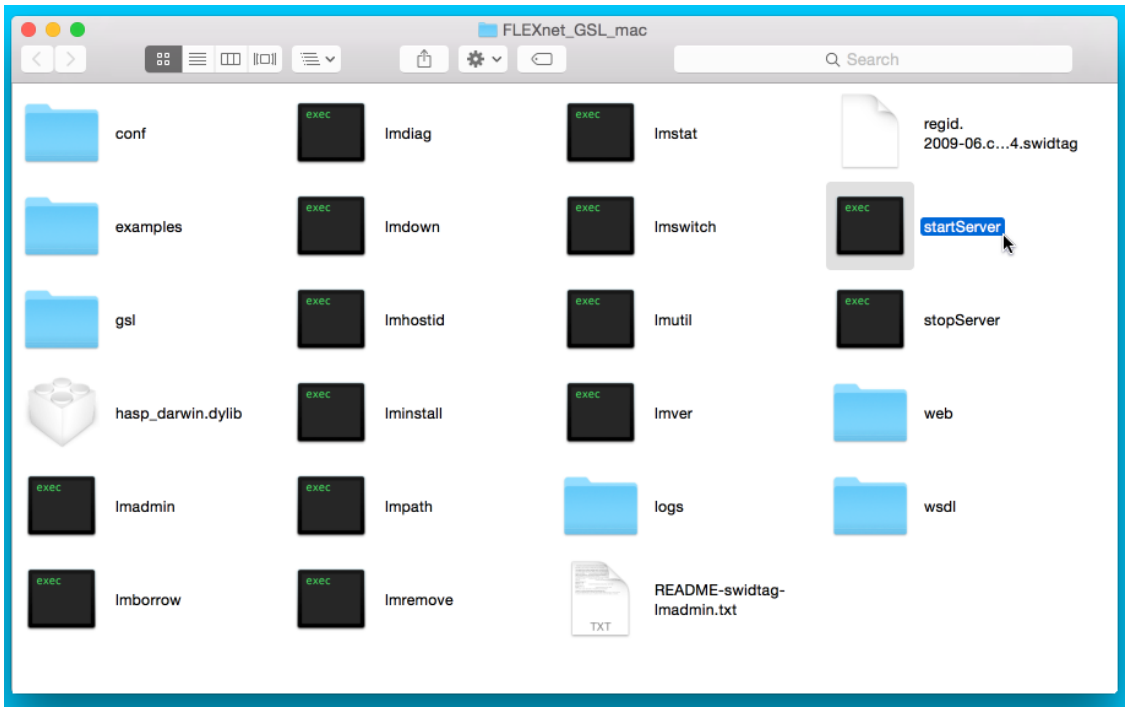

The virtual drive is no longer needed. To eject the virtual drive, right click on the **FLEXnet GSL** drive.

## Review Shared License Usage




To review the current SnapGene license usage, click the **Dashboard** link, then click **Concurrent**. The number of licenses in use (and available) will be shown in the **In Use** column.

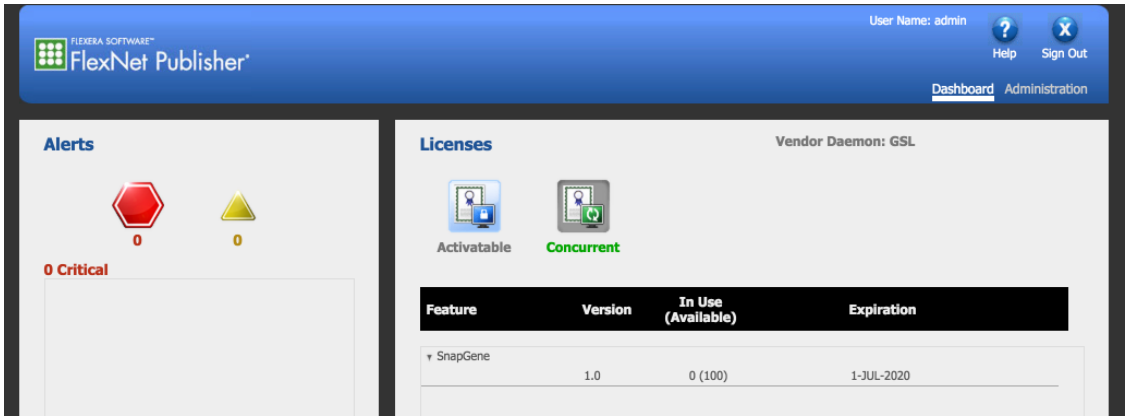
## Restart the Server When Necessary




For the licenses to be accessible, the administrator computer running this server must remain on. If the administrator computer is turned off and then back on, the server must be restarted. For this purpose, open the FLEXnet_GSL_mac folder and double-click the **startServer** file.

## Set Up a License Server on Windows

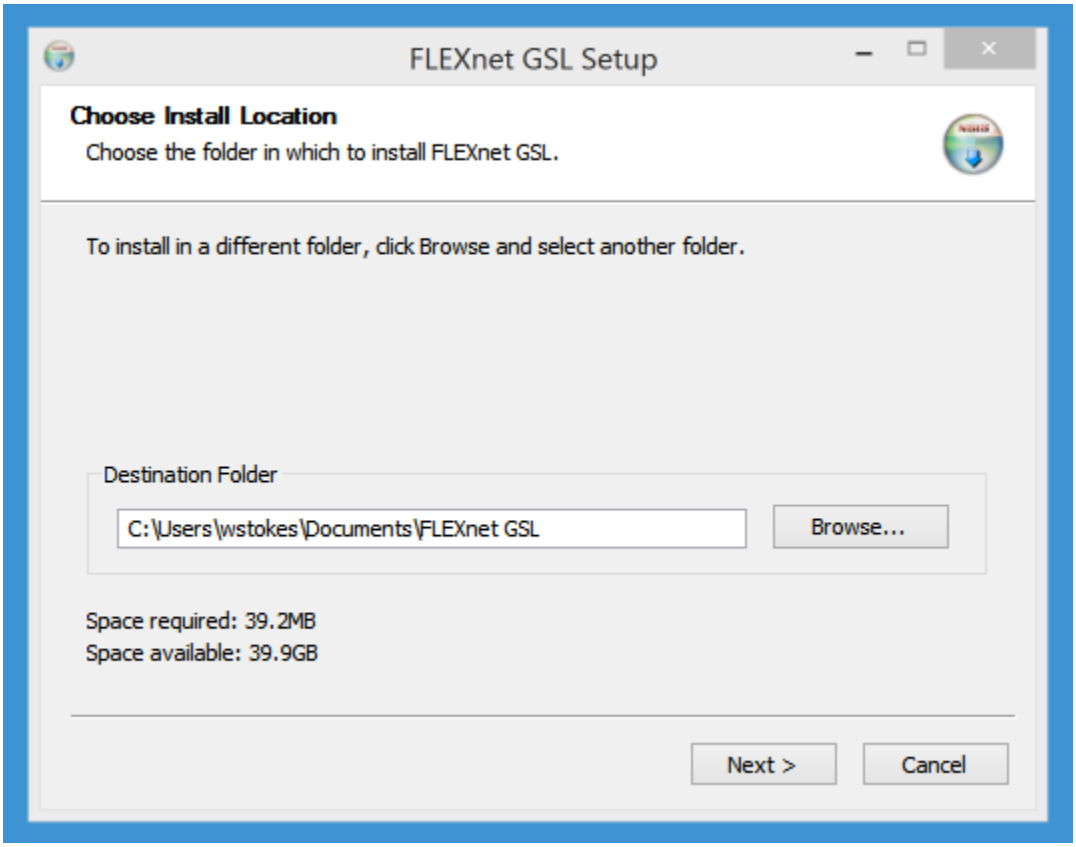
As the account manager, how can I set up a server on a Windows computer for sharing SnapGene licenses among several computers?




To share SnapGene licenses across several computers, first install the FLEXnet License Server on a network computer that will always be available. The SnapGene team will provide a server installation file called **FLEXnet_GSL_win.exe**.

### Install the Server

To install the server, double-click on **FLEXnet_GSL_win.exe**.




Specify a directory to install the server, then click **Next**.

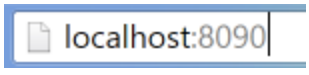
Note: The location must allow the server to write to the directory (e.g., the location should not be Program Files).

Click **Install** in the next window.

## Start the Server

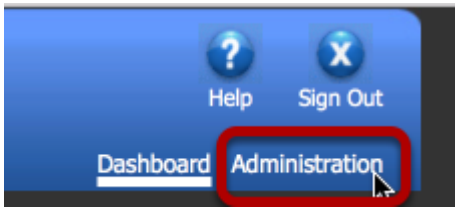
In the start menu, click **Programs** → **FLEXnet GSL** → **Start Server**.

## Access the Server

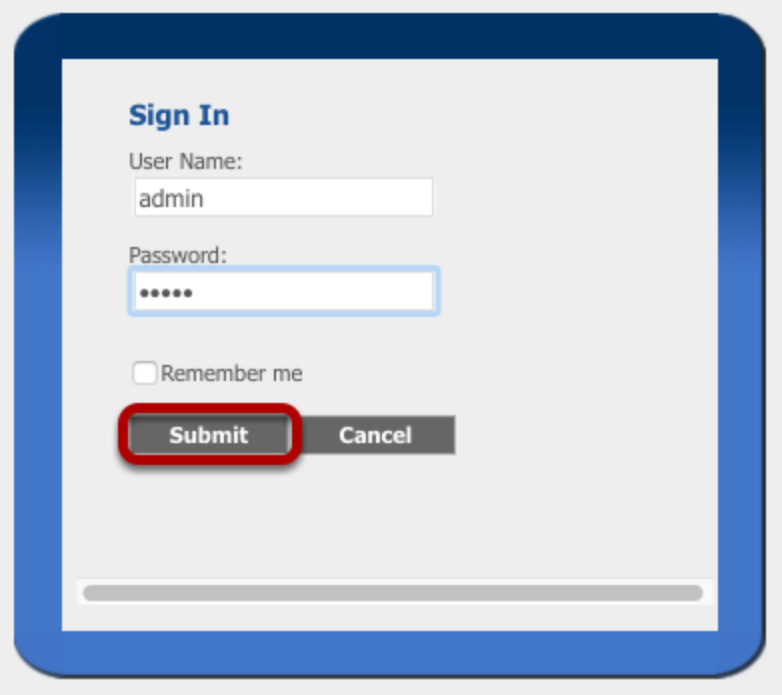



To access the server, open a browser window, then type **localhost:8090**.

## Sign In to the Server

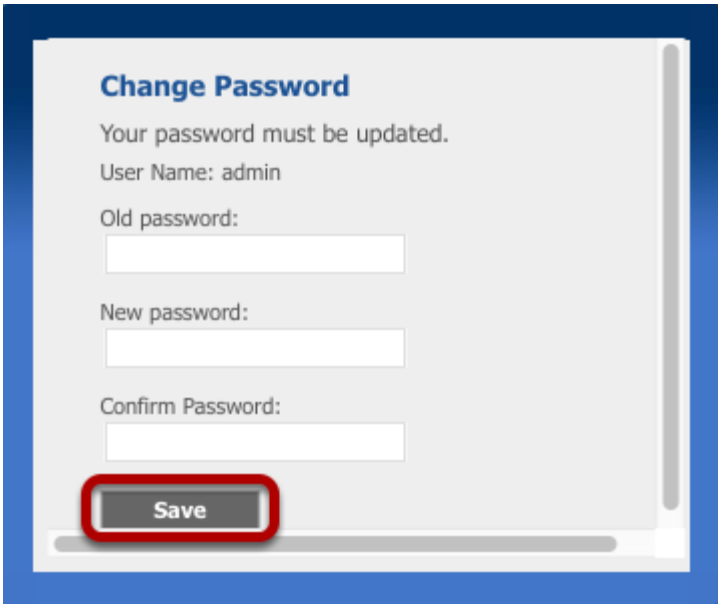



Click the **Administration** link at the top right of the window.



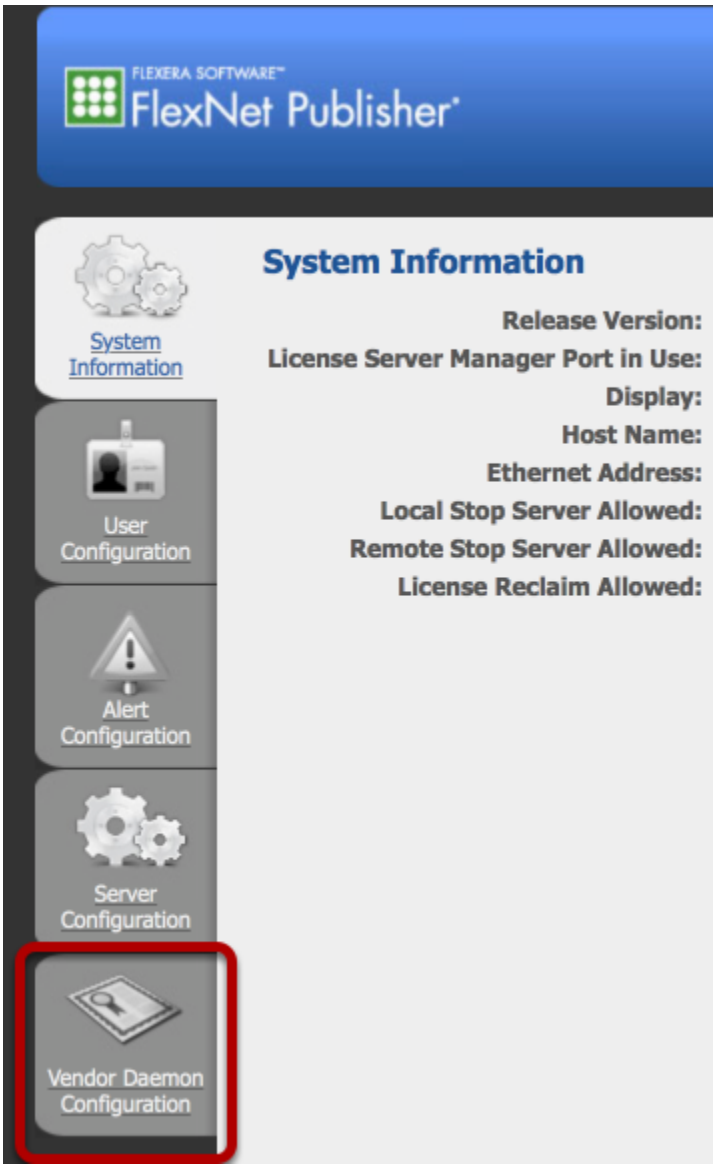

To sign in, type the default User Name: **admin** and Password: **admin**, then click **Submit.**

## Change the Password



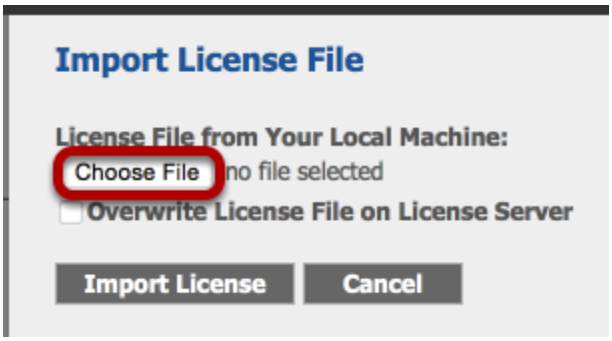

To change the password, type the old and new passwords, then click **Save**.

## Import the SnapGene Vendor Daemon License

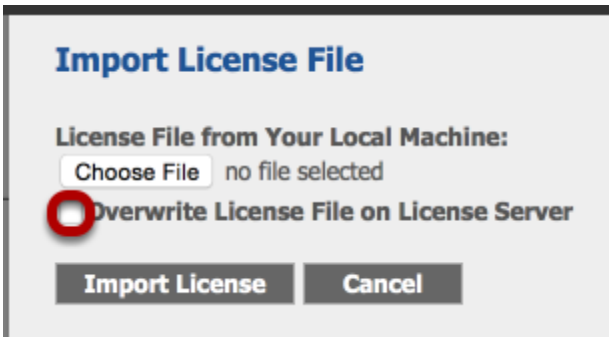



To import the SnapGene Vendor Daemon license, click the **Vendor Daemon Configuration** button.

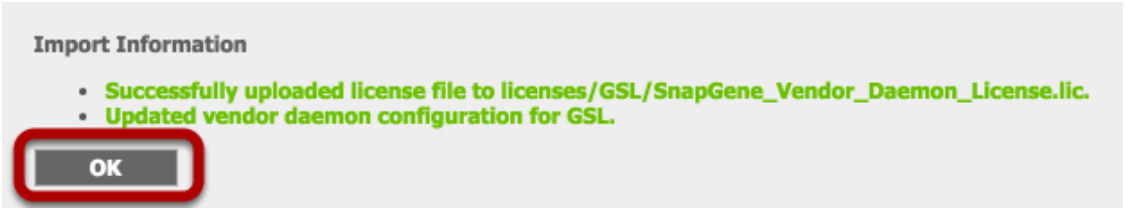
Then click the **Import License** button.




Click **Choose File**, browse to the .lic file sent to you by the SnapGene team, then click **Import License**.

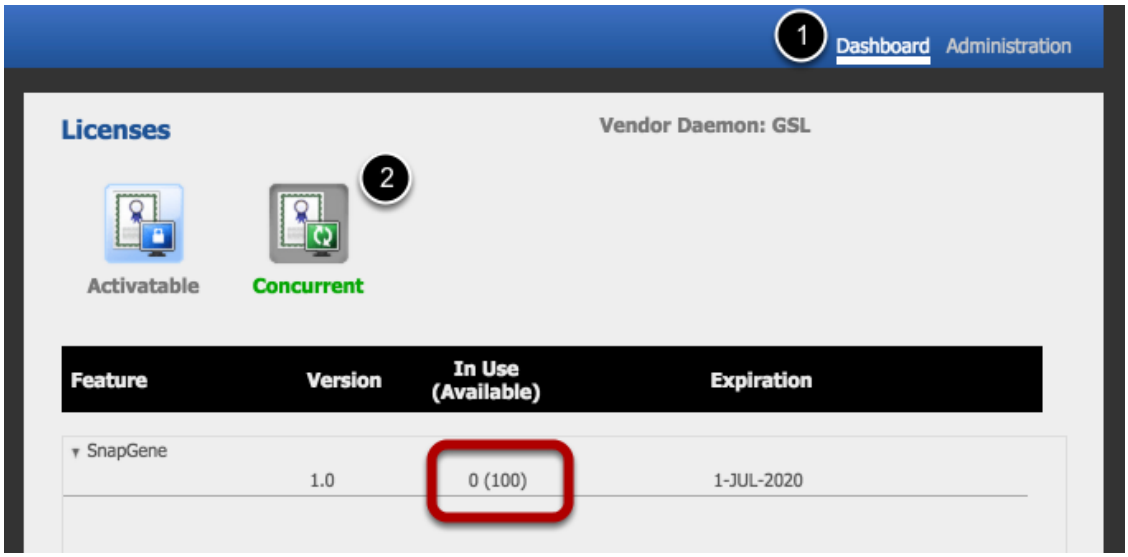



Note: If you received a new license file (e.g., for additional licenses), instead check the box for **Overwrite License File on License Server**, then click **Import License**.




Configuration of the license server is complete. Click **OK.**

## Review Shared License Usage



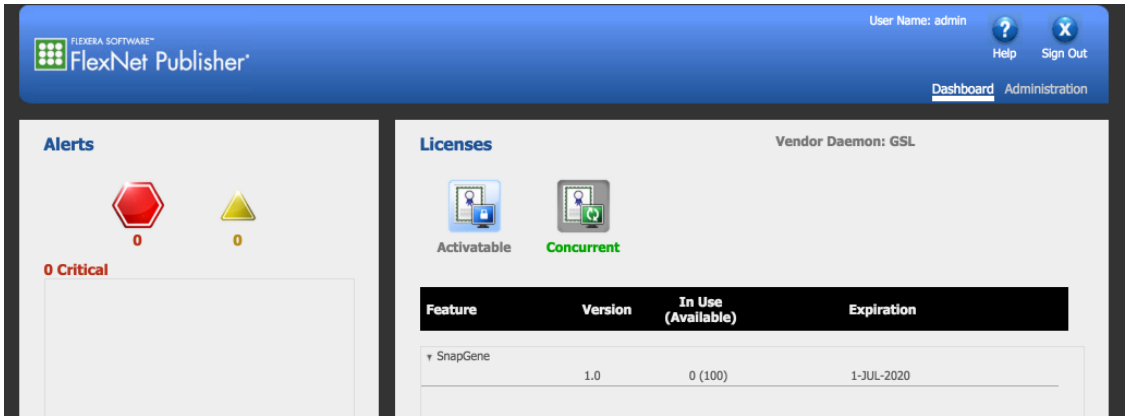

To review the current SnapGene license usage, click the **Dashboard** link, then click **Concurrent**. The number of licenses in use (and available) will be shown in the **In Use** column.

## Restart the Server When Necessary

For the licenses to be accessible, the administrator computer running this server must remain on. If the administrator computer is turned off and then back on, the server must be restarted. For this purpose, in the Start menu, click **Programs** → **FLEXnet GSL** → **Start Server**.

## Set Up a License Server on Linux

As the account manager, how can I set up a server on a Linux computer for sharing SnapGene licenses among several computers?




To share SnapGene licenses across several computers, first install the FLEXnet License Server on a network computer that will always be available.

The SnapGene team will provide a server installation file. Download the file **FLEXnet_GSL_linux.tgz** from your account page at [http://www.snapgene.com/support/myAccount](http://www.snapgene.com/support/myAccount) and save it.

### Install the Server

Select a location for the directory of server files. In this example the parent directory is /usr/local. Commands are preceded by ">>":

Create a parent directory if it does not already exist.

Move the tar file to the parent directory:

```
>> mv FLEXnet_GSL_linux.tgz /usr/local
```

Change to the parent directory:

```
>> cd /usr/local
```

Unpack the archive file:

```
>> tar xzf FLEXnet_GSL_linux.tgz
```

This creates the directory /usr/local/FLEXnet_GSL_linux, which contains all the server files including an INSTALL file with these instructions.

If you have already unpacked the tar file before opening this file, you may simply move the directory FLEXnet_GSL_linux to the desired location.

## Import the SnapGene Vendor Daemon License

Import the SnapGene Vendor Daemon license provided to you by the SnapGene team by running the following command:

```
>> /usr/local/FLEXnet_GSL_linux/lmadmin -import /path/to/FlexNet_SnapGene.lic
-force
```

You should only need to import the license once.

## Start the Server

Start the server using the script in the server directory. DO NOT EXECUTE THIS COMMAND AS ROOT:

```
>> /usr/local/FLEXnet_GSL_linux/startServer
```

This script may be executed at startup. The corresponding stopServer script will stop a running server.

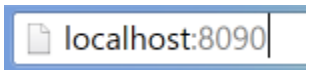
The server must be started on the command line, but once started can be controlled from the web interface.

IMPORTANT NOTE ABOUT 32-BIT LIBRARIES: If starting the server throws an error such as

```
/lib/ld-lsb.so.3: bad ELF interpreter:
```

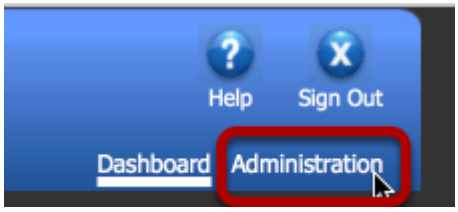
your system is missing some 32-bit libraries. These libraries are obtained by installing the relevant lsb-32 package for the operating system variant. On most Redhat installations the package in question is named redhat-lsb or redhat-lsb.i686

## Administer the Server

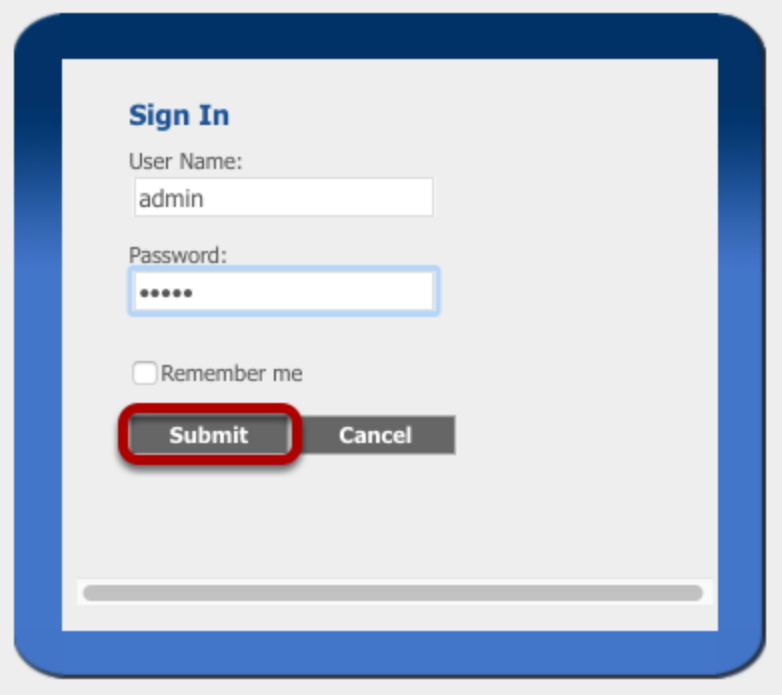



To administer the server, point a browser to **localhost:8090**.

## Sign In to the Server

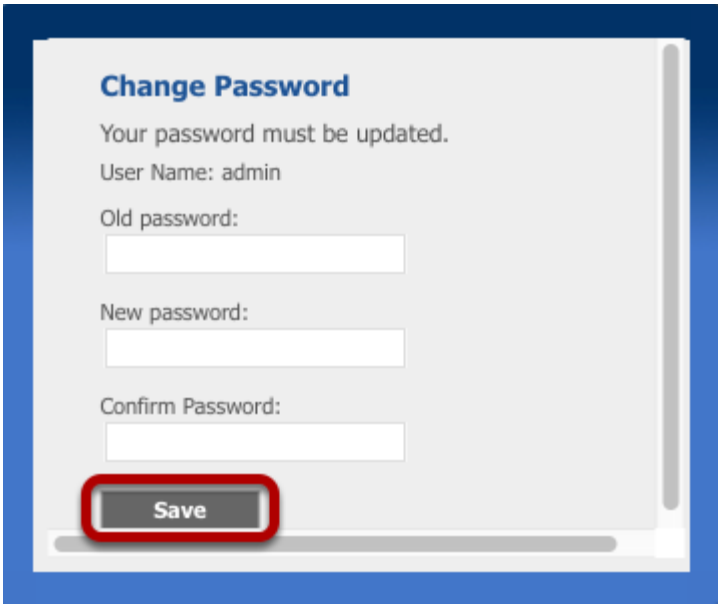



Click the **Administration** link at the top right of the window.



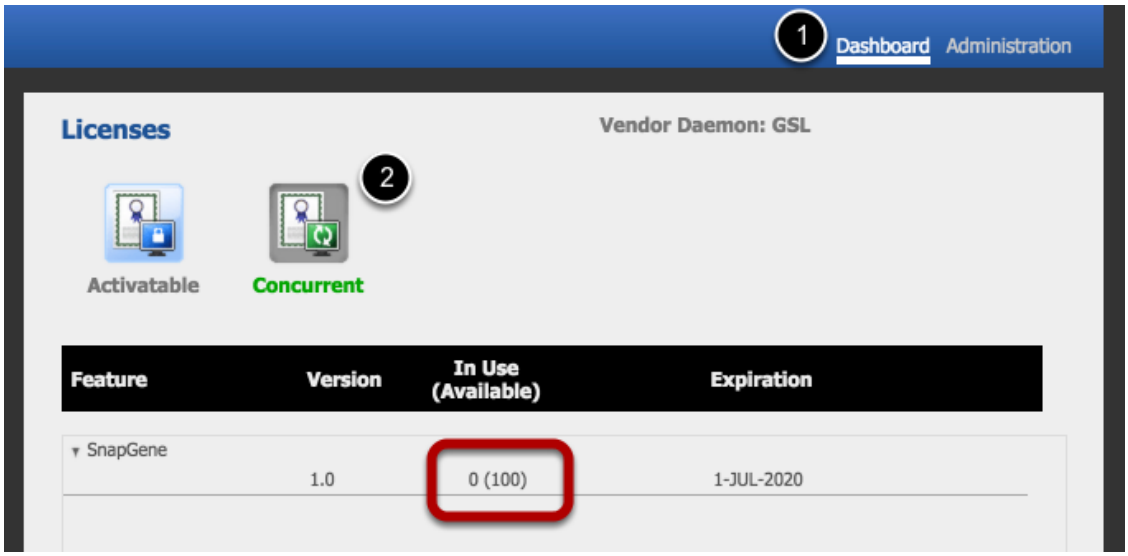

To sign in, type the default User Name: **admin** and Password: **admin**, then click **Submit.**

## Change the Password




To change the password, type the old and new passwords, then click **Save**.

## Review Shared License Usage




To review the current SnapGene license usage, click the **Dashboard** link, then click **Concurrent**. The number of licenses in use (and available) will be shown in the **In Use** column.

The dashboard shows license usage.

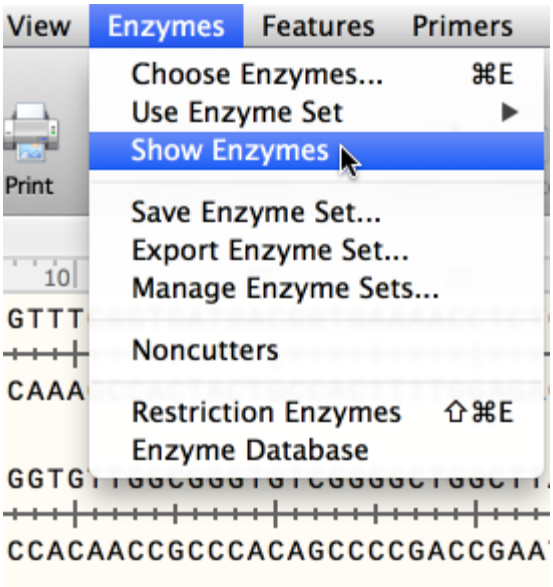
The FlexNet server directory also includes a number of tools for administration of the server and licenses (these should not be needed by most users): lmborrow lmdiag lmdown lmhostid lminstall lmpath lmremove lmstat lmswitch lmutil lmver

# Enzymes

## Show or Hide Enzymes

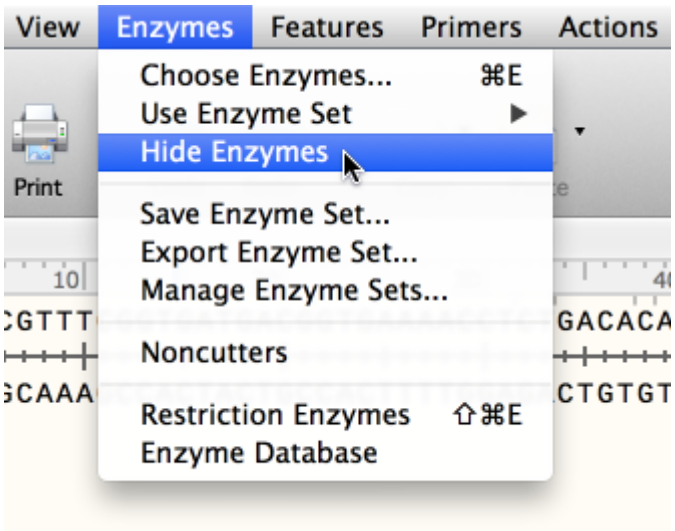
Enzymes displayed in Sequence and Map views can be shown or hidden.

### Show the Enzymes



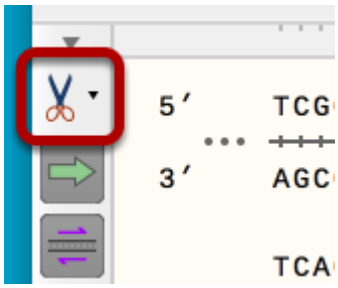

To show the enzymes, click **Enzymes** → **Show Enzymes**.

### Hide the Enzymes




To hide the enzymes, click **Enzymes** → **Hide Enzymes**.

## Toggle the Enyzme Visibility Using the Side Toolbar

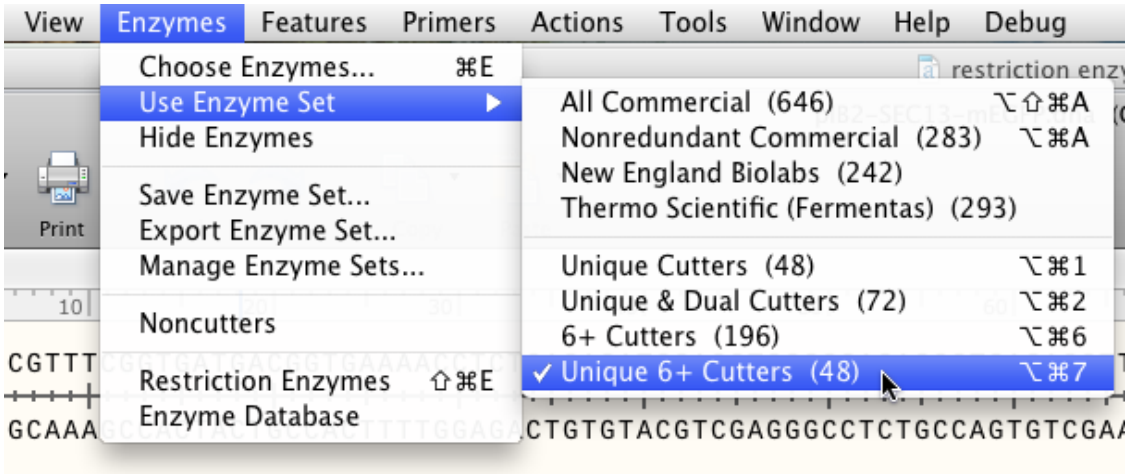



Clicking the "Show enzymes" (scissors) button in the side toolbar can also be used to toggle between showing and hiding all enzymes.

## Choose an Enzyme Set

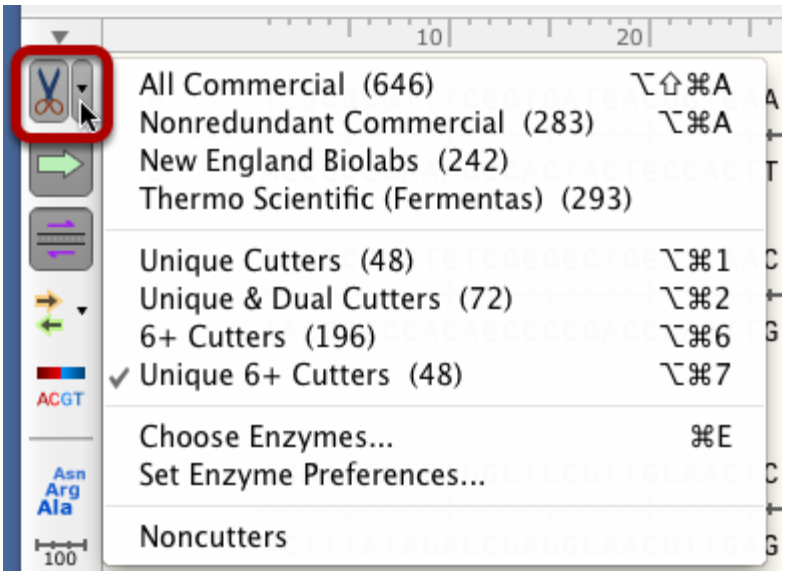
The enzymes shown in Map and Sequence views can be chosen from a list of predefined or custom enzyme sets.

### Choose an Enzyme Set




To choose an enzyme set, click **Enzymes** → **Use Enzyme Set** → **[...]** .

### Choose an Enzyme Set Using the Side Toolbar

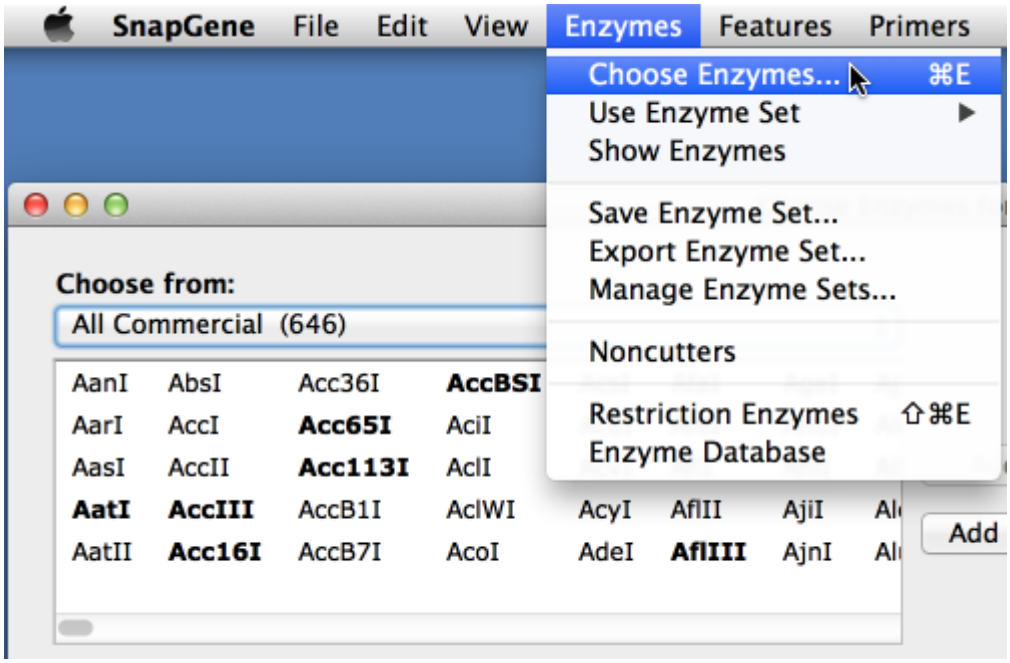



Alternatively, choose an enzyme set by clicking the "**Show enzymes**" (scissors) menu button in the side toolbar.

## Choose Enzymes Manually

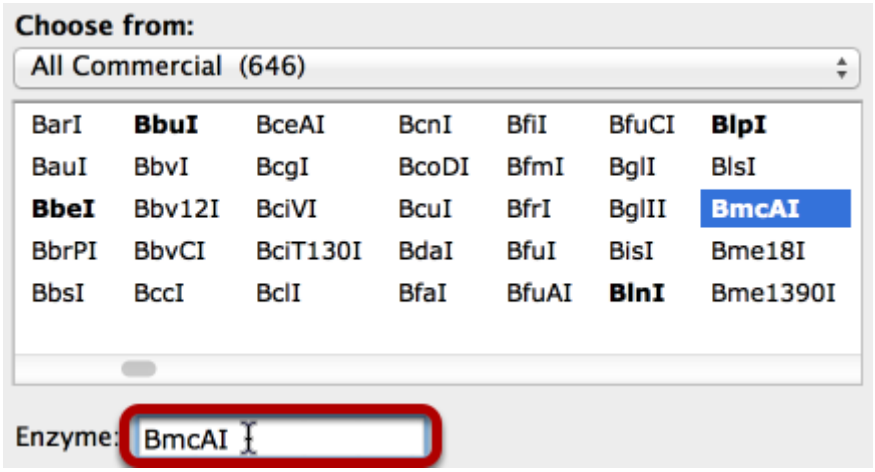
Choose enzymes manually by name, or by editing an existing enzyme set or using search criteria.

### Choose an Enzyme Set




To choose an existing enzyme set, click **Enzymes** → **Choose Enzymes**, then select the enzyme set name in the **Choose from** list.

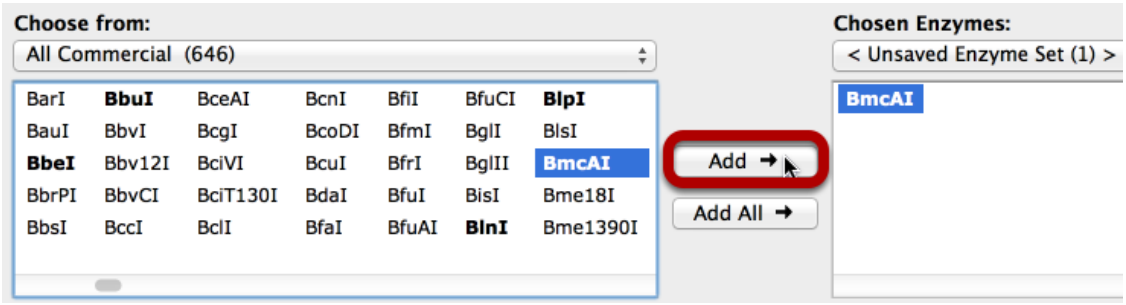
### Choose an Enzyme by Name




To search within the chosen enzyme set, enter the enzyme name in the search box.

Alternatively, click on an enzyme name to select it.

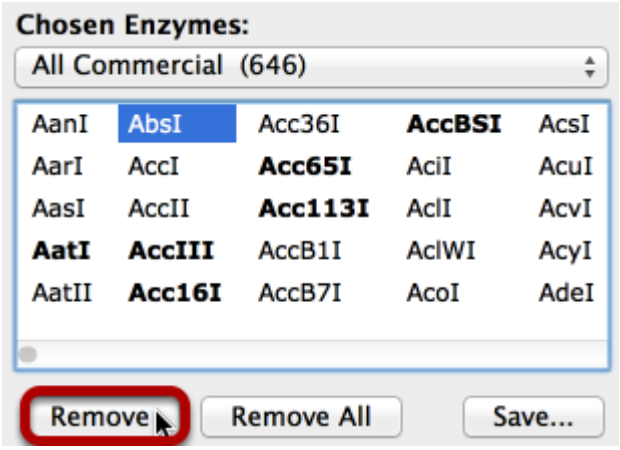
## Add to an Enzyme Set




To add the selected enzyme(s) to the **Chosen Enzymes** set, click the **Add** → button.

To add all of the enzymes from the original set to the **Chosen Enzymes** set, click the **Add All** → button.

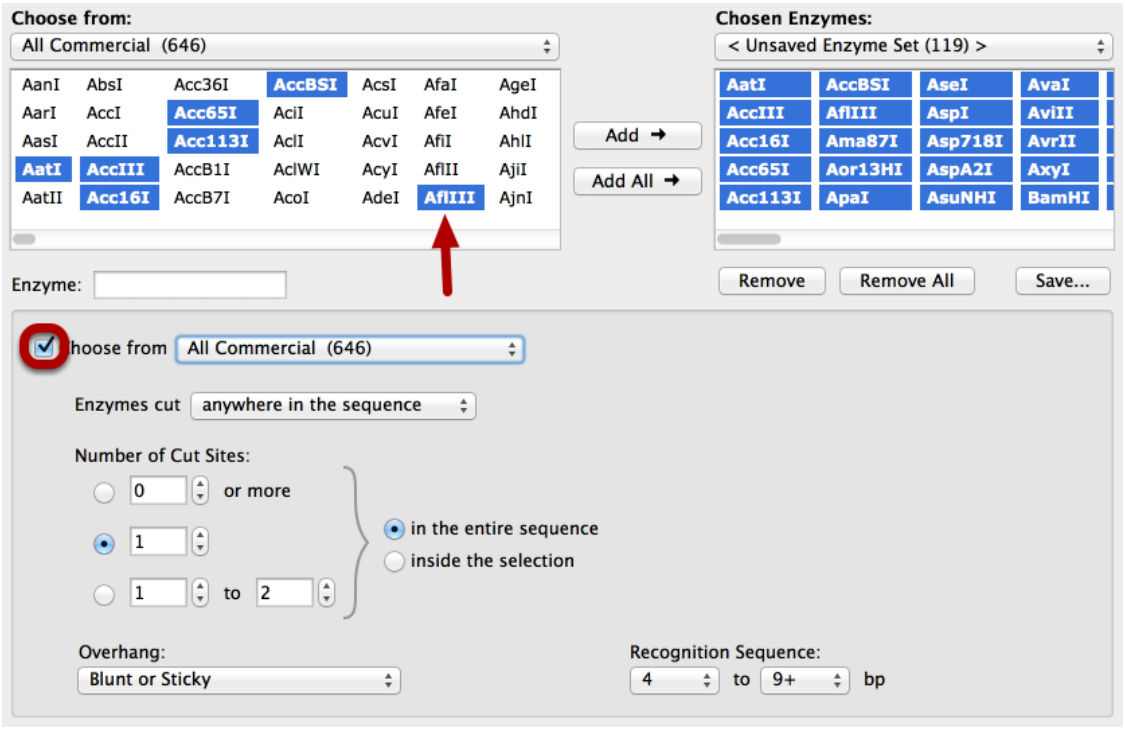
## Remove from an Enzyme Set



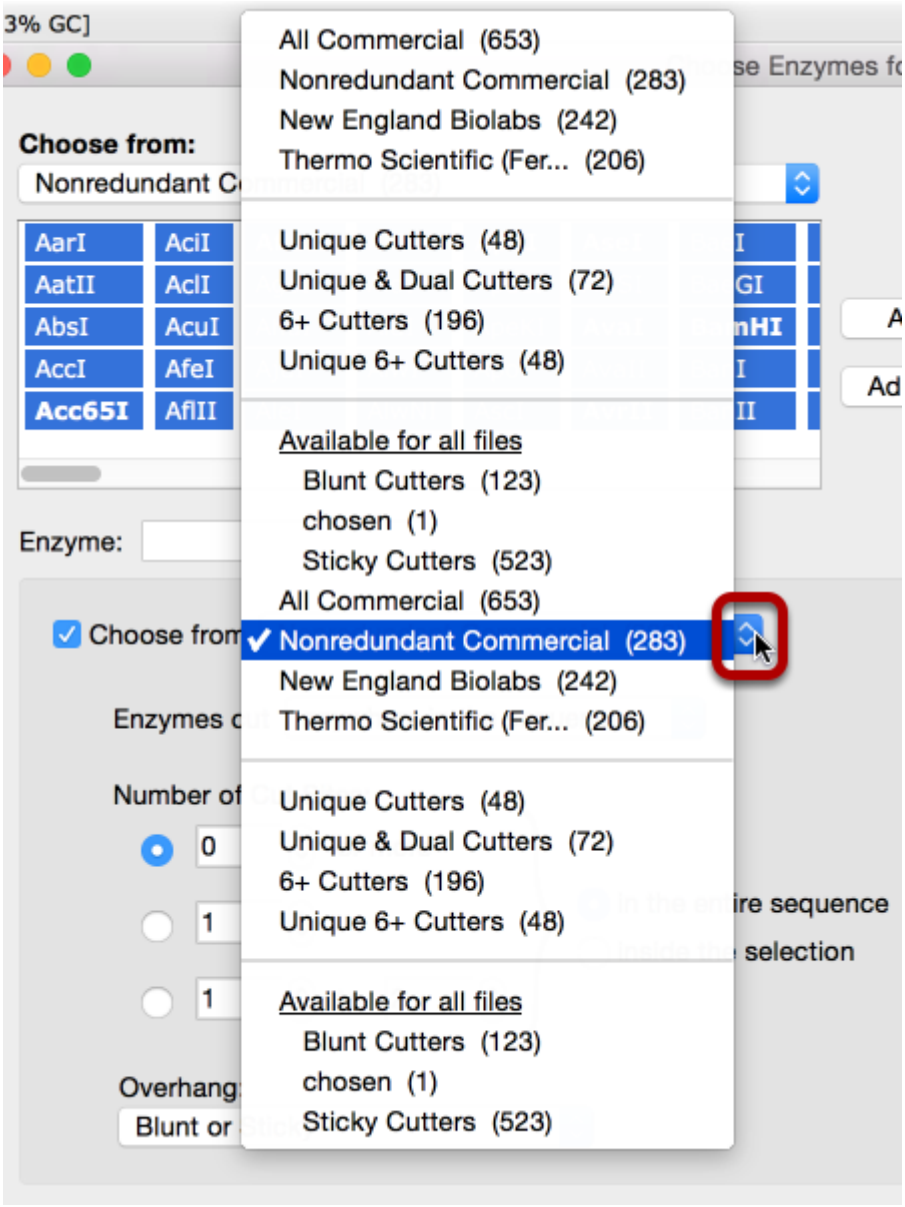

To remove the selected enzyme from the **Chosen Enzymes** set, click the **Remove** button or press the Delete key.

To remove all of the enzymes from the **Chosen Enzymes** set, click the **Remove All** button.

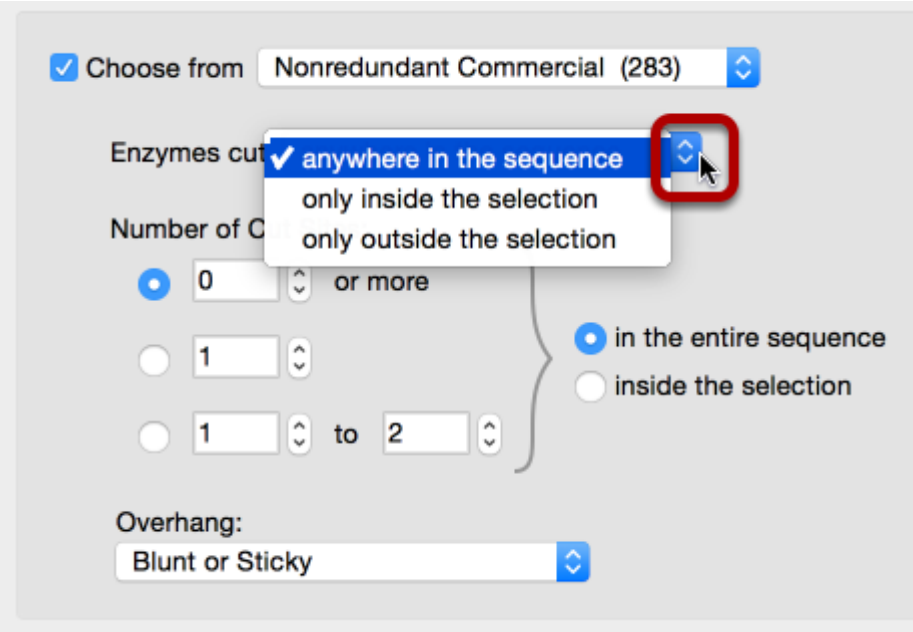
## Choose Enzymes by Criteria



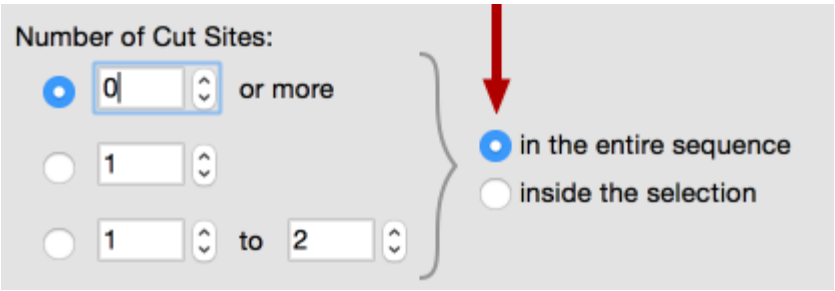

To choose enzymes based on criteria, click the **Choose from** check box. Enzymes that meet the selected criteria will be highlighted in the **Choose from** list.



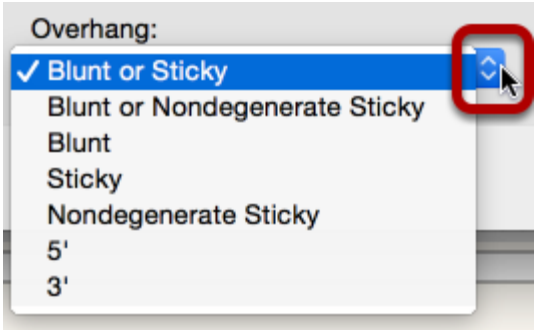

To specify the enzyme set from which the enzymes will be chosen, click the **Choose from** menu button.



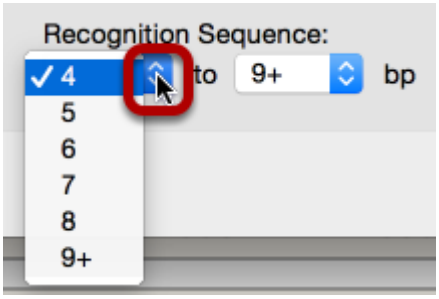

To specify where the chosen enzymes will cut, click the **Enzymes cut** menu button.




To specify the number of cut sites, type a number in the selected box. If not previously specified, the cut site region can be specified on the right.




To specify the enzyme overhang type, click the **Overhang** menu button.

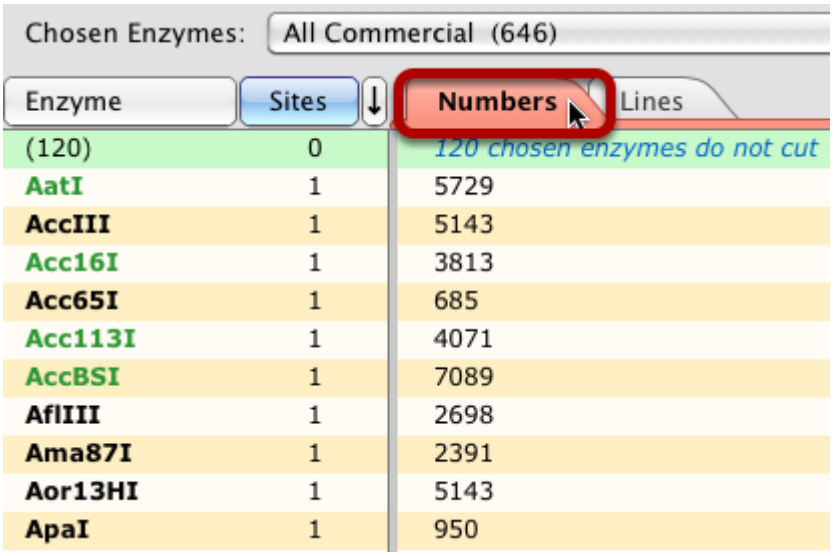



To specify the recognition sequence length, click the **Recognition Sequence** menu buttons.

## View a Restriction Site Overview

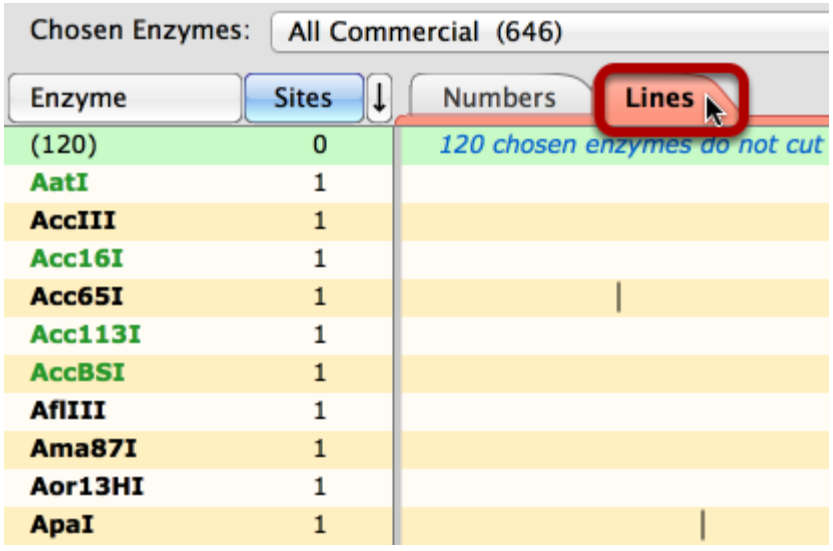
In Enzymes view, the restriction sites can be displayed as lines or numbers. The chosen enzyme set can also be searched or edited.

### Display Restriction Sites as Numbers



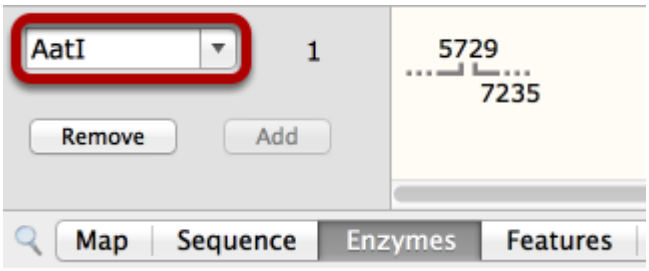

To display restriction sites as numbers, click the **Numbers** tab.

### Display Restriction Sites as Lines



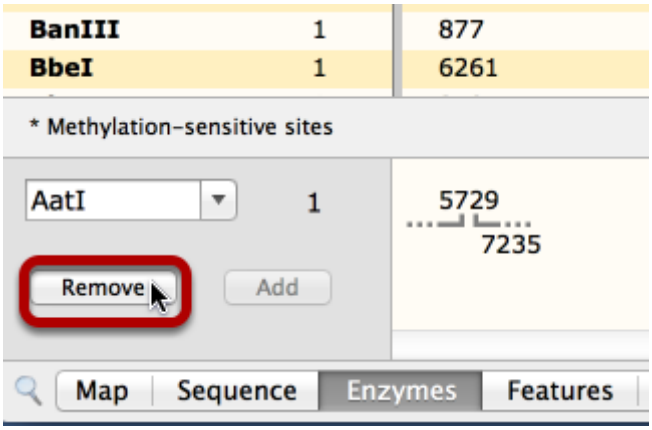

To display restriction sites as lines, click the **Lines** tab.

## Search for an Enzyme



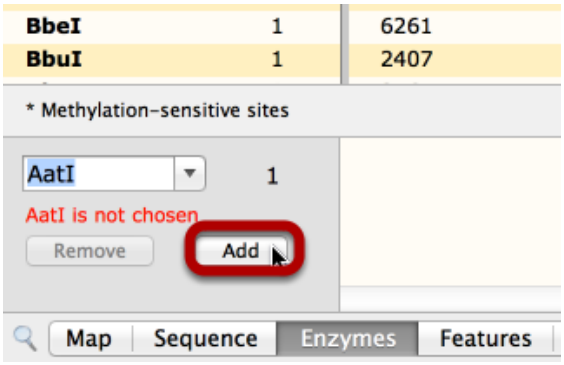

To search for an enzyme within the enzyme set, enter its name in the search box.

## Remove an Enzyme from the Chosen Set




To remove an enzyme from the chosen set, type its name, then click the **Remove** button.

## Add an Enzyme to the Chosen Set

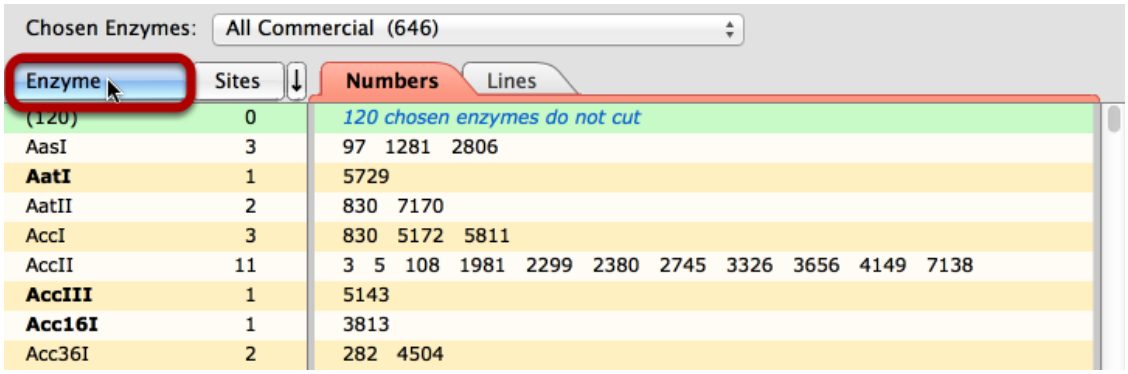



To add an enzyme to the chosen set, type its name, then click the **Add** button.

## Sort Enzymes

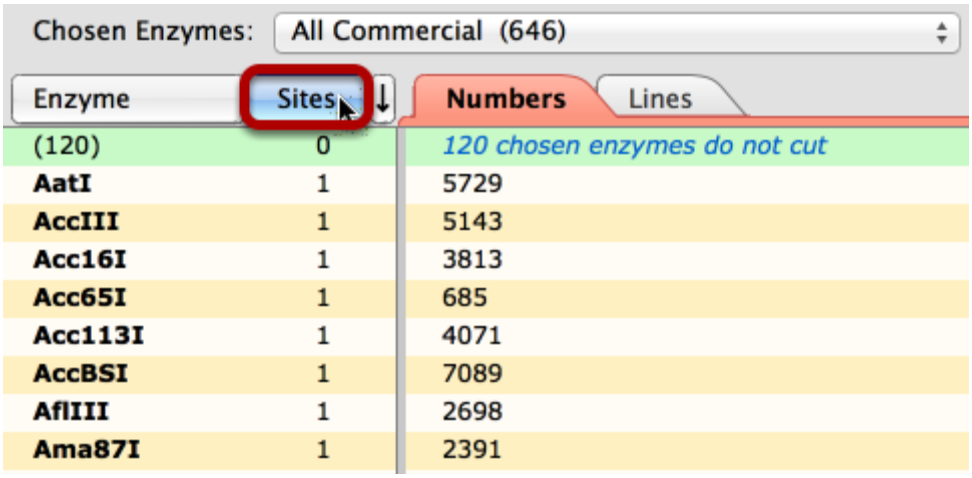
In Enzymes view, the chosen enzyme set can be sorted by name, restriction site count or location, or distance from a selection.

### Sort the Enzyme Set by Name



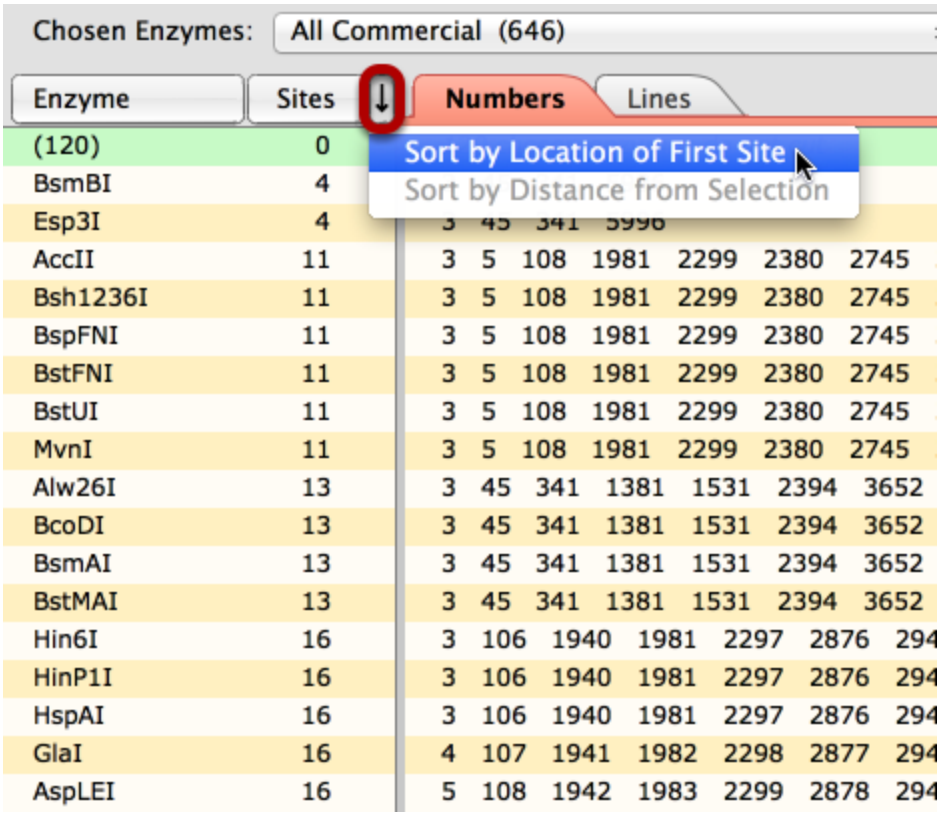

To sort the enzyme set by enzyme name, click the **Enzyme** button.

### Sort the Enzyme Set by Restriction Site Count



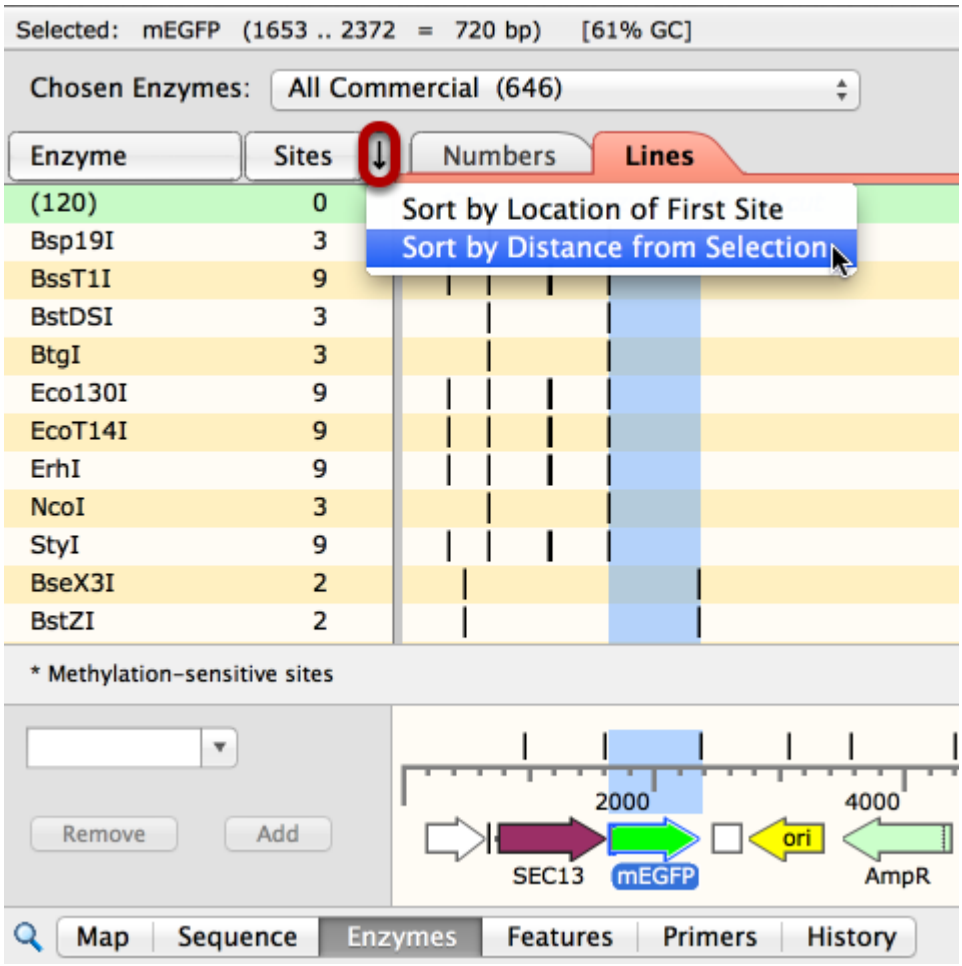

To sort the enzyme set by restriction site count, click the **Sites** button.

## Sort the Enzyme Set by Location of First Site




To sort the enzyme set by restriction site position, click the **Sort Enzyme List** (down arrow) button, then choose **Sort by Location of First Site**.

## Sort the Enzyme Set by Distance from Selection

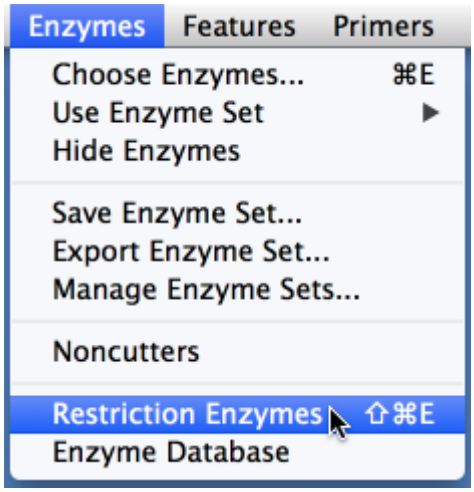



To sort the enzyme set by restriction site proximity to a DNA selection, click the **Sort Enzyme List** (down arrow) button, then choose **Sort by Distance from Selection**. For this purpose, it may be convenient to select a feature by clicking in the minimap at the bottom.

## View Enzyme Information

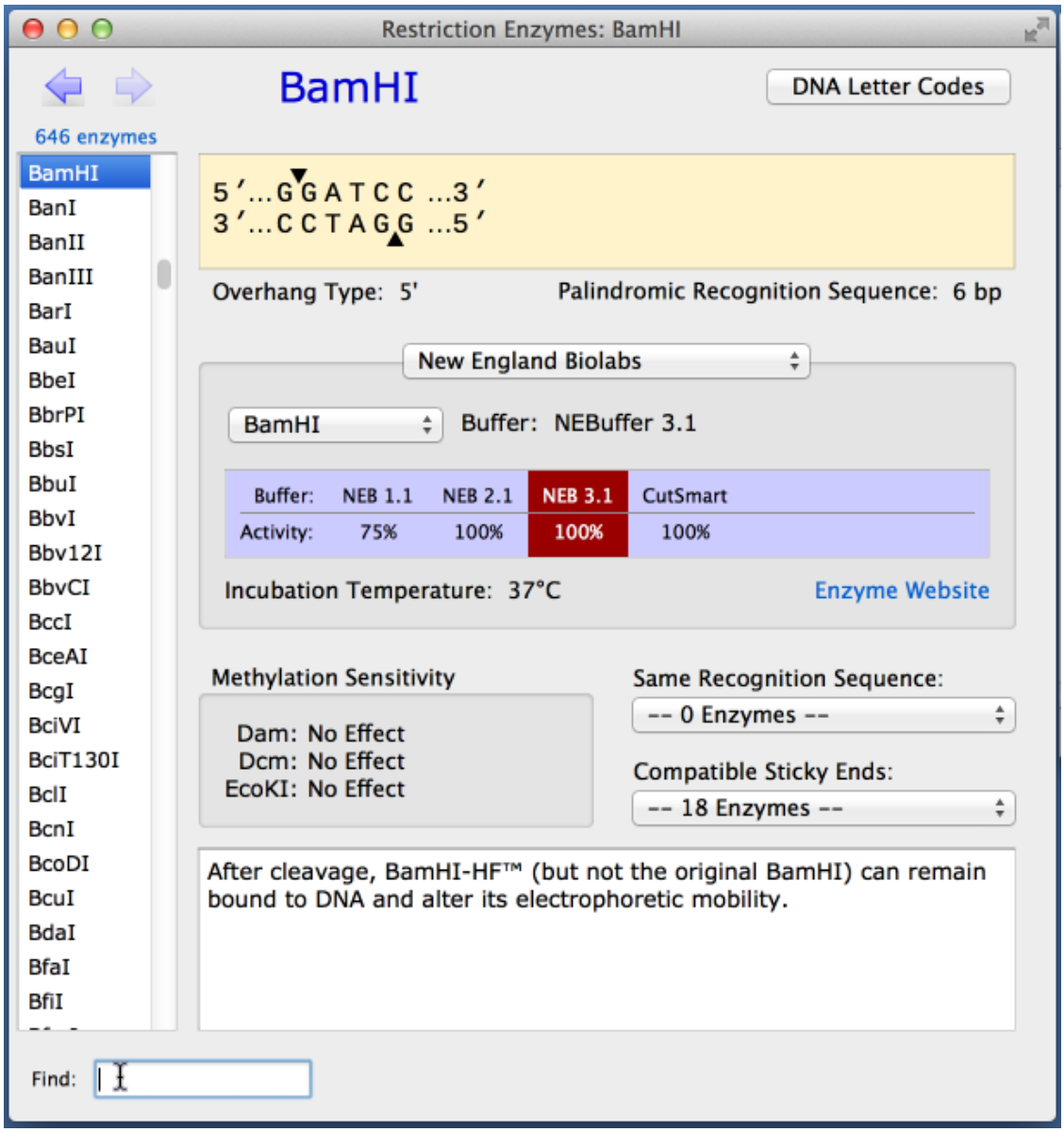
Detailed enzyme information is available in the Restriction Enzymes window by clicking or typing the enzyme name, while a synopsis is available by mousing over the enzyme name.

### Open the Restriction Enzymes Window



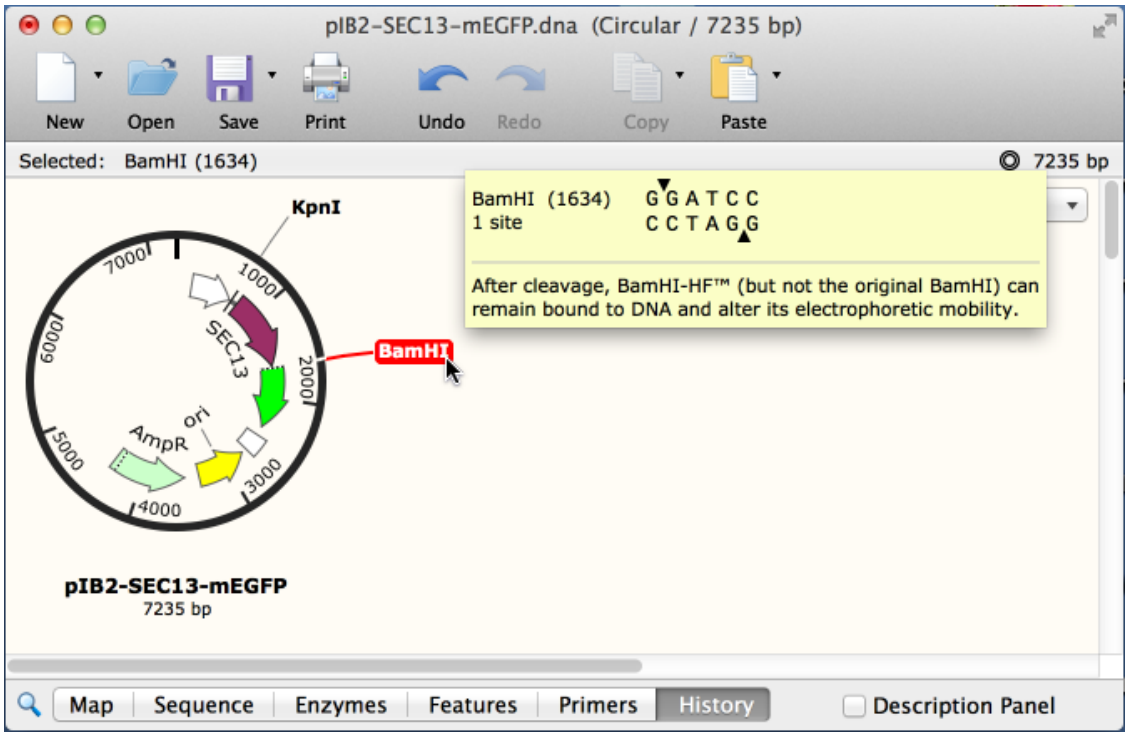

To view the enzyme information, click **Enzymes** → **Restriction Enzymes**.

## View Detailed Enzyme Information




To select the enzyme you want information about, type its name in the **Find** search box. Alternatively, double-click the enzyme name from within a sequence window to display and update the Restriction Enzymes window.

## View a Synopsis of Enzyme Information

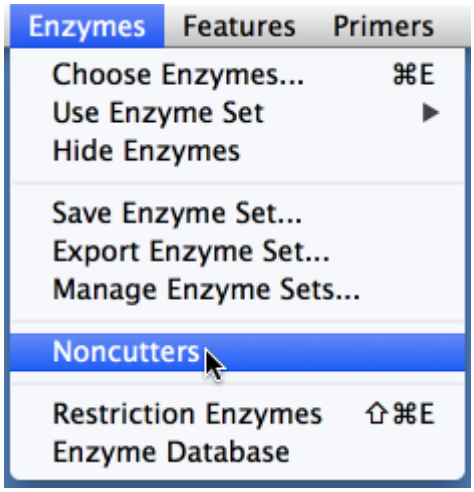



To show a synopsis of enzyme information, mouse over the enzyme name.

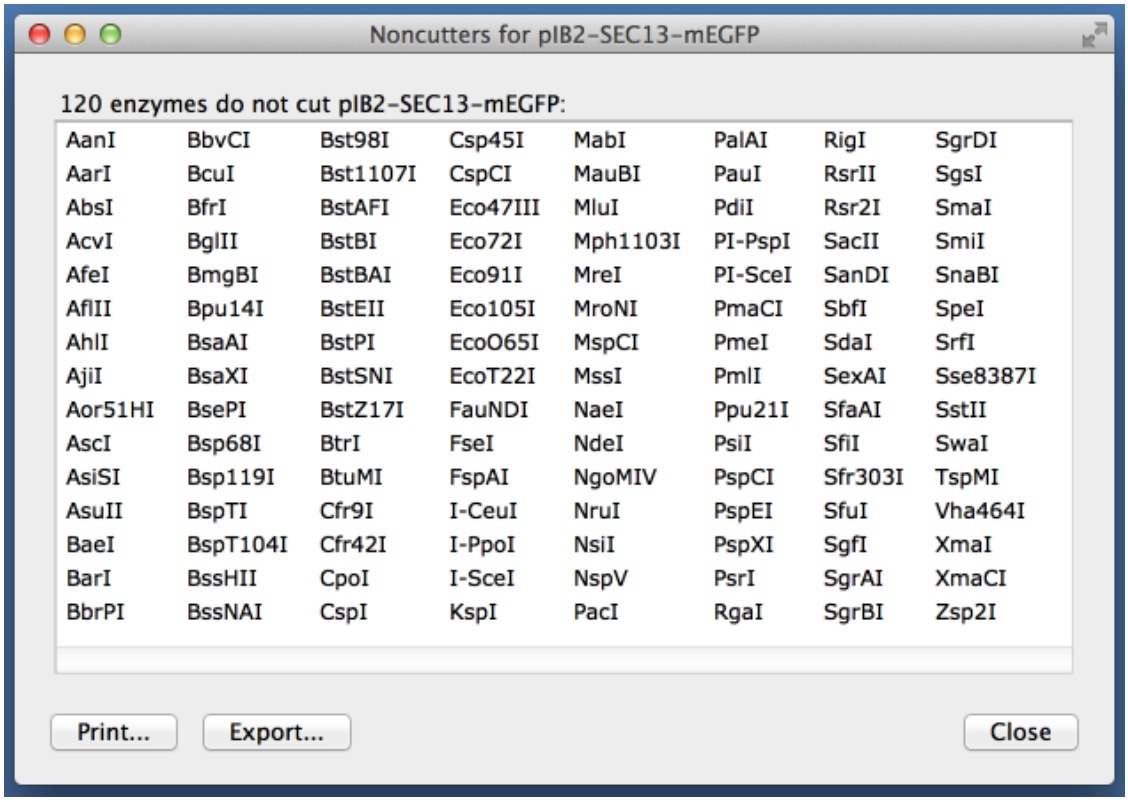
## View Noncutters

To view a list of enzymes that don't cut the sequence, you can use the Enzymes menu, the side toolbar, or the Noncutters button in Enzymes view.

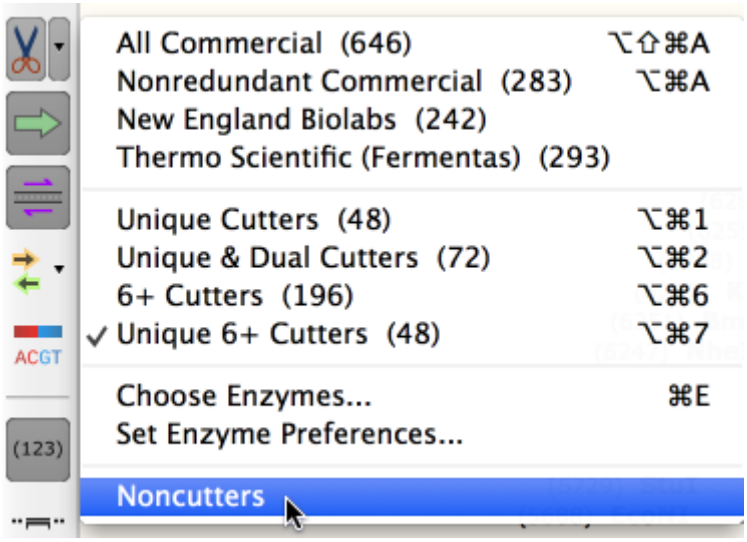
### Open the Noncutters Window




To view the Noncutters, click **Enzymes** → **Noncutters**.

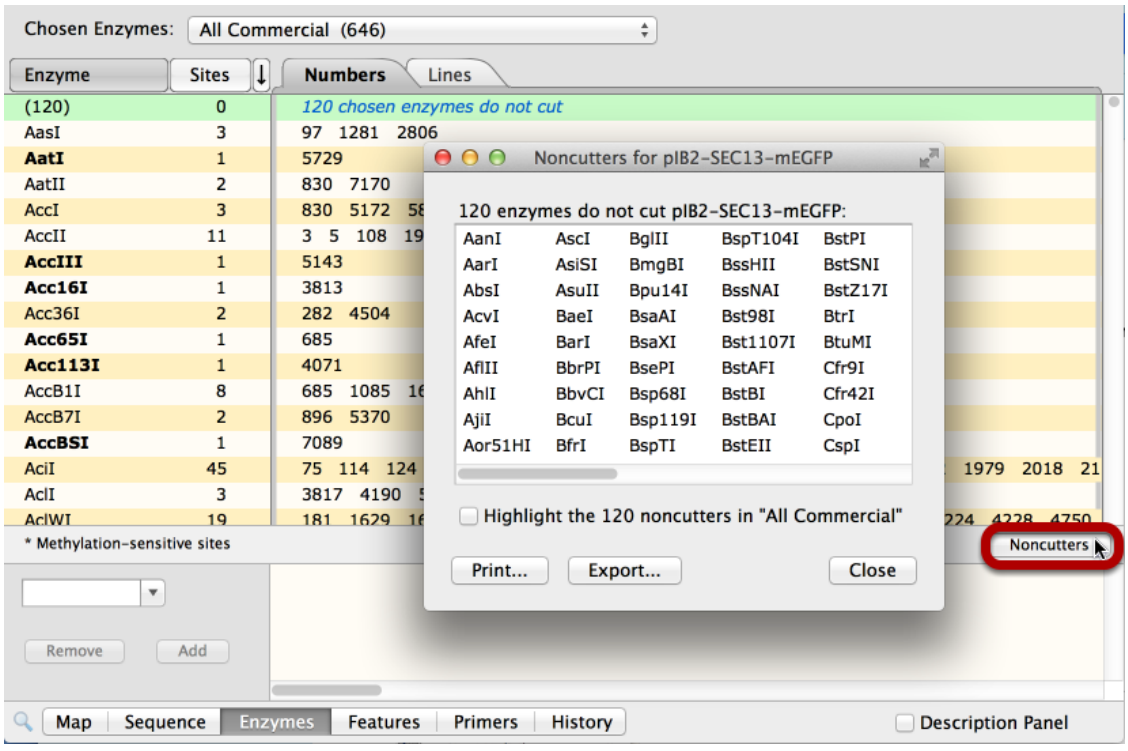


## View Noncutters Using the Side Toolbar




Alternatively, display the **Noncutters** command by clicking the "Show enzymes" (scissors) menu button in the side toolbar.

## View Noncutters in Enzymes view

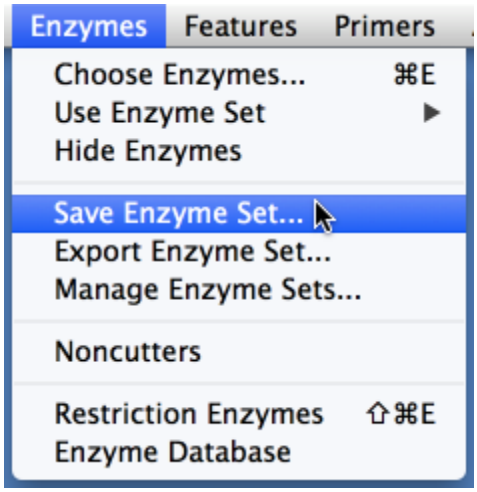



To view the Noncutters from Enzymes view, click the **Noncutters** button.

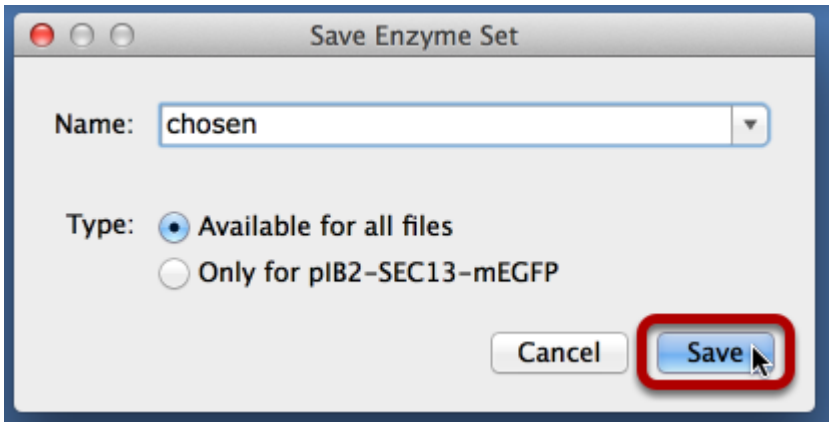
## Save or Export Enzyme Set

Chosen enzymes can be saved for use either with a single sequence, or with all sequences. Alternatively, enzymes can be exported for transfer to another computer.

### Save an Enzyme Set

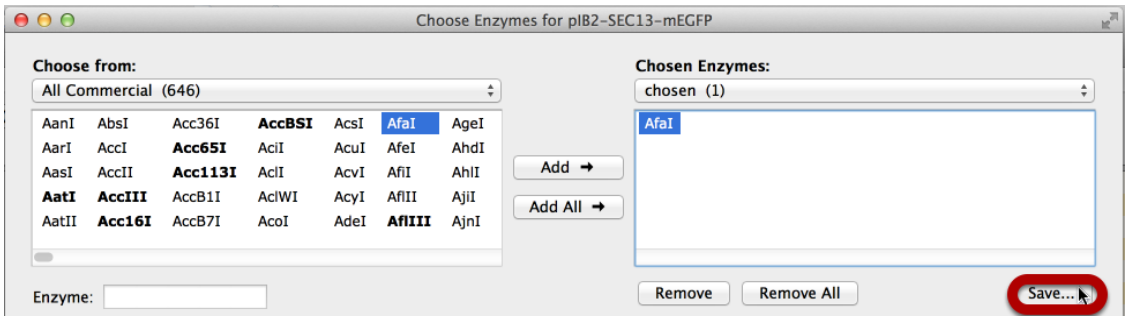



To save the currently chosen enzymes, click **Enzymes** → **Save Enzyme Set...** .



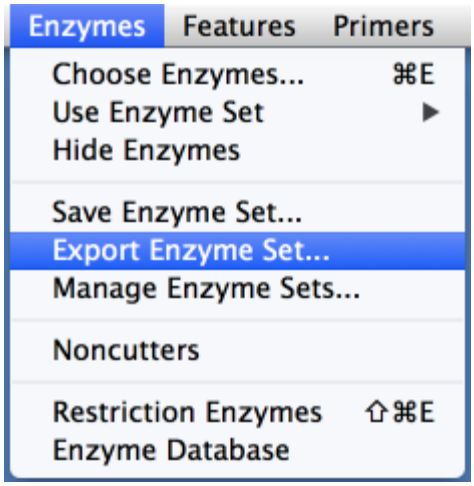

Type the file name, choose whether the set should be available for this sequence only or for all files, and then click **Save**.

## Save an Enzyme Set from the Choose Enzymes Dialog

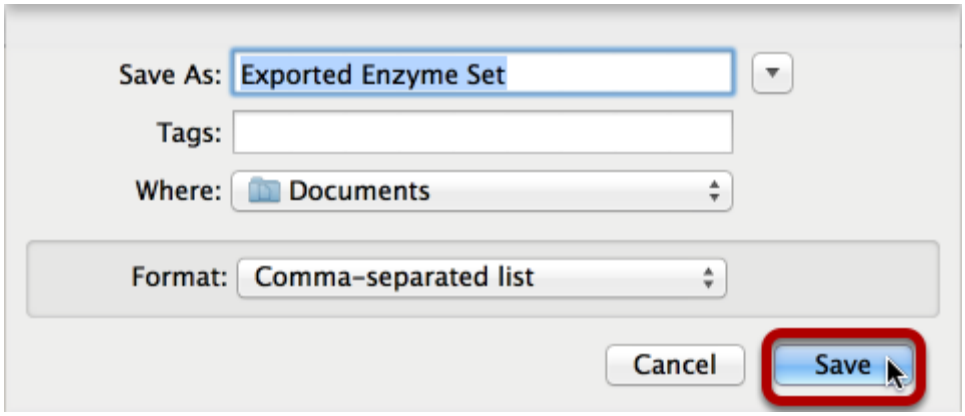



Alternatively, to save the currently chosen enzymes, click **Enzymes** → **Choose Enzymes...** and then click **Save...** .

## Export an Enzyme Set




To export the currently chosen enzymes, click **Enzymes** → **Export Enzyme Set...** .

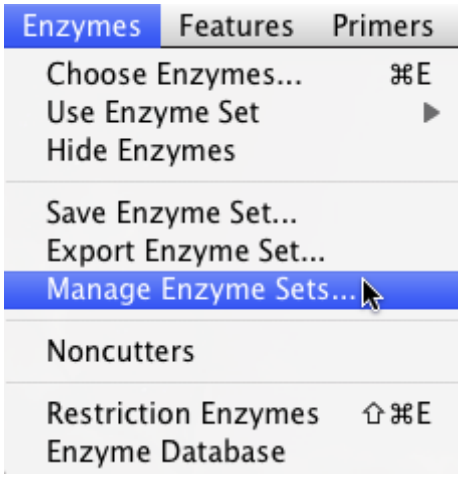



Type the file name and click **Save**.

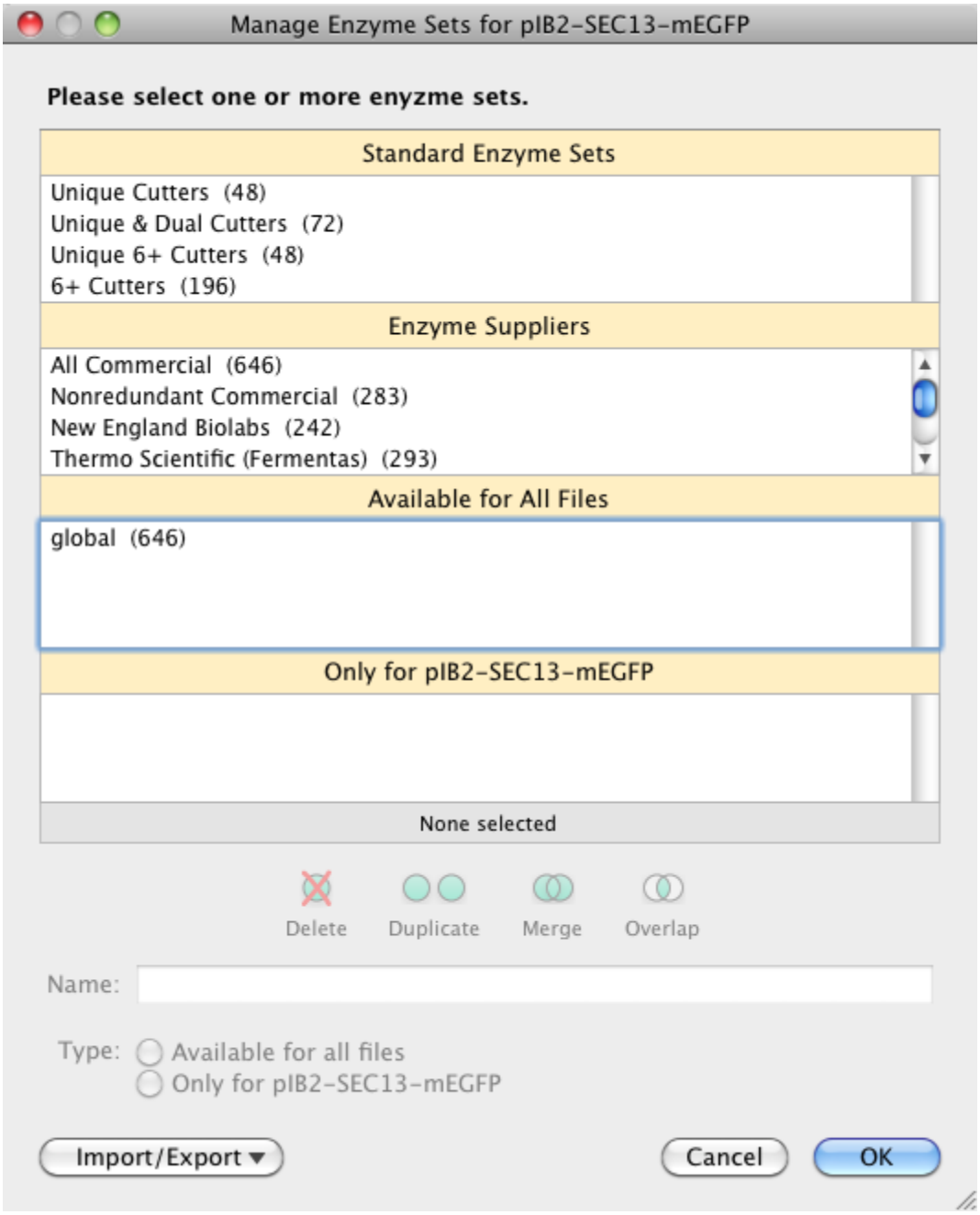
# Manage Enzyme Sets

Custom enzyme sets can be managed by changing their name or type, or by combining, deleting, duplicating, importing, or exporting existing enzyme sets.

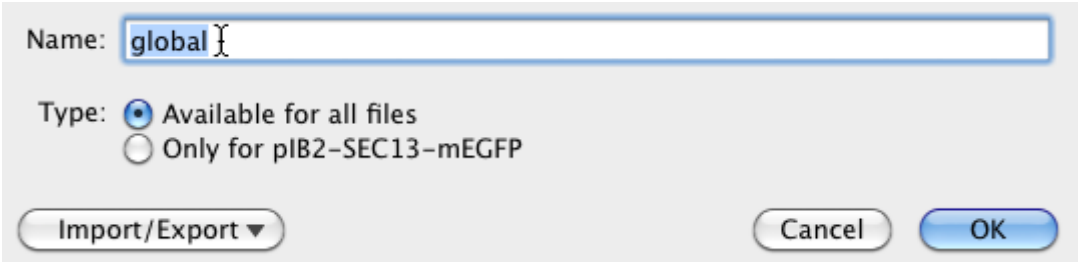
## Manage Enzyme Sets




To manage enzyme sets, click **Enzymes** → **Manage Enzyme Sets...** .

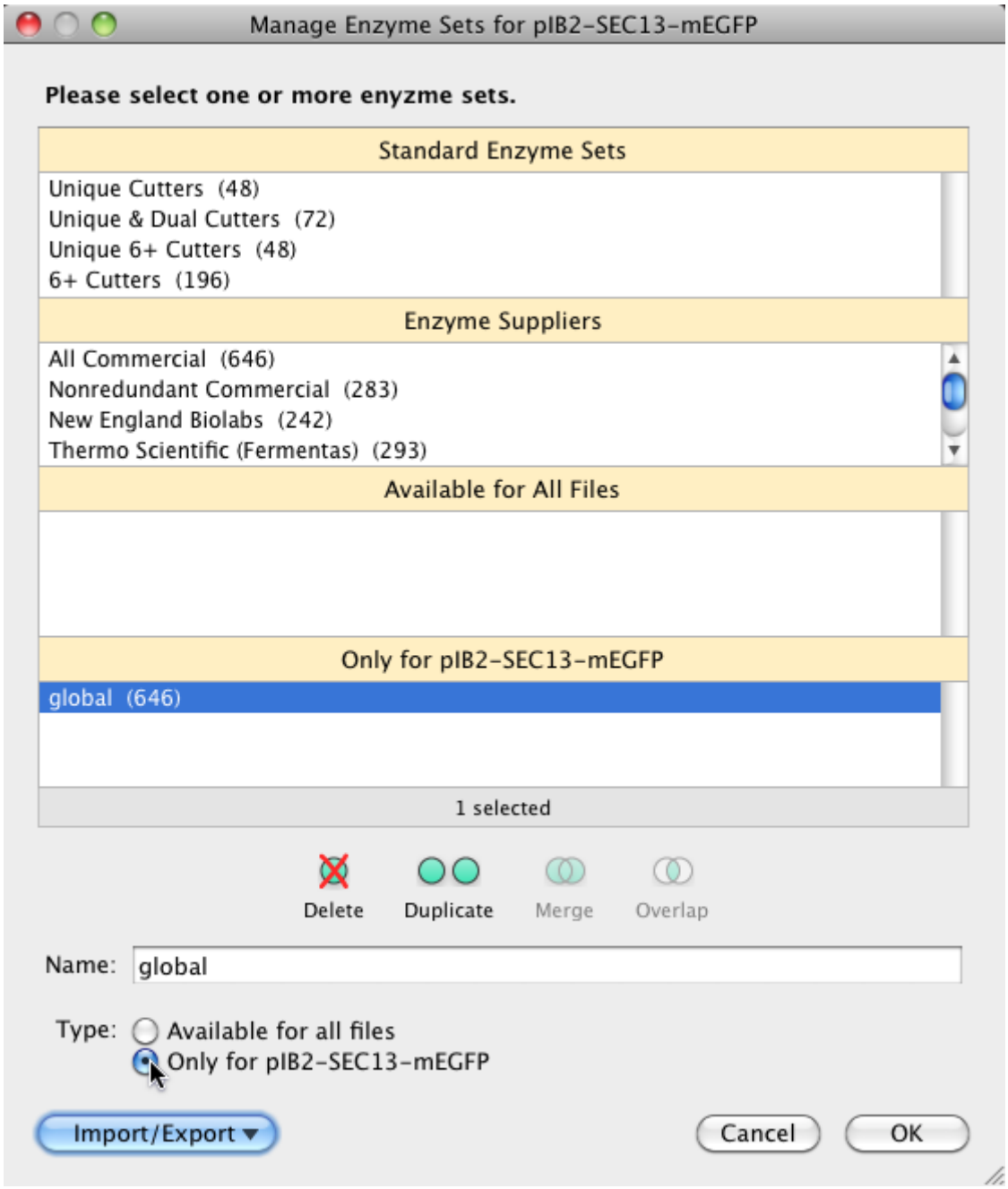



## Rename the Selected Enzyme Set



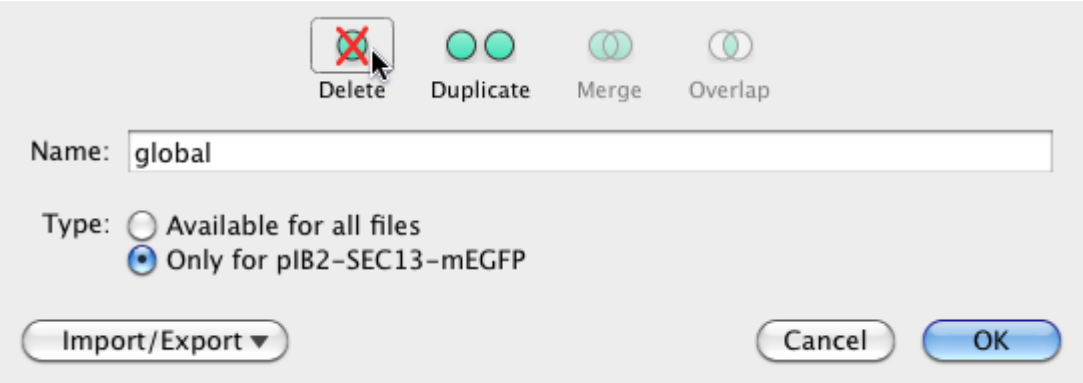

To rename the selected enzyme set, edit its name in the **Name** text box, then click **OK**.

## Change the Enzyme Set Type



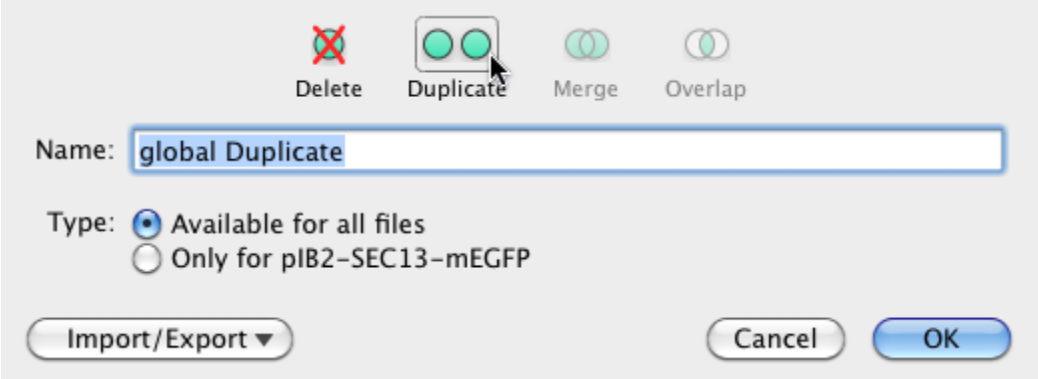

To change the type of the selected enzyme set, choose the type, then click **OK**.
Alternatively, drag and drop the enzyme set name from one of the two bottom list boxes to the other.

## Delete the Selected Enzyme Set



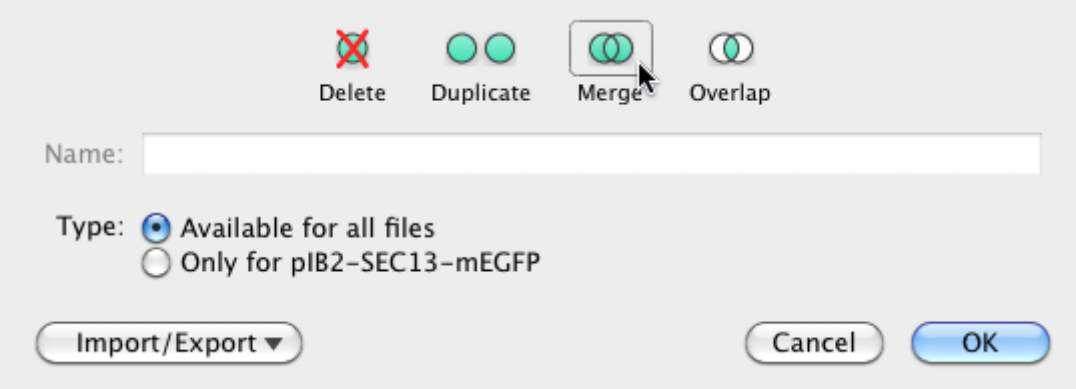

To delete the selected enzyme set, click the **Delete** button, then click **OK**.

## Duplicate the Selected Enzyme Set



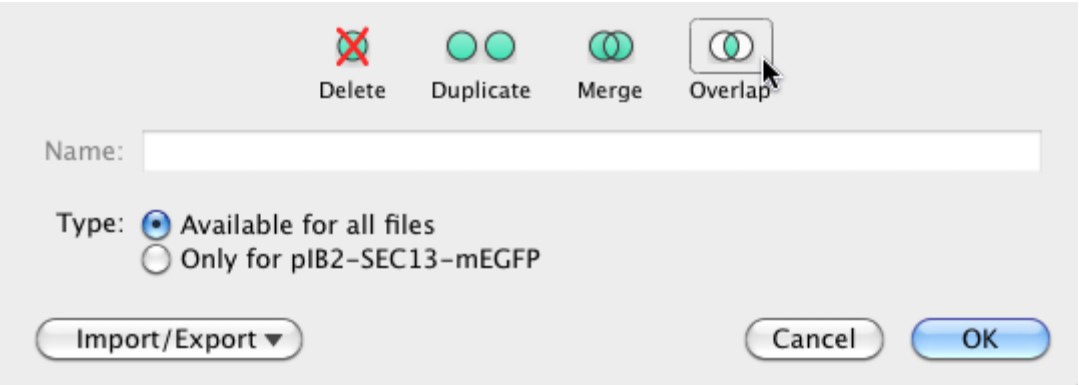

To duplicate the selected enzyme set, click the **Duplicate** button, then click **OK**.

## Merge Enzyme Sets



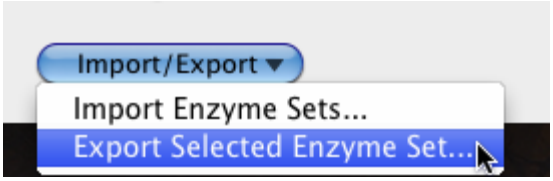

To merge the selected enzyme sets, click the **Merge** button, then click **OK**.

## Create an Overlap Enzyme Set



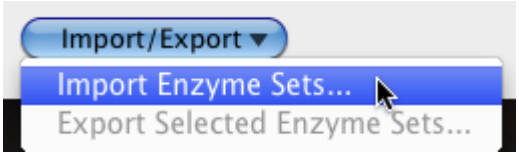

To create an enzyme set that contains only the enzymes the selected enzyme sets have in common, click the **Overlap** button, then click **OK**.

## Export the Enzyme Set




To export the selected enzyme set, click the **Import/Export** menu button, then click **Export Selected Enzyme Set...** , type the filename, and click **Save**.

## Import the Enzyme Set

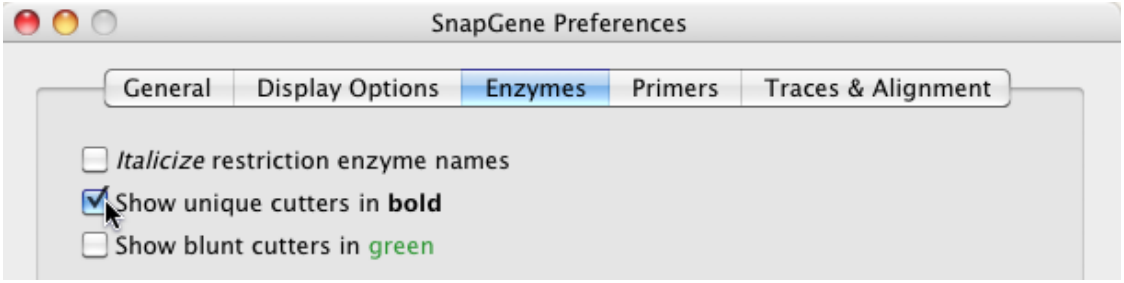



To import an enzyme set, click the **Import/Export** menu button, then click **Import Enzyme Sets...** , select the [Exported Enzyme Set].txt file, and click **Open**. Alternatively, drag and drop an enzyme set .txt file to either of the bottom two list boxes. The imported enzyme set name will be shown in the **Available for All Files** list box.

## Set Enzyme Preferences

SnapGene will remember customized preferences for displaying enzymes and enzyme sets, as well as the order of preference for enzyme suppliers.

### Set the Enzyme Display Preferences



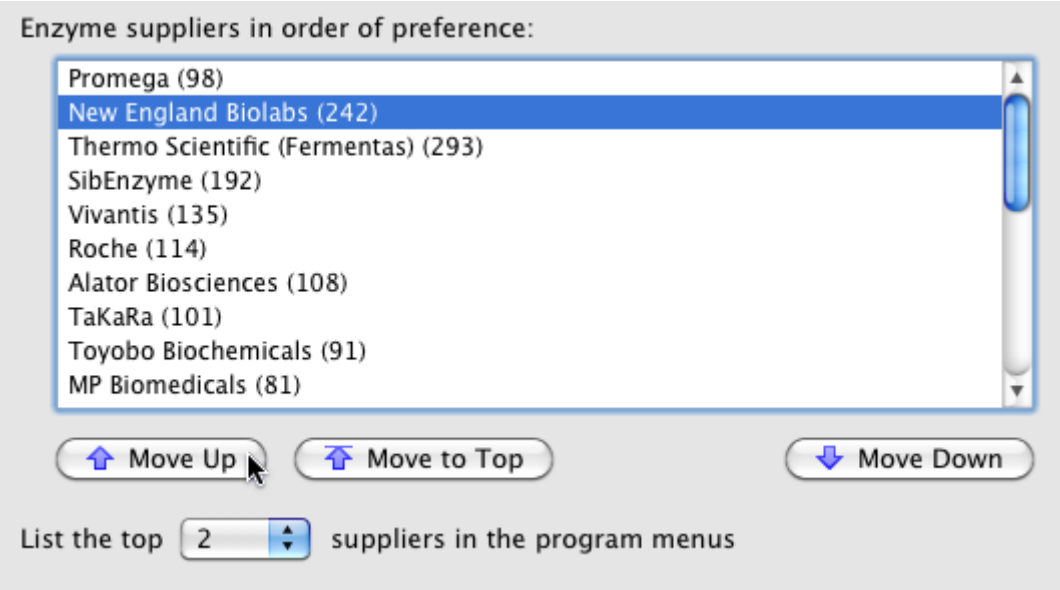

On a Mac Computer: To view the enzyme Preferences, click **SnapGene** → **Preferences**, then click the **Enzymes** tab.
On a Windows Computer: To view the enzyme Preferences, click **Edit** → **Settings**, then click the **Enzymes** tab.

Click the check boxes to customize the way enzyme names are shown.

### Set the Restriction Site Display Preferences

Click either the **Numbers** or **Lines** tabs to set the default way restriction sites are shown in Enzymes view.

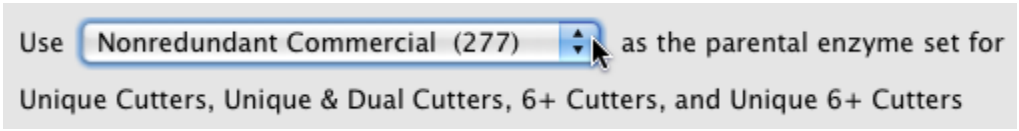
## Change the Supplier List Order




Click the **Move Up**, **Move to Top**, and **Move Down** buttons to change the order of preference for enzyme suppliers. Enzyme sets for the selected number of highest-ranking suppliers will be listed within the program menus.

The Nonredundant Commercial enzyme set is created based on the enzyme suppliers ranking. For equivalent enzymes from multiple suppliers, only the enzyme from the highest ranking supplier is used in the Nonredundant Commercial enzymes set.

## Choose the Parental Enzyme Set



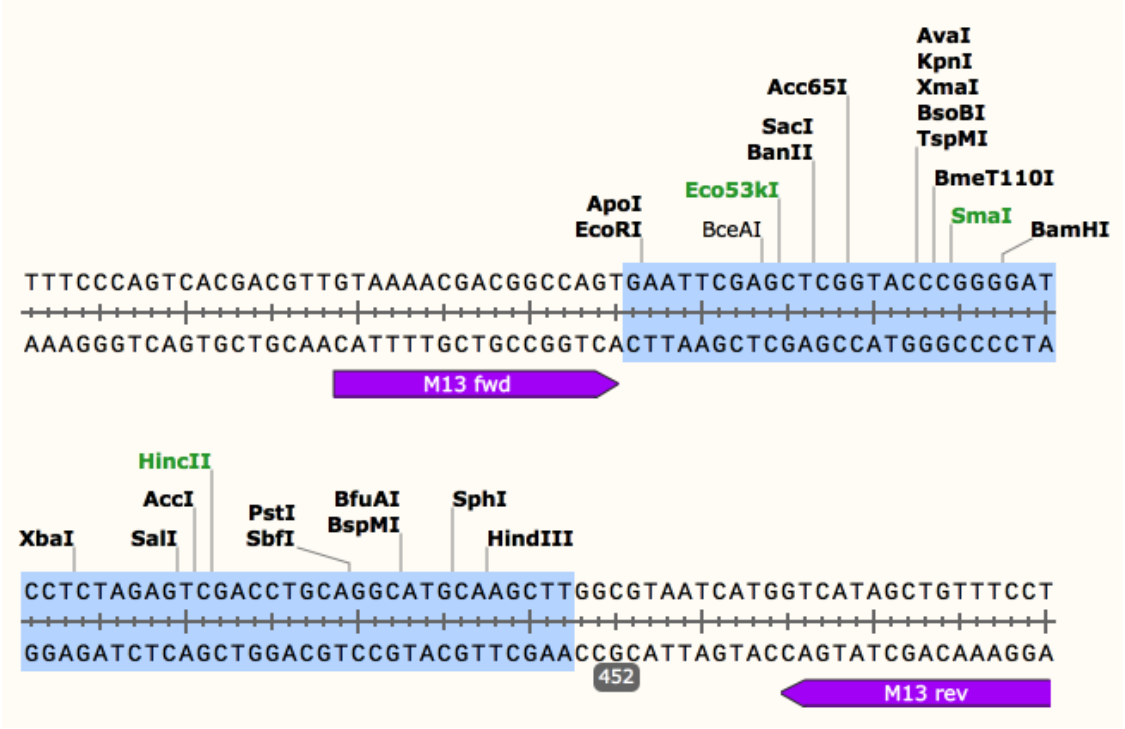

Click the menu to select which enzymes can be included in the following enzyme sets: Unique Cutters, Unique & Dual Cutters, 6+ Cutters, and Unique 6+ Cutters.

# Features

## Create a Simple Feature

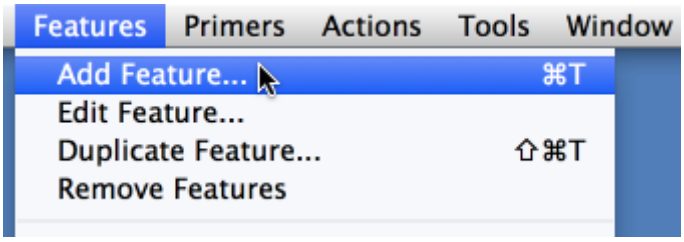
How can I designate a feature within a DNA sequence?

### Select the Sequence Region



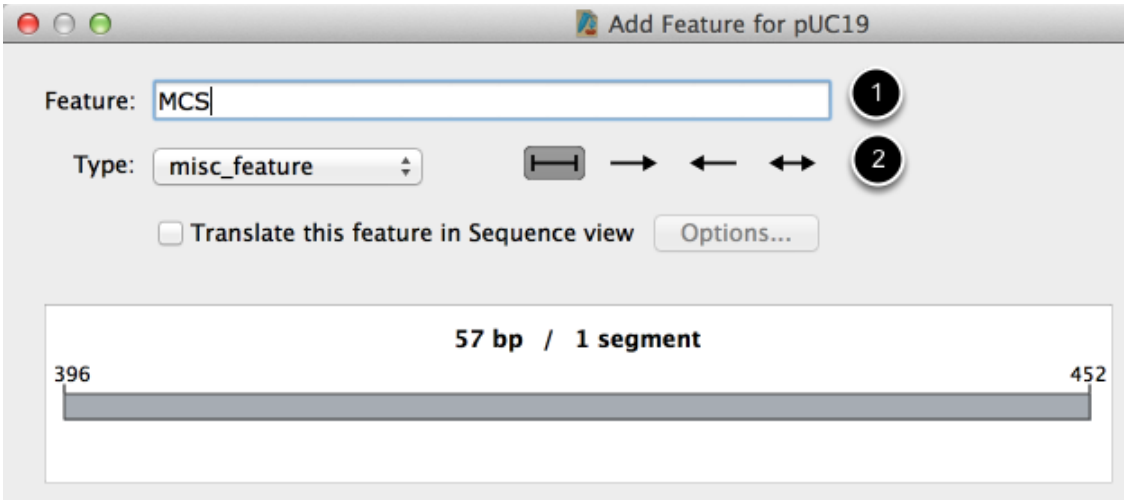

Select the sequence region that will be labeled with a feature. In this example, a multiple cloning site (MCS) has been selected in Sequence view.

### Add a Feature




To add a feature, click **Features** → **Add Feature...** .

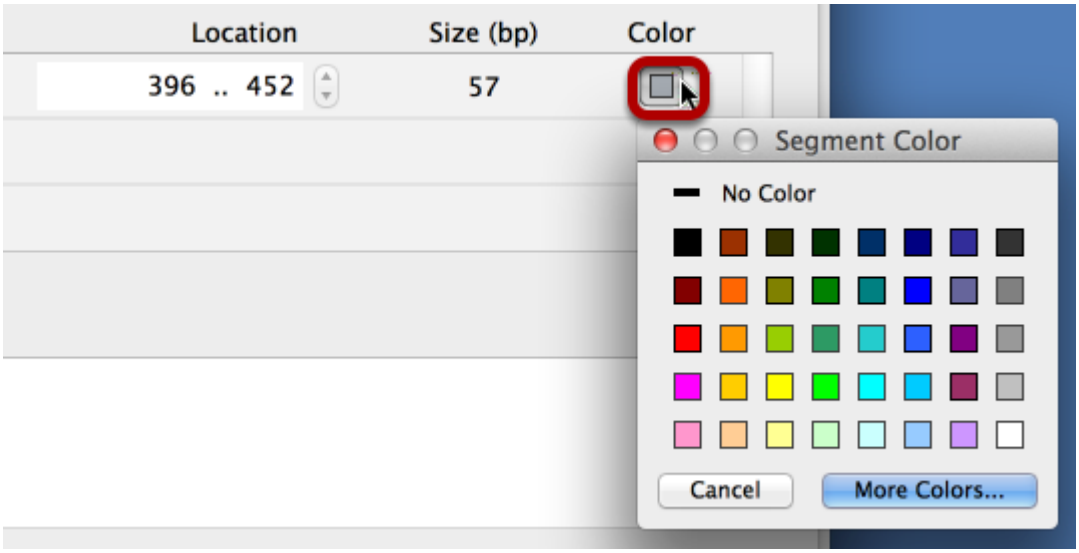
## Specify the Feature Name, Type, and Directionality




(1) Type the name of the new feature.

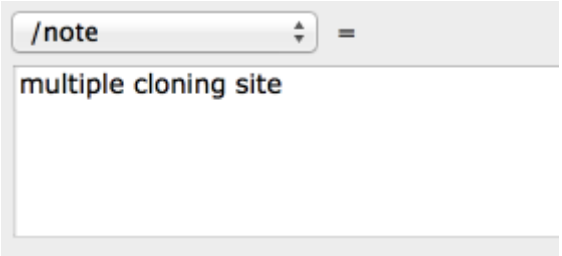
(2) Specify the feature type (according to GenBank conventions) and directionality (nondirectional, forward, reverse, or bidirectional).

## Choose the Feature Color



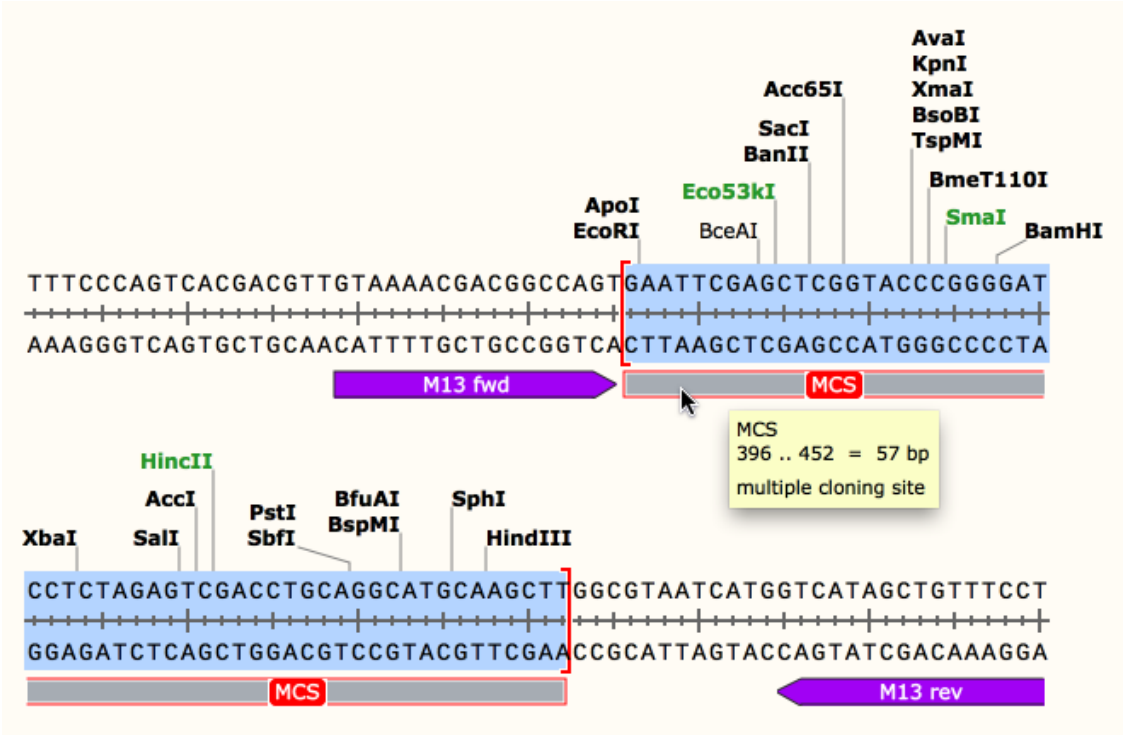

Click the "Color" button. Choose a standard color, or click **More Colors...** . To display the feature as a thin black line, click **No Color**.

## Enter a Description




If desired, enter a description in the "/note" field. Once the feature has been created, the contents of this field will appear in a tooltip if you mouse over the feature.

## View the Feature

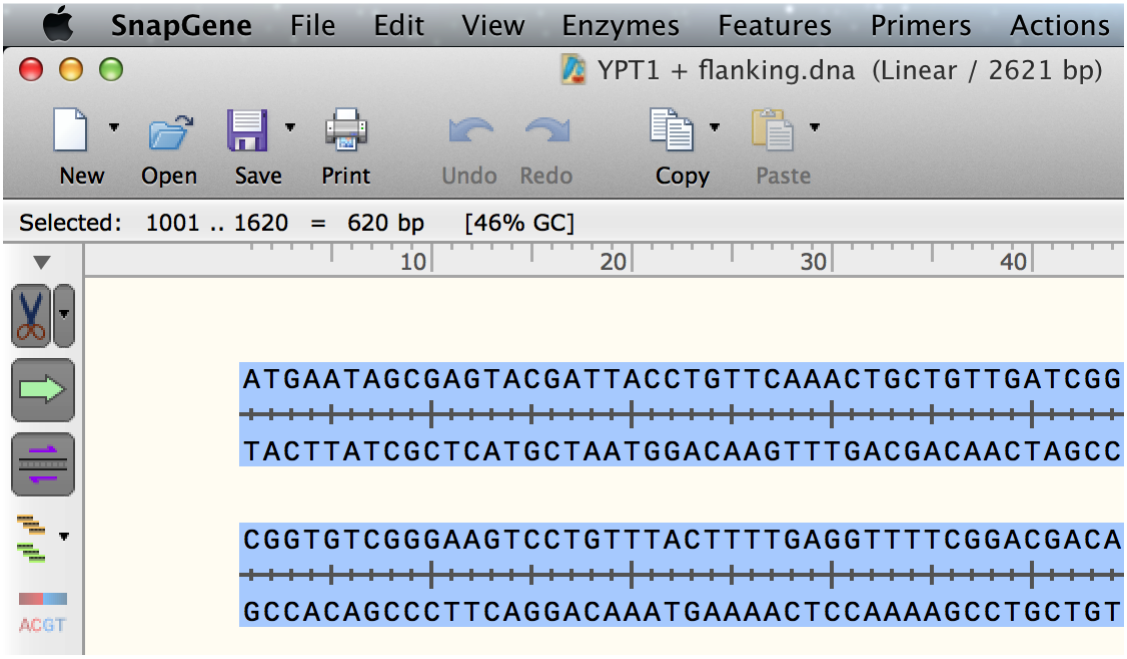



Click **OK** to see the new feature.

## Create a Translated Feature

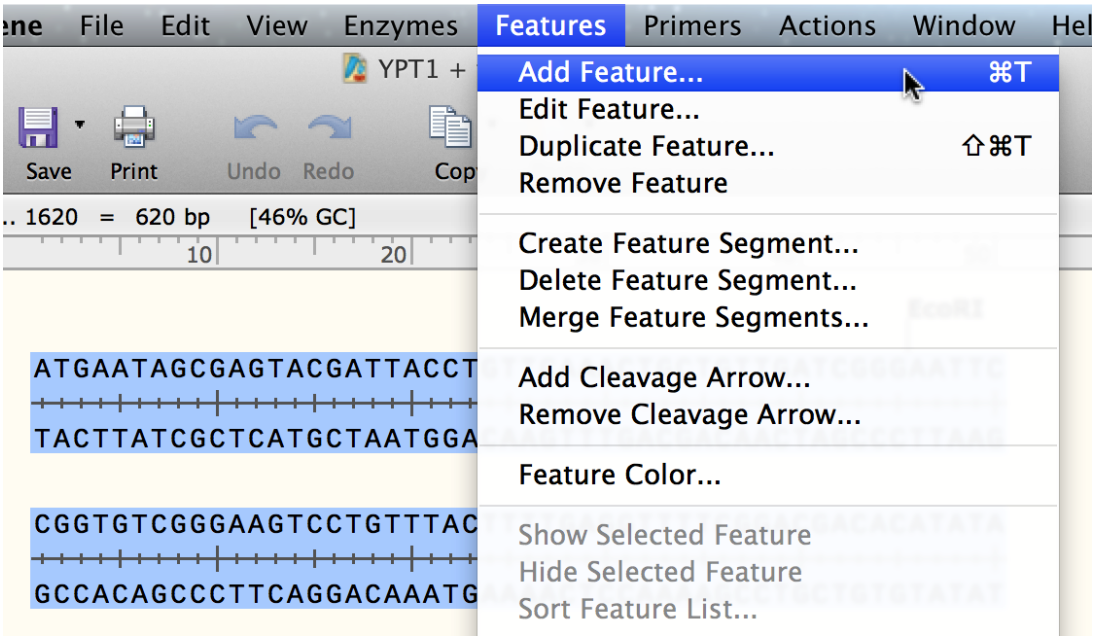
How can I create a feature that will be translated in Sequence view?

### Select the Translated Sequence Region



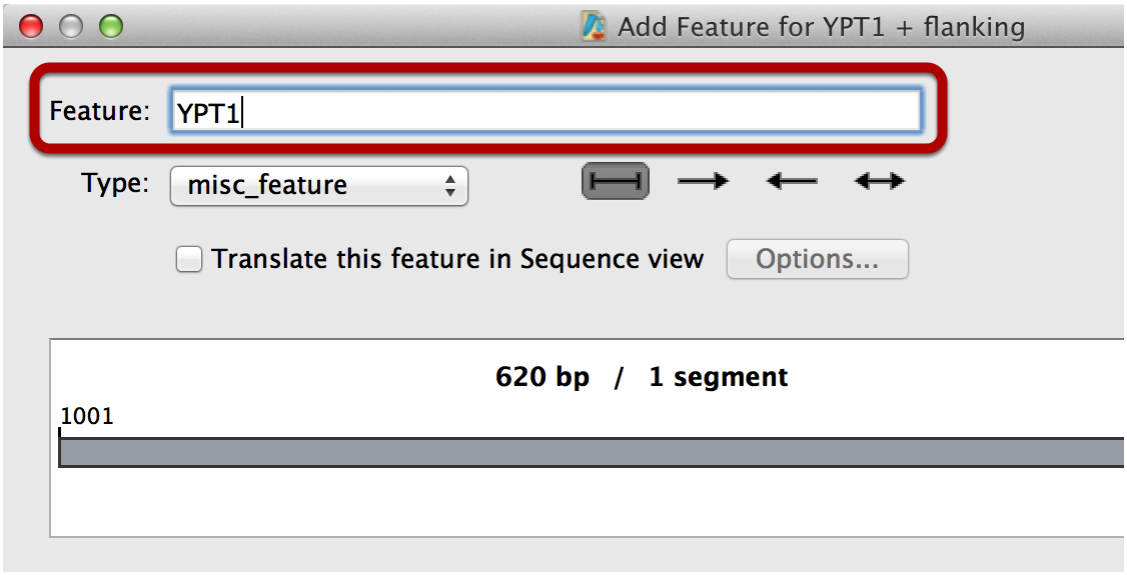

Select the sequence region that will be annotated with a translated feature.

### Add a Feature



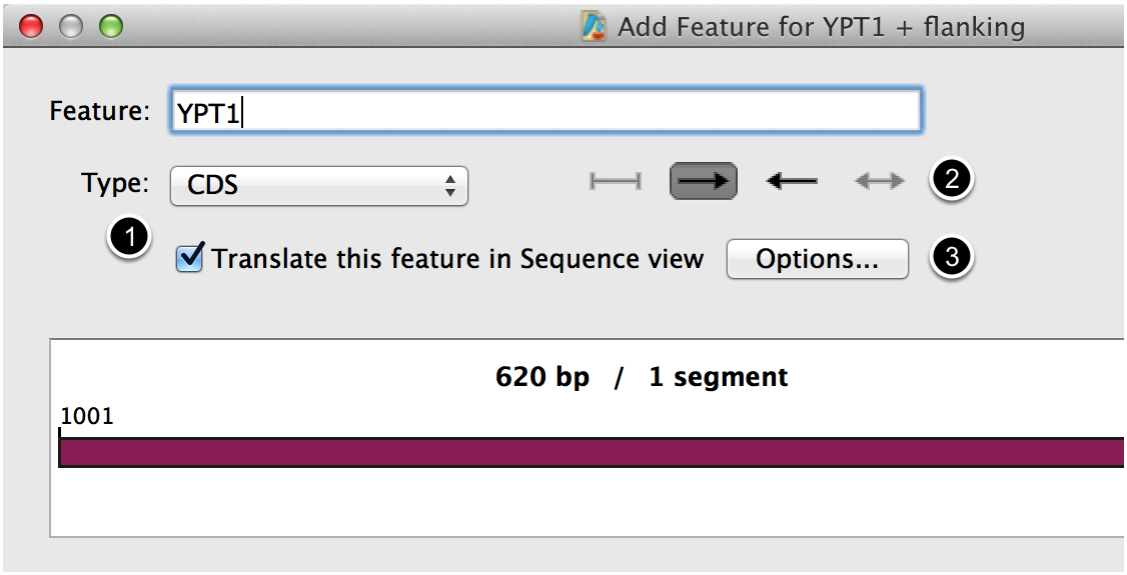

To add a feature, click **Features** → **Add Feature...** .

## Name the Feature




Type the name of the new feature.

## Turn on Translation

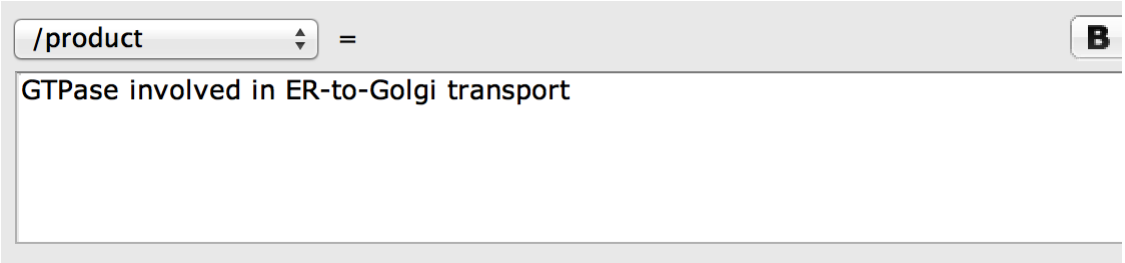



(1) To turn on translation, either click the check box, or choose "CDS" (coding sequence) as the feature type.

(2) Specify the directionality of translation as either forward (top strand) or reverse (bottom strand).

(3) Click "Options" to change the reading frame or genetic code or translation numbering, or to ensure that the first codon is translated as methionine.

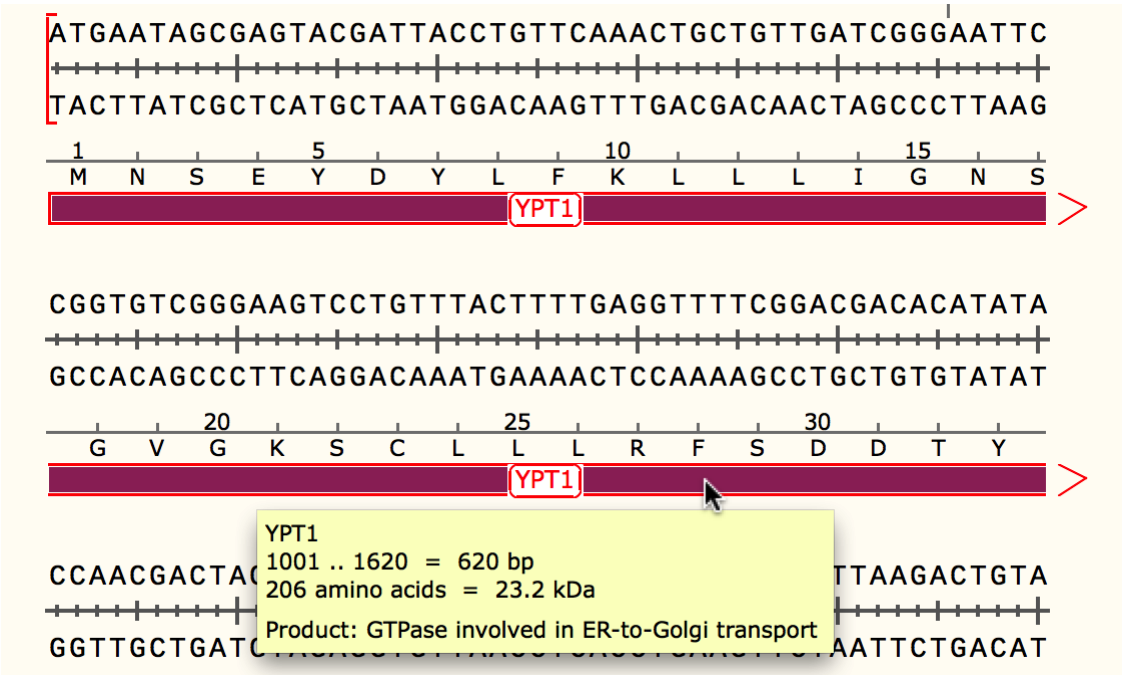
## Enter a Description




If desired, enter a description of the protein in the "/product" field. You may want to provide additional information in the "/note" field.

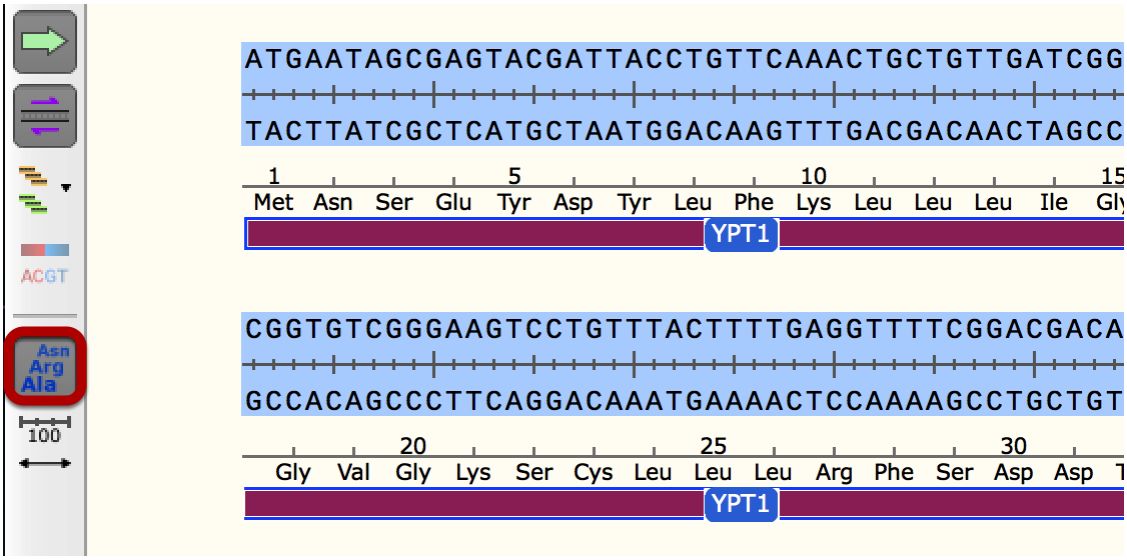
Once the feature has been created, the contents of both fields will appear in a tooltip if you mouse over the feature.

## View the Translation




Click **OK** to see the translated feature in Sequence view.

## Choose 1- or 3-Letter Amino Acid Codes

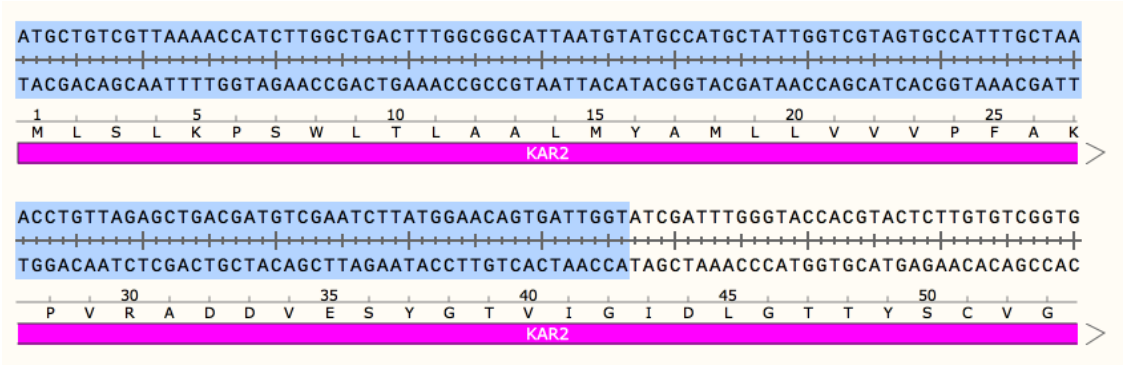



If desired, click the button in the side toolbar to switch between 1- and 3-letter amino acid codes.

# Create a Feature Segment

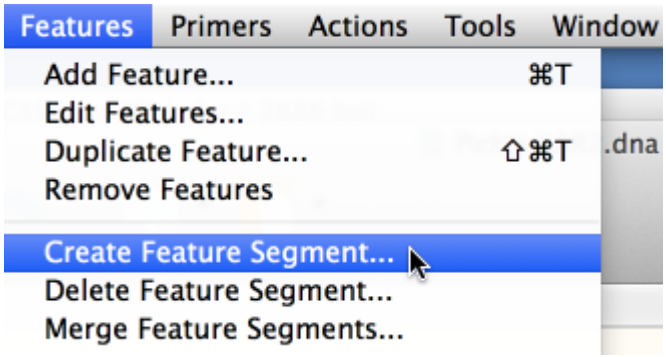
How can I designate part of a feature as a segment?

## Select the Feature Segment Region



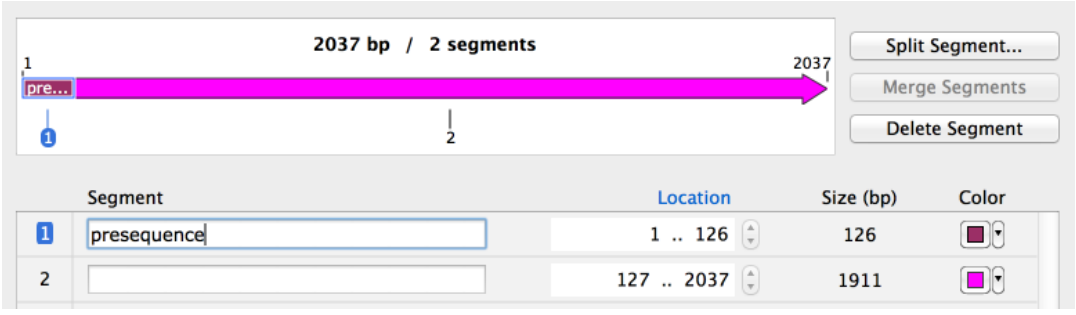

Select the DNA sequence region that will be labeled with a feature segment.

## Create a Feature Segment



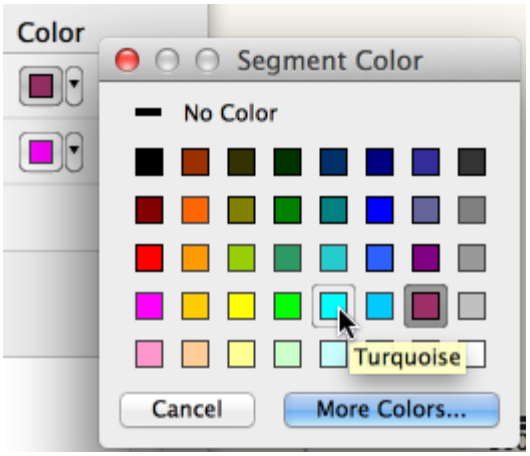

To create a feature segment, click **Features** → **Create Feature Segment...** .

## Name the Segment



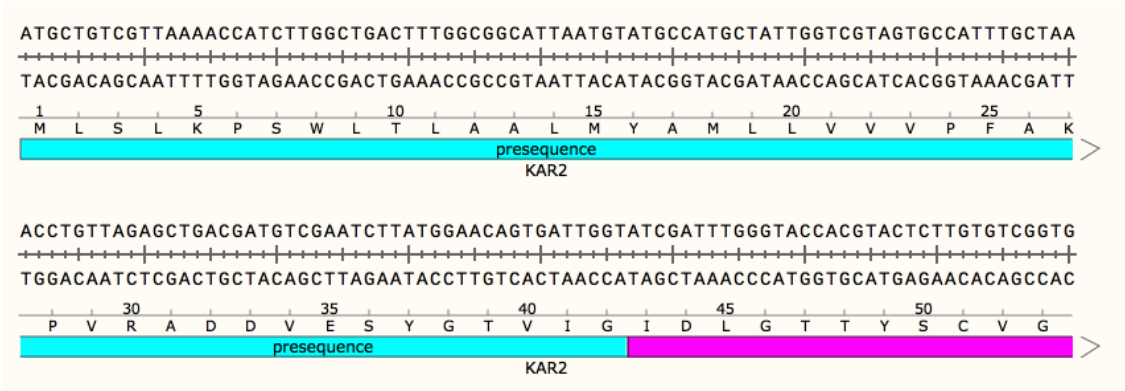

Type the segment name if desired.

## Choose the Segment Color




Choose a different color for the segment if desired.

## View the Segment

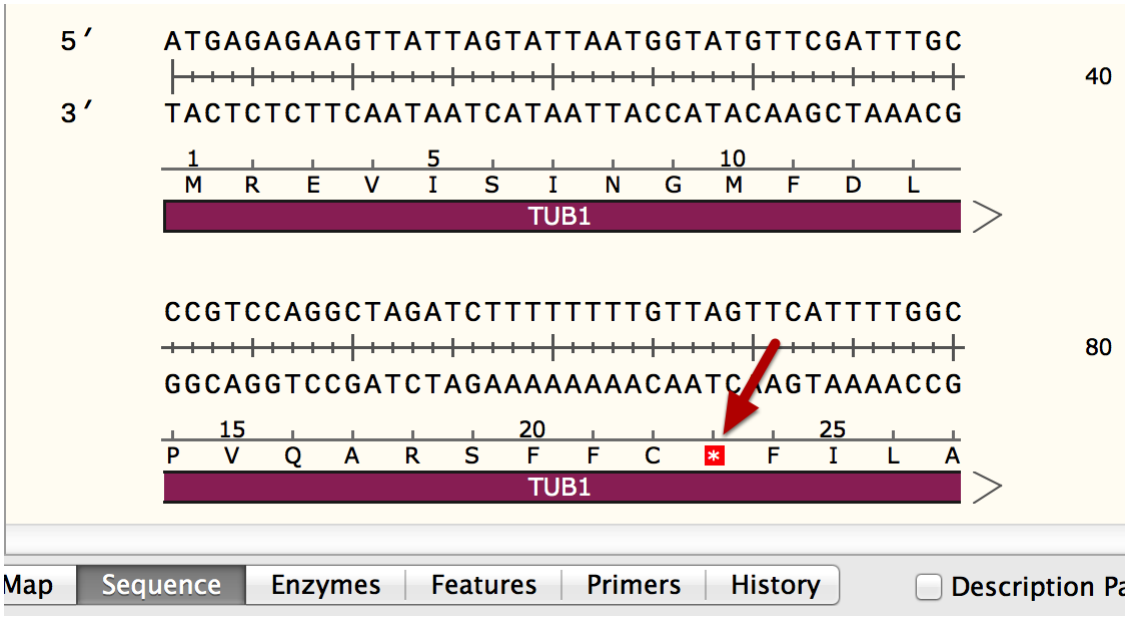



Click **OK** to see the segmented feature.

## Annotate an Intron

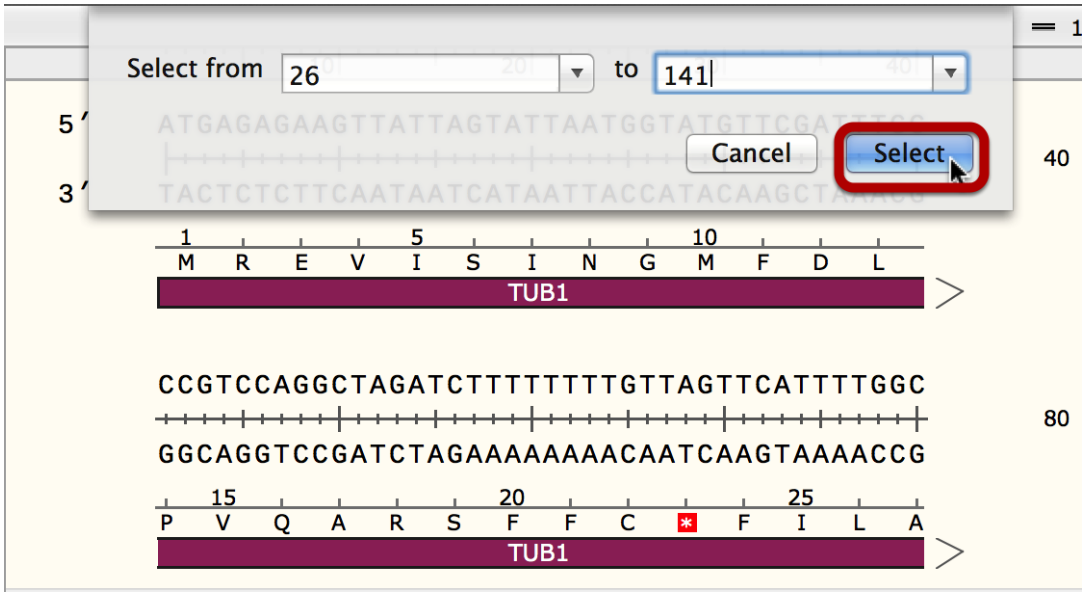
How can I add an intron annotation to a translated feature?

### View the Translated Feature



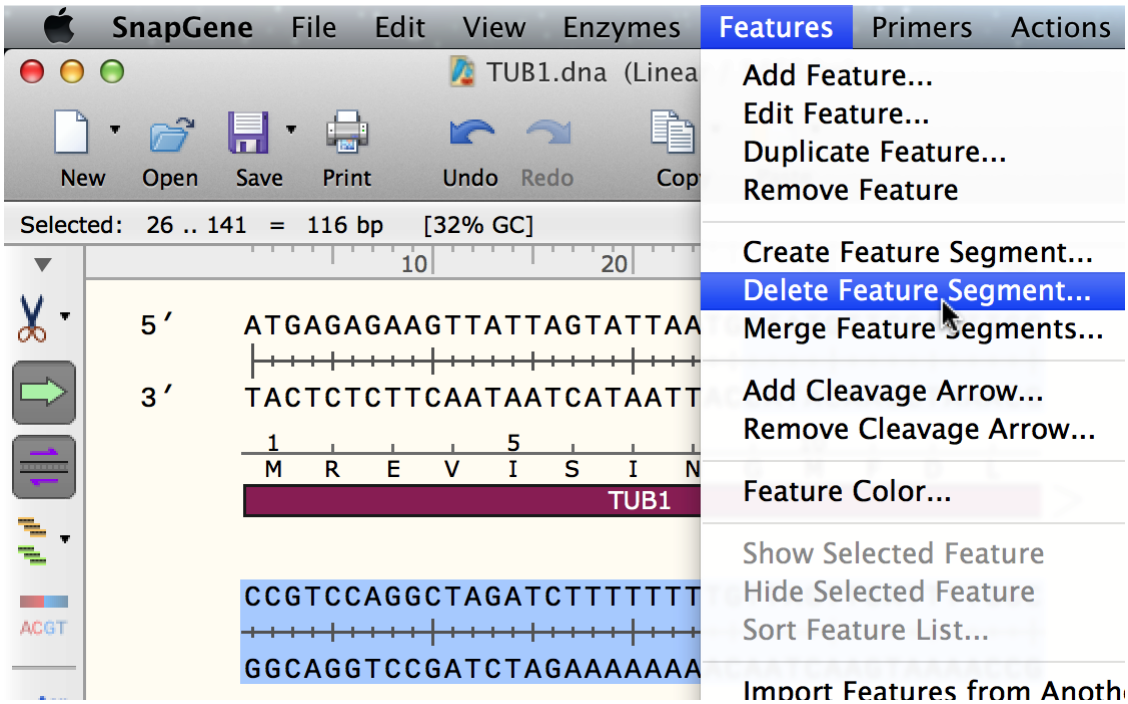

In this example, a *Saccharomyces cerevisiae* tubulin gene has been annotated as a translated feature. The intron has not yet been specified, and as a result, the translation contains stop codons.

### Select the Intron Sequence




According to the *Saccharomyces* Genome Database, the intron lies between bases 26 - 141. To specify such a numerical range click **Edit** → **Select Range...** , type the range, then click **Select**.

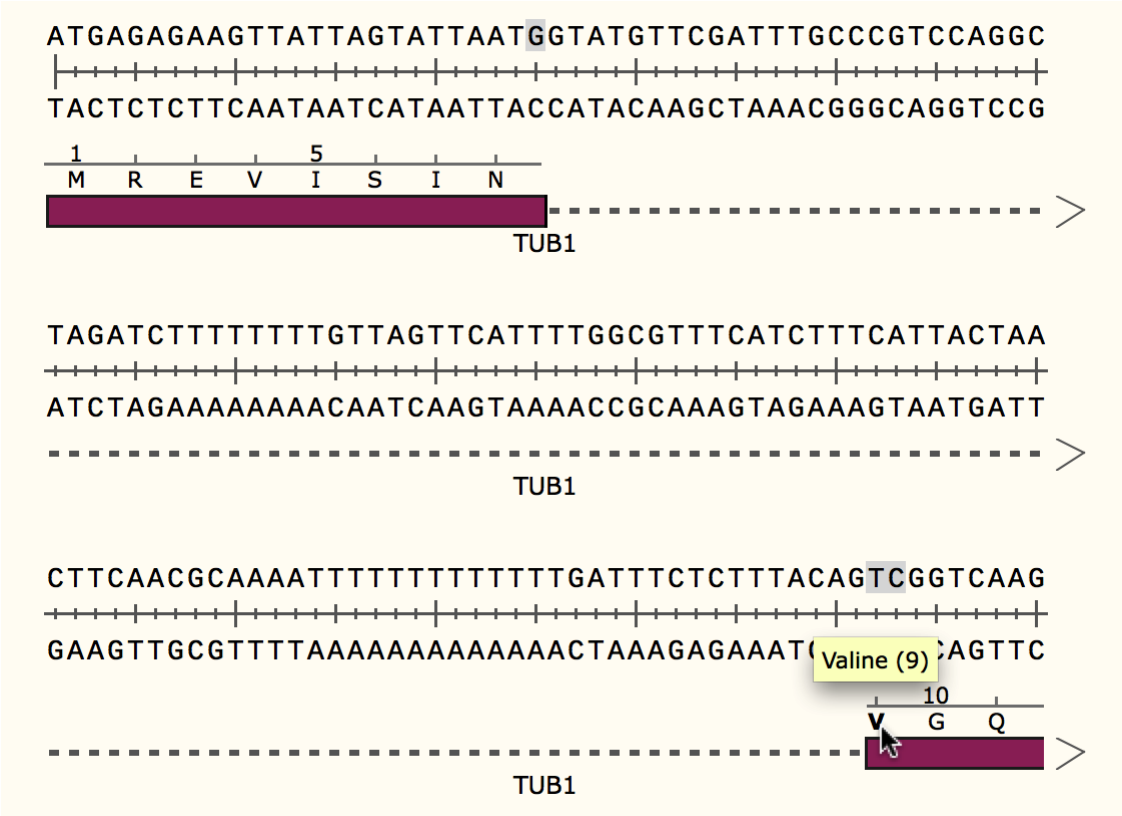
## Delete the Selected Feature Segment




To remove the intron from the feature, click **Features → Delete Feature Segment...**.

When the Edit Feature dialog appears, click **OK**.

## View the Intron

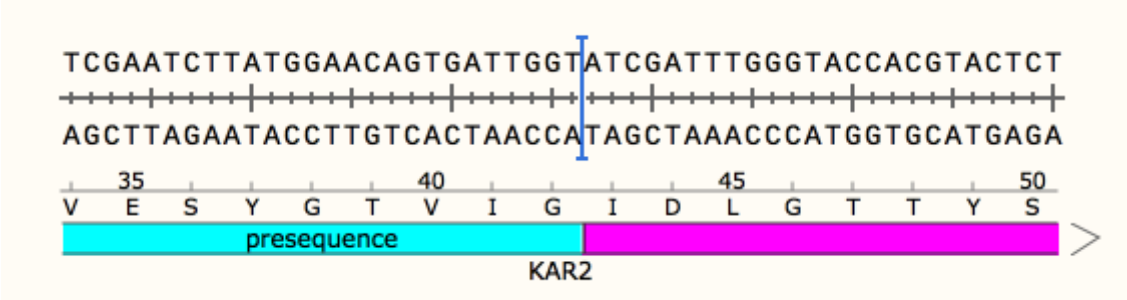



A dashed line will now mark the intron. The translation will adjust to span the intron boundaries. In this example, the codon for Valine-9 is split between the two exons.

# Add a Cleavage Arrow

How can I display a cleavage arrow in a feature?

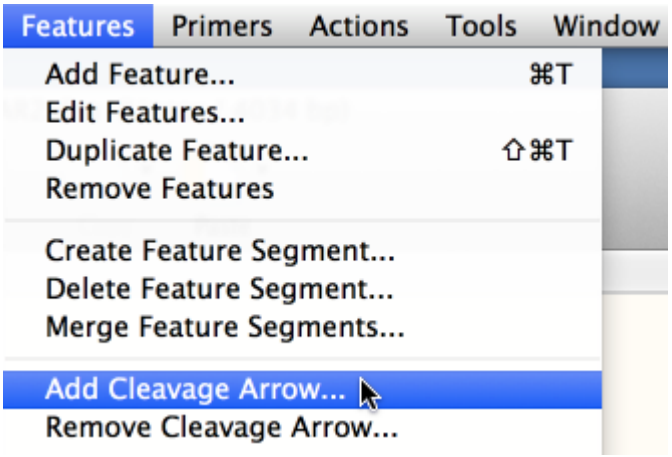
## Specify an Insertion Point




The following procedure can be performed when you first create a feature, but the lesson will assume that the feature already exists.

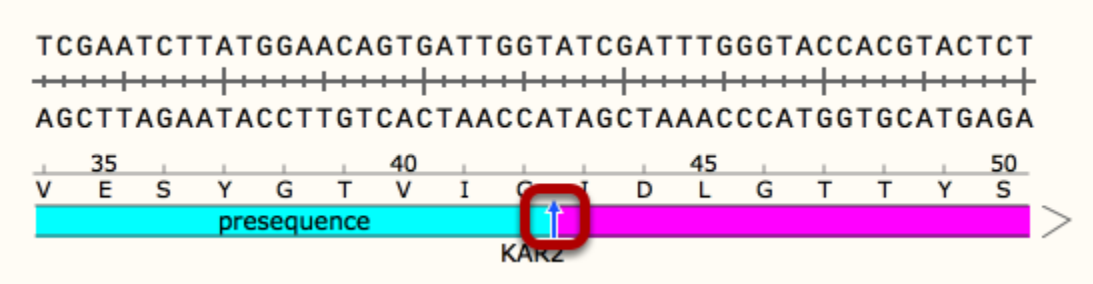
To specify the cleavage arrow location, click within the DNA sequence in Sequence view to place an insertion point cursor. In this example, the cleavage arrow will be at the boundary between two feature segments, but a cleavage arrow can be placed anywhere in a feature.

## Add a Cleavage Arrow



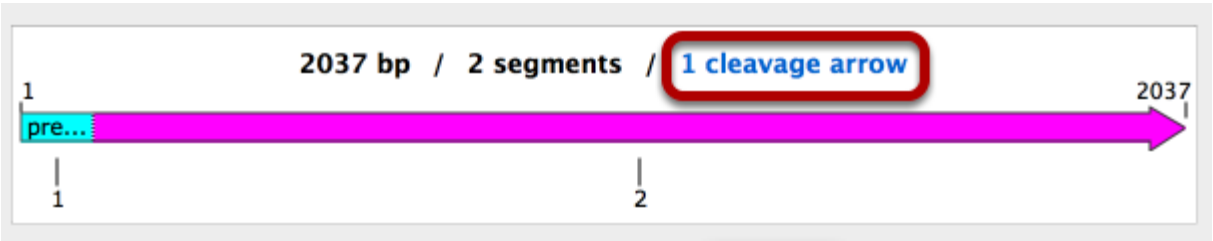

To add a cleavage arrow, click **Features** → **Add Cleavage Arrow...** .

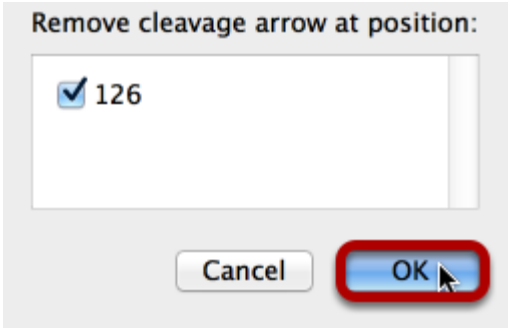
## View the Updated Feature




Click **OK** to return to Sequence view. The cleavage arrow will be displayed.

## Remove a Cleavage Arrow



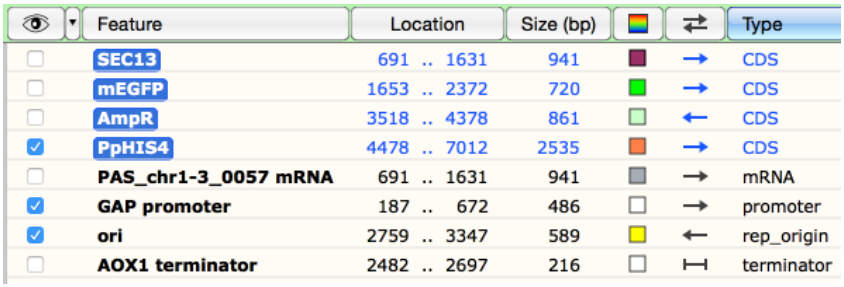

To remove a cleavage arrow, double-click the feature to open the Edit Feature dialog. Then click the blue control text above the feature diagram to list the cleavage arrows that can be removed. Click **OK** to remove cleavage arrows at the selected positions.

Alternatively, select the feature, then click **Features → Remove Cleavage Arrow...** .

# Edit Multiple Features

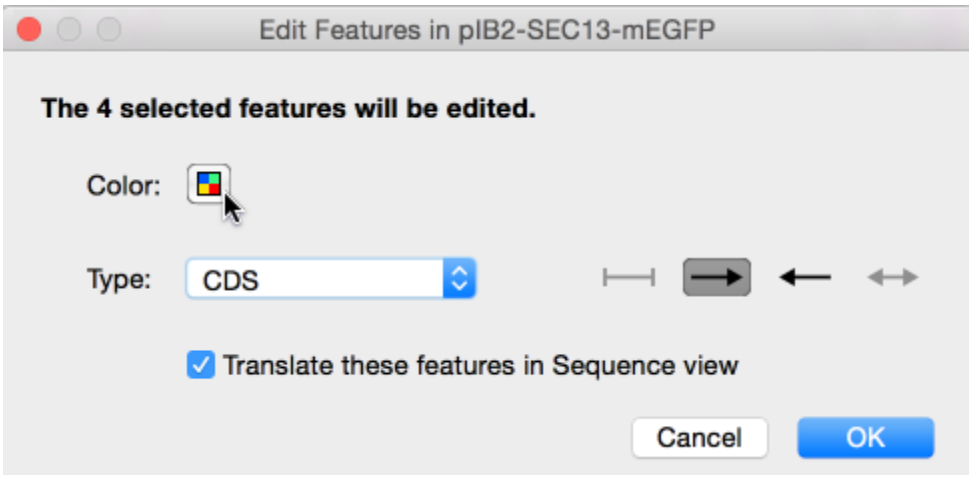
How can I edit multiple features' properties all at once?

## Select the Features



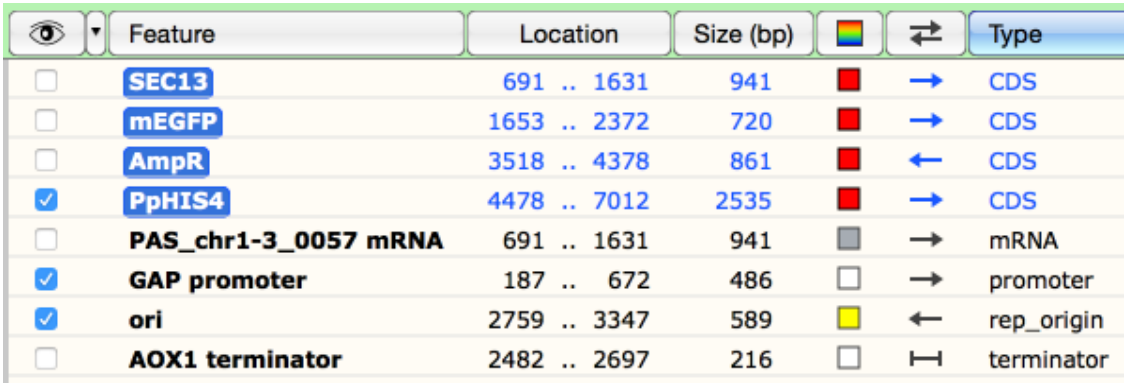

Select multiple features. To open the Edit Features dialog, click **Features** → **Edit Features**.

## Specify the Changes




Choose color, type, directionality and whether to "Translate these features in Sequence view" for all of the selected features, then click **OK**.

## View the Updated Features



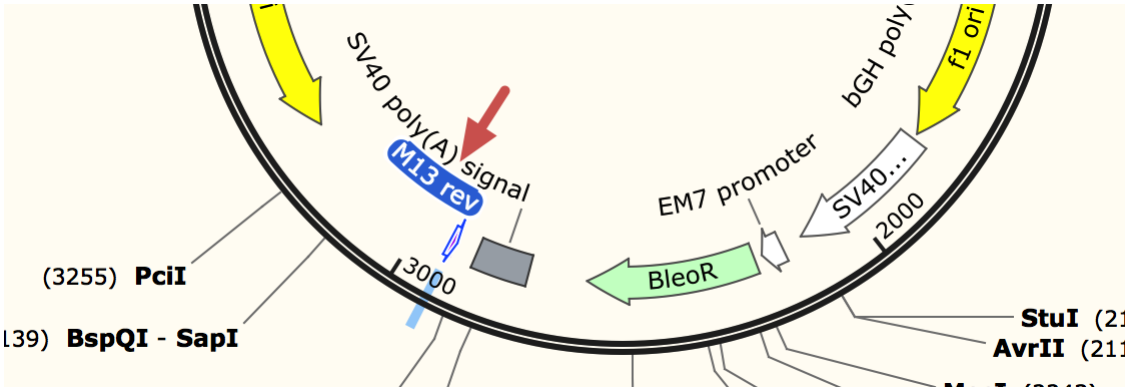
| Feature | Location | Size (bp) | Type |
|---|---|---|---|
| SEC13 | 691 .. 1631 | 941 | CDS |
| mEGFP | 1653 .. 2372 | 720 | CDS |
| AmpR | 3518 .. 4378 | 861 | CDS |
| PpHIS4 | 4478 .. 7012 | 2535 | CDS |
| PAS_chr1-3_0057 mRNA | 691 .. 1631 | 941 | mRNA |
| GAP promoter | 187 .. 672 | 486 | promoter |
| ori | 2759 .. 3347 | 589 | rep_origin |
| AOX1 terminator | 2482 .. 2697 | 216 | terminator |

All of the selected features will now have the specified changes. In this example, the color of the CDS features were all changed to red.

## Delete Features

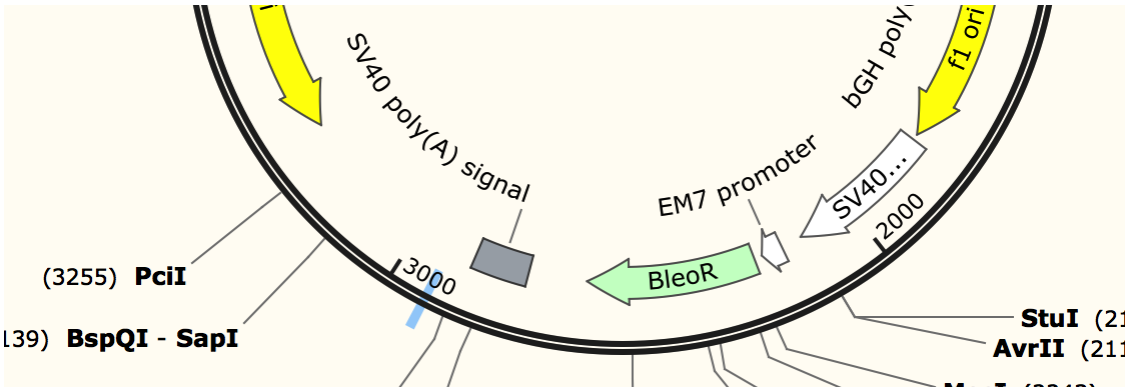
How can I delete a feature without removing the underlying DNA?

### Select the Feature




Click on the feature to select it in Map, Sequence, or Features view.

### Delete the Feature




Press the "**Delete**" key, or choose **Features → Remove Feature**. The feature will be deleted. The underlying DNA will be unchanged, and will remain selected.

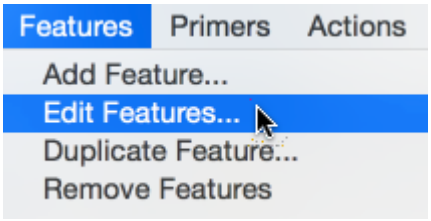
To delete the underlying DNA, press "**Delete**" again.

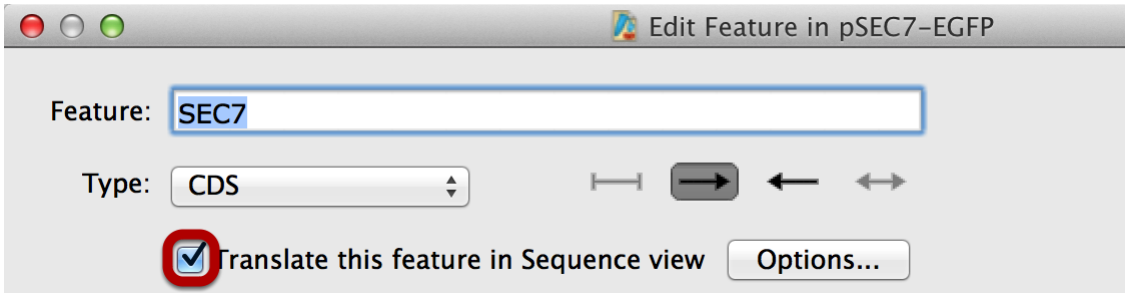
**Tip:** To delete a feature and its underlying DNA with one keystroke, hold down the Opt (Mac) or Alt (Windows) key while pressing "**Delete**".

## Confirm an In-Frame Gene Fusion

How can I confirm that two coding sequence features are in the same reading frame?

### Ensure that Both Features are Translated



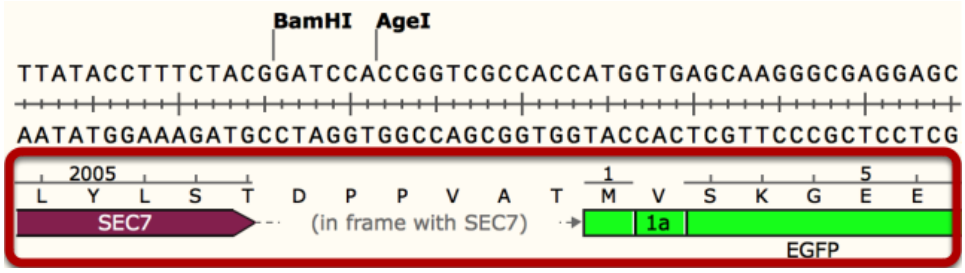

To display each feature translation in Sequence view, double-click the feature or select the feature and choose **Features** → **Edit Feature...** . Then click the **Translate this feature in Sequence view** check box.

### View the Translation Junction in Sequence View

In Sequence view, scroll to the junction of the feature translations. If the two features are in frame, they will be vertically aligned.

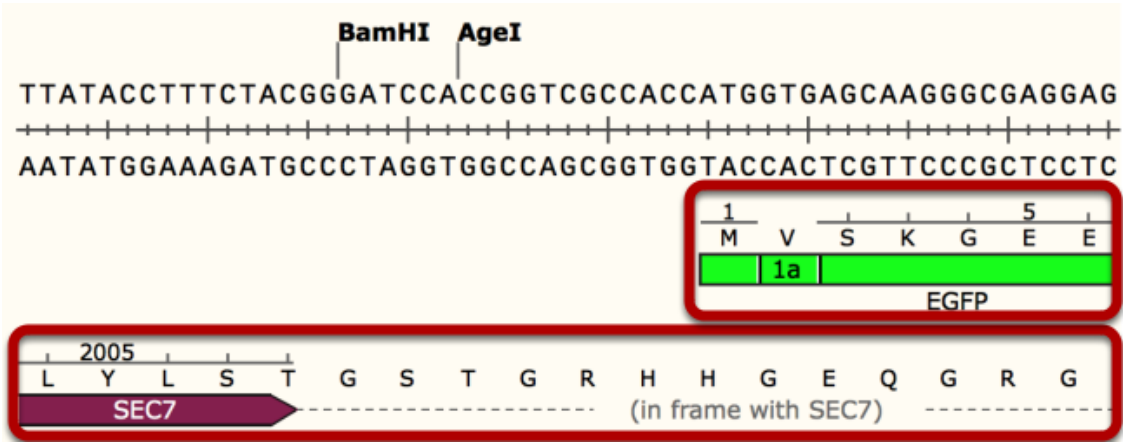
Note: The upstream feature should lack a stop codon to ensure that its translation extends beyond the end of the feature.

### In-Frame Example




If the two features are in frame, the extended translation from the upstream feature will be on the same line as the translation of the downstream feature.

## Out-of-Frame Example

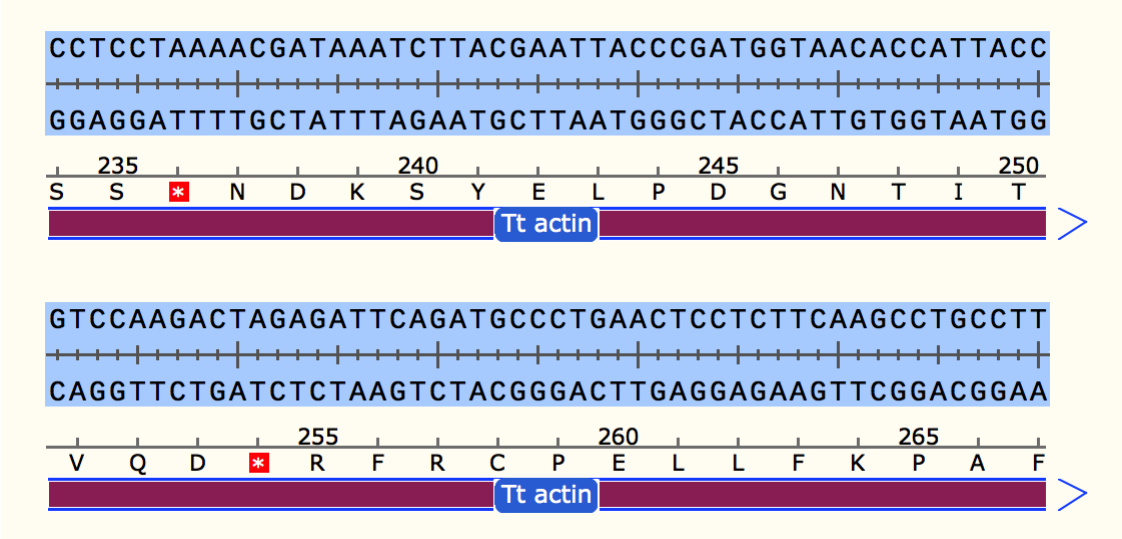



If the two features are not in frame, the extended translation from the upstream feature will not be on the same line as the translation of the downstream feature.

## Set the Genetic Code for a Feature

How can I specify a nonstandard genetic code for a translated feature?

### Select the Feature

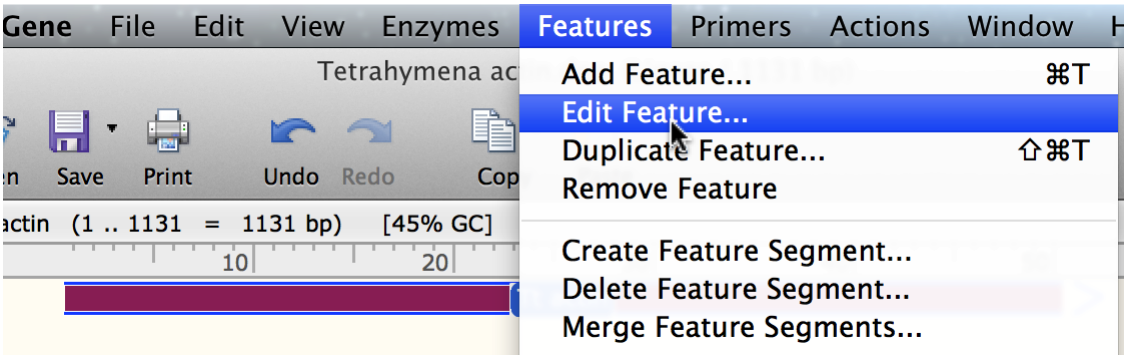



**The following procedure can be performed when you first create a translated feature, but the lesson will assume that the translated feature already exists.**

In this example, the feature is actin from the ciliate *Tetrahymena thermophila*, and stop codons are present because the standard genetic code is being used.

Click the translated feature to select it.

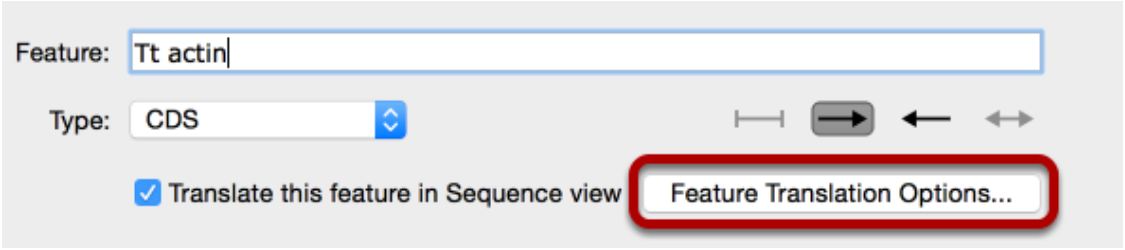
### Open the Edit Feature Dialog




To open the Edit Feature dialog, click **Features** → **Edit Feature...** .

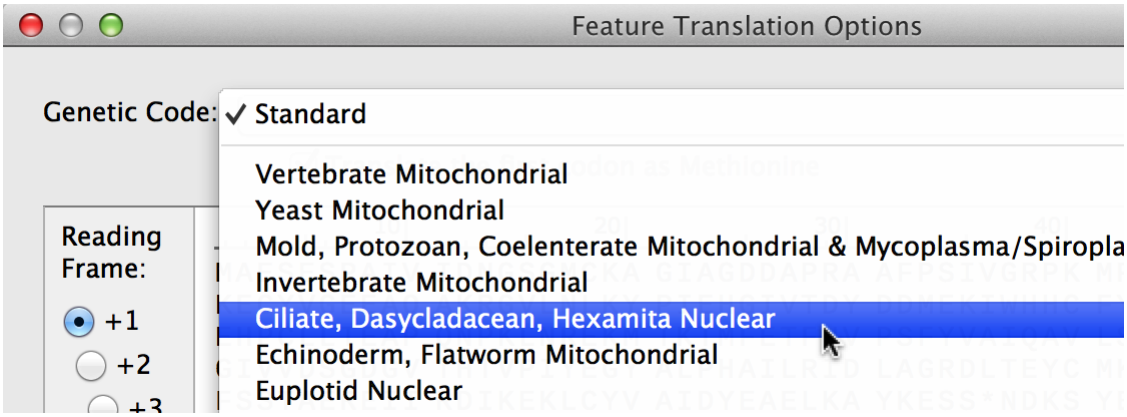
Alternatively, you can double-click the feature.

## Open the Feature Translation Options Dialog



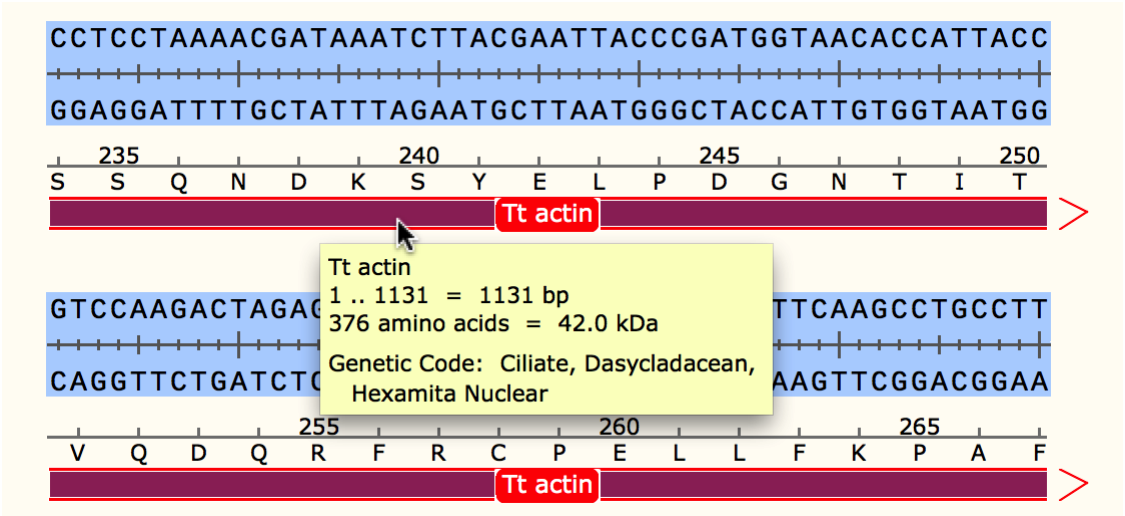

To open the Feature Translation Options dialog, click **Options...** .

## Choose the Genetic Code




Expand the **Genetic Code** menu, then choose the desired genetic code.

## View the Updated Feature Translation

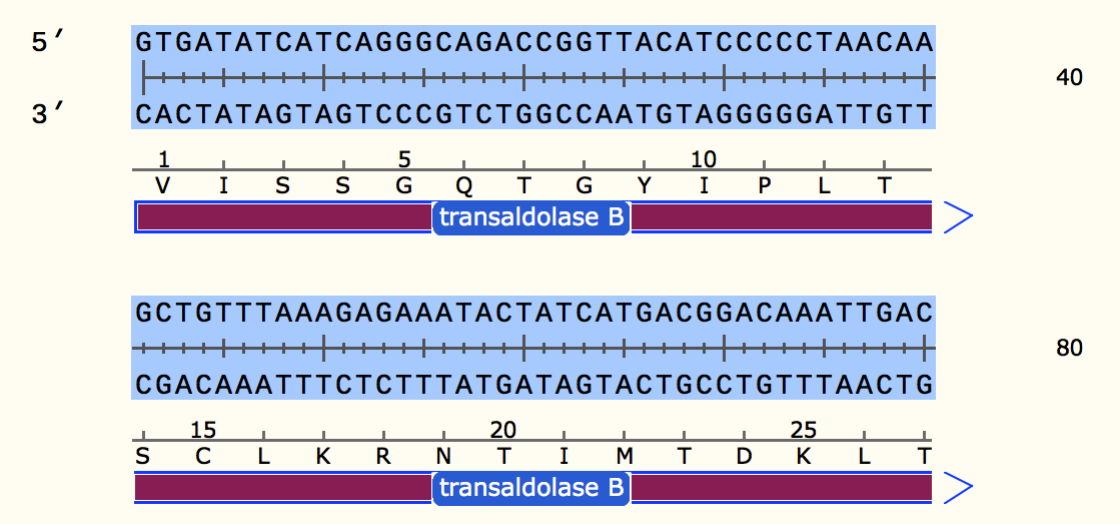



Close both dialogs by clicking **OK**. The updated genetic code will now be used in the translation.

## Designate a Non-ATG Start Codon

For a translated feature, how can I ensure that a non-ATG start codon is translated as methionine?

### Select the Feature

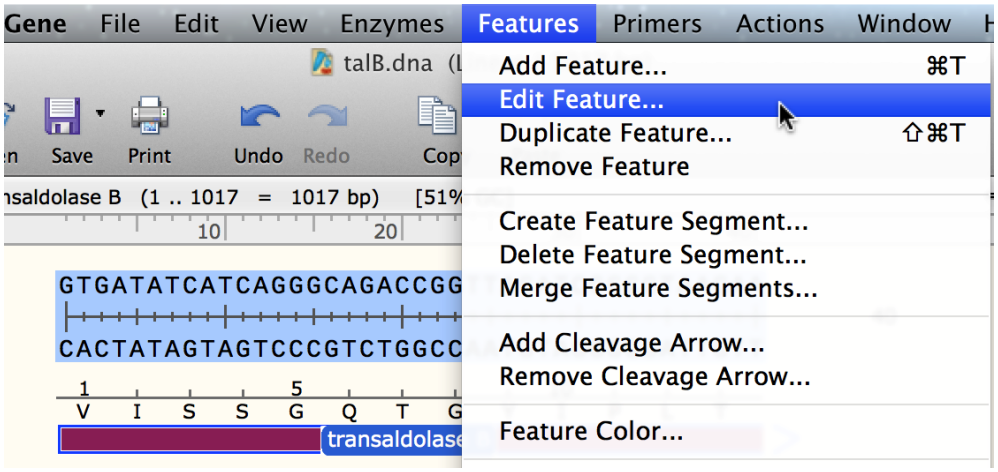



**The following procedure can be performed when you first create a translated feature, but the lesson will assume that the translated feature already exists.**

In this example, the feature is *E. coli* transaldolase B, which has a GTG start codon.

Click the translated feature to select it.

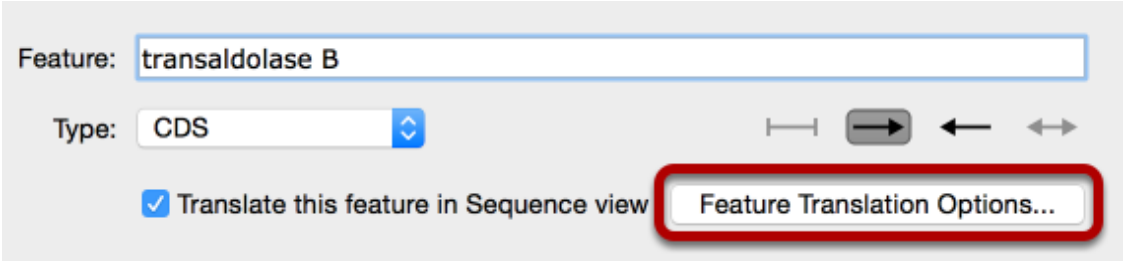
### Open the Edit Feature Dialog



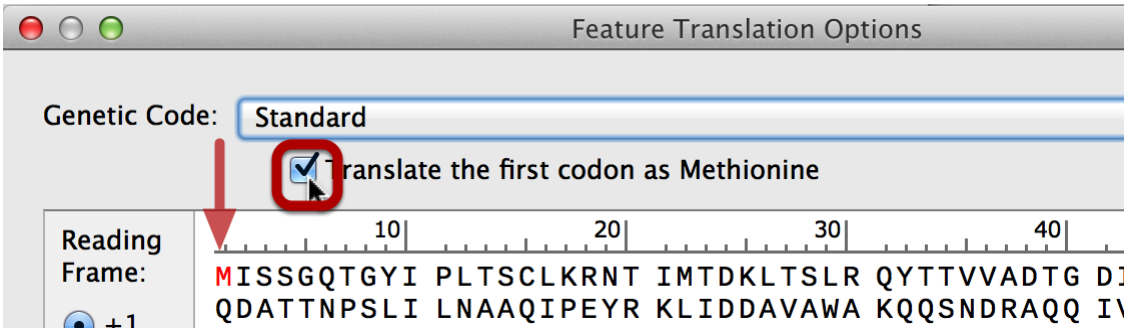

To open the Edit Feature dialog, click **Features** → **Edit Feature...** .

Alternatively, you can double-click the feature.

## Open the Feature Translation Options Dialog

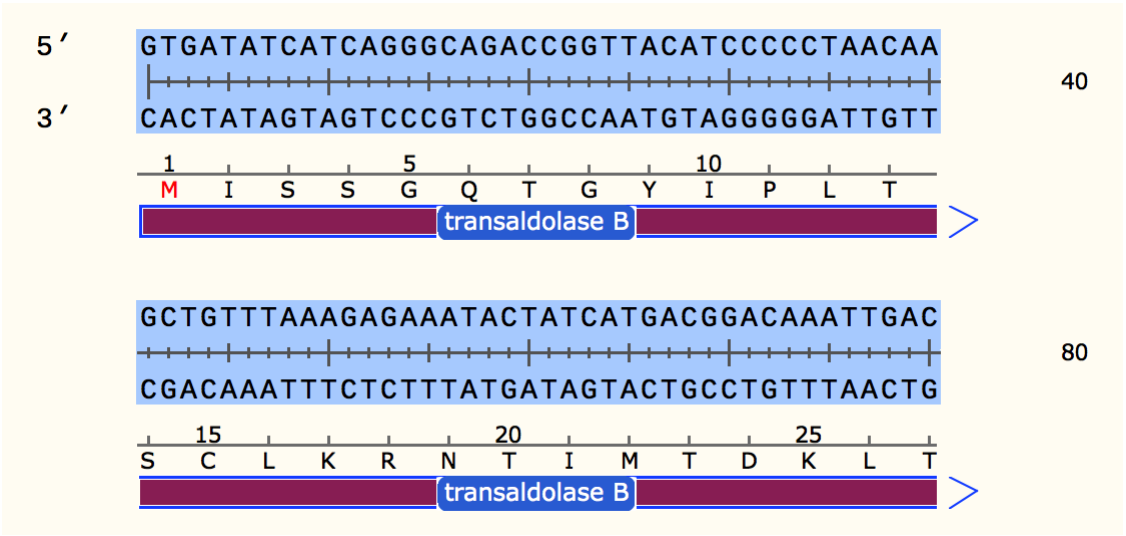



To open the Feature Translation Options dialog, click **Options...** .




Click the **Translate the first codon as Methionine** check box. The first amino acid will change to an "M", and will be shown in red.

## View the Updated Feature

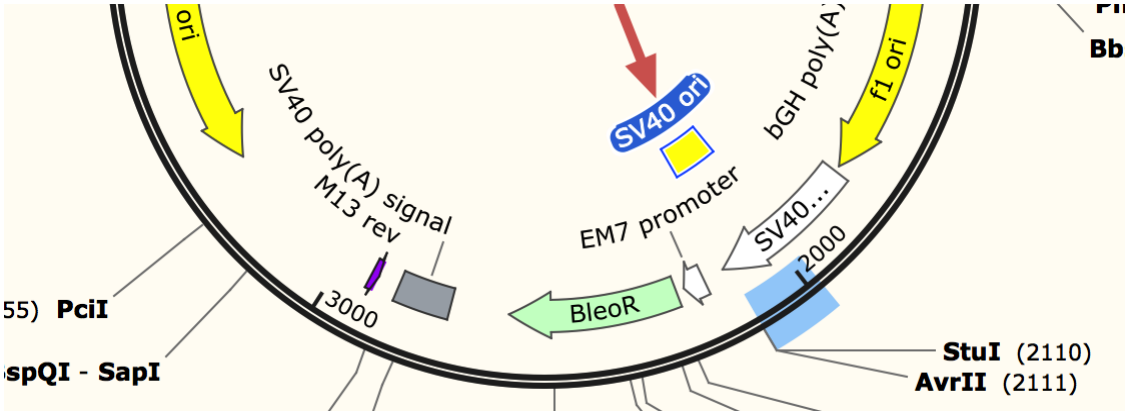



Close both dialogs by pressing **OK**. The start codon will now be translated as methionine.

## Hide or Show Features

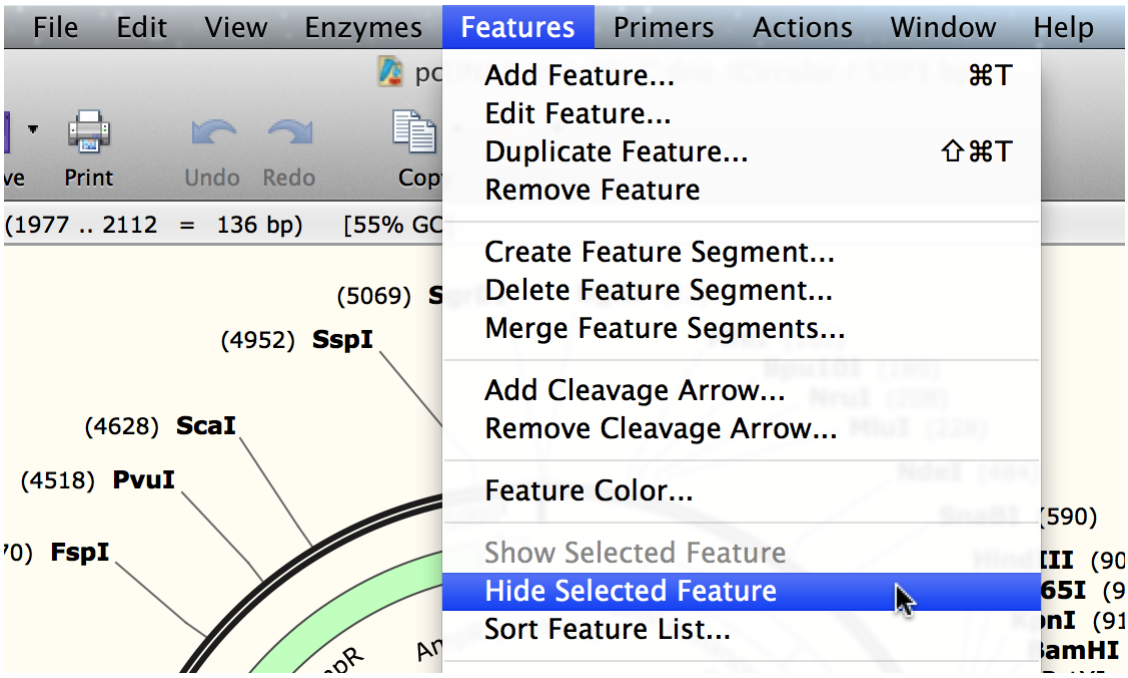
How can I change the visibility of a feature?

### Select the Feature




Click on the feature to select it in Map, Sequence, or Features view.

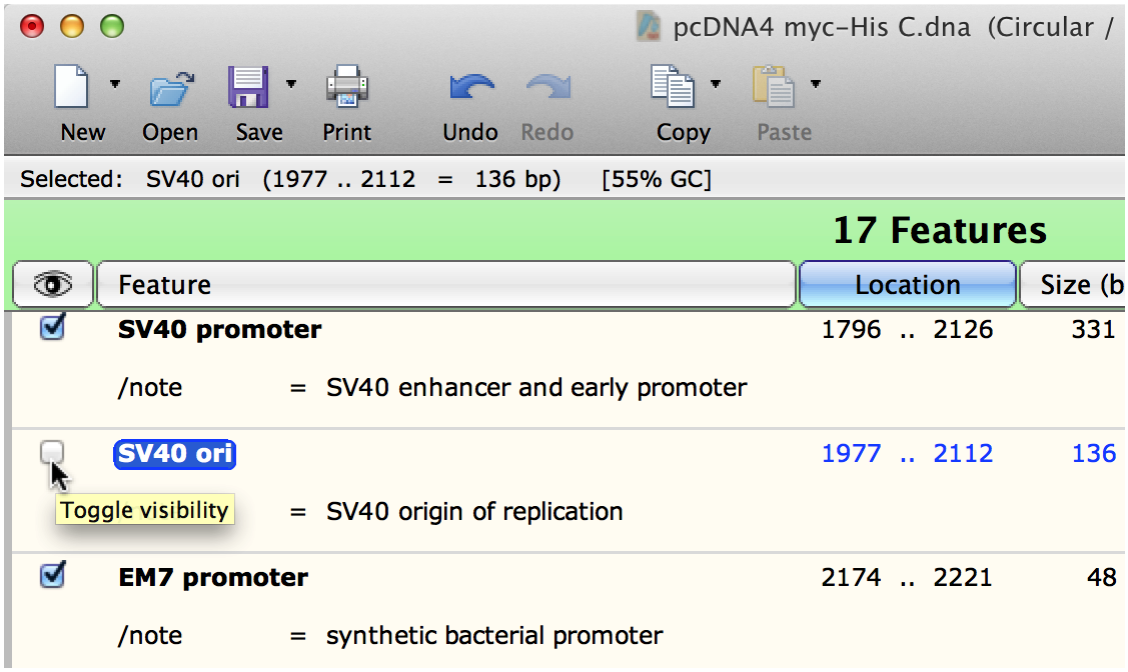
### Hide the Feature




Choose **Features → Hide Selected Feature**.

If multiple features were selected, you can use **Features → Hide Selected Features** to hide all of them.

## Show the Feature



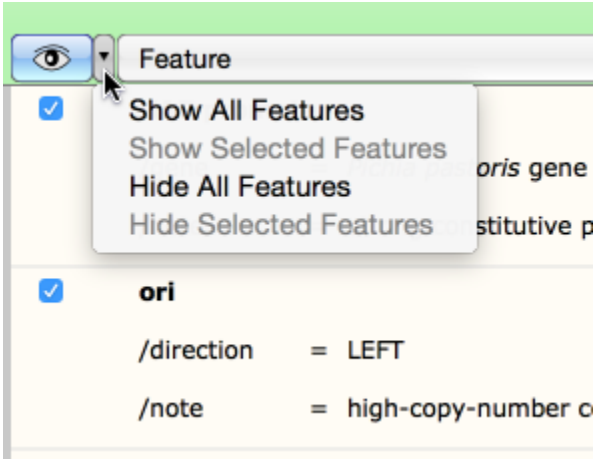

To show a hidden feature, switch to Features view. Click the empty check box to the left of the feature name to make the selected feature visible in the other views.

Alternatively, you can click on the feature name to select it, and then choose **Features → Show Selected Feature**.

## Change Multiple Features' Visibility

To change the visibility of multiple features, select the features in Features view, then click **Features → Show Selected Features**.

## Change Feature Visibility Using the Sorting "Eye" Button




To change the visibility of features, choose from the menu options next to the "Eye" sorting button.

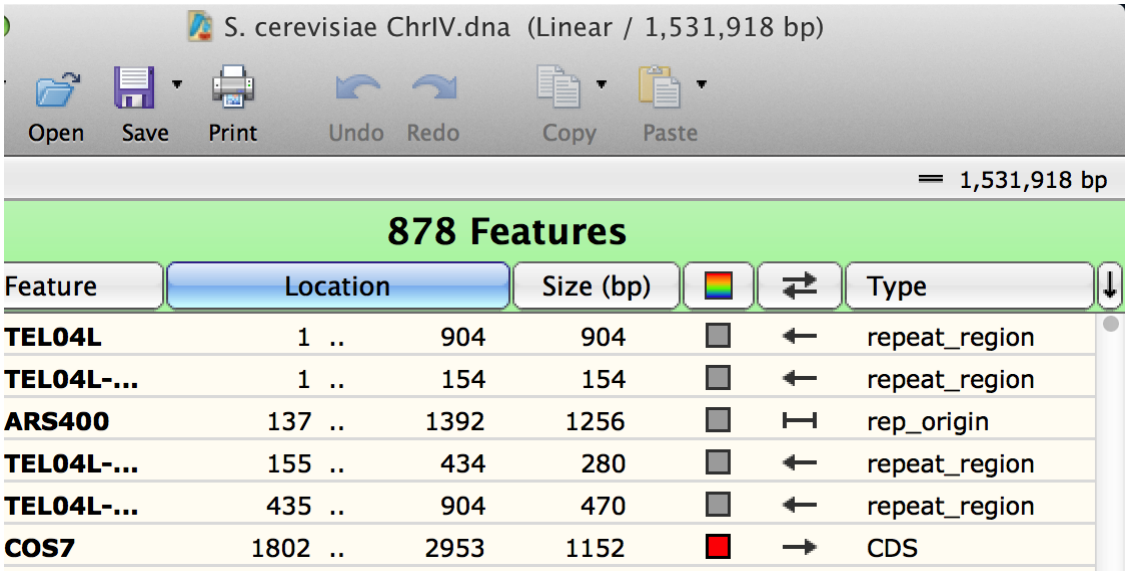
## Toggle Feature Visibility Using the Side Toolbar

In Map or Sequence view, clicking the "Show features" (green arrow) button in the side toolbar toggles the display of the features whose check boxes are checked in Features view.

## Sort the Feature List

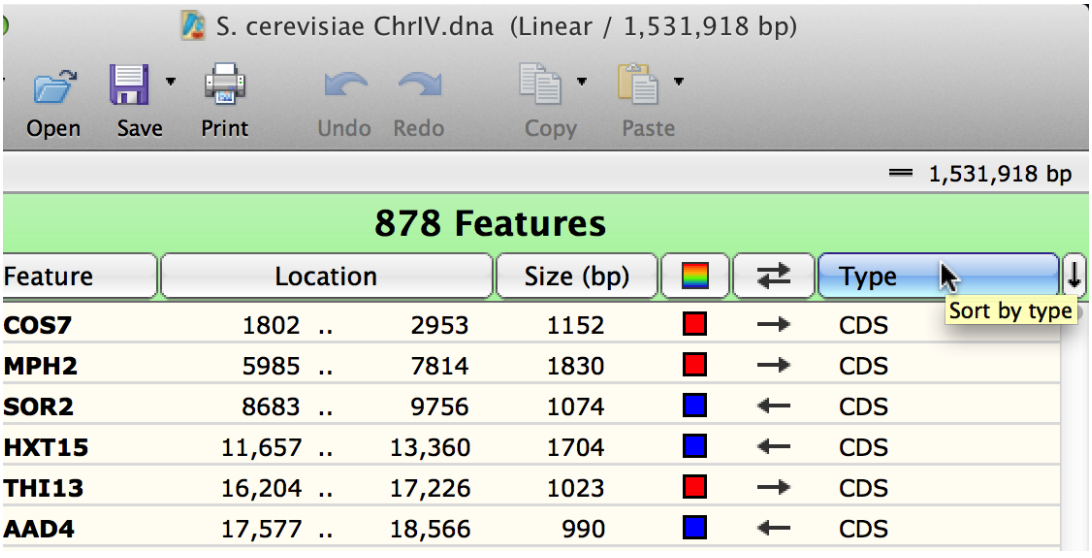
How can I sort the list in Features view?

### View the Feature List




The column headers in Features view are also sorting buttons. By default, features are sorted primarily by **Location**.

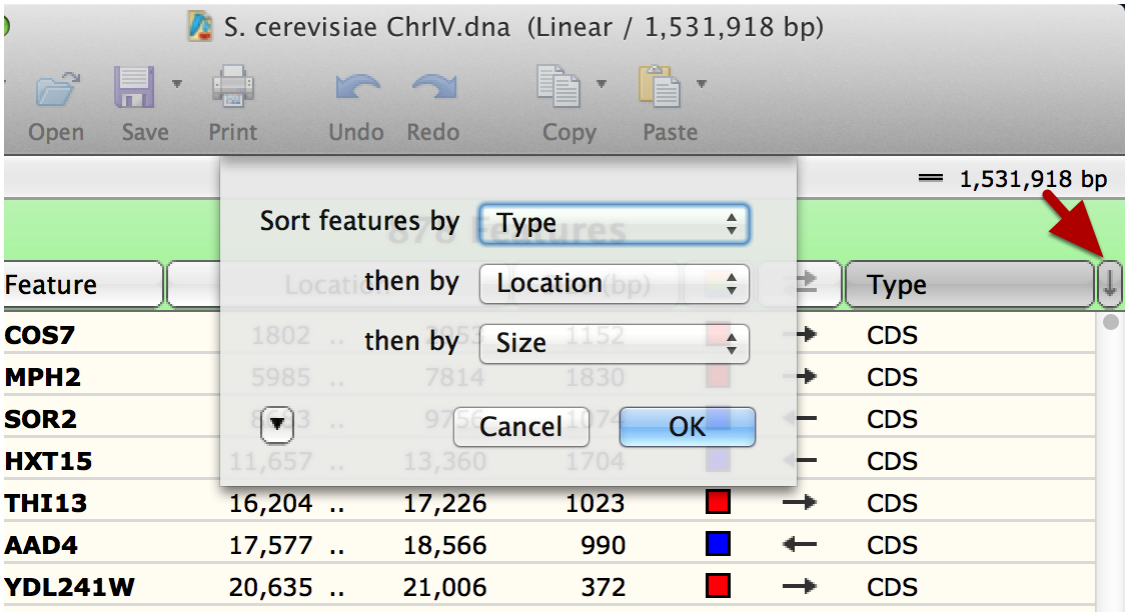
### Change the Primary Sorting Parameter




To sort primarily by another parameter, press the appropriate column button.

In this example, the features are now sorted primarily by **Type**, then by **Location**.

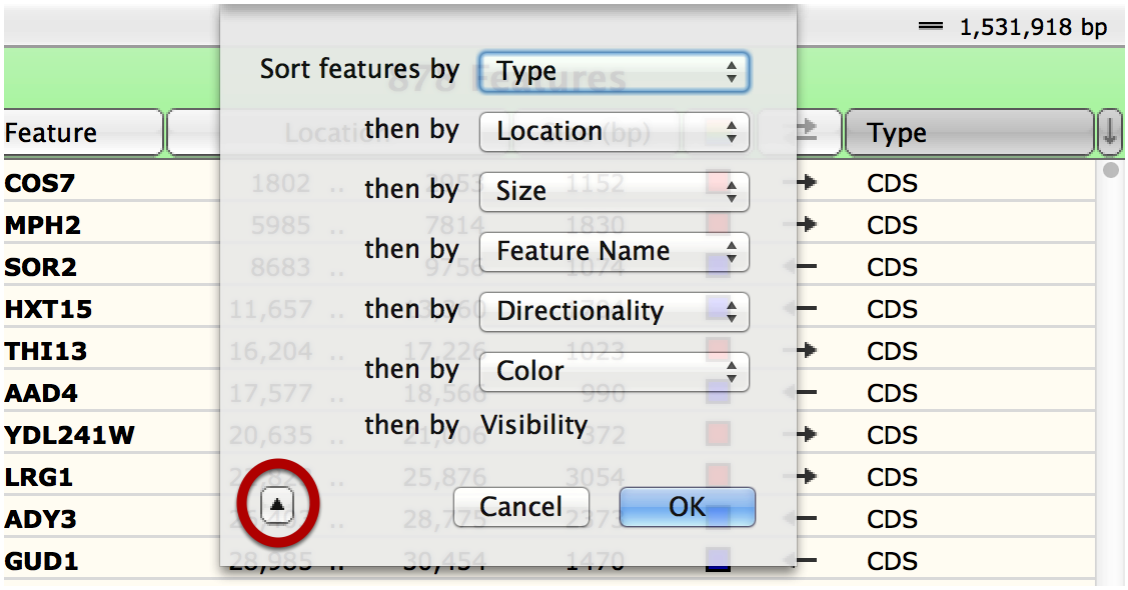
## Specify Additional Sorting Parameters




To specify secondary and tertiary sorting parameters, press the button at the far right side of Features view to open the Sort Feature List dialog.

Alternatively, you can click **Features** → **Sort Feature List...** .

## Completely Specify the Sorting Parameters

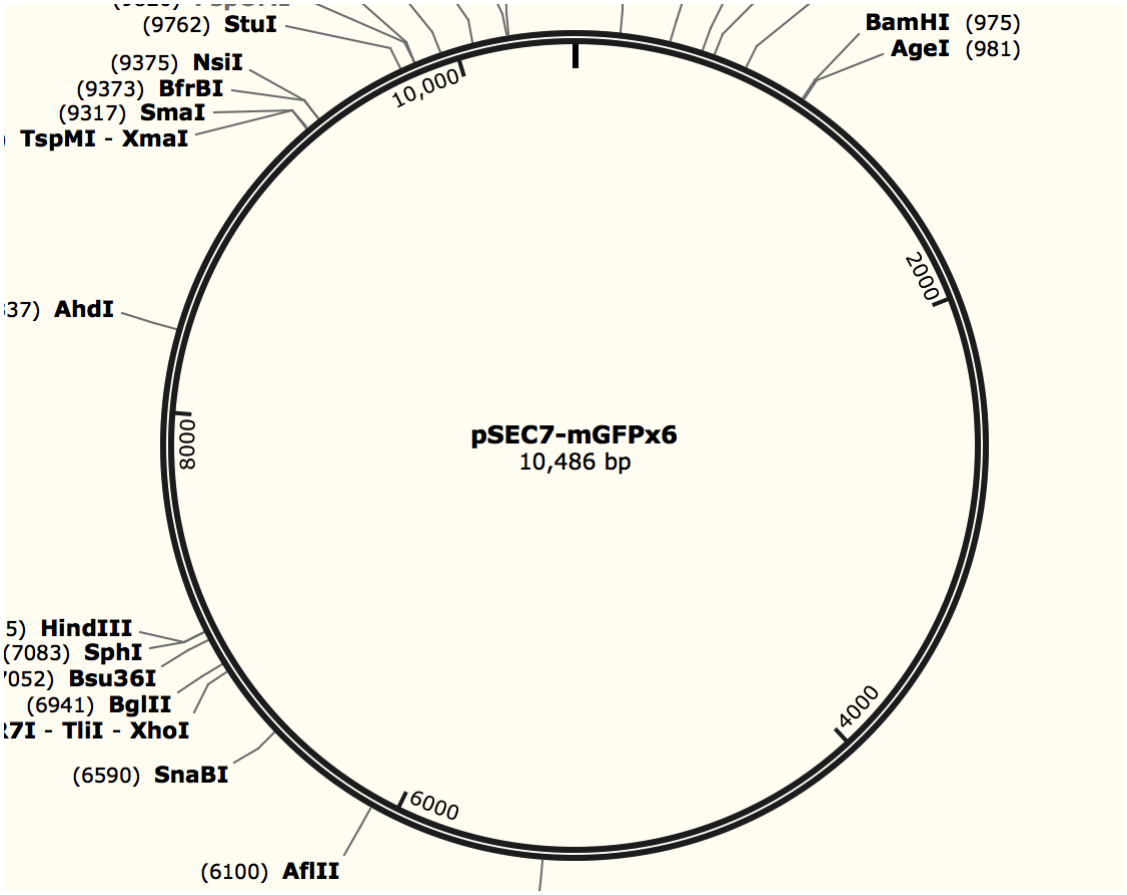



For complete control of the sorting parameters, press the disclosure button in the lower left corner of the dialog.

## Import Feature Data

How can I copy feature annotations from one file to another?

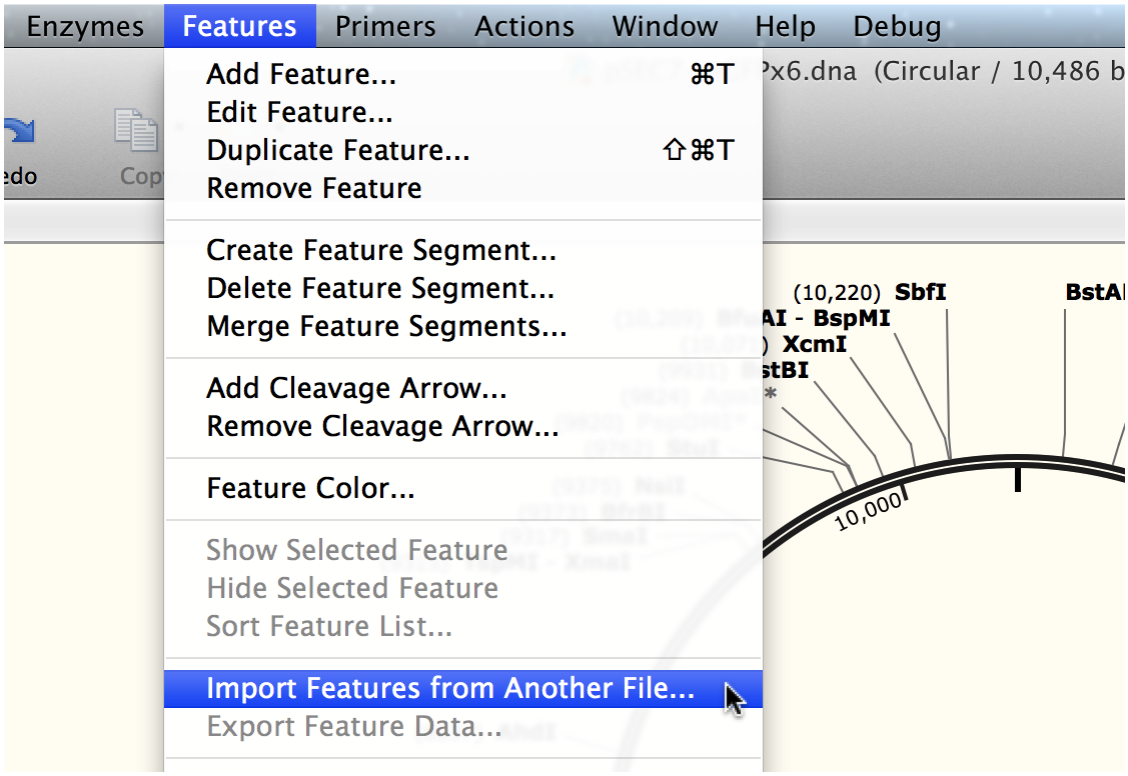
### Open the Destination File



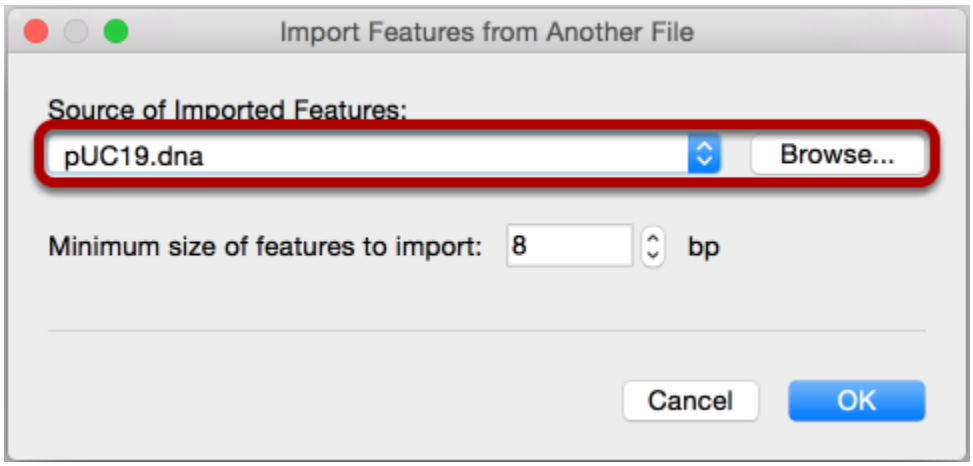

Open the file that will receive the imported features.

In this example, the destination file has no features. If the destination file already has features, SnapGene will not duplicate existing features by default, but will offer the option of importing duplicate features.

## Specify the Source File

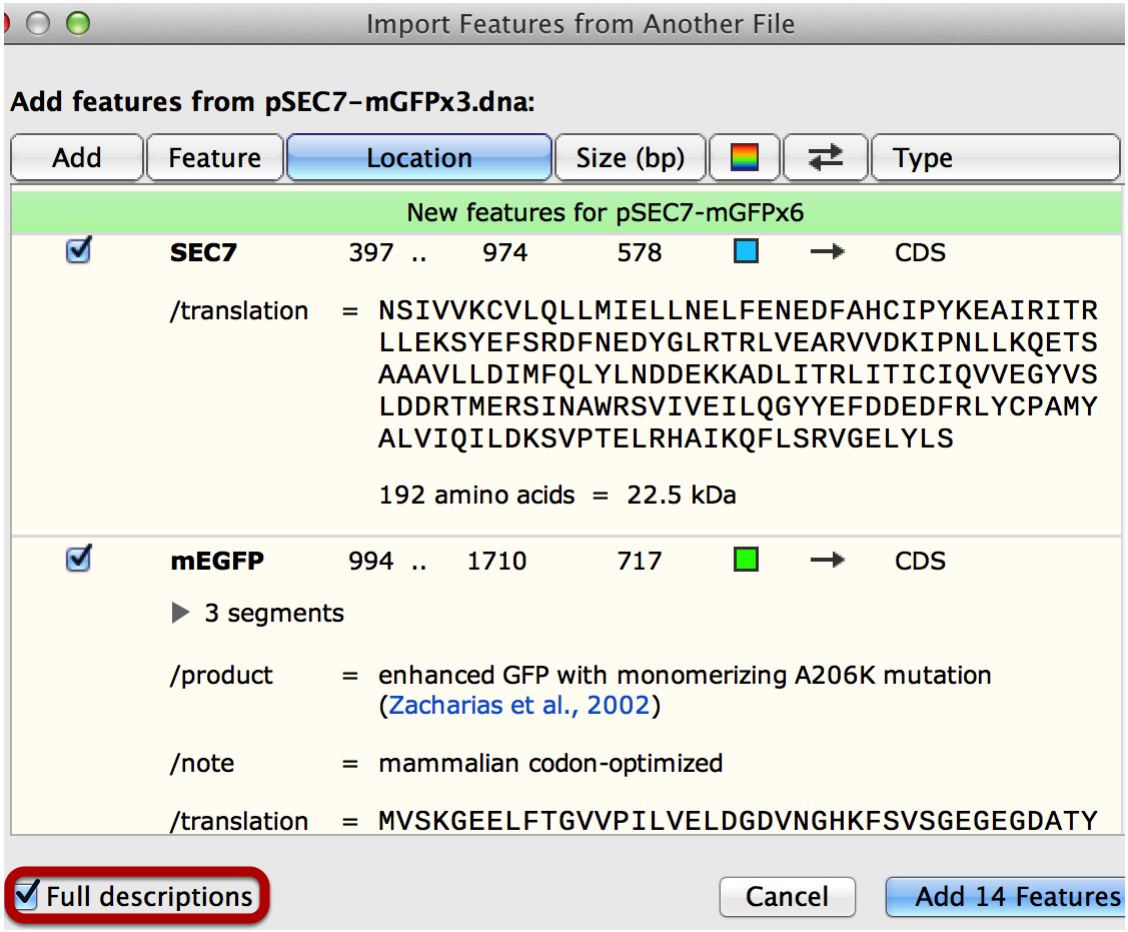



To specify the source file, click **Features** → **Import Features from Another File...** .



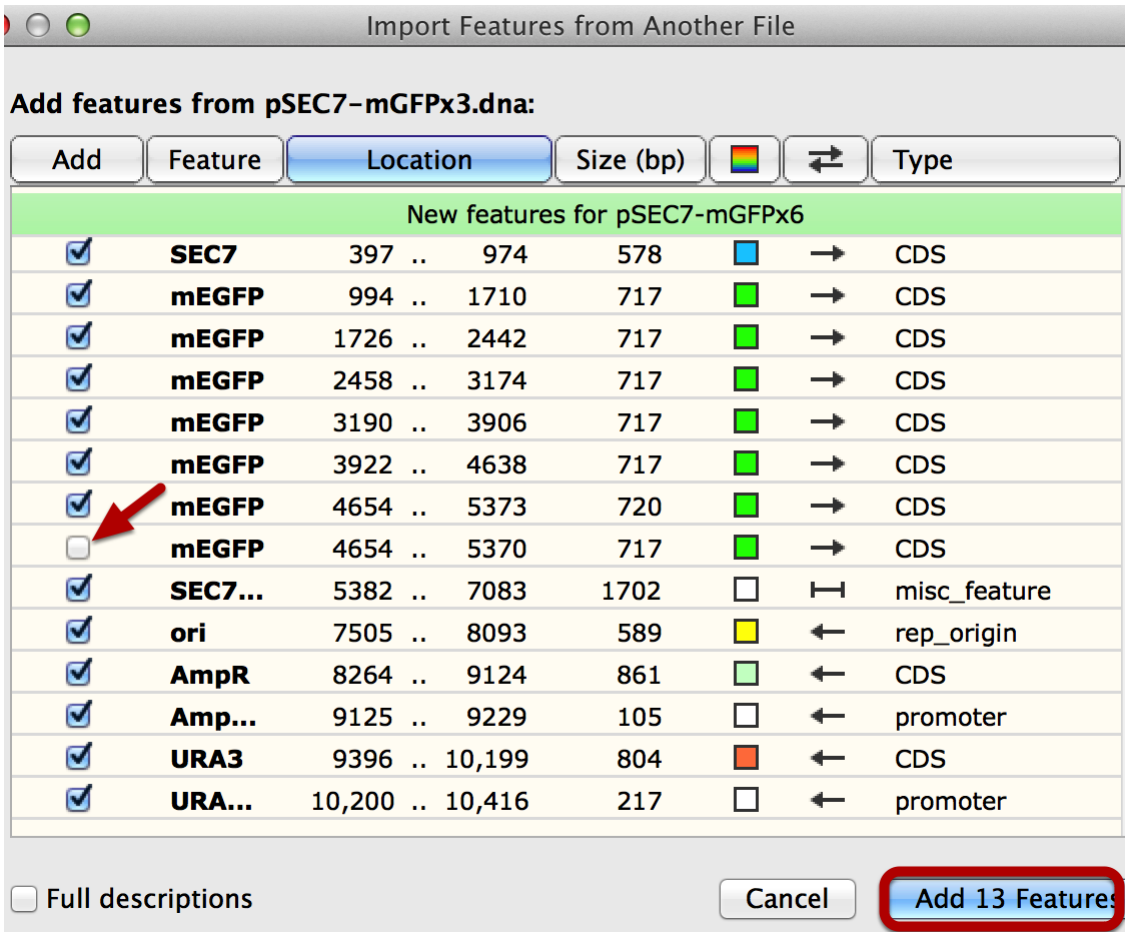

In the Import Features from Another File dialog, choose a *.dna source file that has the desired feature annotations, then click **OK**.

## View the Available Features List




A second dialog will list the features that are to be imported. If desired, click the check box at the lower left to make the list more compact.

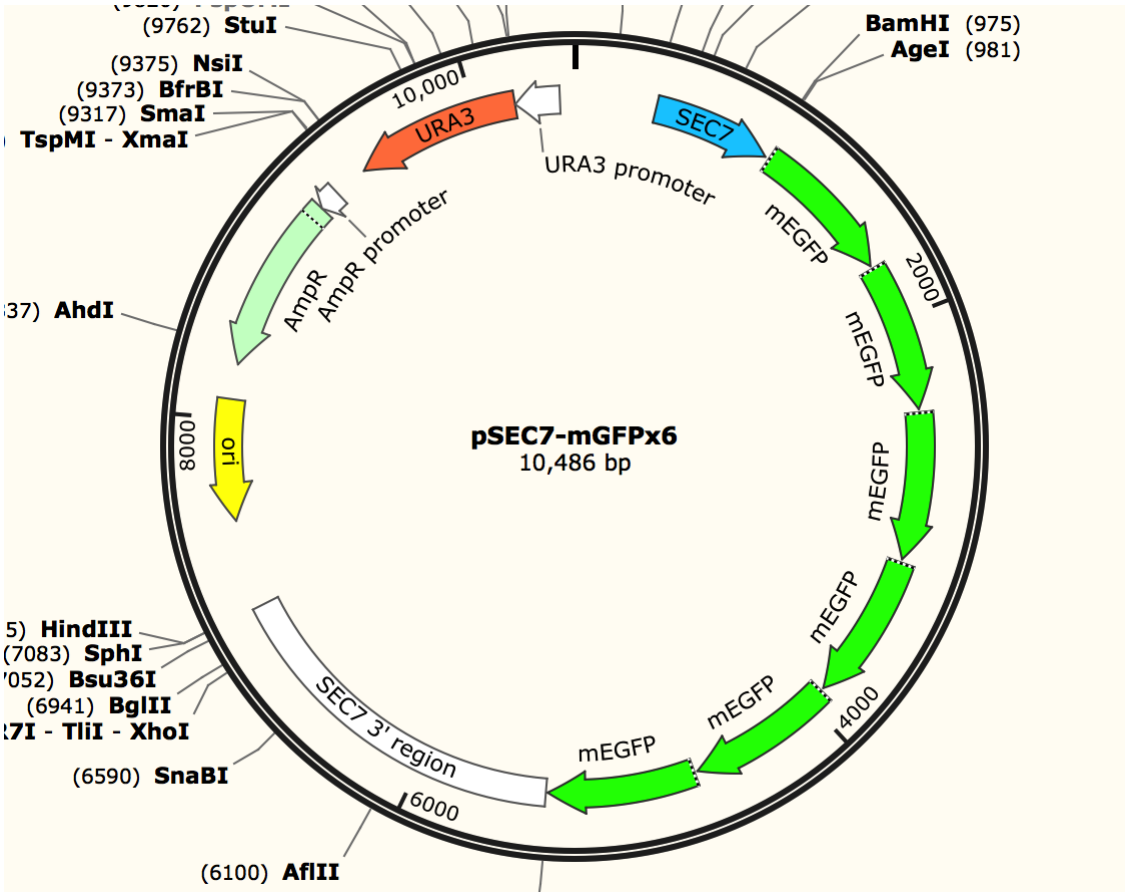
## Specify Which Features Will Be Added




The **Add** column indicates which features will be imported. To change the default choices, click the relevant check boxes, then click **Add [N] Features**.

In this example, the final mEGFP feature would have been annotated twice, once with and once without the stop codon. The check box for the version without the stop codon was manually unchecked.

## View the Imported Features

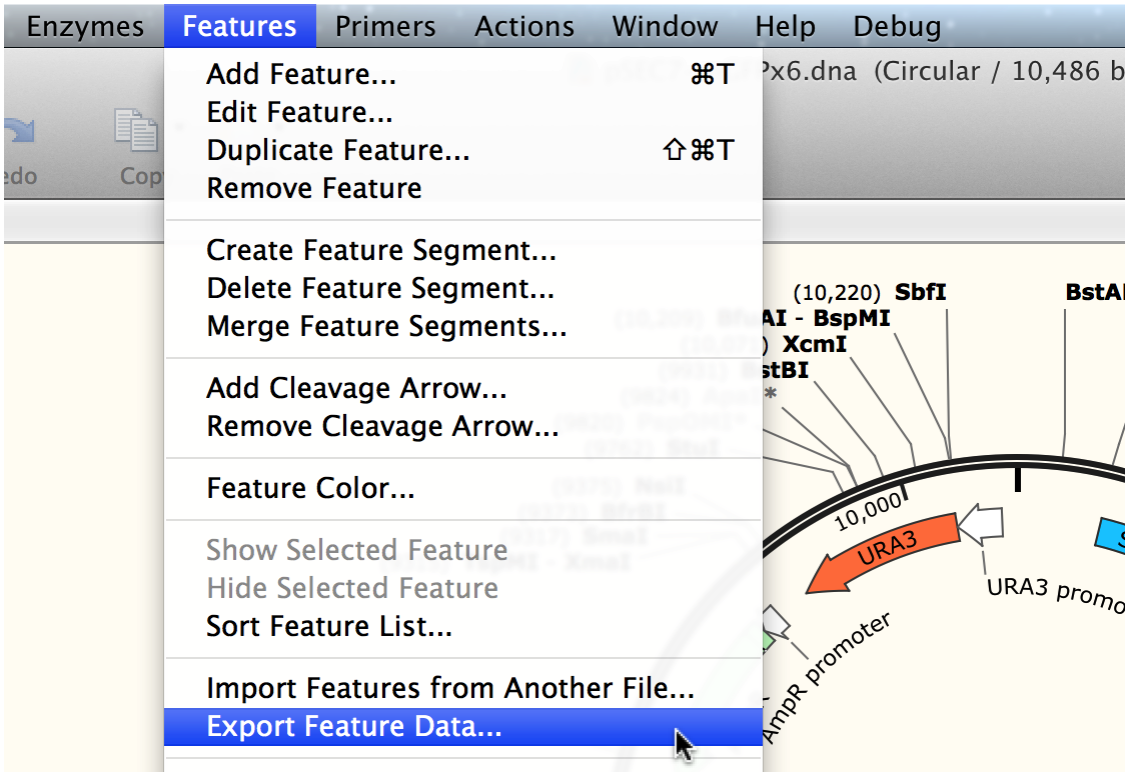



The imported features will be added to the destination file.

# Export Feature Data

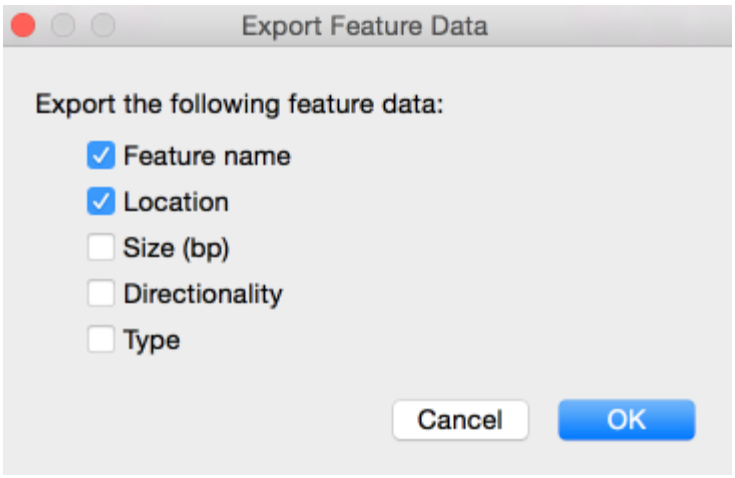
How can I export feature names, positions, and other data to a text file?

## Specify the Data to Be Exported



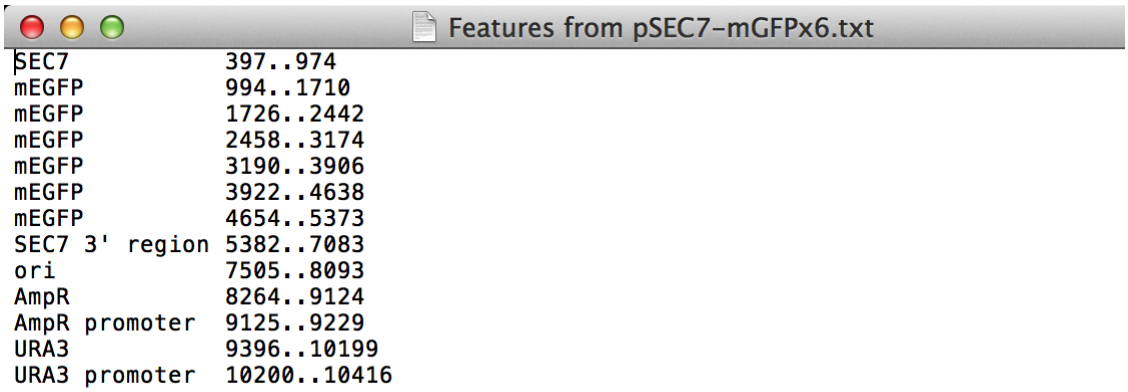

Open the file containing the feature data of interest. Choose **Features** → **Export Feature Data...** .

## Specify the Feature Data to Export




In the resulting dialog, specify which properties of the features will be exported, then click **OK**.

## View the Exported Feature Data

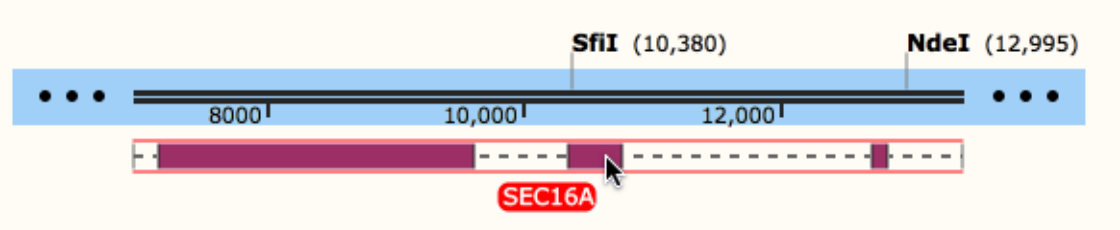



The exported data will be listed in a legibly formatted, text file. This text file is intended for use in other applications, and is not suitable for importing feature data into SnapGene.

## Splice to Remove Introns

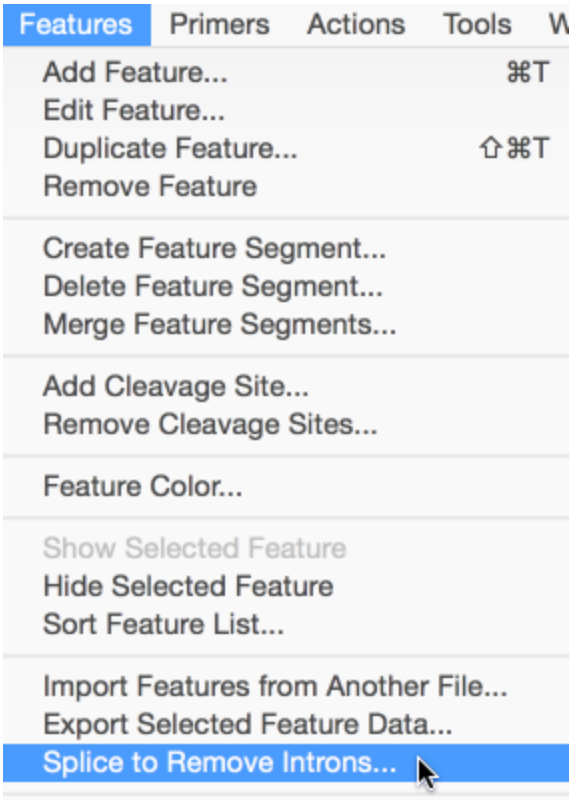
How can I remove introns from a feature to obtain the cDNA sequence?

### Select the Feature

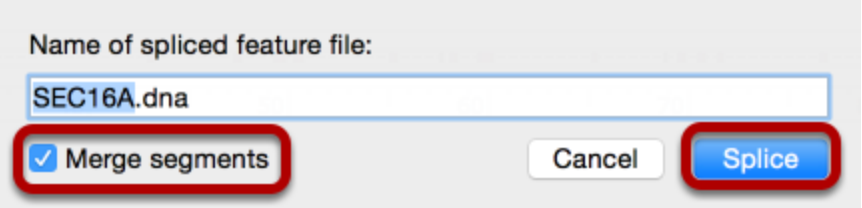



In Map, Sequence, or Features view, click to select a feature that contains introns.

### Splice to Remove Introns

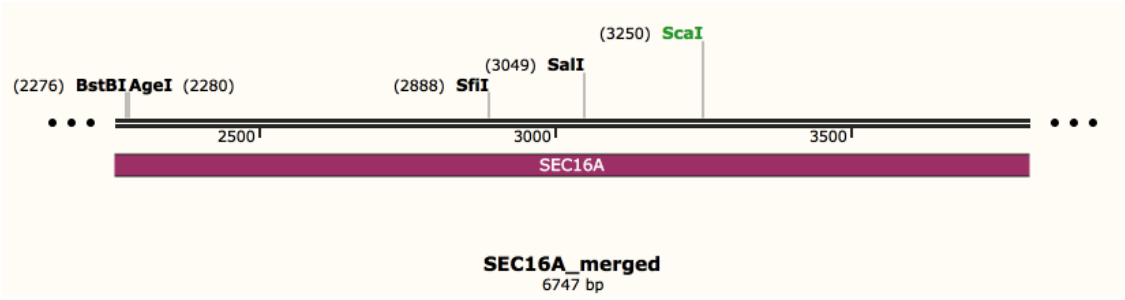



To remove introns from the feature, click **Features** → **Splice to Remove Introns...** .

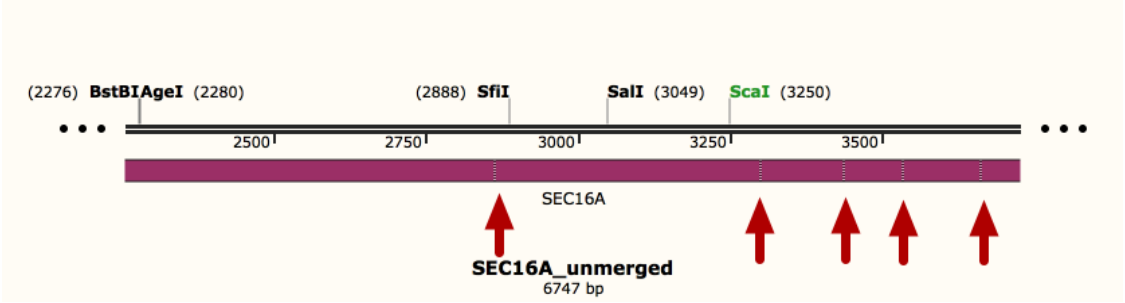



If desired, adjust the file name of the new sequence. To merge the segments from the original feature, click the check box. When you are ready to make the new file, click **Splice**.

## View the Sequence




If the segments were merged, you will see a seamless spliced feature.

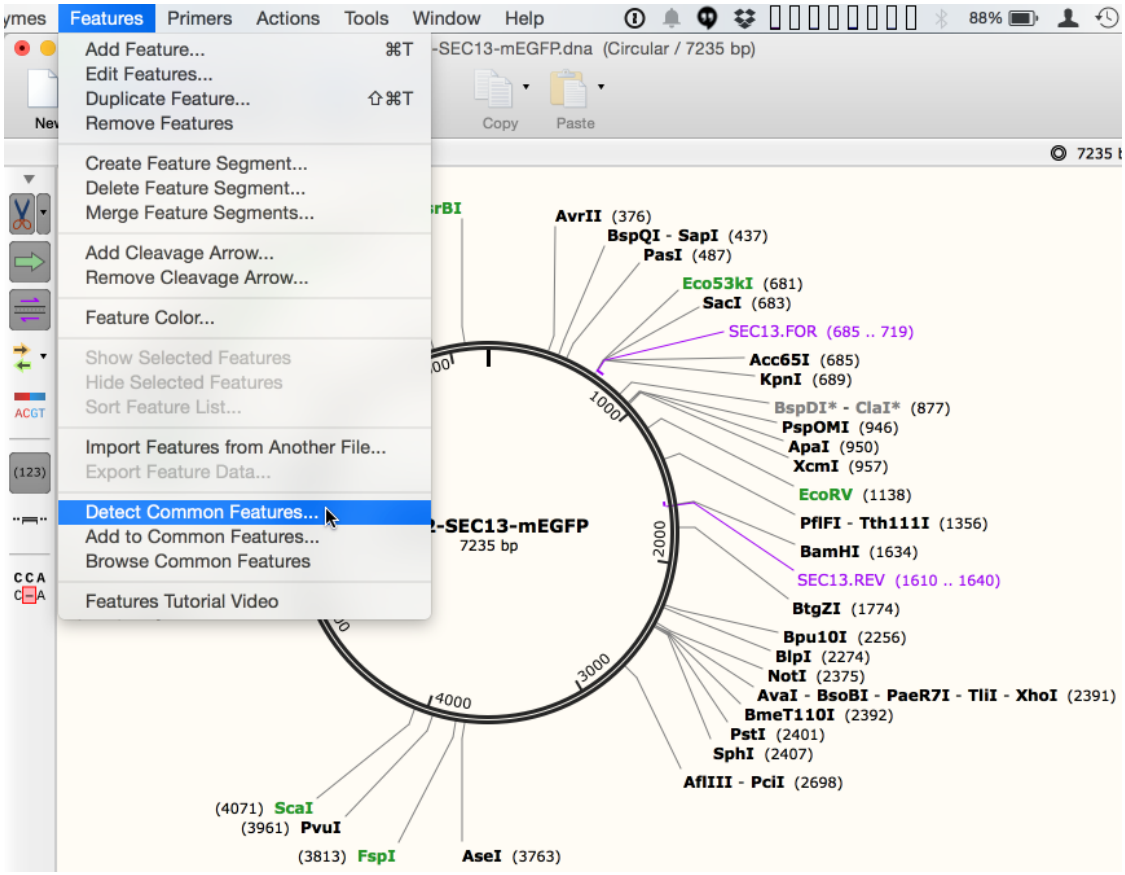



If the segments were not merged, the former intron boundaries will be visible as lines in Map or Sequence view.

# Detect Common Features

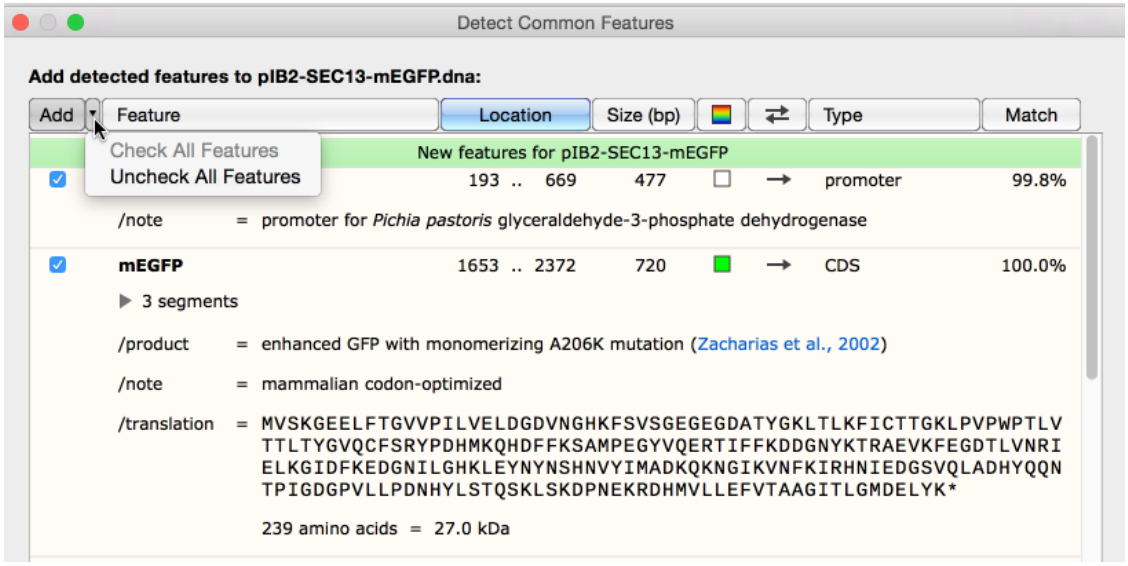
How can I detect the common features within a DNA sequence?

## Detect the Common Features



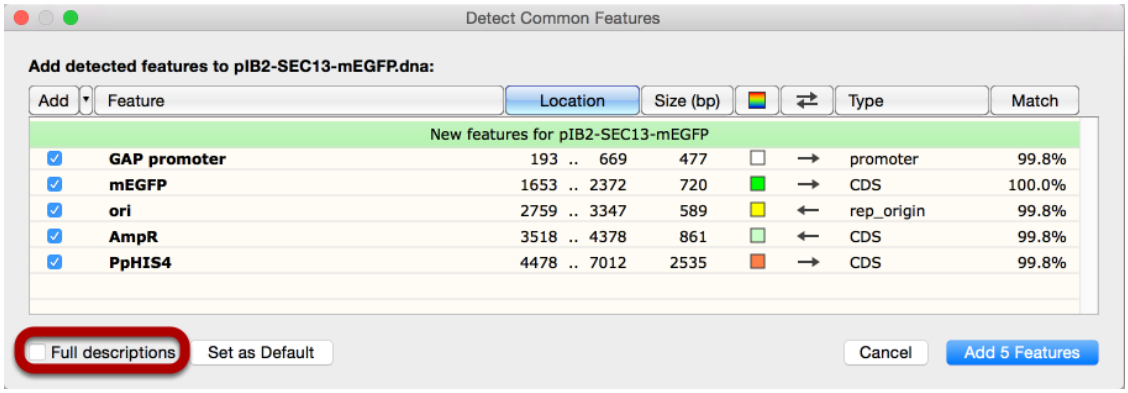

To detect the common features within a DNA sequence click **Features → Detect Common Features...** .

## Choose the Features



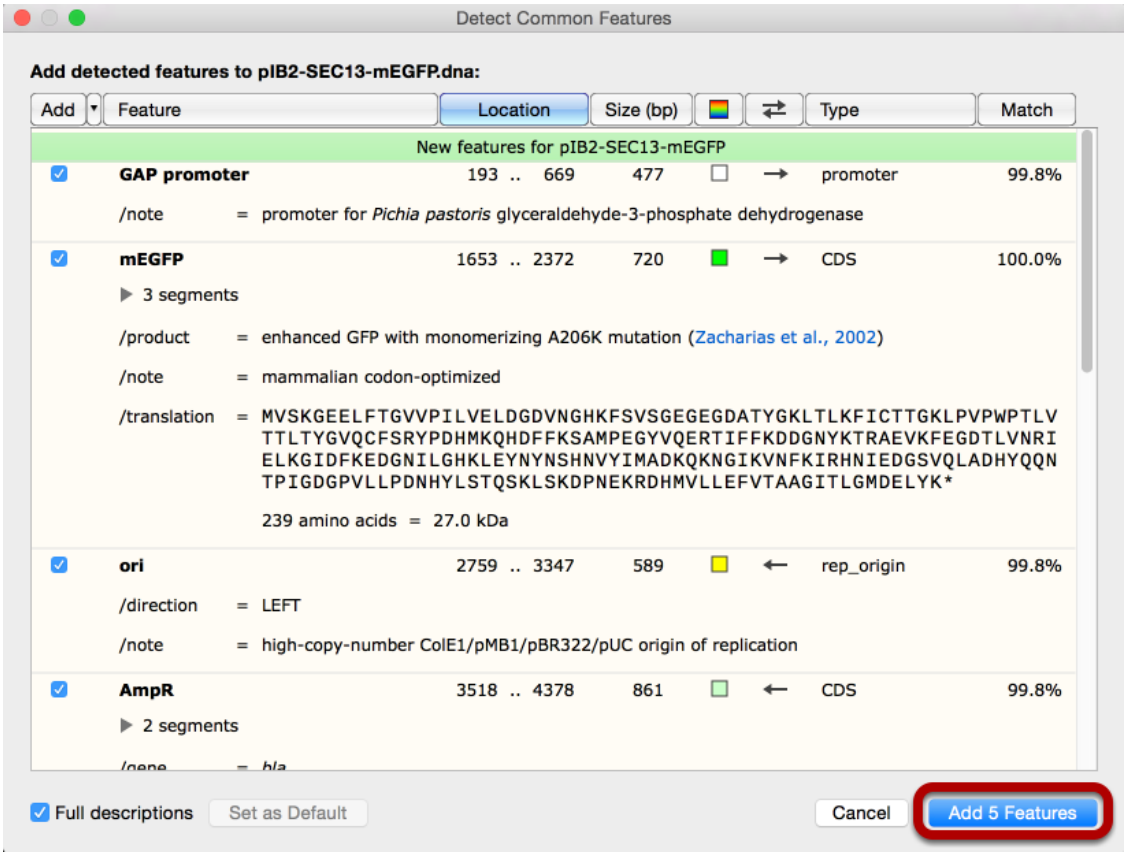

Uncheck any features that are not desired. The **Add** menu provides options for quickly checking or unchecking all of the features.

## Hide the Full Descriptions



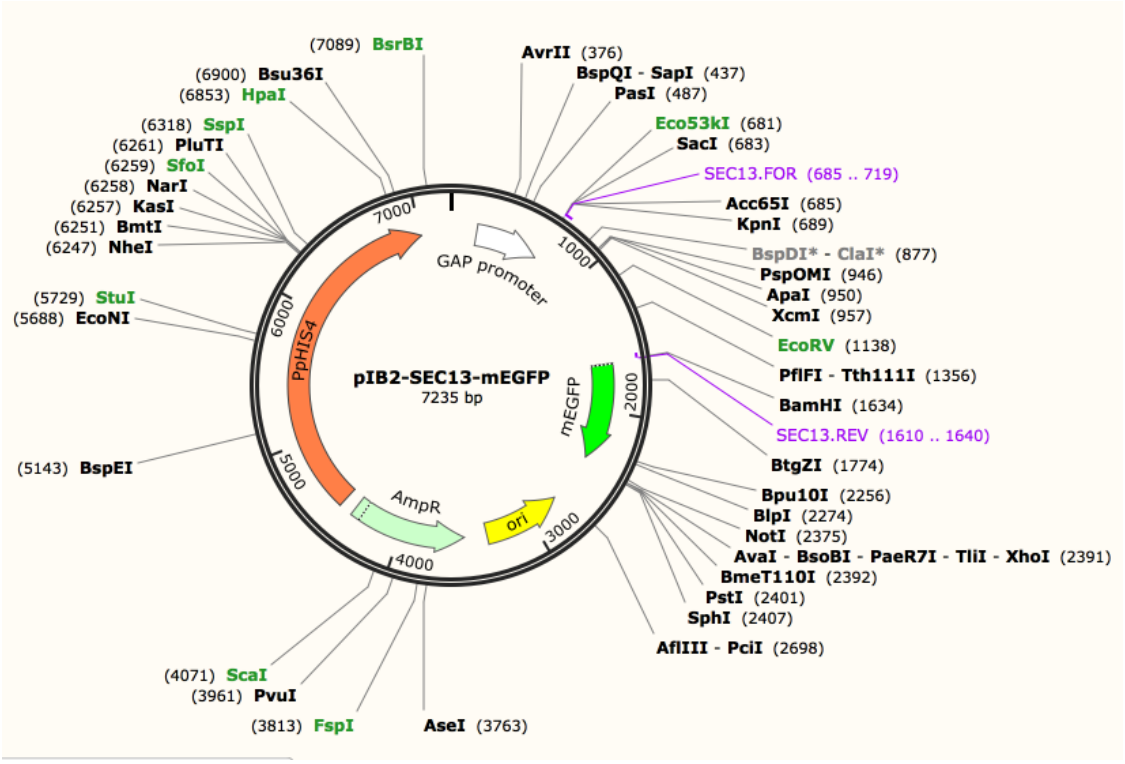

To see each feature in a single line, click to uncheck the **Full descriptions** check box. To hide the full descriptions by default, uncheck **Full descriptions**, then click **Set as Default**.

## Add the Features




To add the selected features to the DNA sequence, click **Add [N] Features**.

## View the Annotated Sequence

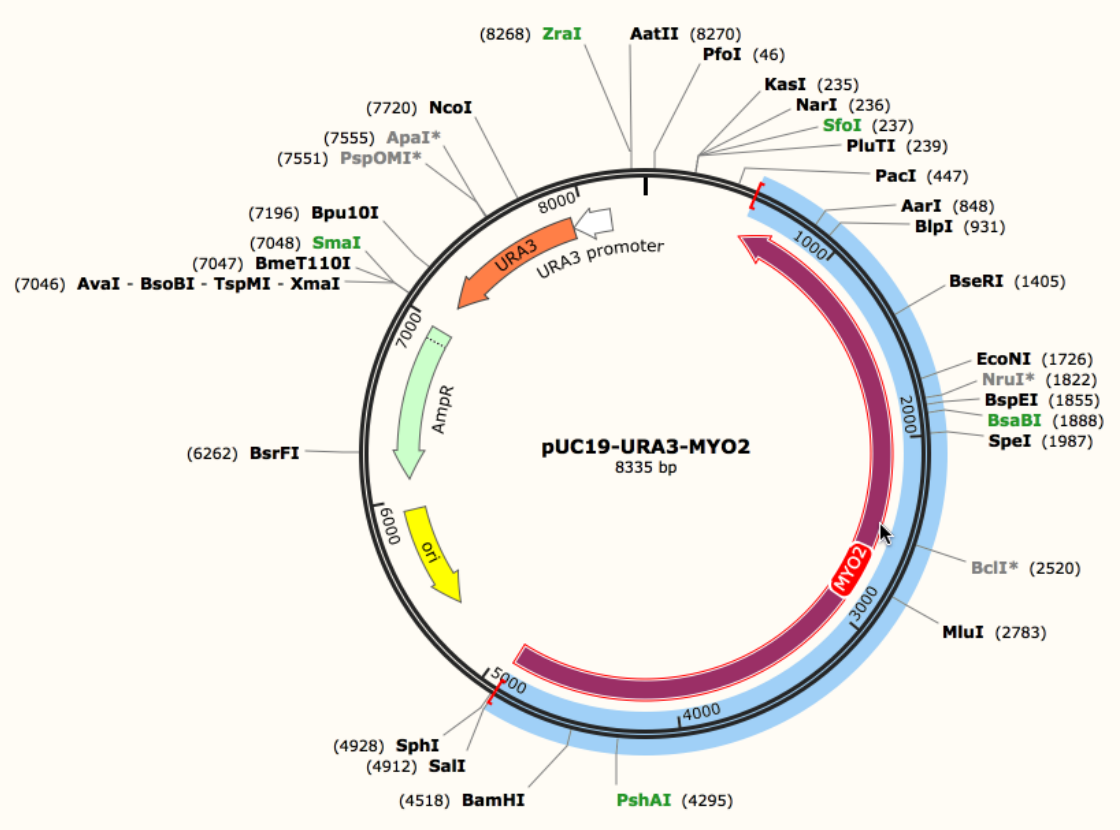



The DNA sequence now includes the selected features.

## Detect Custom Features

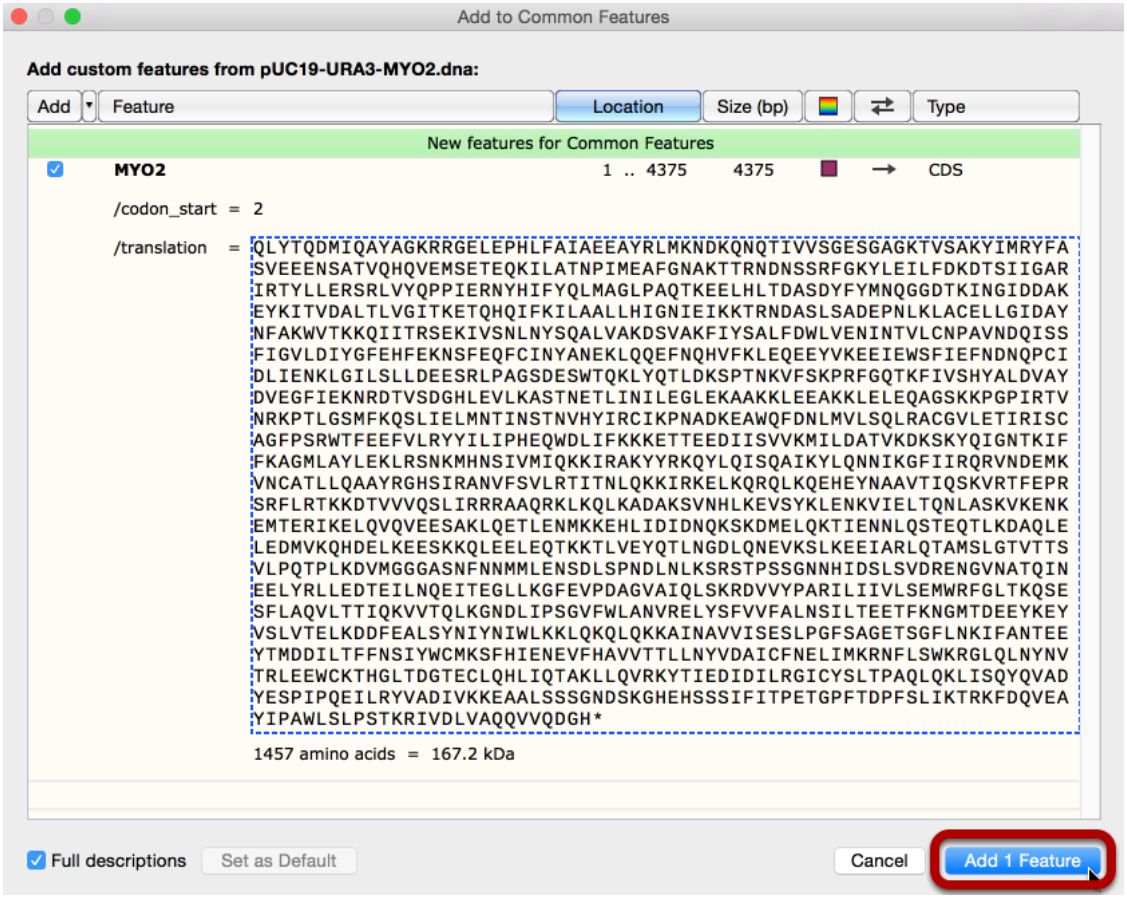
How can I detect custom features within a DNA sequence?

### Select a Feature for Addition to the List



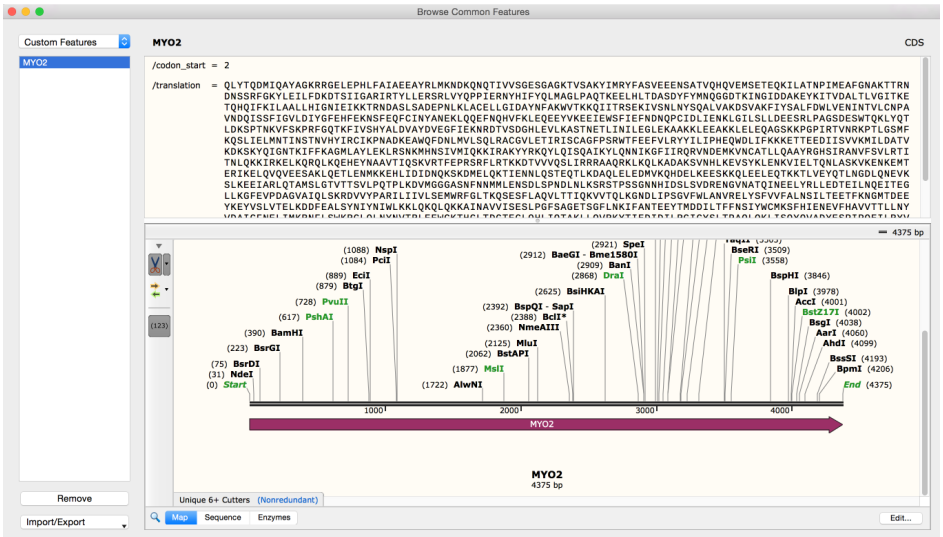

To detect a custom feature, it must first be added to the common features list. Select the feature to be added, then click **Features → Add to Common Features...** .

## Add the Feature



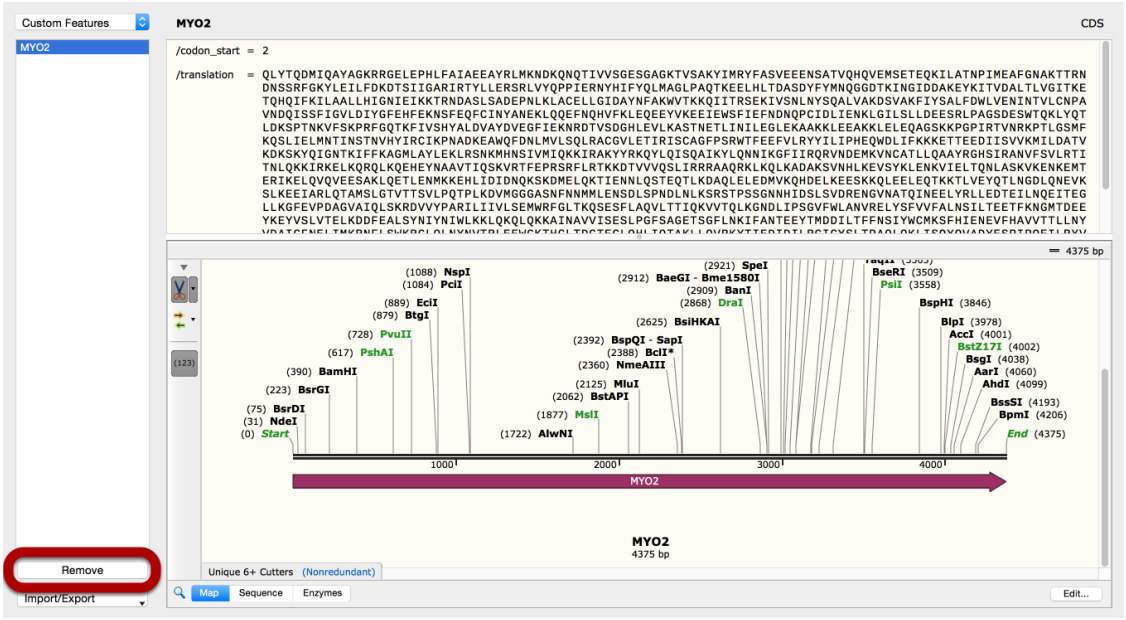

To add the selected feature to the common features list, click **Add 1 Feature**.

## View the Custom Features List



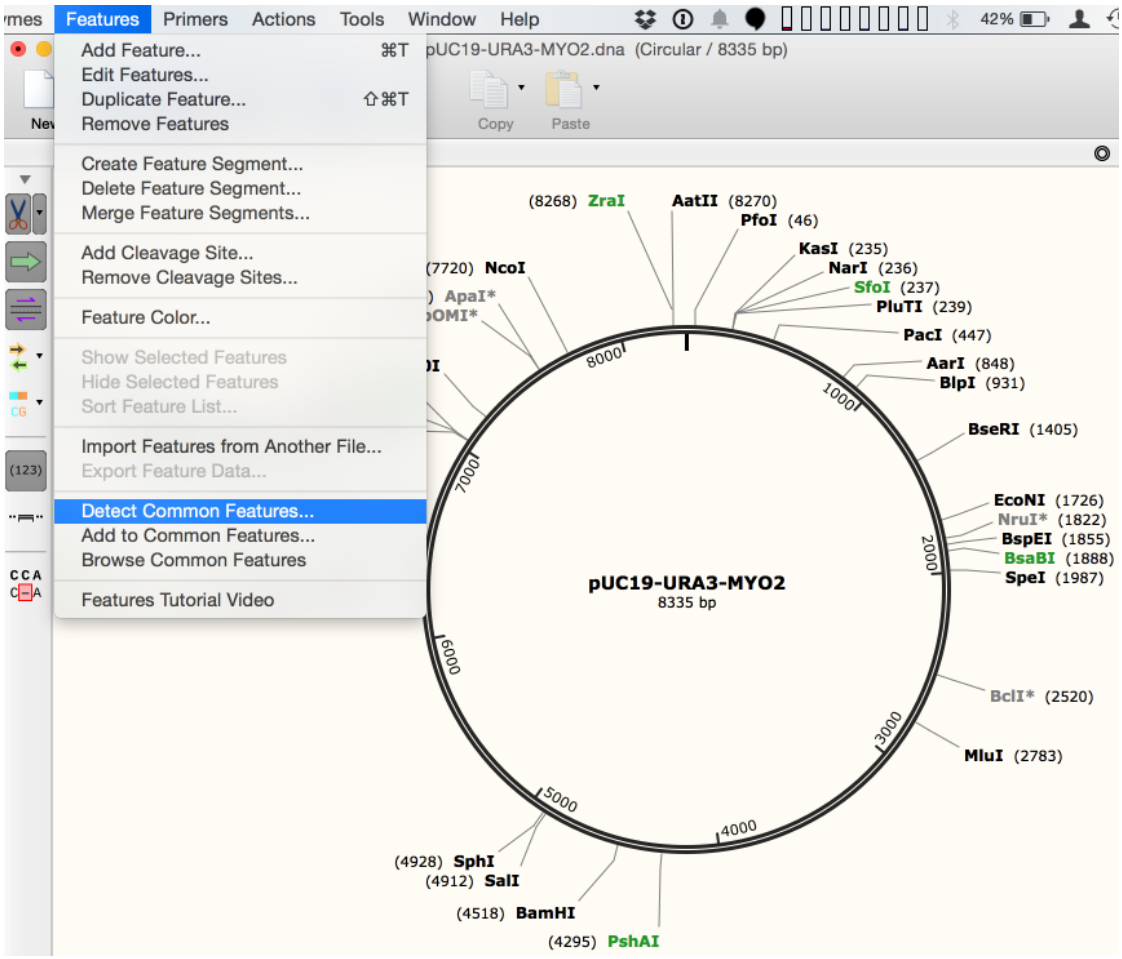

The selected feature will be added to the **Custom Features** list within the Browse Common Features dialog.

# Remove a Custom Common Feature

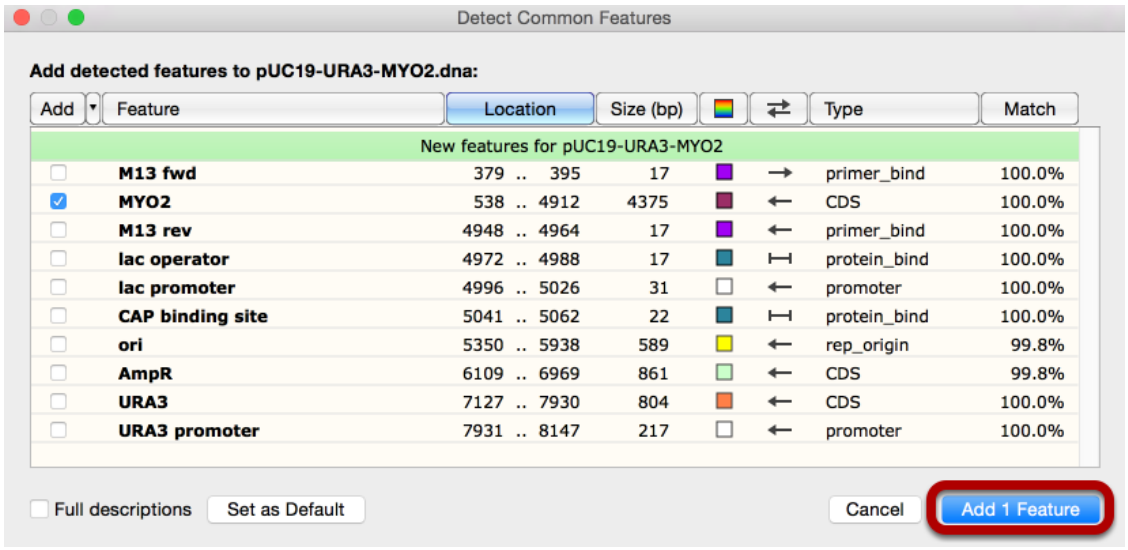



To remove a feature from the custom features list, click in the list box to select the feature, then click **Remove**.

## Detect the Custom Feature



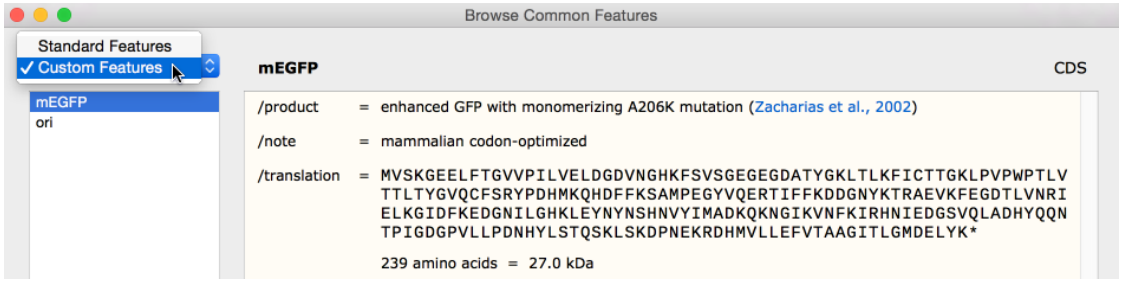

To detect the custom features within a DNA sequence, click **Features → Detect Common Features...** , check the desired feature, then click **Add 1 Feature**.



# Edit Common Features

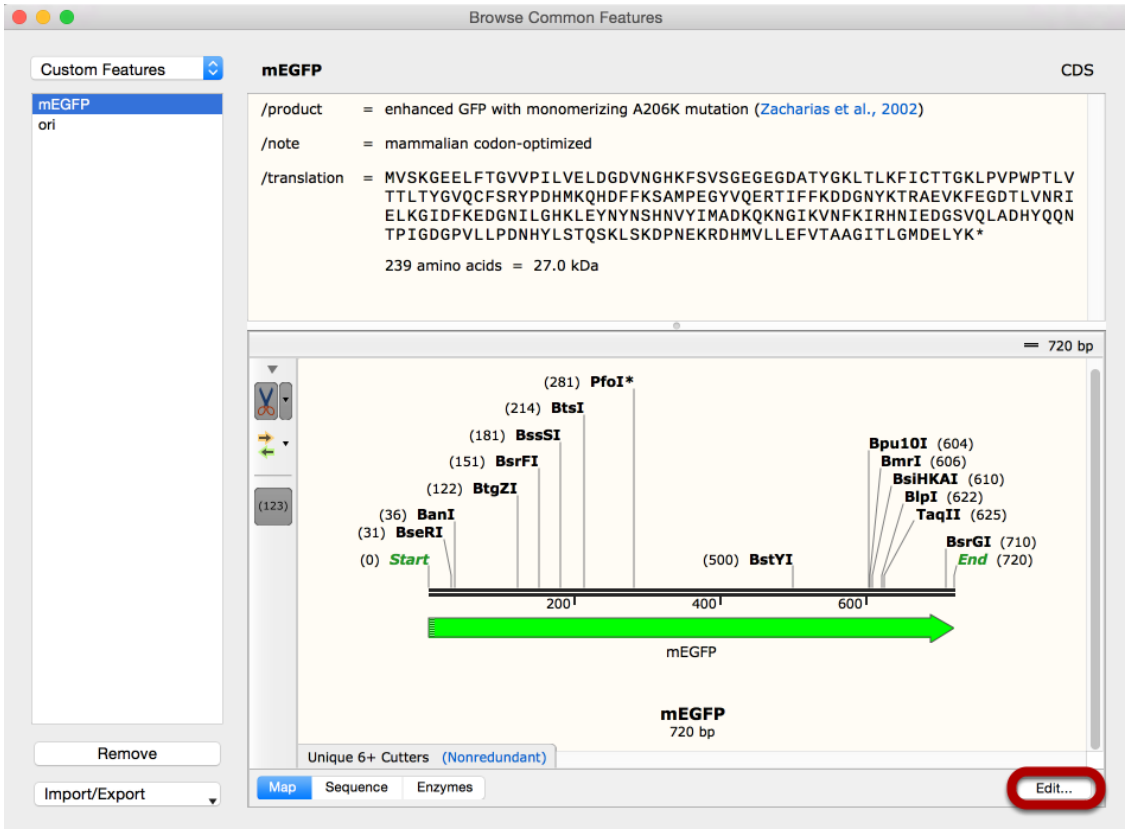
How can I edit common features?

## Choose the Common Features List



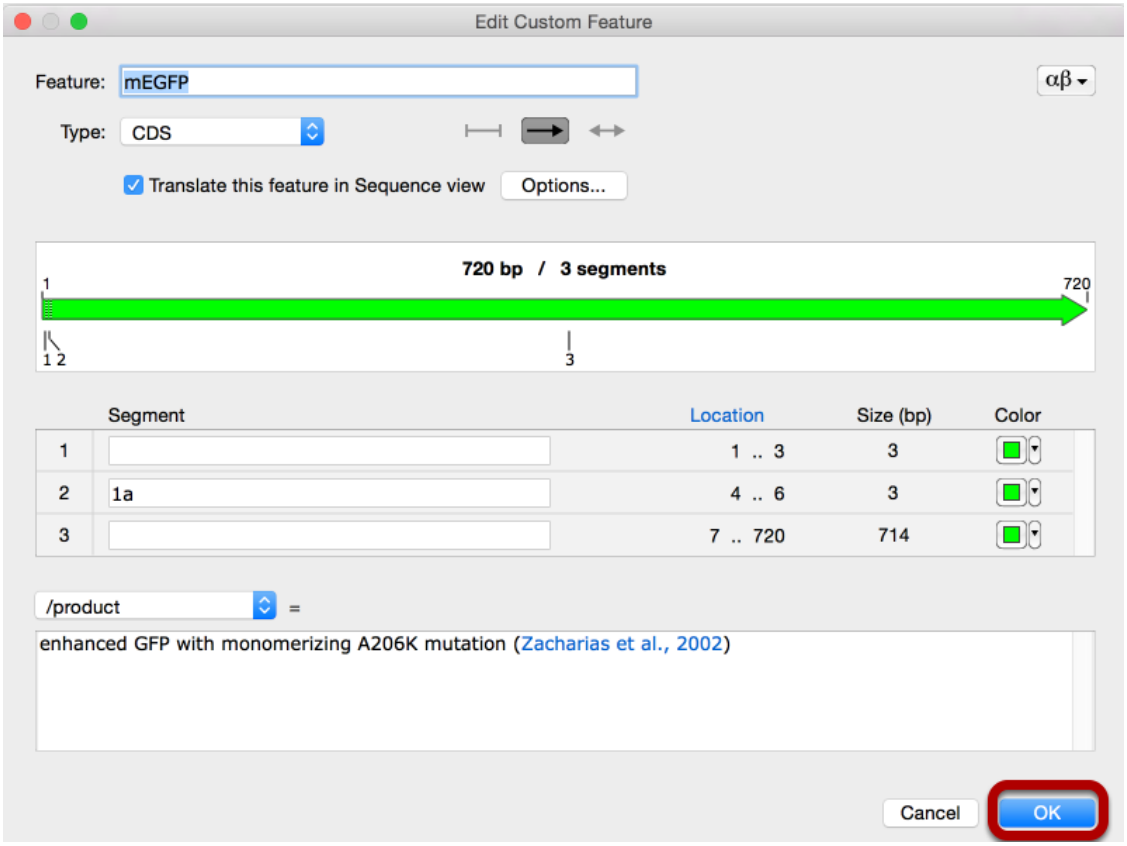

To open the Browse Common Features dialog, click **Features → Browse Common Features**. Choose the desired common features list (Standard or Custom) from the menu.

## Select the Feature



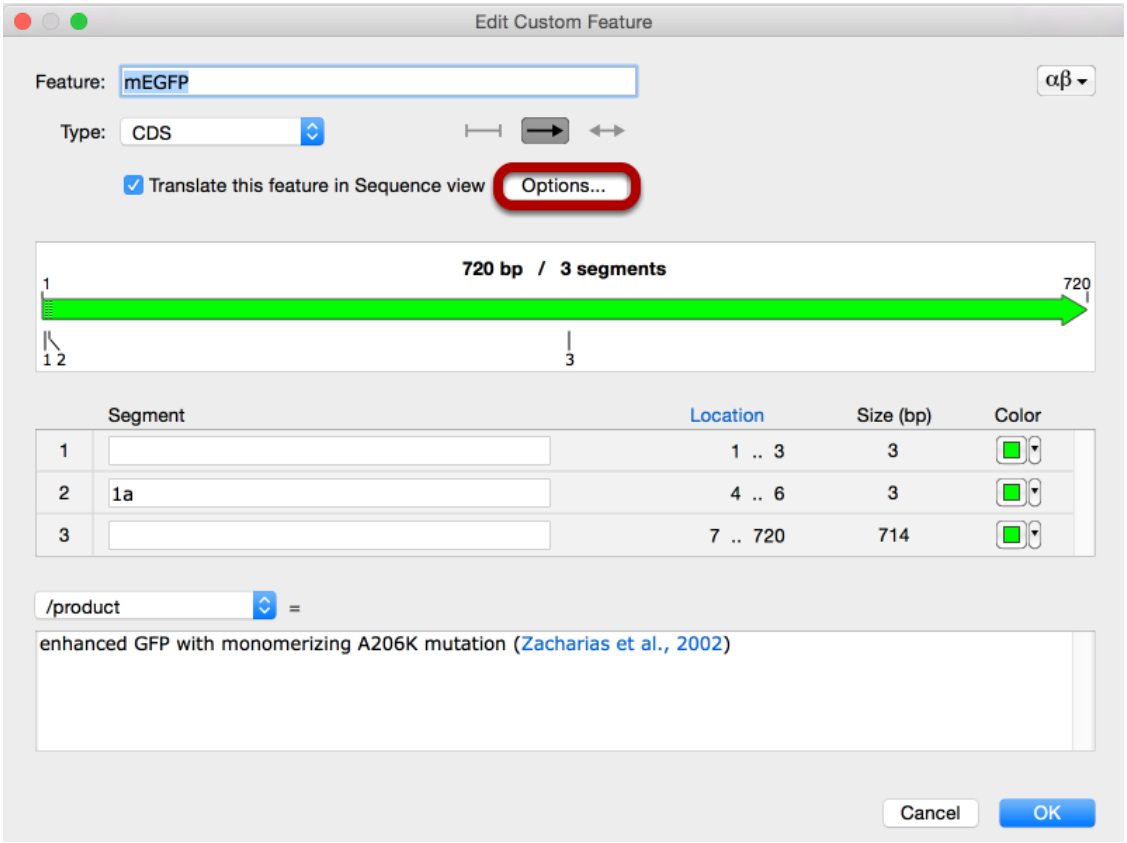

Click the feature name, then click **Edit...** .

## Edit the Feature

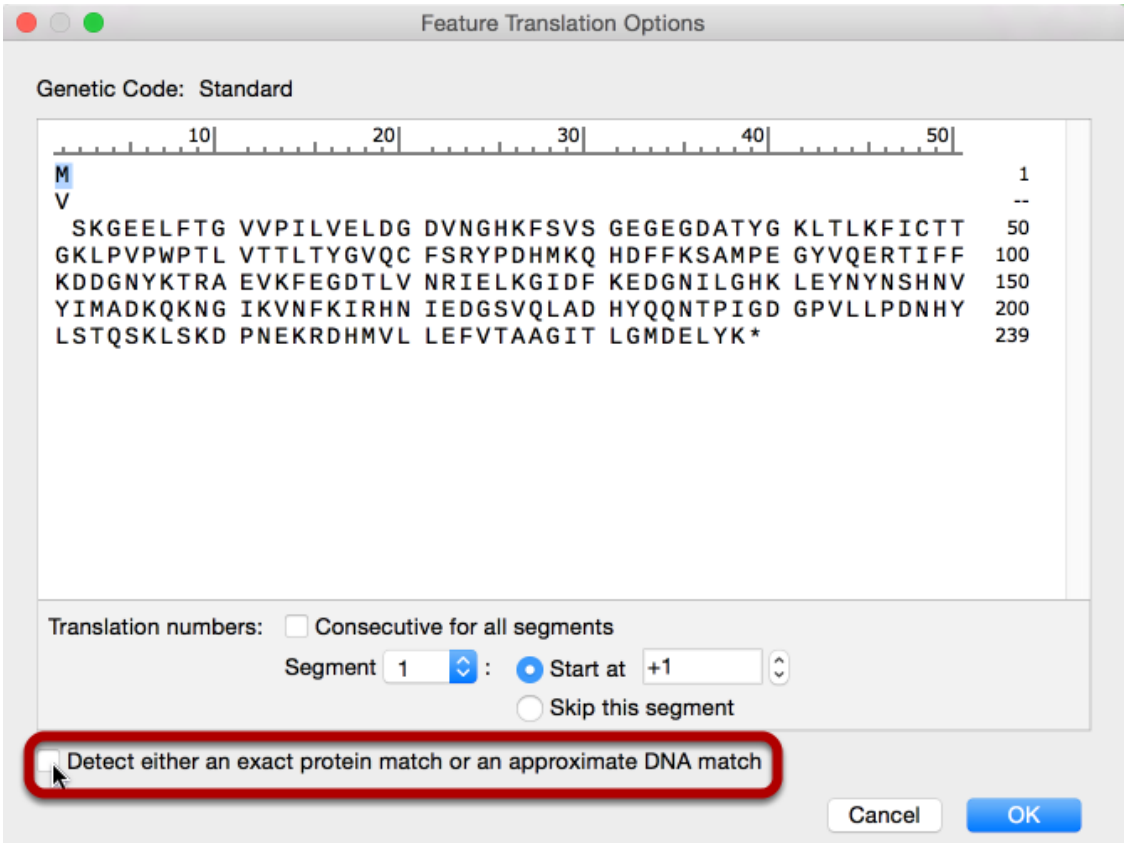



Edit the feature details, then click **OK**.

## Edit the Feature Translation Options

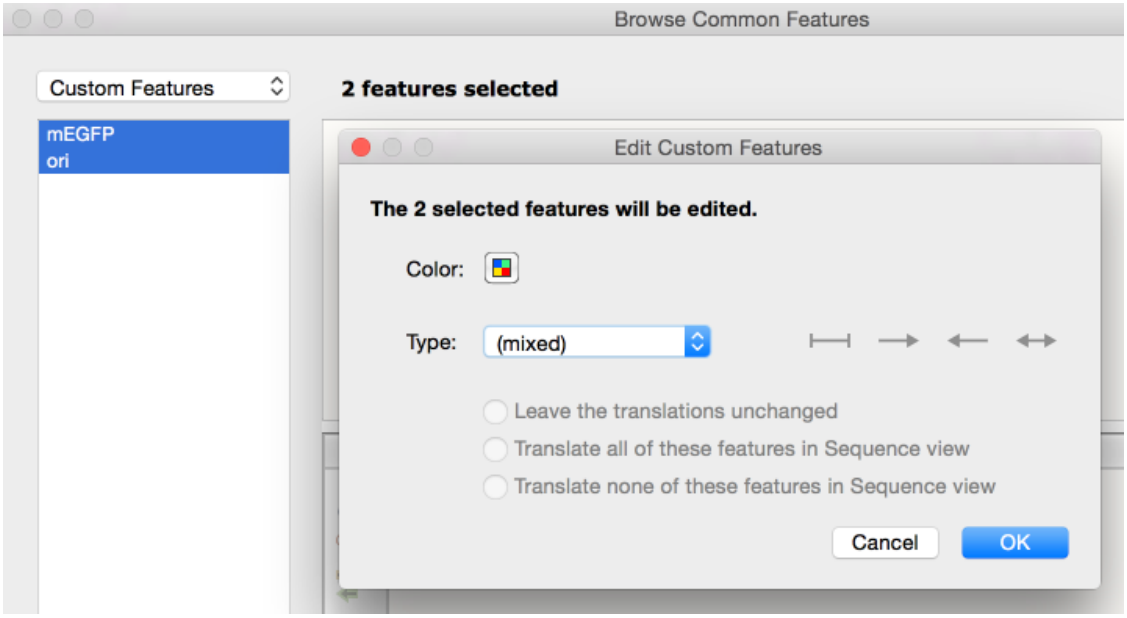



To edit the feature translation options, click the **Options...** button.




To allow the detection of a translated feature regardless of the codons used, check the **Detect either an exact protein match or an approximate DNA match** check box.

## Edit Multiple Common Features

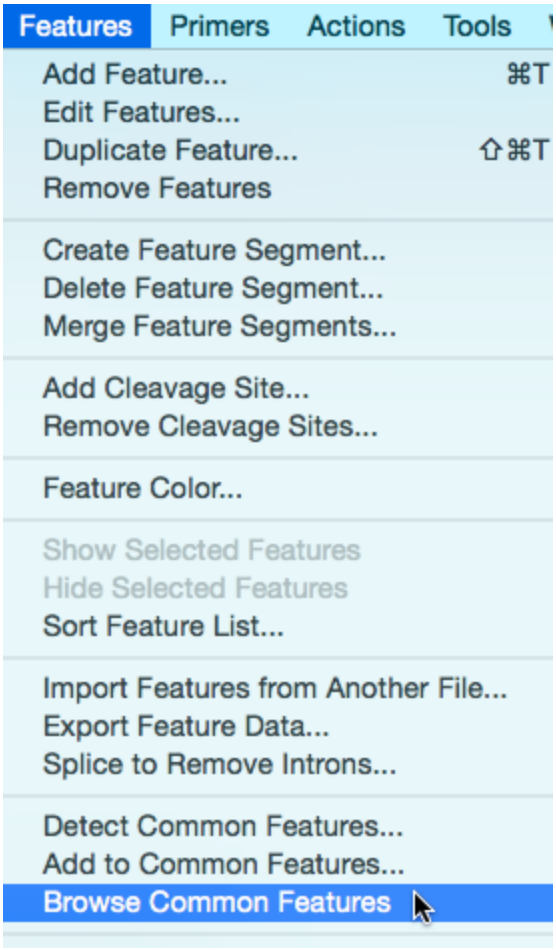



To simultaneously edit multiple features' color or type, select the features in the list box, click **Features** → **Edit Features**, then click **OK**.

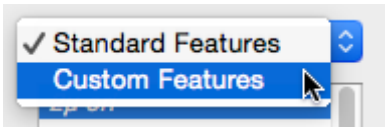
## Share Custom Feature List

How can I share my list of custom common features with a colleague?

### Browse the Custom Features

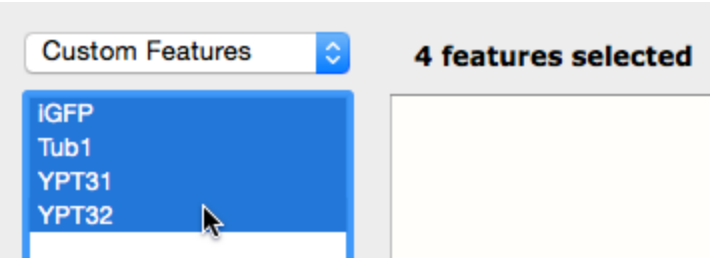



To browse the Custom Features, click **Features** → **Browse Common Features**.




Then switch the menu control at the upper left to **Custom Features**.

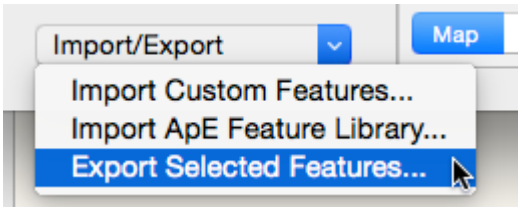
## Select the Custom Features




Click a custom feature name to select the desired feature, then Shift-click or Command-click to select additional features.

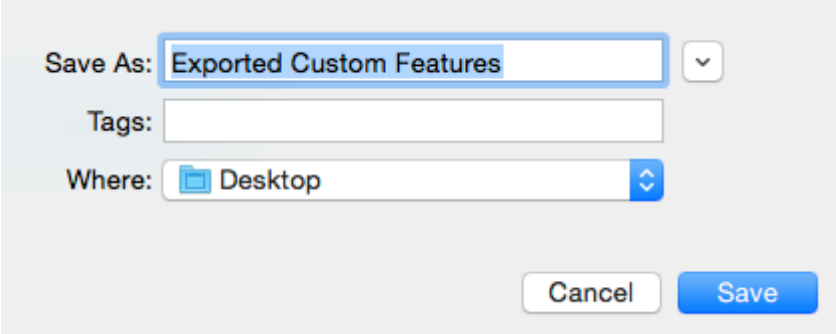
Alternatively, click in the list of custom features and then choose **Edit** → **Select All**.

## Export the Selected Custom Features

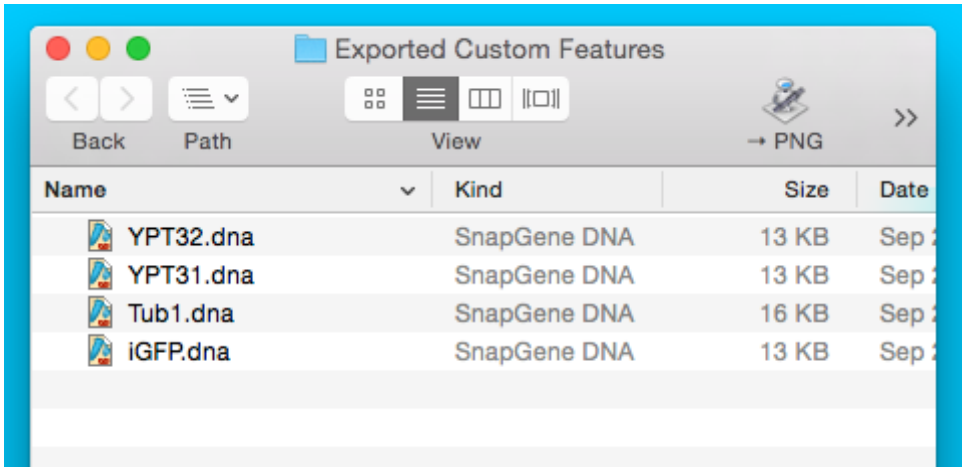



From the menu at the lower left, choose **Export Selected Features...** .

## Save the Features

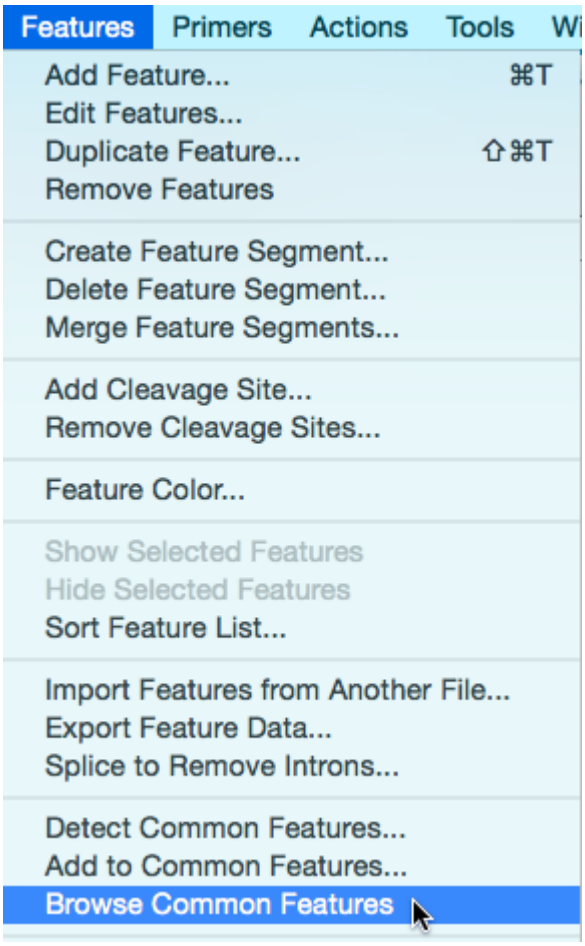



Specify the folder name and destination, then click **Save**.

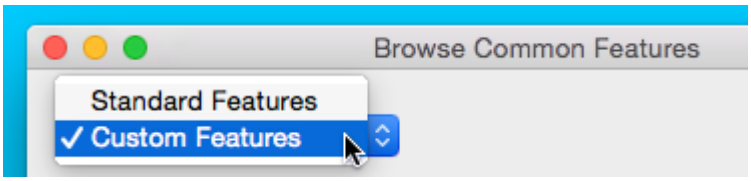



If more than one feature is selected, the result will be a folder containing a .dna file for each exported feature.

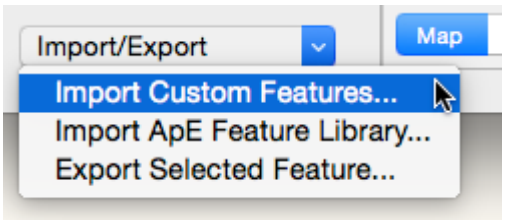
## Import the Custom Features




To import a list of custom features, your colleague should click **Features** → **Browse Common Features** and then follow the remaining steps below.

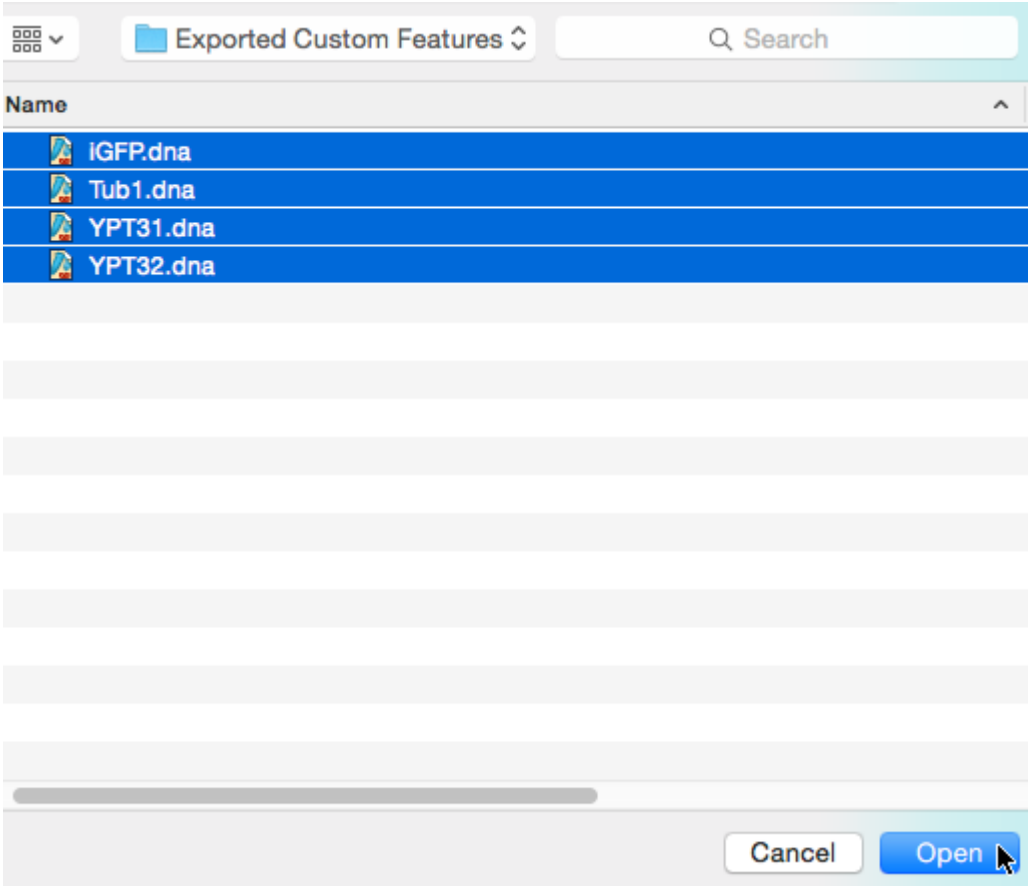



Switch the menu control at the upper left to **Custom Features**.



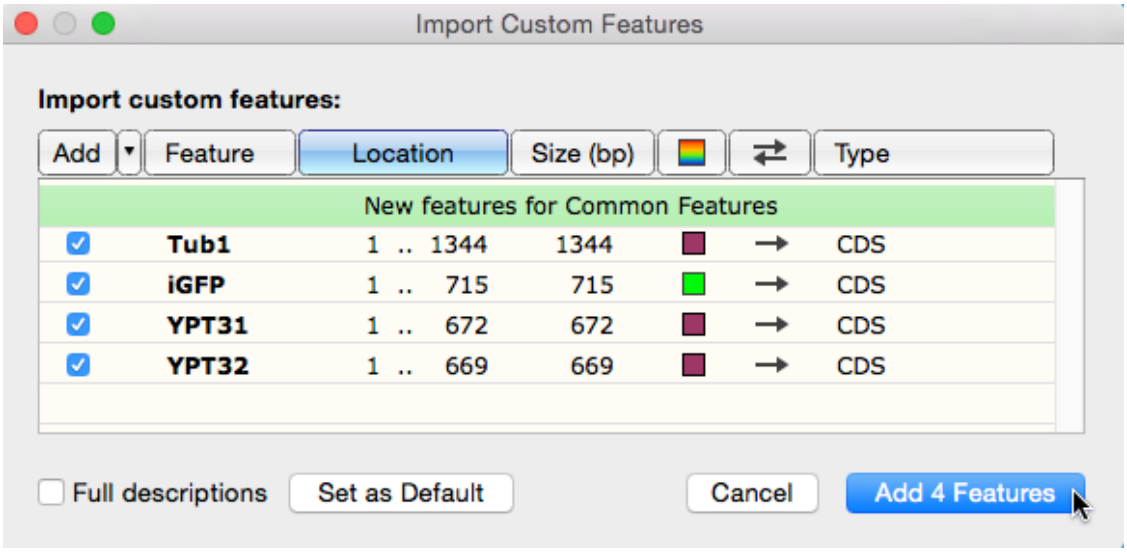

Choose **Import Custom Features...** from the menu at the lower left.

## View the List of Available Features




Browse to the folder of exported custom features, select all of the features within that folder, then click **Open**.

## Specify Which Features To Add



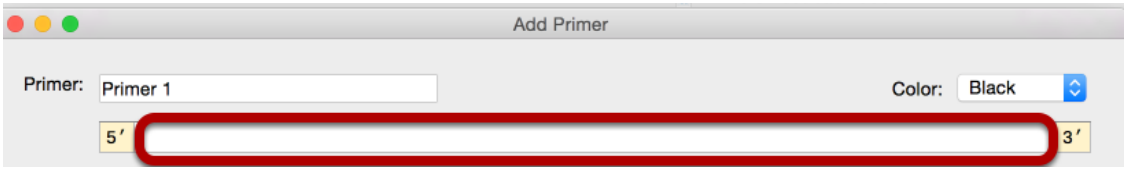

The Add column indicates which features will be imported. Click the relevant check boxes, then click **Add [N] Feature(s)**. The imported features will be added to the custom common features list.

# Primers

## Create a Primer

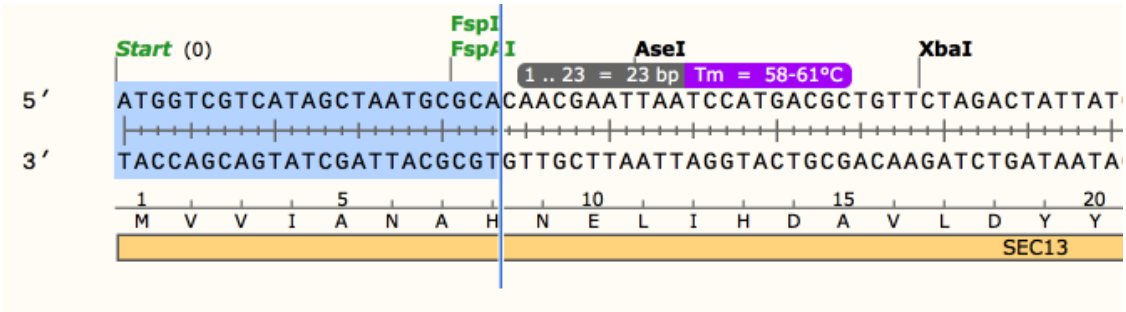
How can I create a primer?

### Paste a Primer Sequence




To create a primer, click **Primers** → **Add Primer...** , then copy and paste a sequence.

### Select the Binding Site (optional)

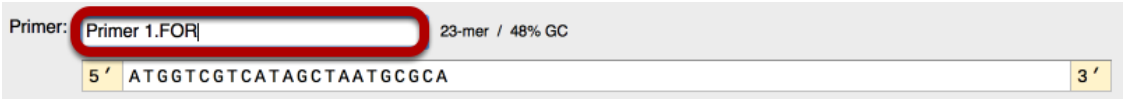



Alternatively, you can begin by selecting the desired binding site. If you click the mouse and drag to select, the melting temperature of a corresponding primer will be shown.

### Specify the Selected Strand (optional)

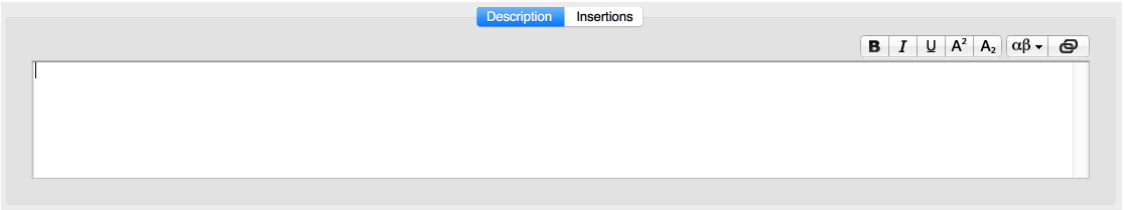
If the primer will be made from a selected binding site, specify whether the **Top Strand** or **Bottom Strand** of the selection should be used.

### Name the Primer




If desired, type the primer name.

## Edit the Description

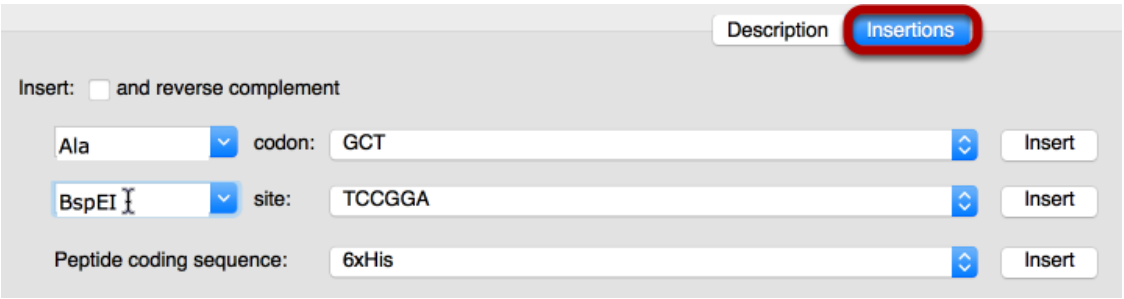



If desired, type a description. The **Description** tab is open by default.

## Modify the Primer (optional)

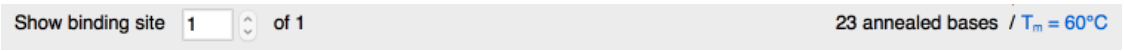
If desired, modify the primer to add a 5’ extension or introduce a mutation. There are two options.

One option is to manually edit the primer sequence in the text box.



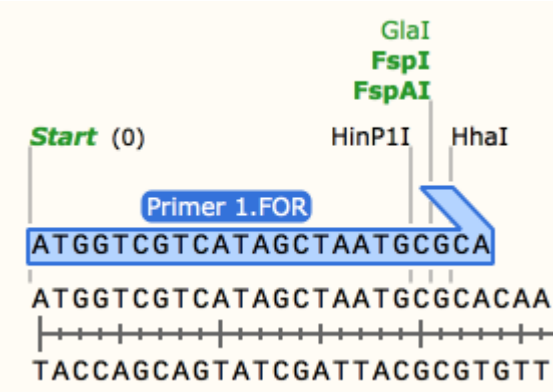

The other option is to click in the primer sequence at the location of the change, and then click the **Insertions** tab. Use the dialog controls to add a desired codon, restriction site, or peptide coding sequence.

## View the Binding Sites and Melting Temperature




Note the number of binding sites and the calculated melting temperature. To see a summary of the melting temperature calculation methods, click the blue text at the lower right corner of the dialog.

## View the Primer

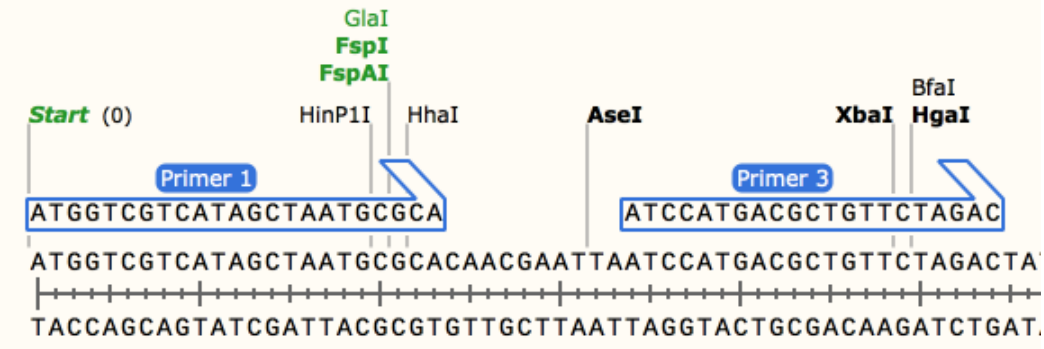



Click **Add Primer to Template** to see the new primer.

## Edit Multiple Primers

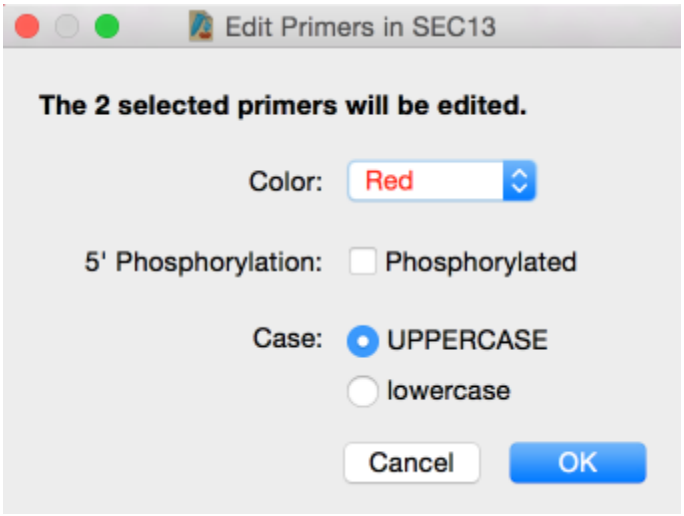
How can I edit multiple primers at the same time?

### Select the Primers



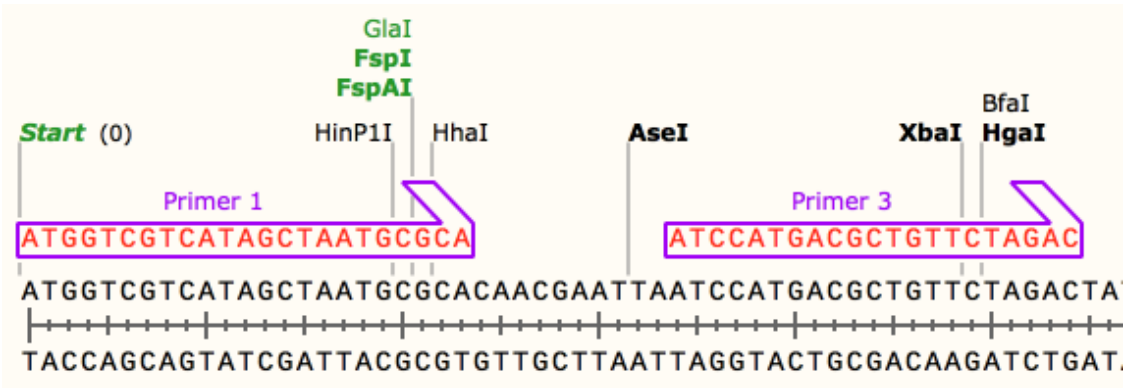

Select multiple primers. To open the Edit Primers dialog, click **Primers** → **Edit Primers**.

### Specify the Changes

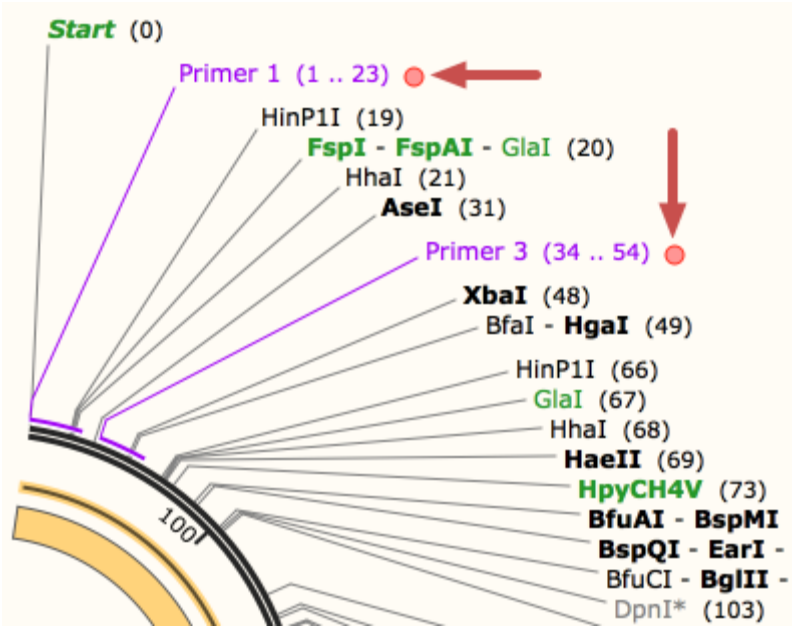



Choose the color, phosphorylation, and font case for all of the selected primers, then click **OK**.

## View the Updated Primers




All of the selected primers will now have the specified changes. In this example, the color of the primers is changed to red, as shown in Sequence view.

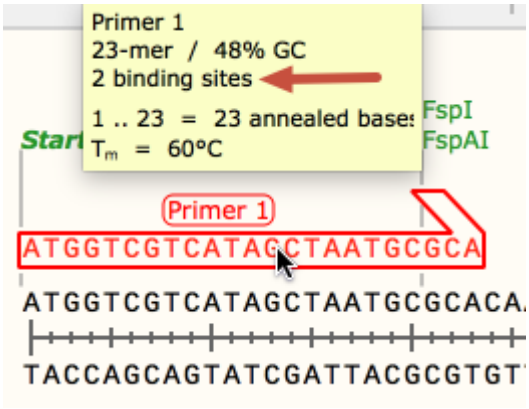



In Map view, a circle shows the updated color of the primers.

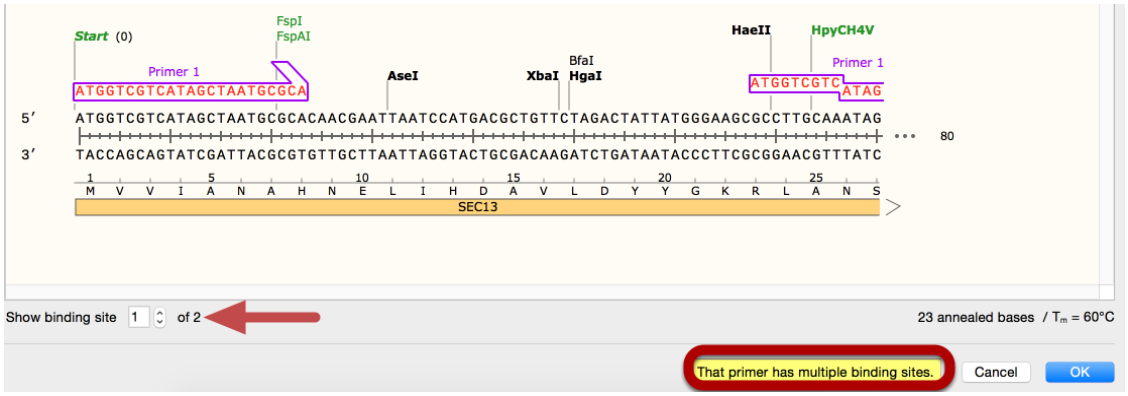
## Adjust Primer Hybridization Parameters

How can I adjust the hybridization parameters to see more or fewer primer binding sites?

### Review the Primer Binding Sites

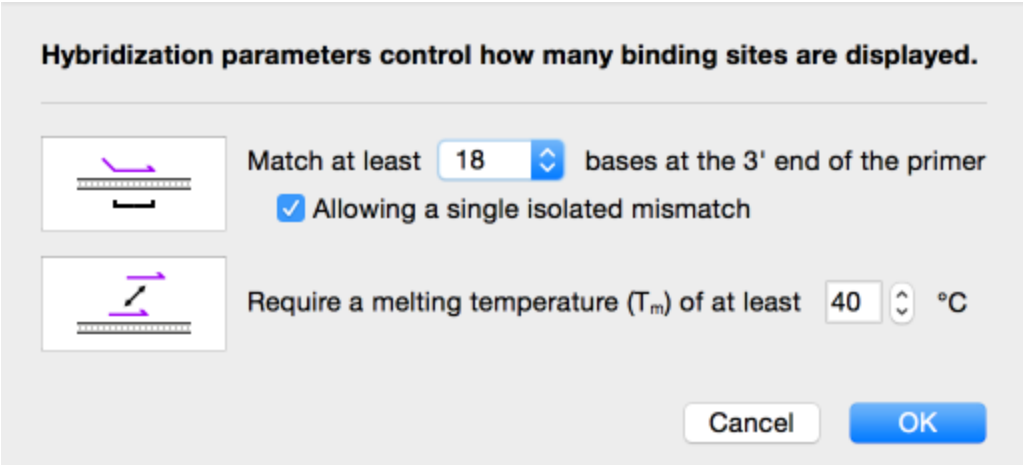



To show a synopsis of primer information, mouse over the primer name. Note the number of primer binding sites.




Alternatively, double-click the primer to open the Edit Primers dialog. If a primer has multiple binding sites, a warning message will be shown in a yellow box. The total number of binding sites is also shown. You can switch between different primer binding sites using the control button.

## Adjust the Hybridization Parameters

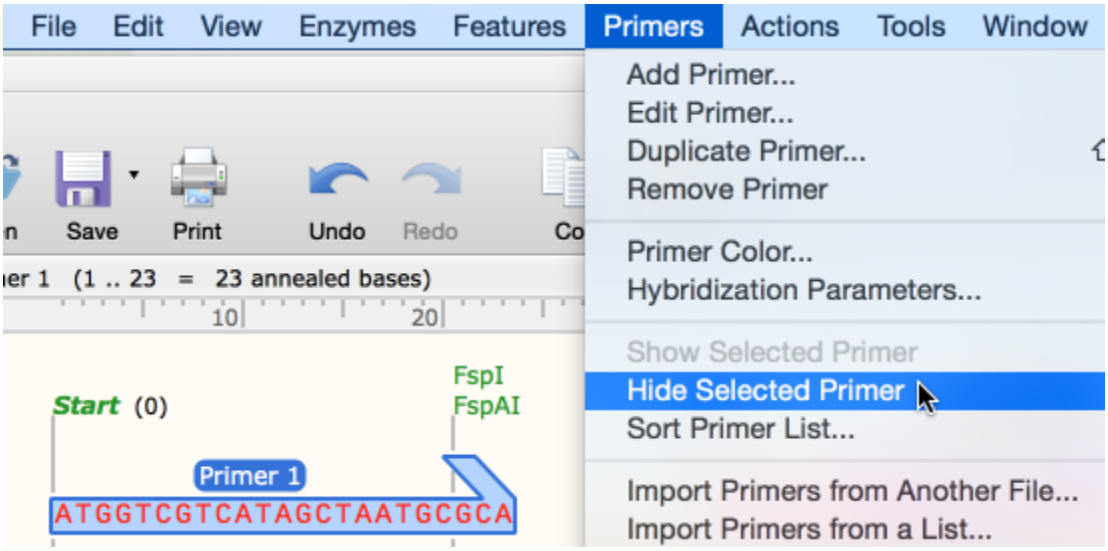



To adjust the primer hybridization parameters, click **Primers** → **Hybridization Parameters**. The resulting dialog provides controls for adjusting the number of matching bases at the 3’ end of the primer as well as the minimum melting temperature.

# Hide or Show Primers

How can I change the visibility of a primer?

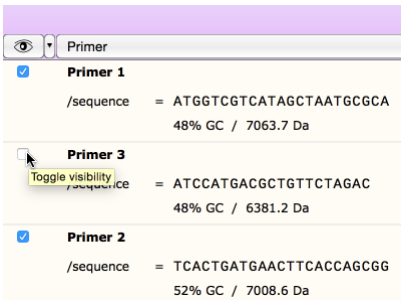
## Hide the Primer




To hide a selected primer that is visible, click **Primers** → **Hide Selected Primer**.

If multiple primers are selected, you can use **Primers** → **Hide Selected Primers** to hide all of them.

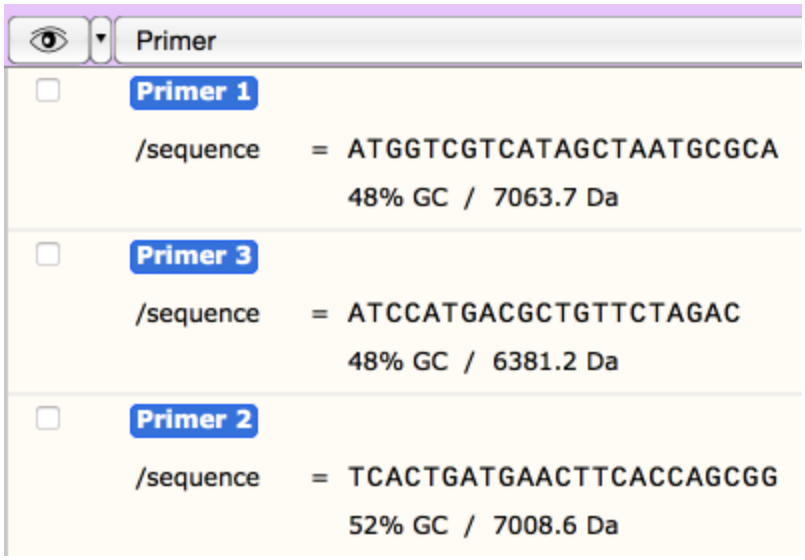
## Show the Primer




To show a hidden primer, switch to Primers view, then click the empty check box to the left of the primer name. The selected primer will now be visible in the other views.

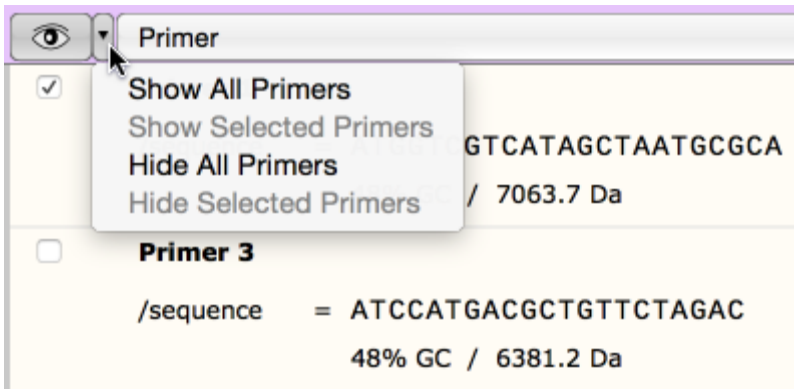
Alternatively, select the primer name, then click **Primers** → **Show Selected Primer**.

## Show Multiple Primers




To show multiple primers that are hidden, select the primers in Primers view, then click **Primers** → **Show Selected Primers**.

## Change Primer Visibility Using the Sorting "Eye" Button




To change the visibility of primers, choose from the menu options next to the "Eye" sorting button.

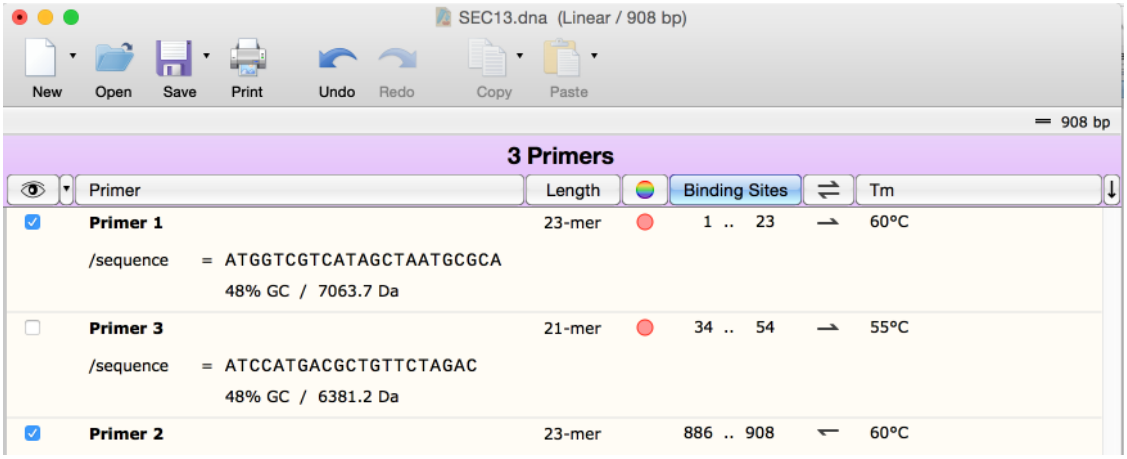
## Toggle Primer Visibility Using the Side Toolbar

In Map or Sequence view, clicking the "Show primers" button in the side toolbar toggles the display of the primers whose check boxes are checked in Primers view.

# Sort the Primers List

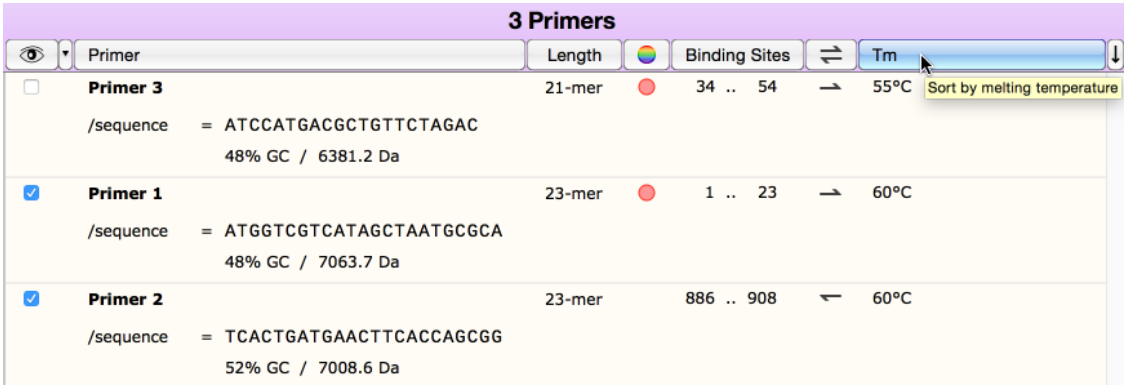
How can I sort the list in Primers view?

## View the Primers List




The column headers in Primers view are also sorting buttons. By default, the **Binding Sites** button is chosen, meaning that primers are sorted by location.

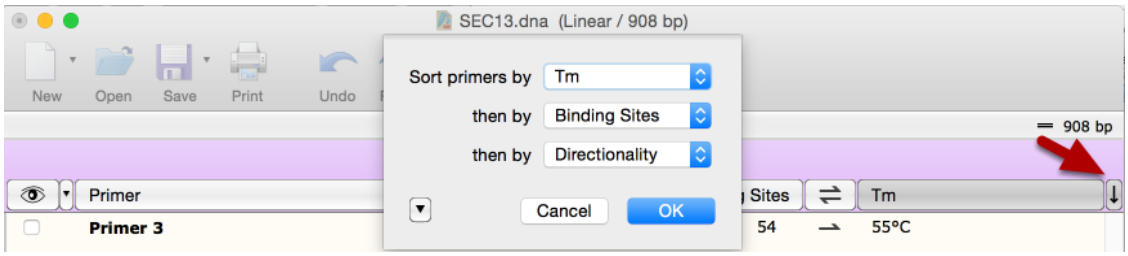
## Change the Primary Sorting Parameter




To sort primarily by another parameter, press the appropriate column button.

In this example, the primers are now sorted primarily by **Tm**, then by **Binding Sites**.

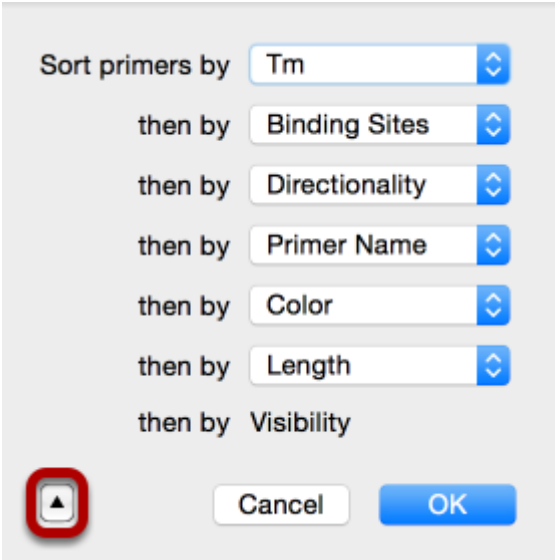
## Specify Additional Sorting Parameters




To specify secondary and tertiary sorting parameters, press the button at the far right side of Primers view to open the Sort Primers List dialog.

Alternatively, you can click **Primers** → **Sort Primer List...** .

## Completely Specify the Sorting Parameters

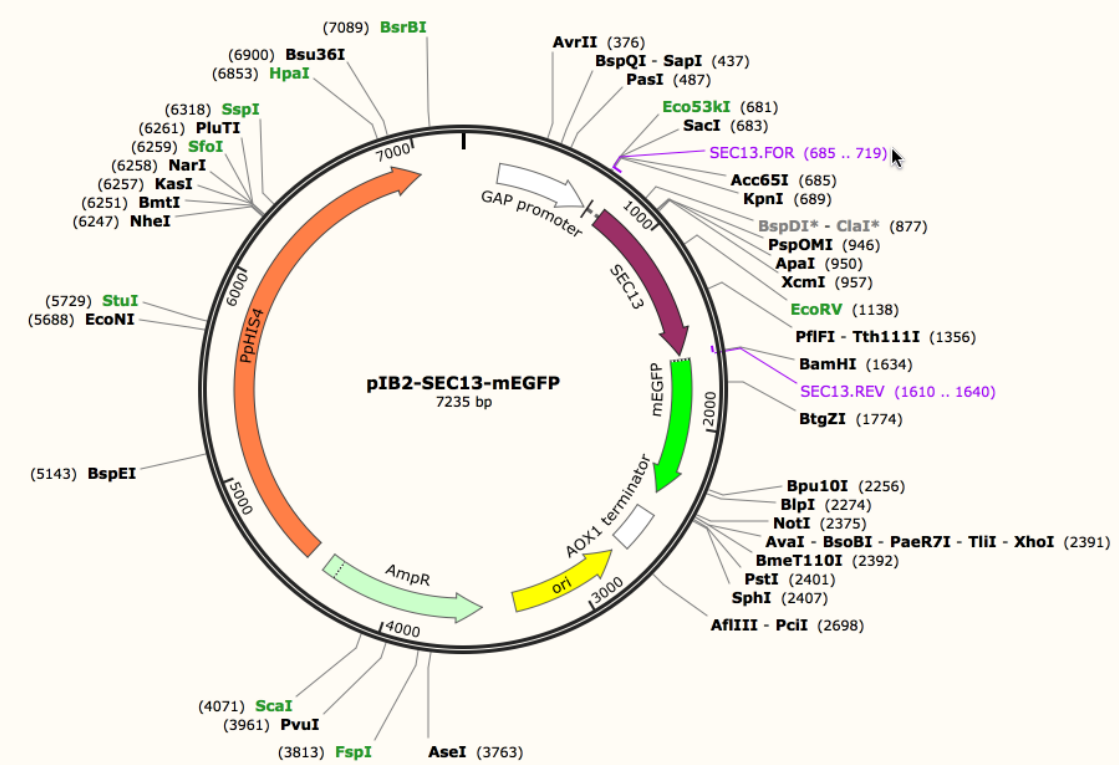



For complete control over the sorting parameters, press the disclosure button in the lower left corner of the dialog, then click **OK**.

# Import Primers

How can I import primers from one file into another?

## Open the Destination File



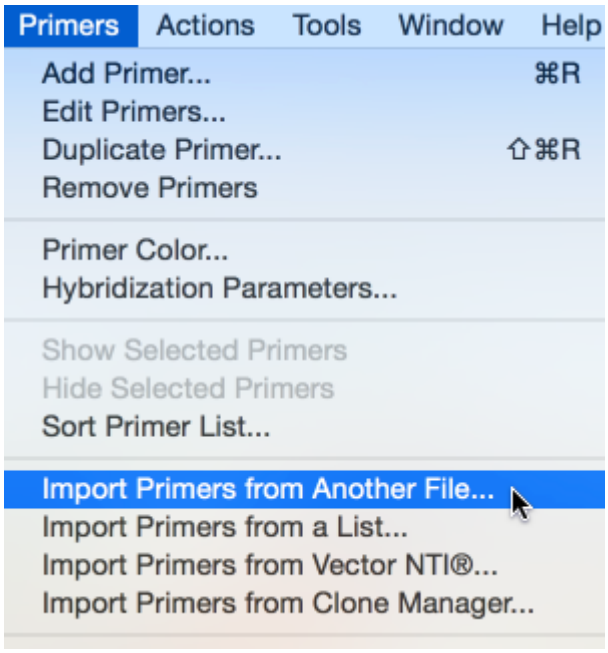

Open the file that will receive the imported primers.

SnapGene will import only primers that can hybridize with the destination file sequence. If the destination file already has primers, SnapGene will not duplicate existing primers by default, but will offer the option of importing duplicate primers.

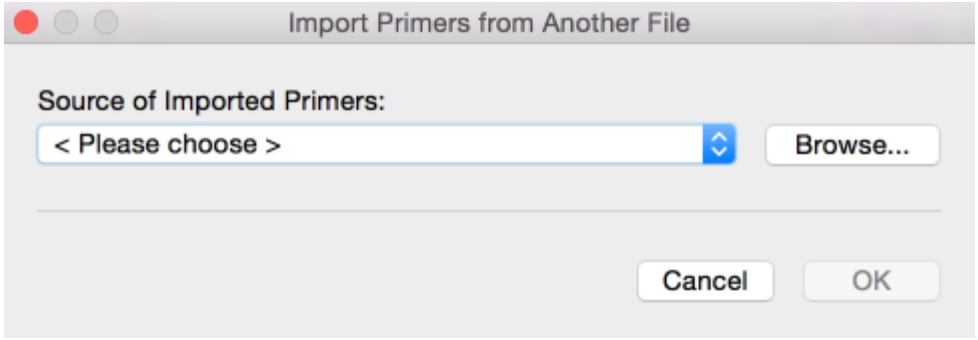
Primers can be imported from:

- a source *.dna file
- a list in *.txt or *.rtf format
- a Vector NTI® database or oligo archive
- a Clone Manager primer archive

## Specify a *.dna File

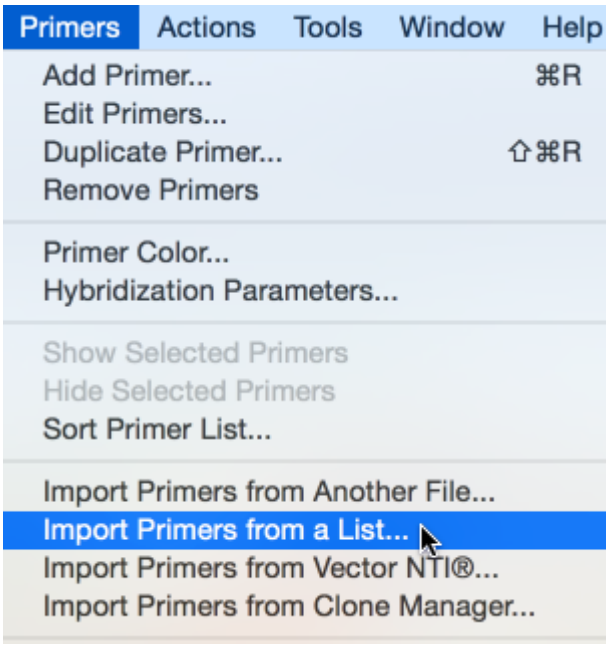



To specify the source file, click **Primers** → **Import Primers from Another File...** .

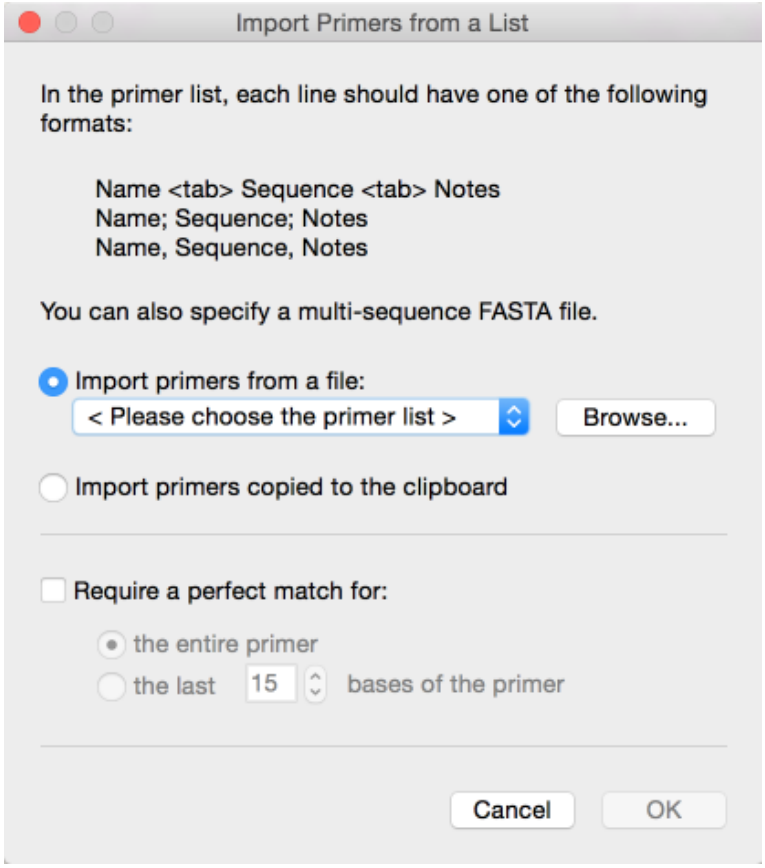



In the Import Primers from Another File dialog, choose a *.dna source file, then click **OK**.

## Specify a Source List

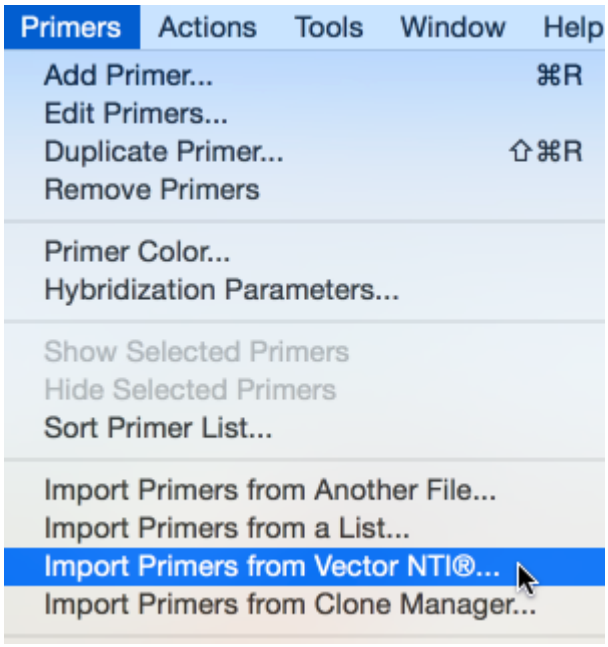



To specify the source list, click **Primers** → **Import Primers from a List...** .

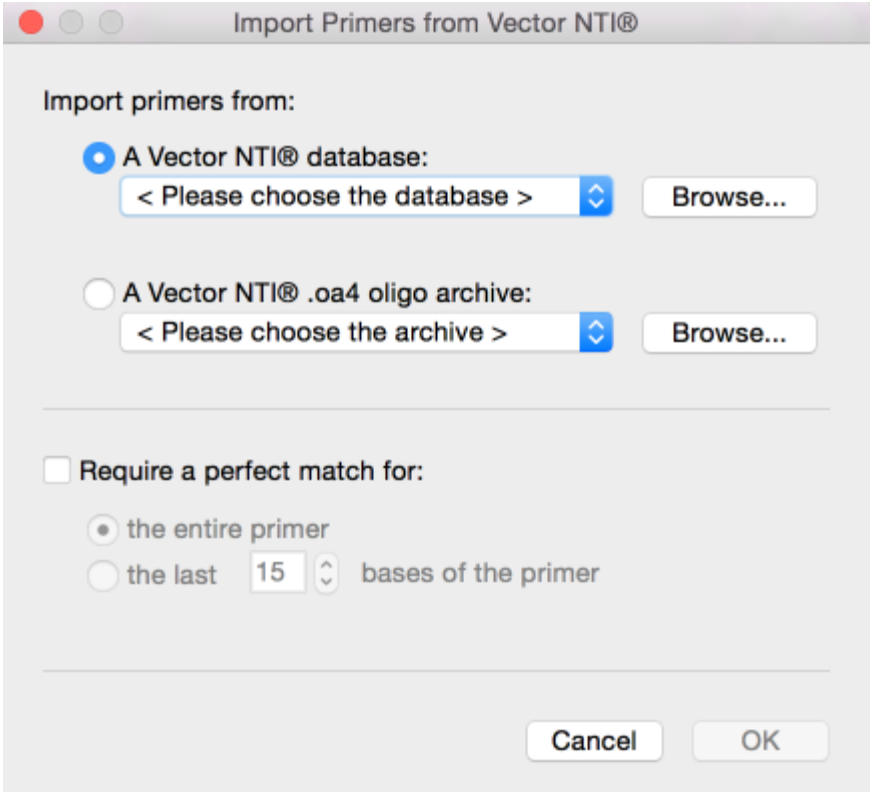



In the Import Primers from a List dialog, choose a *.txt or *.rtf source file, or choose **Import primers copied to the clipboard**, then click **OK**.

## Specify a Vector NTI® File

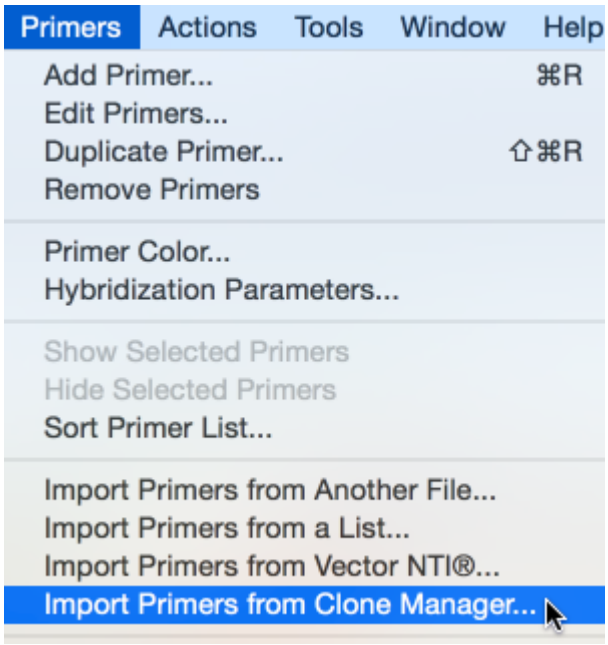



To specify the source Vector NTI® file, click **Primers** → **Import Primers from Vector NTI®...** .

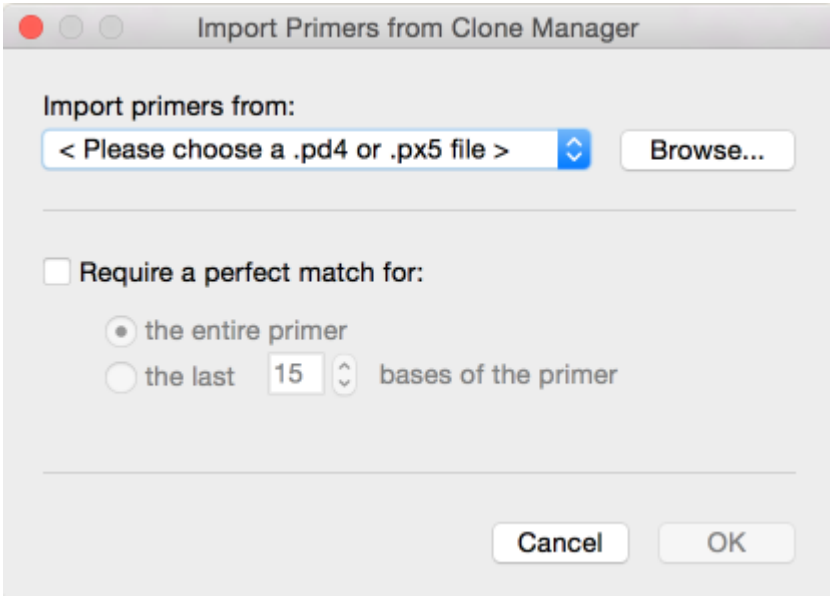



In the Import Primers from Vector NTI® dialog, choose a database or oligo archive, then click **OK**.

## Specify a Clone Manager File

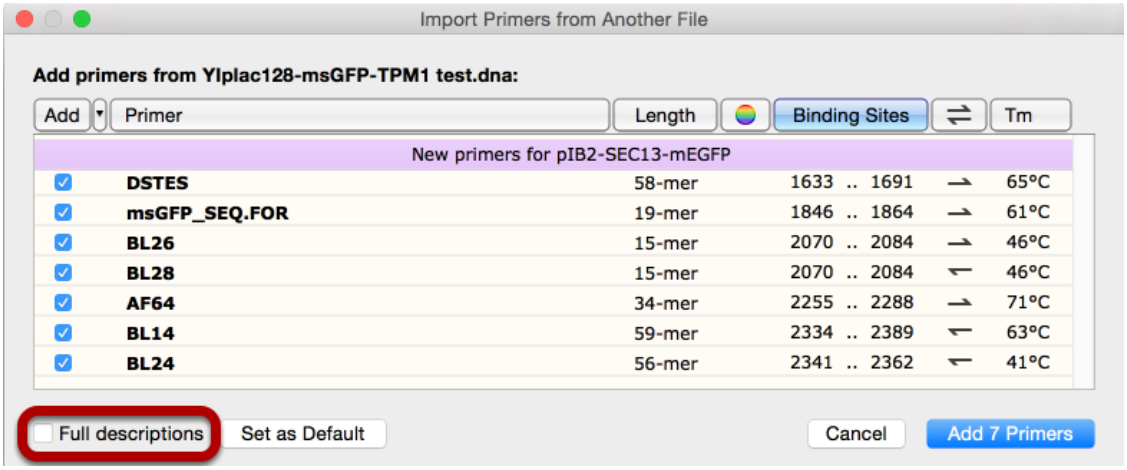



To specify the source Clone Manager file, click **Primers** → **Import Primers from Clone Manager...** .



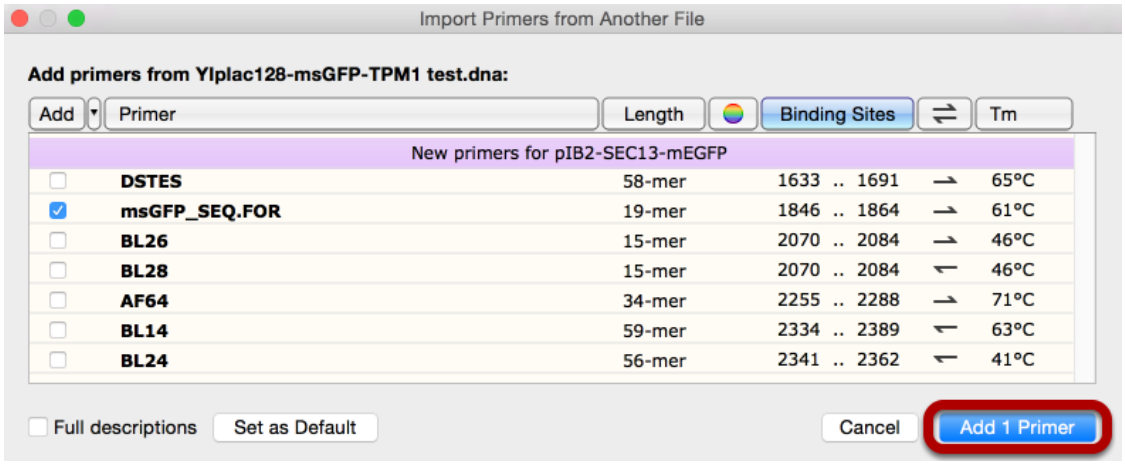

In the Import Primers from Clone Manager dialog, choose a *.pd4 or *.px5 file, then click **OK**.

## View the Available Primers List



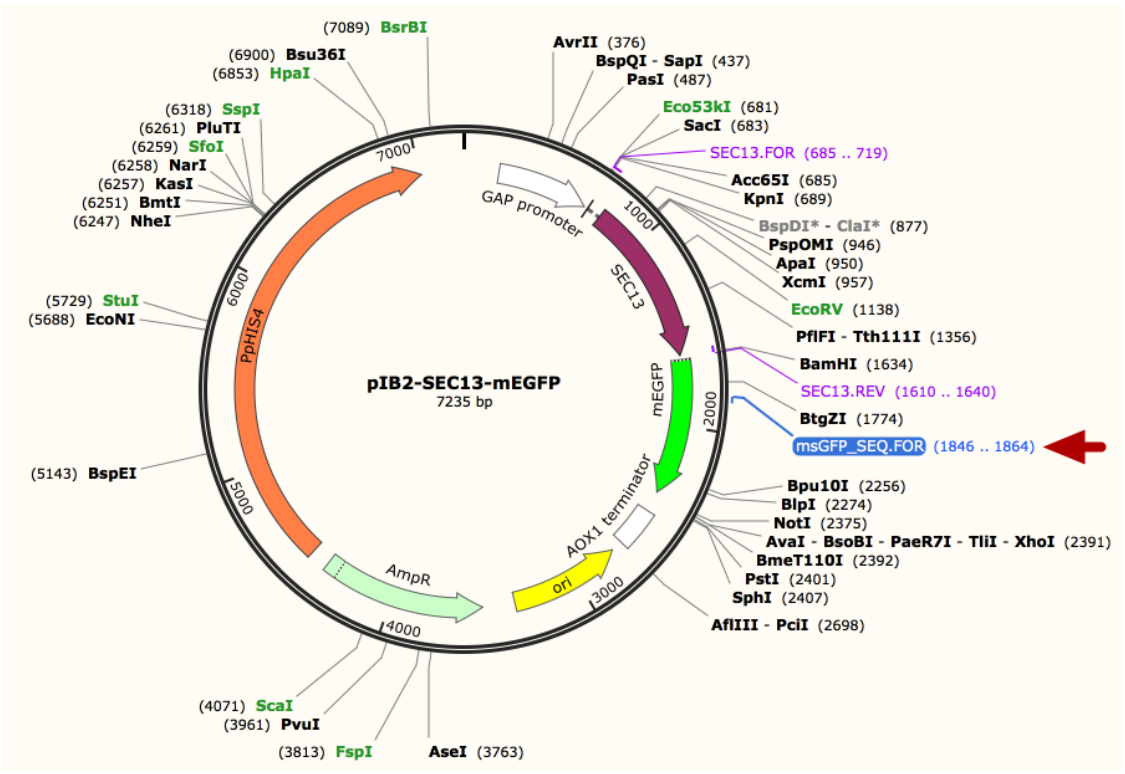

A second dialog will list the primers that are to be imported. If desired, click the check box at the lower left to show the descriptions.

## Specify Which Primers Will Be Added




The **Add** column indicates which primers will be imported. To change the default choices, click the relevant check boxes, then click **Add [N] Primer(s)**.

## View the Imported Primers

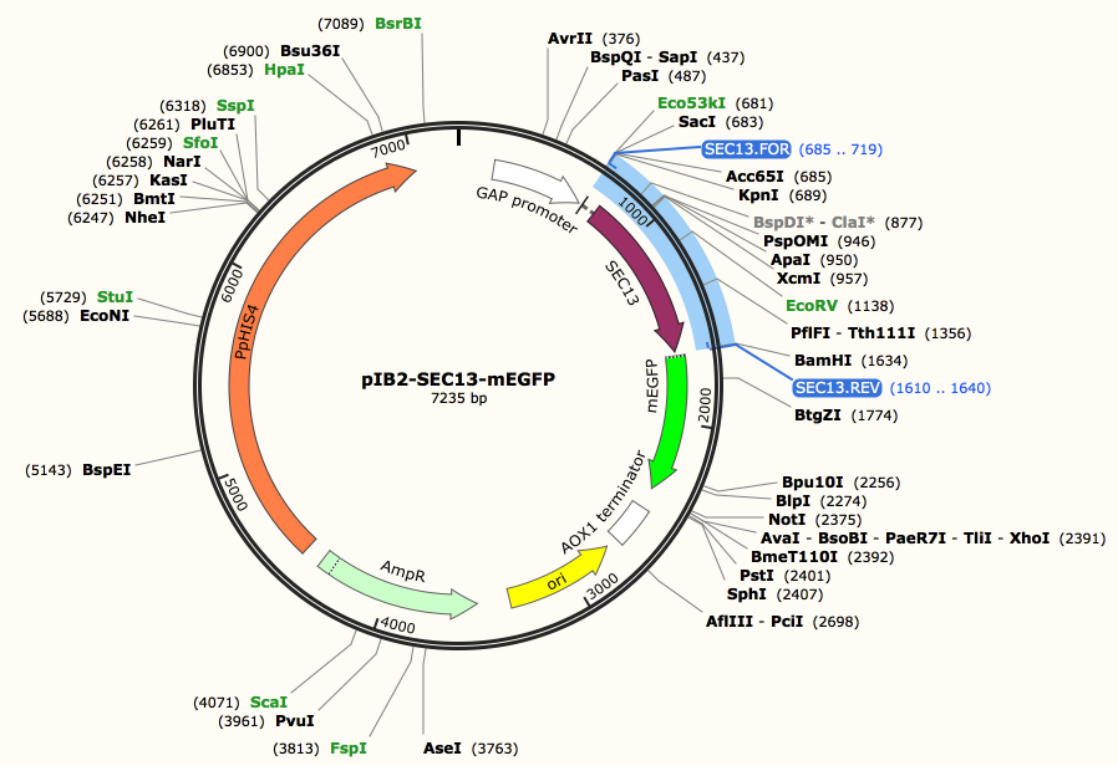



The imported primers will be added to the destination file.

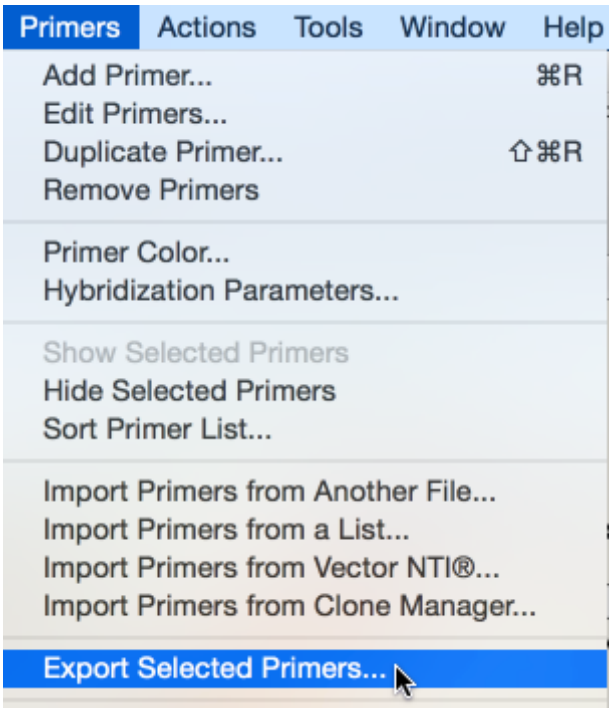
## Export Primers

How can I export primers to another file?

### Select the Primers to Be Exported

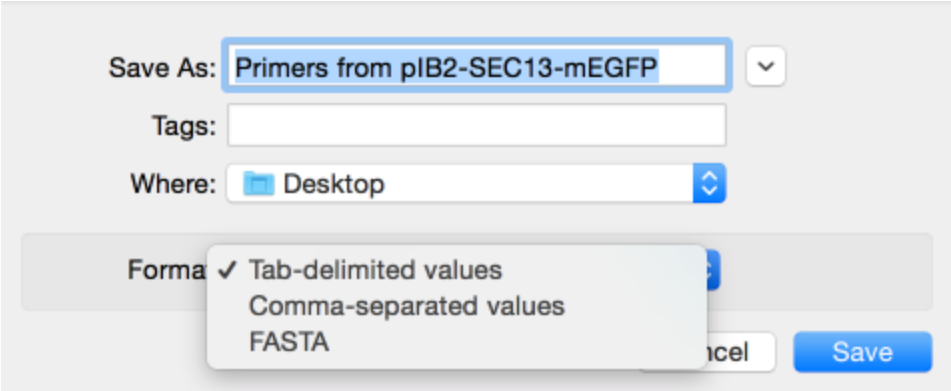



Select the primers of interest.



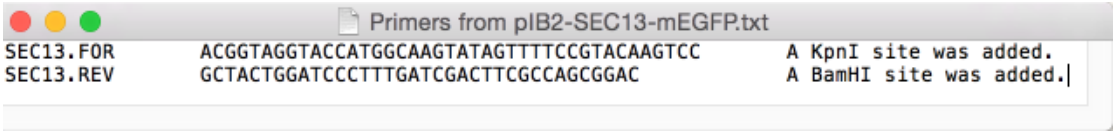

Choose **Primers** → **Export Selected Primers...** .

## Save the Primers




Specify the file name, destination, and file format, then click **Save**.

## View the Exported Primers



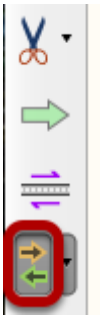

The exported data will be listed in a legibly formatted text file. This text file is intended for ordering oligos or importing primers into another SnapGene file.

# Translations

# Show or Hide Translations and ORFs

Translations displayed in Map and Sequence views can be shown or hidden.

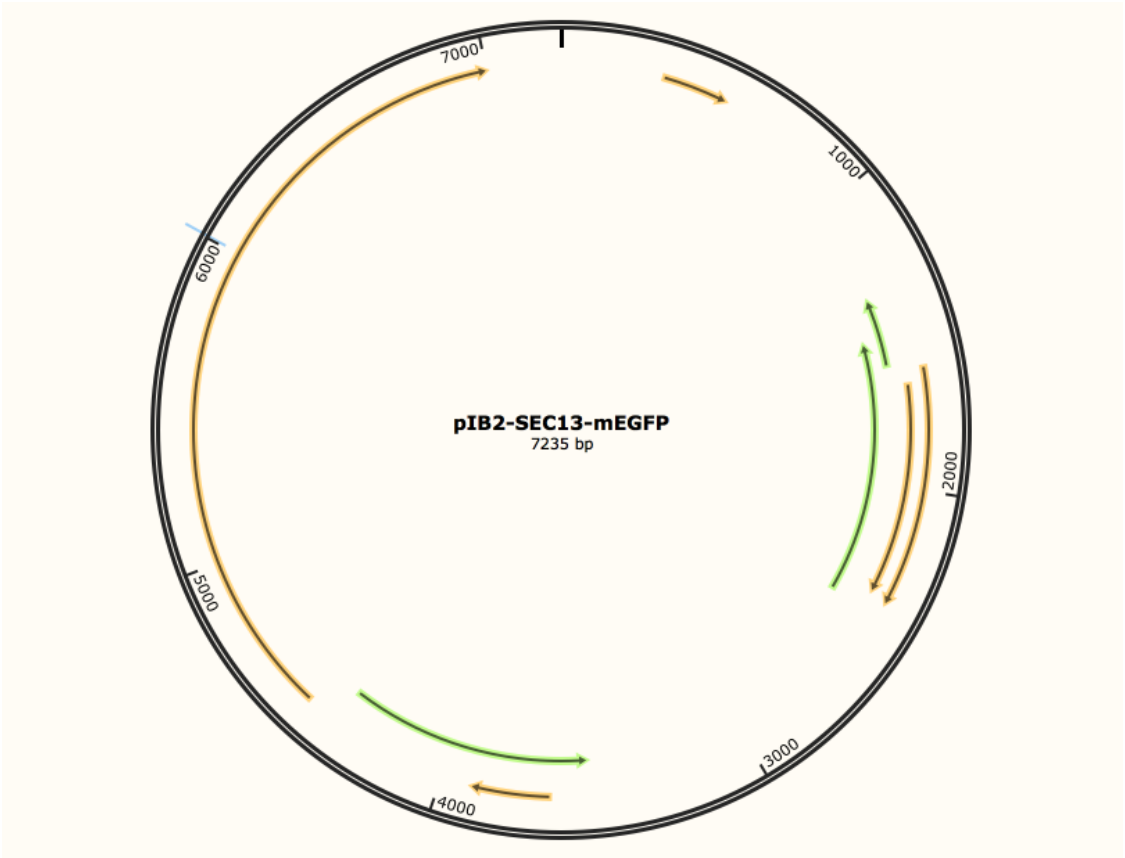
## Show or Hide Translations and ORFs



To show or hide translations and ORFs, click the "Show translations" button in the side toolbar.

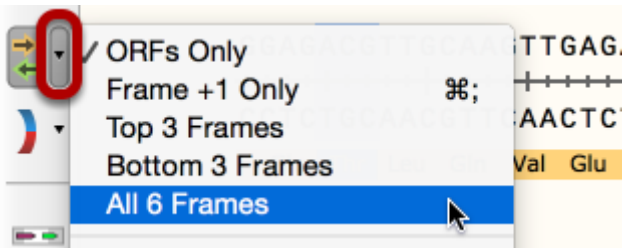
## View the Translations

In Sequence view, individual amino acids will be displayed. ORFs will be highlighted in orange and green.




In Map view, top-strand ORFs are highlighted in orange, and bottom-strand ORFs are highlighted in green.

## Choose the Translation Options

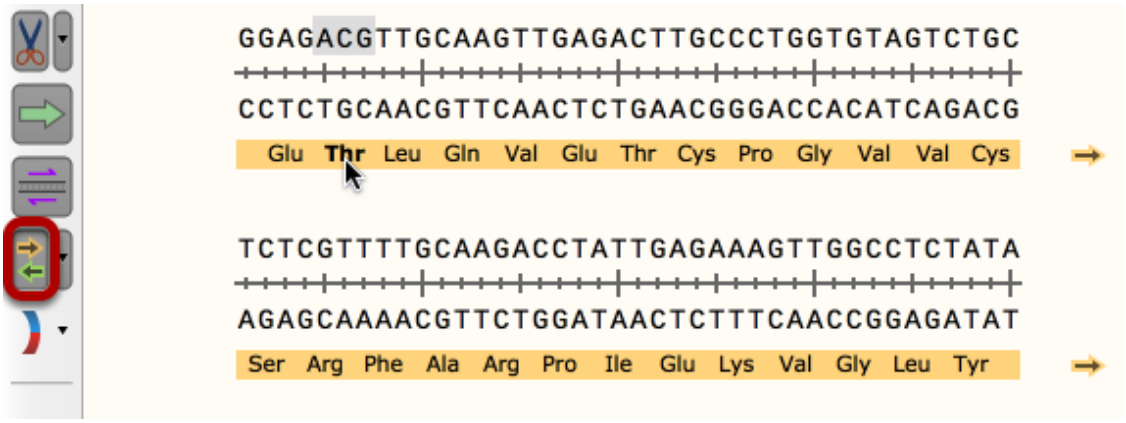



As described in the accompanying lessons, use the "Choose translated sequences" menu button in the side toolbar to select the ORFs and translations that are displayed.

## Choose the Translation Frames

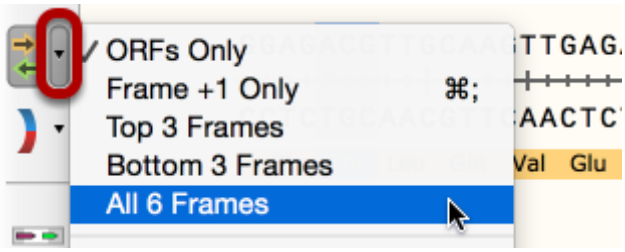
How can I choose which translation reading frames are displayed in Map and Sequence view?

### Show Translations



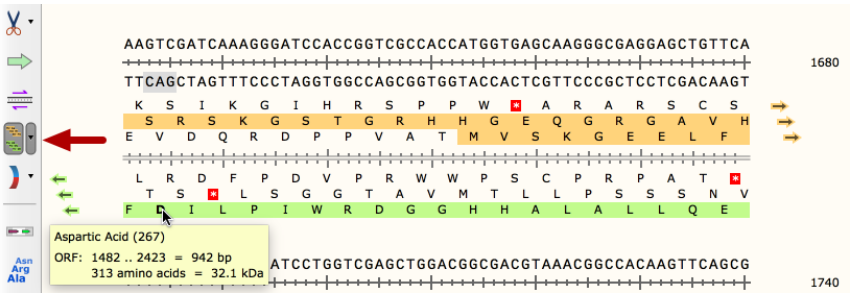

If translations are not visible, click the "Show translations" button in the side toolbar.

### Select the Translation Reading Frames




Click the side toolbar "Choose translated sequences" menu button to select the translation reading frames.

### View the Updated Translation

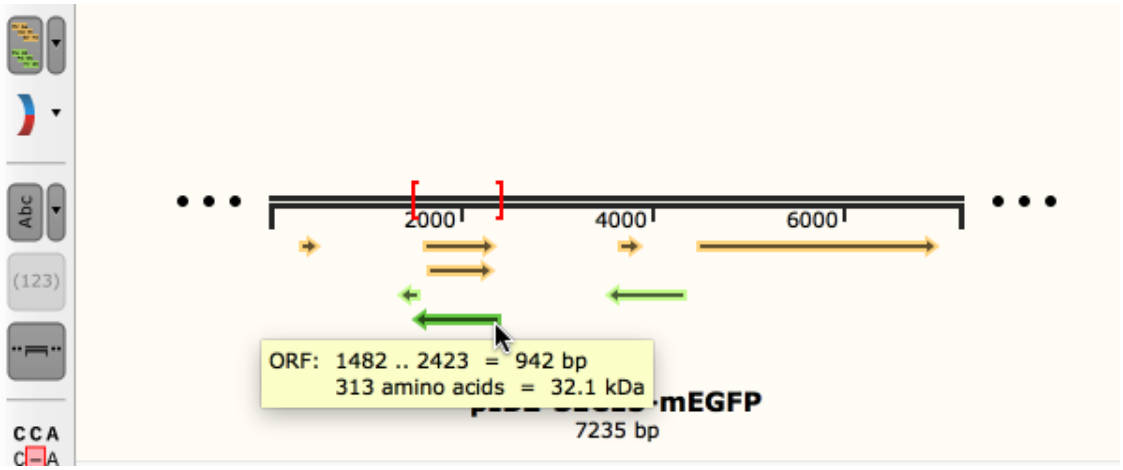



In this example, the updated Sequence view shows the "All 6 Frames" selection.

Top-strand ORFs are highlighted in orange, and bottom-strand ORFs are highlighted in green.

Small arrows at the end of each row remind the viewer of the translation frame and directionality.

Note: The **Show translations** button appearance changed to reflect the "All 6 Frames" selection.

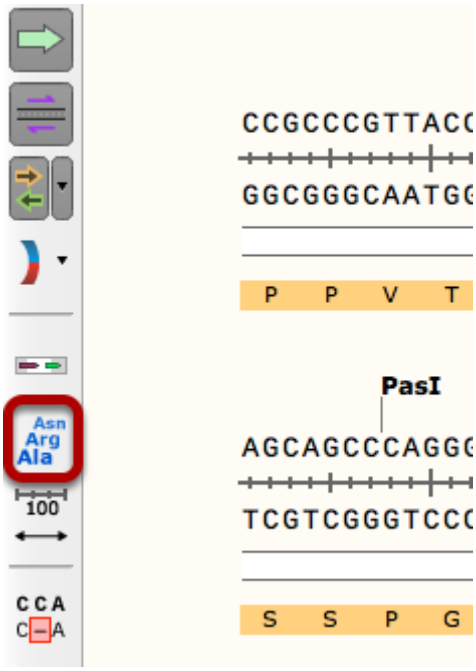



In Map view, the ORFs within the selected translation reading frames are shown.

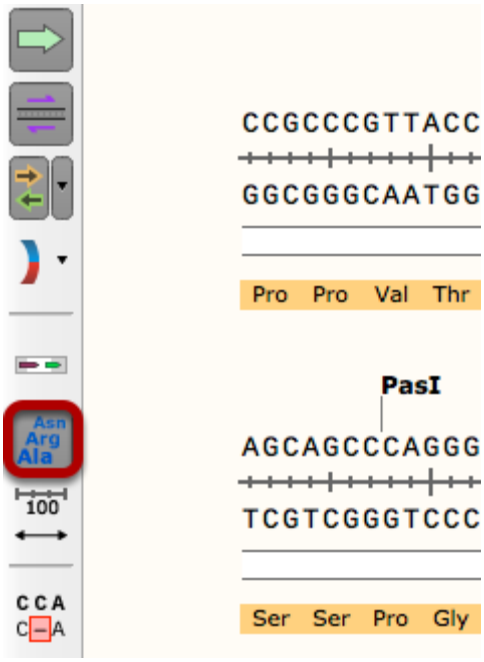
## Choose the Translation Format

How can I choose the translation letter code format shown in Sequence view?

### Choose 1- or 3-Letter Amino Acid Codes

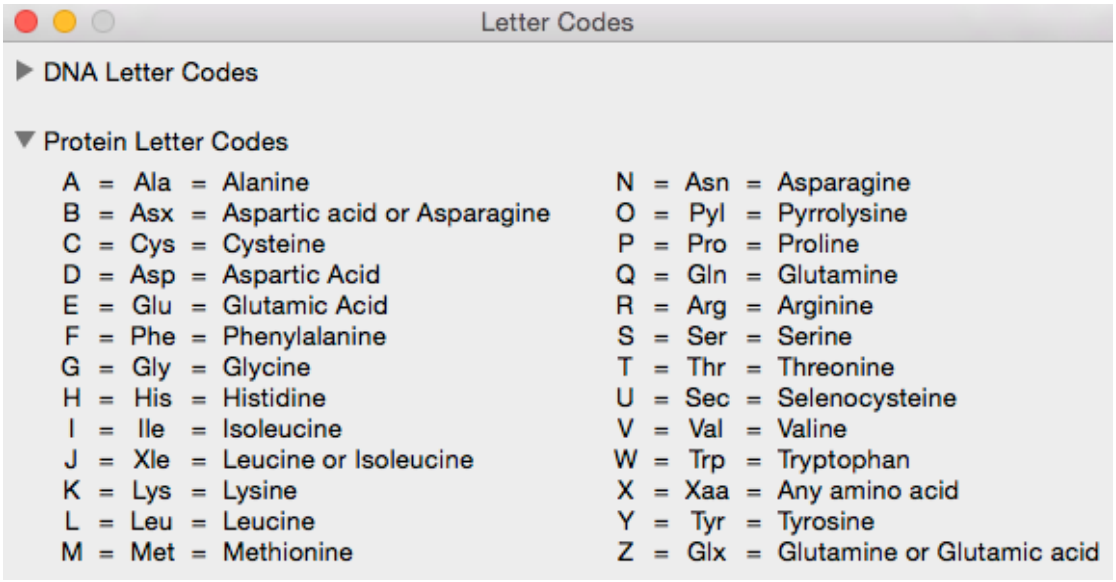



Click the "Use 3-letter amino acid codes" button in the side toolbar to change the translation letter code format from 1-letter to 3-letter codes.

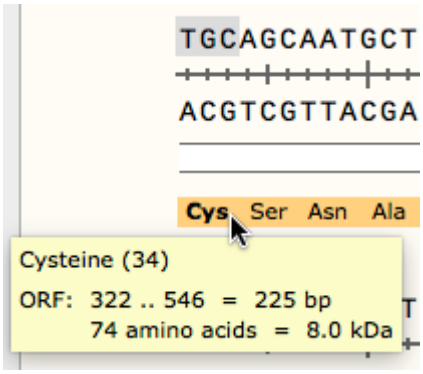



The "Use 3-letter amino acid codes" button background is gray when 3-letter codes are shown.

## Review the Letter Codes




To review the protein letter code abbreviations, click **Tools → Letter Codes.**

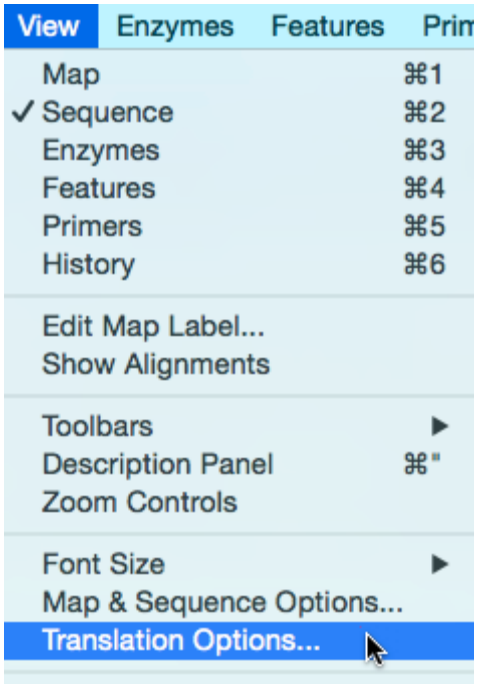



Alternatively, mouse over the amino acid letter code to review the letter code abbreviation, the position of the amino acid in the ORF, and a synopsis of the ORF data.

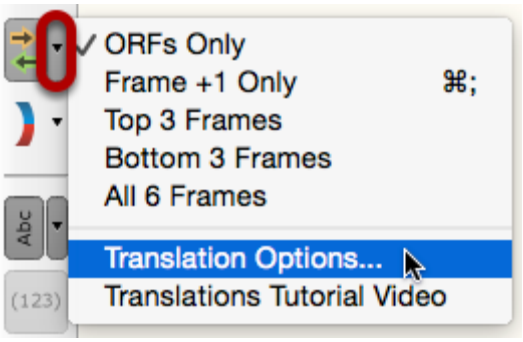
## Choose the Translation Genetic Code

How can I specify a nonstandard genetic code for translations?

### Open the Translation Options Dialog

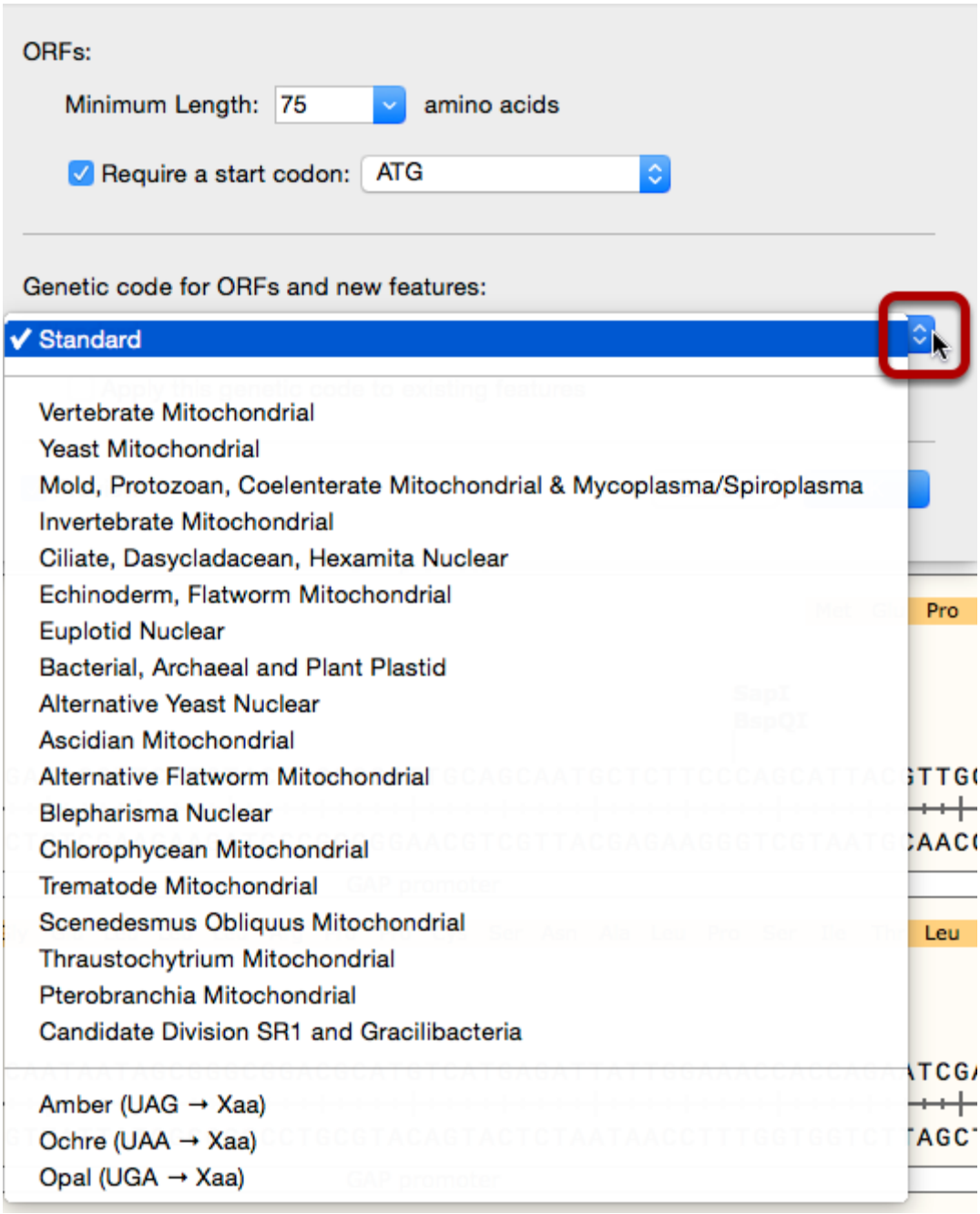



To open the Translation Options dialog, click **View** → **Translation Options...** .

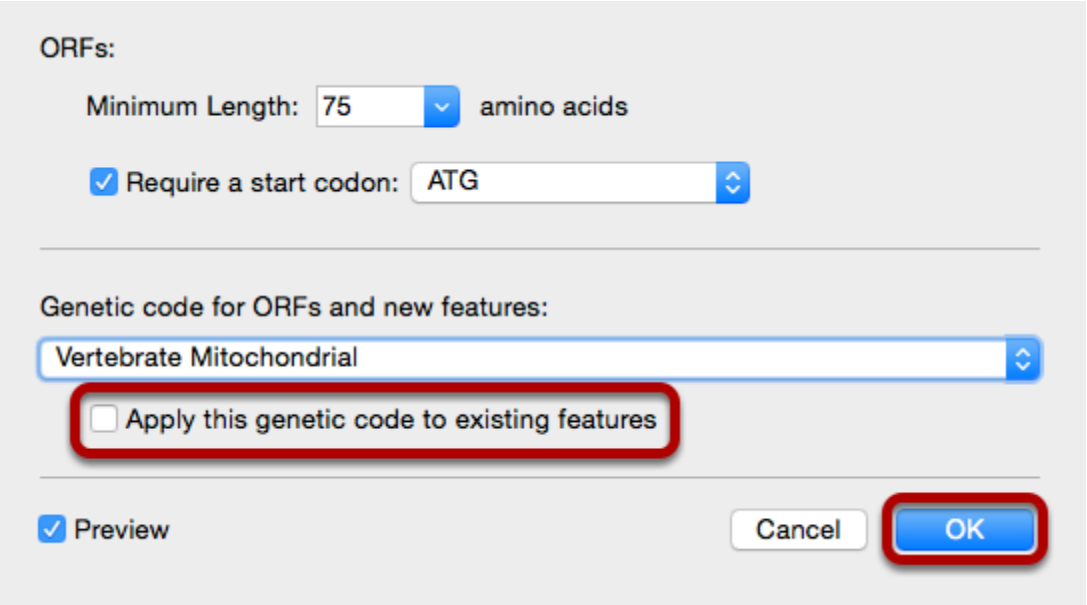



Alternatively, click the side toolbar "Choose translated sequences" menu button, then click **Translation Options...** .

## Choose the Translation Genetic Code




Expand the **Genetic code for ORFs and new features** menu, then choose the desired genetic code.



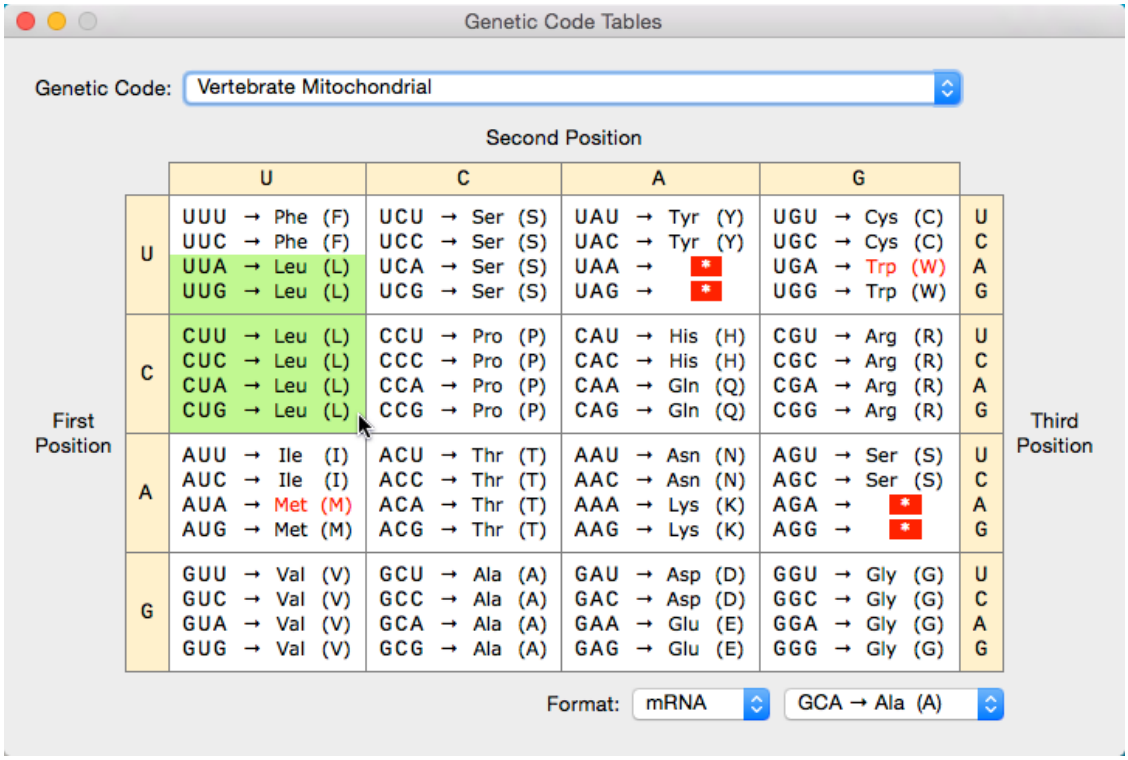

To apply the selected genetic code to existing features, click the check box. Close the dialog by clicking **OK**.

## View the Updated Translation

The updated genetic code will now be used in the translations.

Note: The nonstandard genetic code name is highlighted in yellow.

## Review the Genetic Code Table

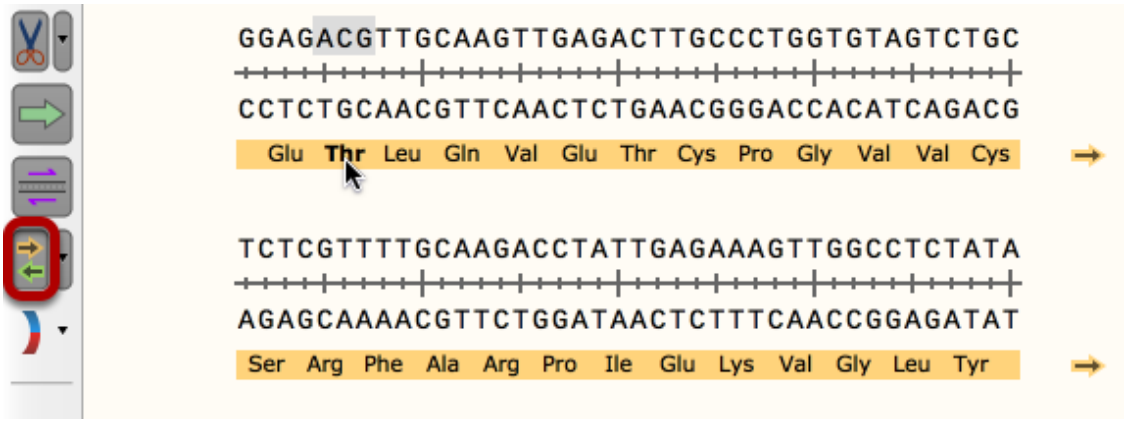



To review the selected genetic code translation table, click **Tools** → **Genetic Code Tables.** Mouse over the window matrix to highlight redundant translations.

# Display Open Reading Frames

How can I adjust the display of translations in Map and Sequence views?

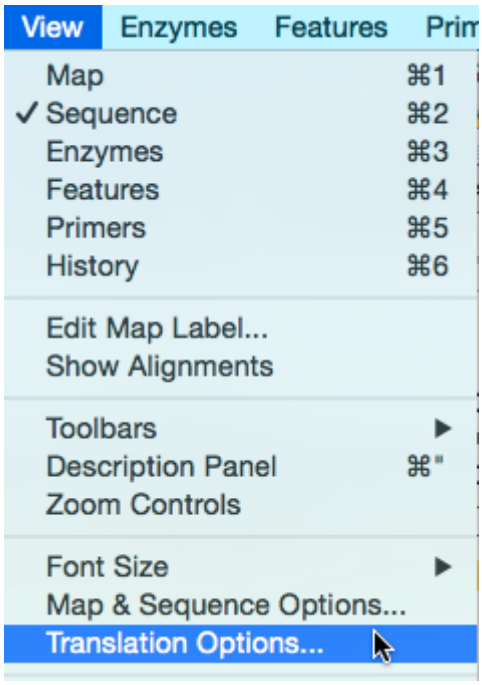
## Show Translations



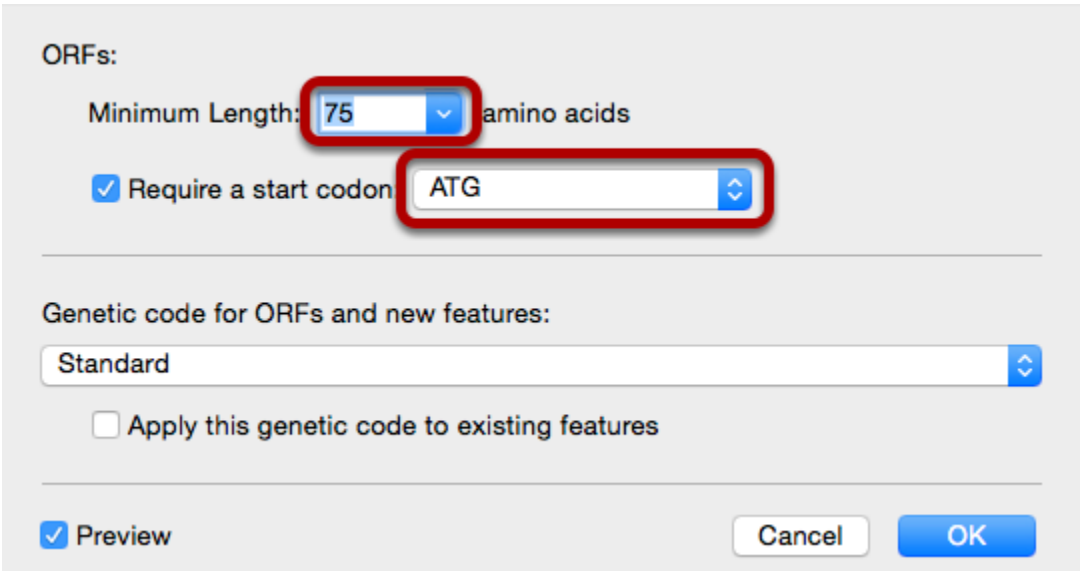

If ORFs are not visible, press the "Show translations" button in the side toolbar.

In Map view, top-strand ORFs are highlighted in orange, and bottom-strand ORFs are highlighted in green.

## Change Translation Options




To open the Translation Options dialog, click **View** → **Translation Options...** .




To adjust the criteria for showing ORFs, specify the minimum ORF length and the start codon options, then click **OK**.

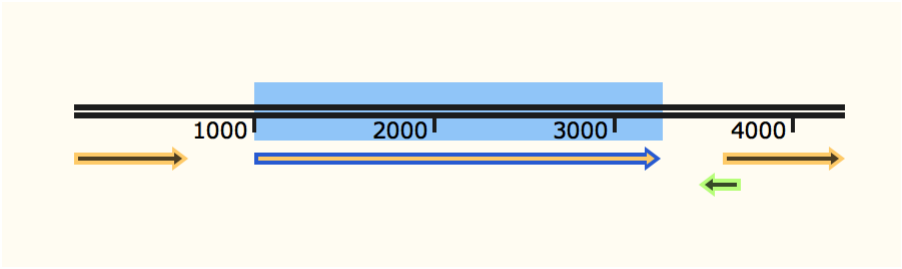
## Convert an ORF to a Feature

How can I convert an ORF to a translated feature?

### Display ORFs

If ORFs are not visible, press the "Show translations" button in the side toolbar to show ORFs.

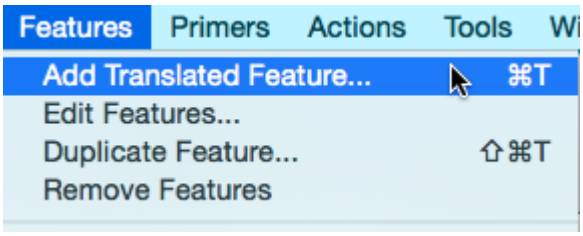
### Select the ORF




Click on the relevant ORF arrow to select it.

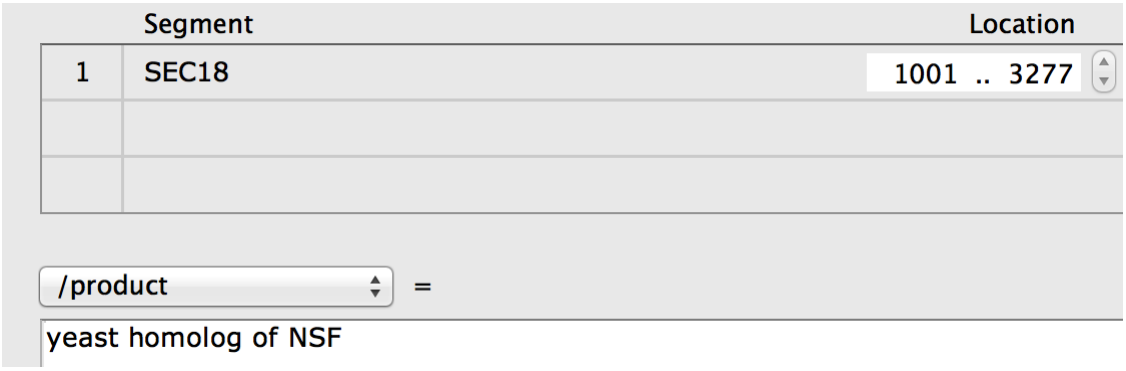
In Sequence view, double-click on one of the amino acids to select the ORF.

### Add a Translated Feature

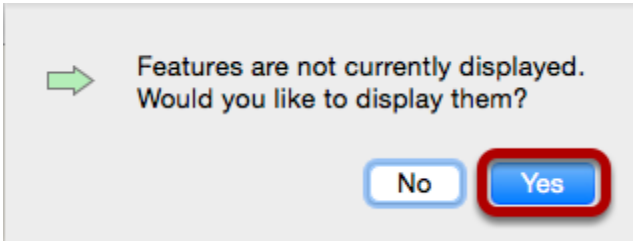



Click **Features** → **Add Translated Feature...** .

## Specify the Feature Details

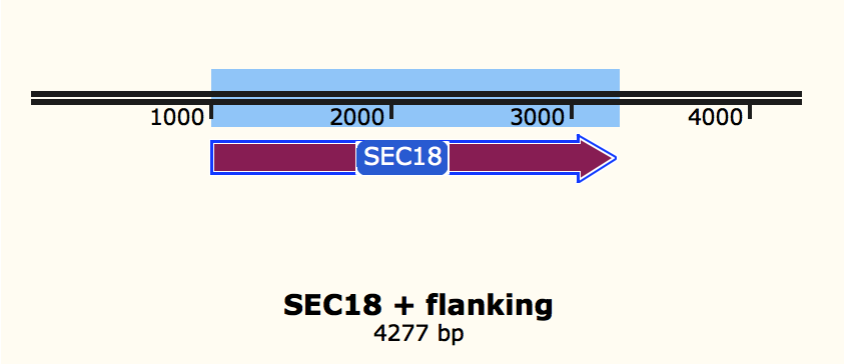



In the Add Feature dialog, type the feature name, and specify additional properties as desired, then click **OK**.




If necessary, click **Yes** in the subsequent dialog.

## View the Feature



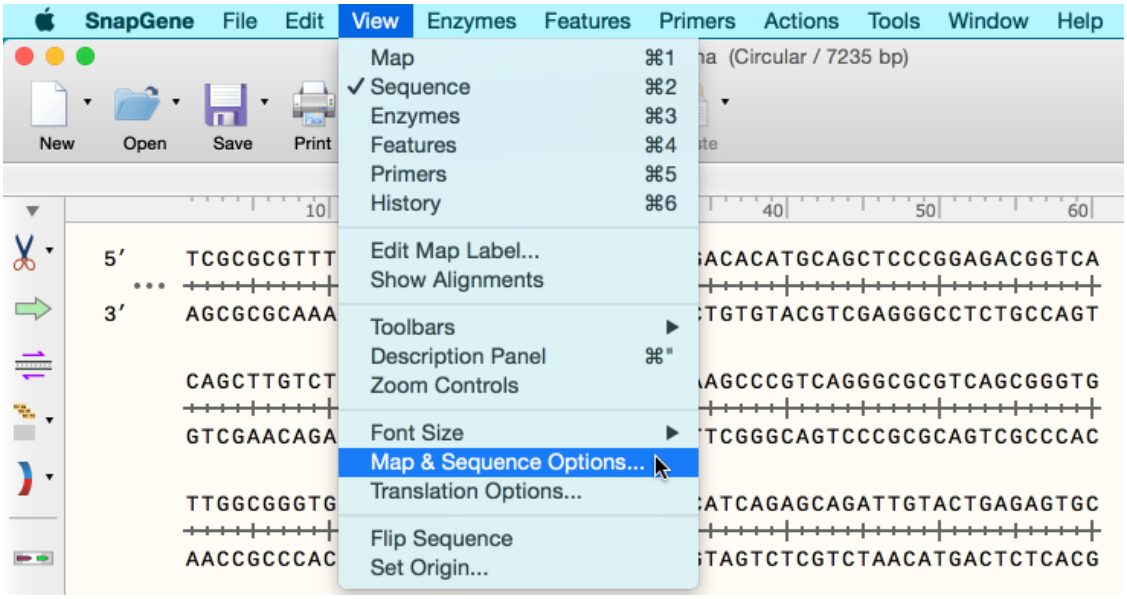

Press the "Show translations" button to hide ORFs, leaving the translated feature visible.

# Colors

## Show or Hide Colors

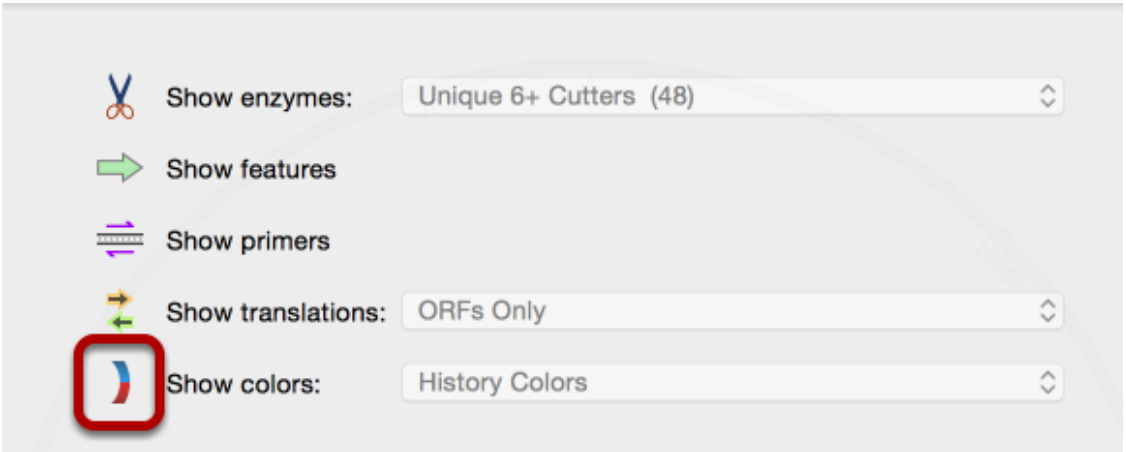
The sequence colors shown in Map and Sequence views can be shown or hidden.

### Open the Map and Sequence Options Dialog




To open the Options dialog, click **View** → **Map & Sequence Options...** .

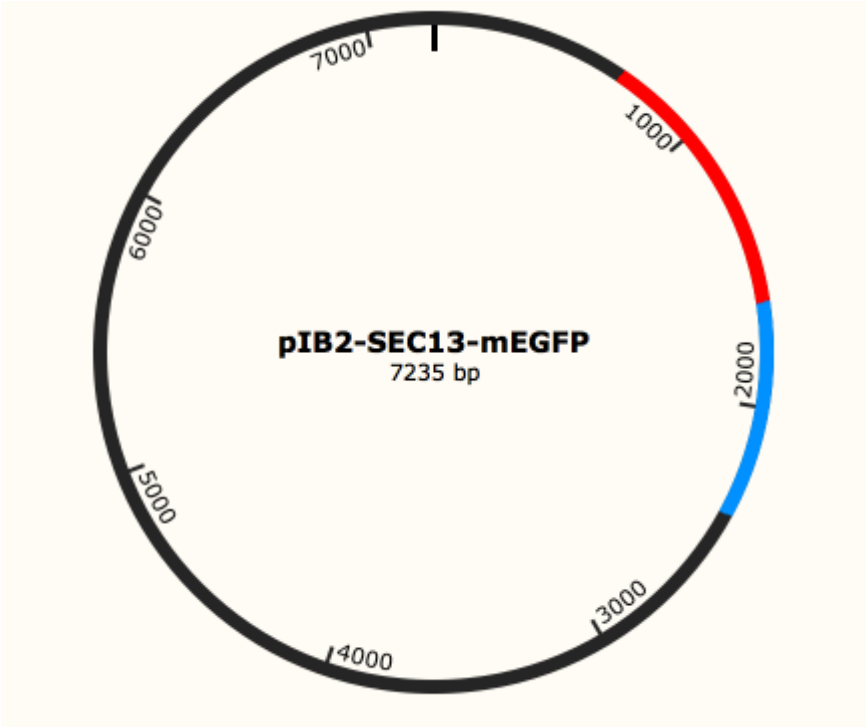
### Show the Sequence Colors




To show the sequence colors, click the "Show colors" button, then click **OK**.

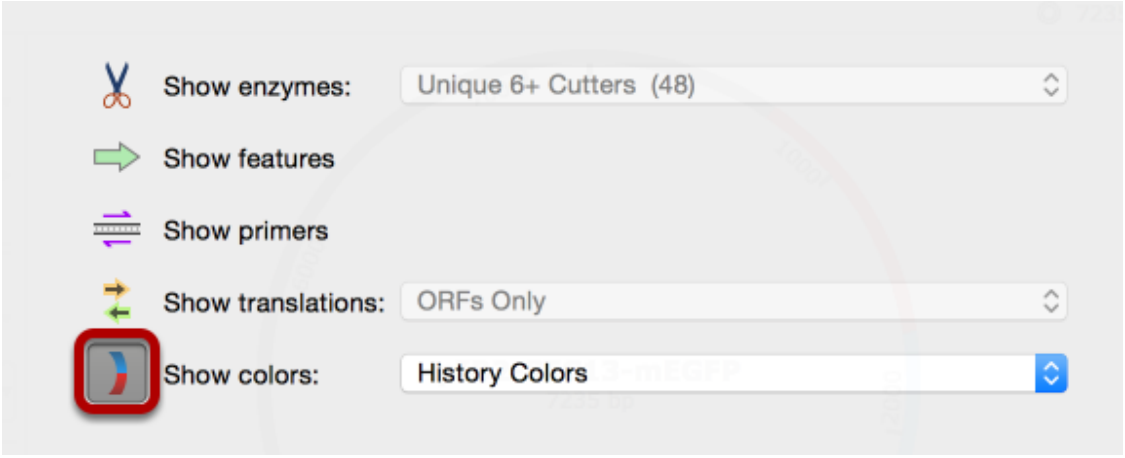
This button can also be used to toggle between showing and hiding colors.

## View the Sequence Colors



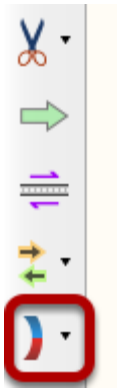

Sequence colors will now be shown in Map and Sequence views.

## Hide the Sequence Colors




To hide the sequence colors, click **View** → **Map & Sequence Options...** , click the "Show colors" button, then click **OK**.

## Show or Hide Sequence Colors Using the Side Toolbar

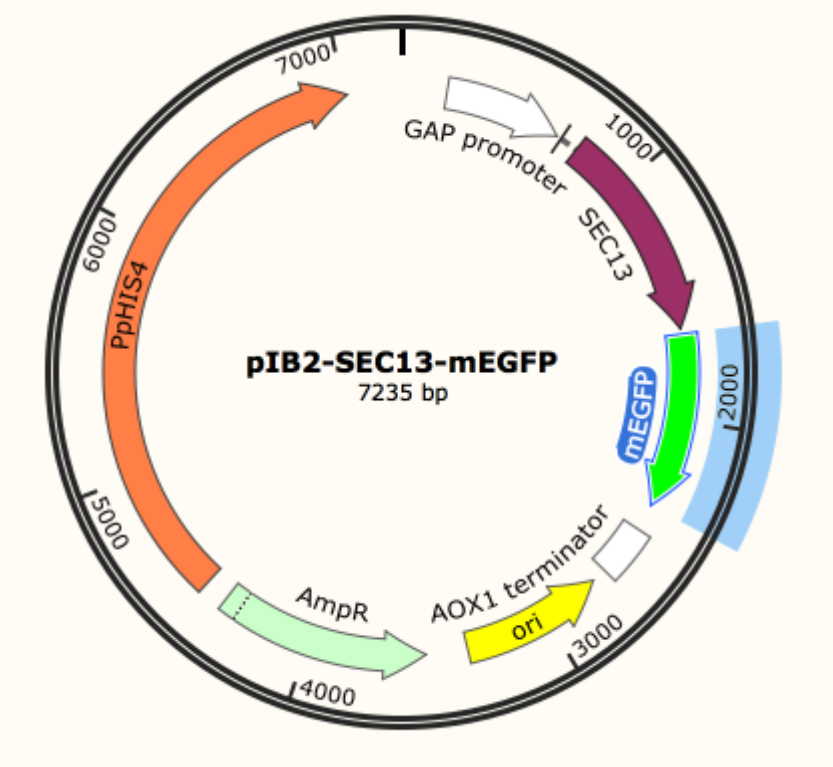


Alternatively, click the "Show colors" button in the side toolbar to toggle between showing and hiding sequence colors.

## Customize the Sequence Colors

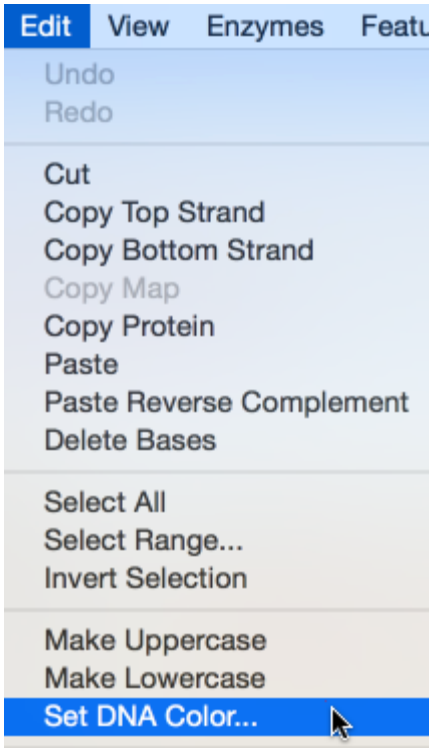
The sequence colors shown in Map and Sequence views can be customized.

### Select a Sequence Region

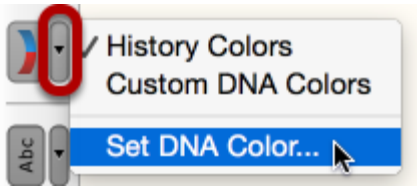



Select the sequence region that will be labeled with a color.

## Customize the Sequence Color

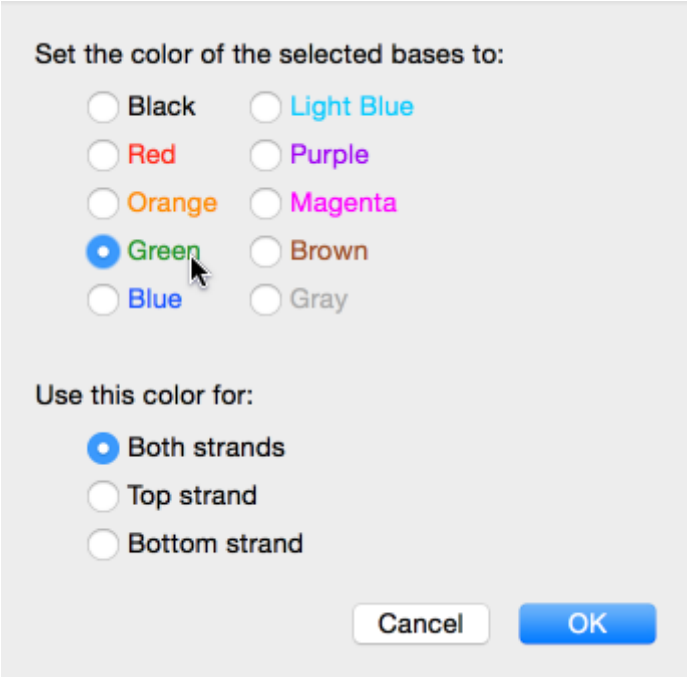



To customize the sequence color, click **Edit** → **Set DNA Color...** .



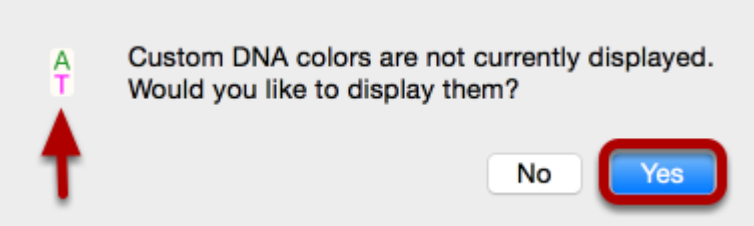

Alternatively, to customize the sequence color using the side toolbar, click the "Show colors" menu button from the side toolbar, then click "Set DNA Color..." .

## Customize the Sequence Color




Choose the color and specify which strand(s), then click **OK**.

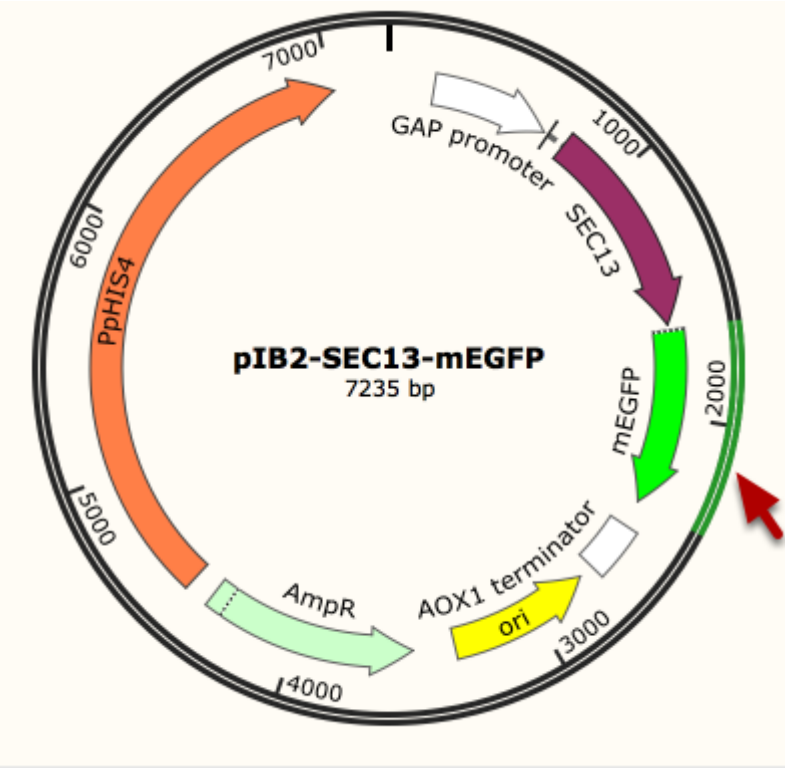
## Switch Color Modes




Click **Yes** to show Custom DNA colors (note the icon change).

Alternatively, to show previously customized DNA colors, open the Options dialog by clicking **View → Map & Sequence Options...** . Click the "Show colors" menu button to select "Custom DNA Colors", then click **OK**.

## View the Custom Sequence Colors

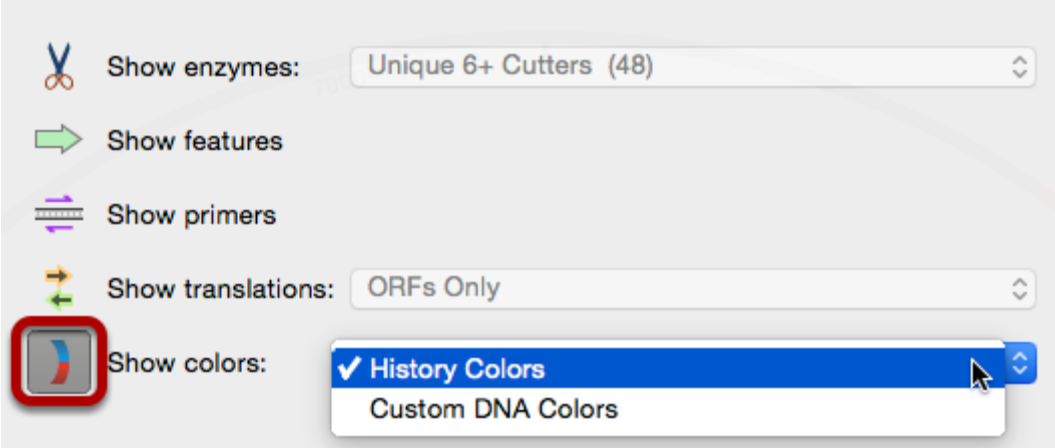



The customized sequence colors will now be shown in Map and Sequence views.

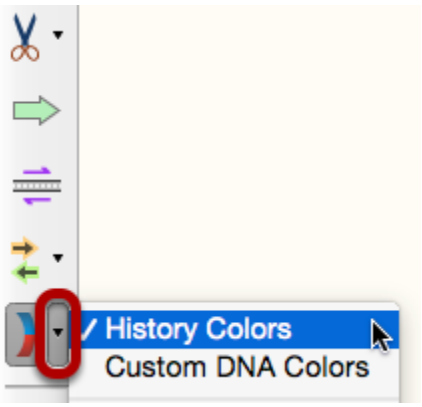
## Choose a Color Mode

Sequence colors can be displayed in two modes: history colors (default) and custom colors.

### History Colors Mode

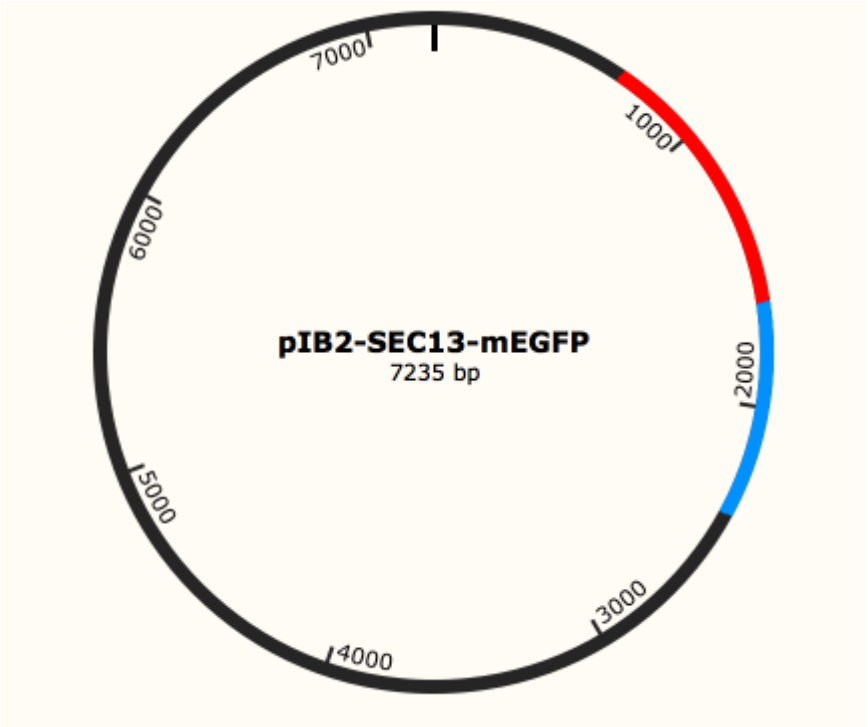



To view history colors, open the Options dialog by clicking **View → Map & Sequence Options...** . Click the "Show colors" button, click the **Show colors** menu button, select "History Colors", then click **OK**.



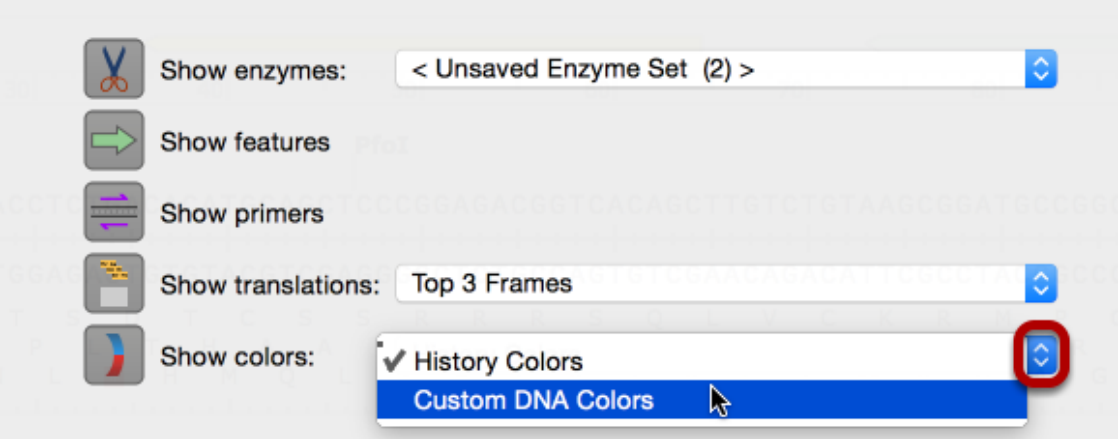

Alternatively, click the "Show colors" menu button from the side toolbar, then click "History Colors."

## View History Colors

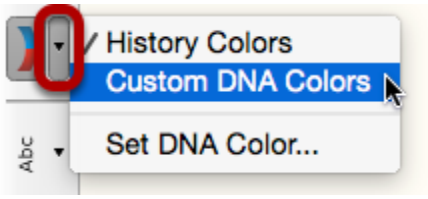



History colors will now be shown in Map and Sequence views.

## View Custom DNA Colors

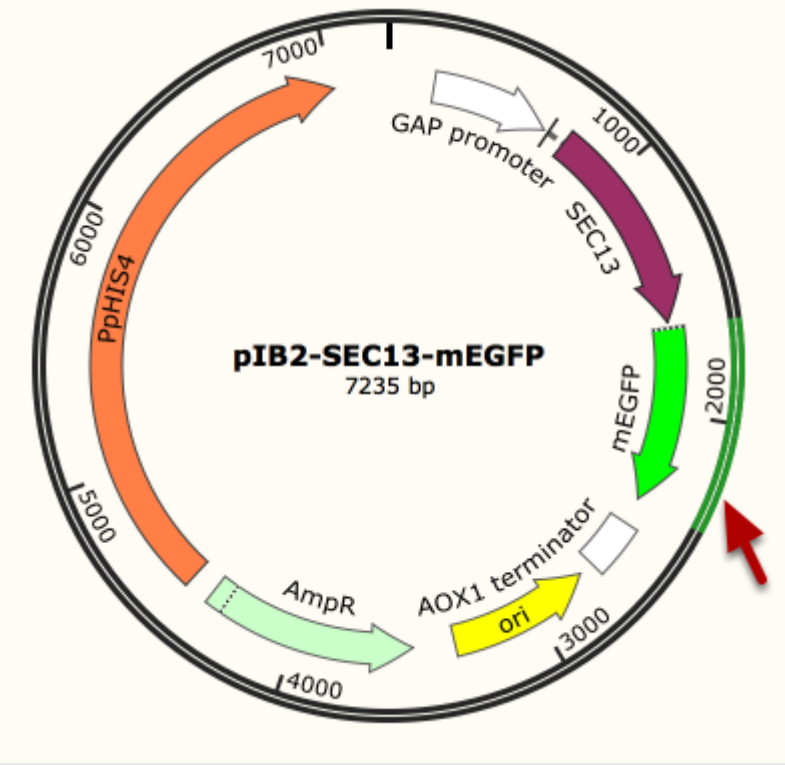



To switch from displaying history colors to previously defined custom colors, open the Options dialog by clicking **View** → **Map & Sequence Options...** . Click the **Show colors** menu button, select "Custom DNA Colors", then click **OK**.




Alternatively, click the "Show colors" menu button from the side toolbar, then click "Custom DNA Colors".

## View Custom DNA Colors

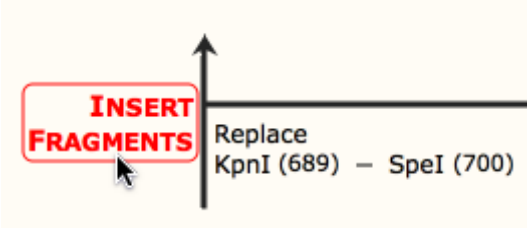



Customized sequence colors will now be shown in Map and Sequence views.

## Show Colors in History View

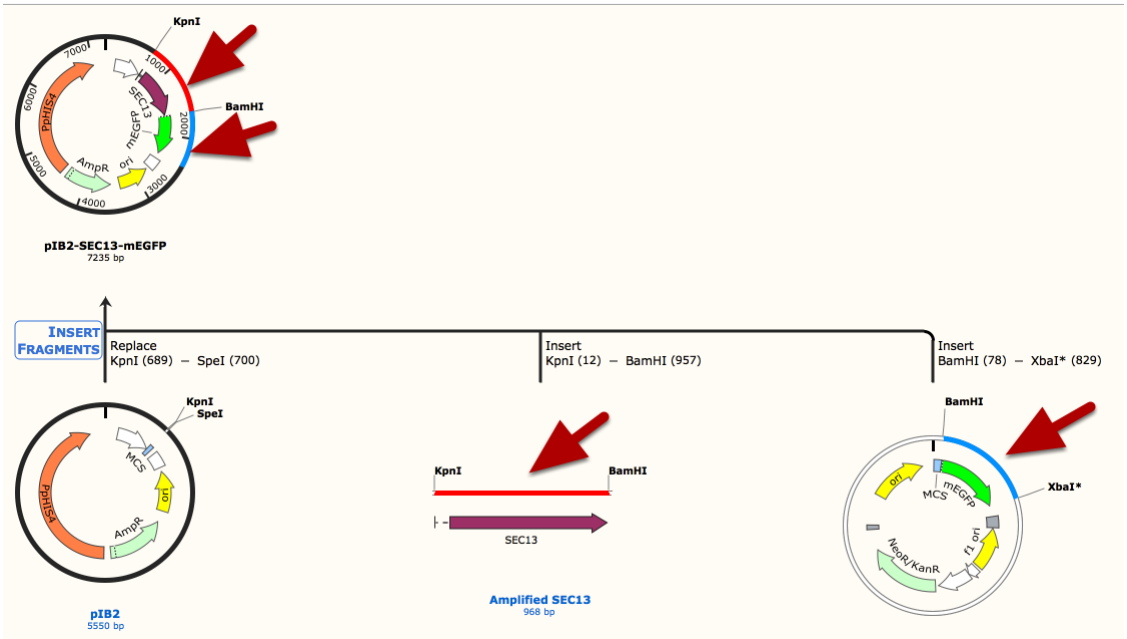
The sequences relevant to an operation can be colored in History view.

### Select the Operation



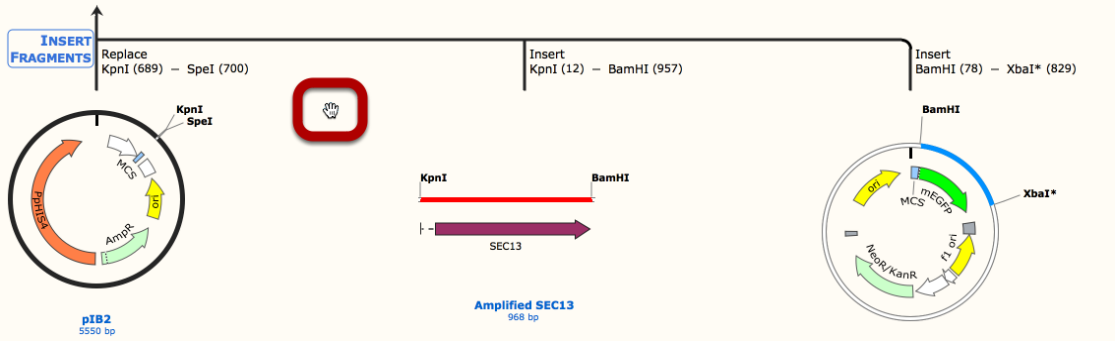

Click the operation name in History view.

### View the Highlighted Regions




The relevant regions of the sequences are now colored in History view.

## Hide the History Colors



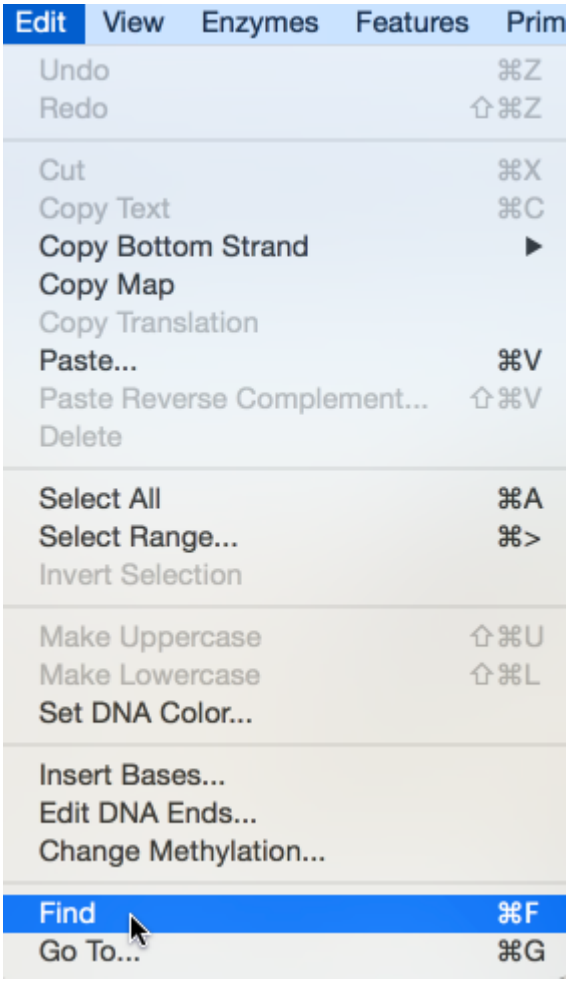

To hide the history colors, click anywhere in the white area.

# Searching

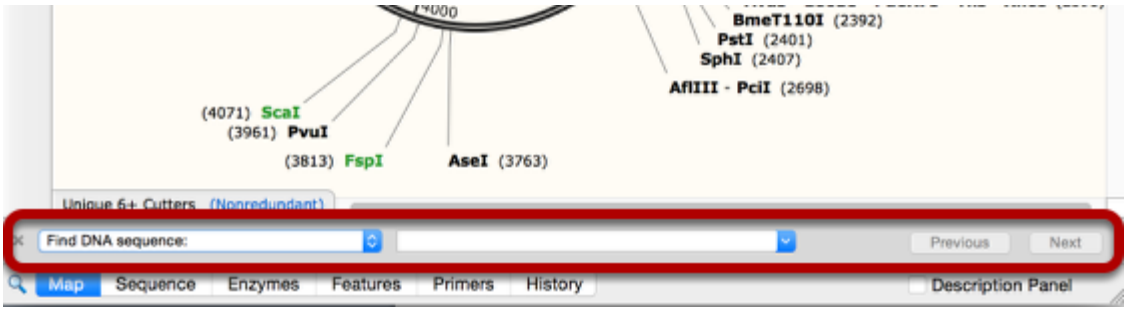
# Search for a DNA Sequence

How do I search for a DNA sequence?

## Show the Find Controls

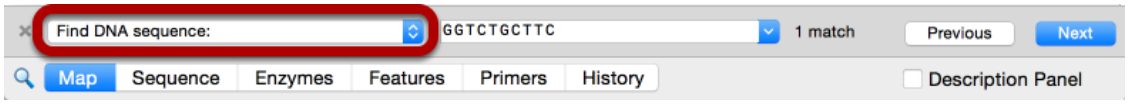



To show the Find controls, click **Edit** → **Find**.




The Find controls will appear near the bottom of the window.

## Search for a DNA Sequence




**Find DNA sequence:** will be chosen by default.

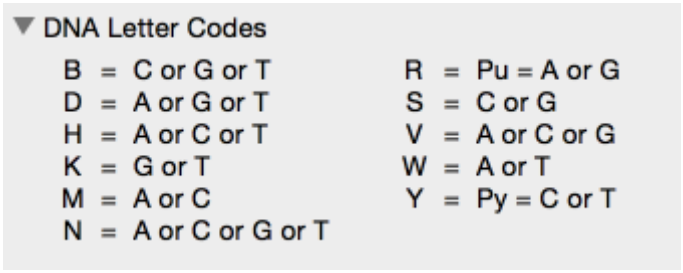
To search for a specific DNA sequence, type the query in the search box, then click **Next**.

## Search with Ambiguous Bases

To allow multiple bases at a specific position in the query sequence, use parentheses to enclose the set of allowed bases.

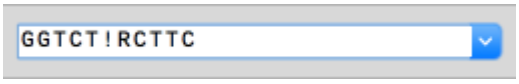
In the example shown in the screenshot, either A or G will be allowed at the designated position. SnapGene will find matches to either GGTCT**A**CTTC or GGTCT**G**CTTC.

Alternatively, to allow multiple bases at a position, use a degenerate DNA letter code such as R.

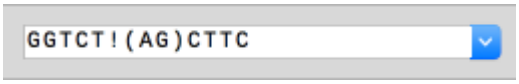



To view a list of DNA letter codes, click **Tools** → **Letter Codes**.

## Search with Excluded Bases




To exclude bases at a particular position, use an exclamation point in front of the forbidden bases. For example, **!**R means not A, G, or R.

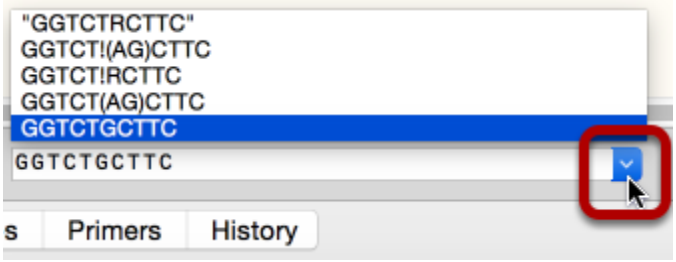



Alternatively, enclose the excluded bases in parentheses. For example, **!(**AG**)** means not A, G, or R.

## Search for a Degenerate Base

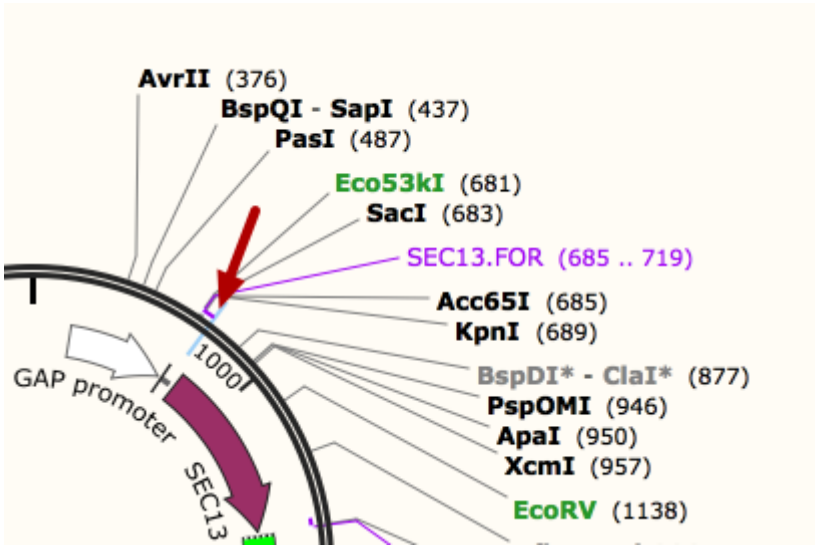
To search for a degenerate base, enclose the query in quotes. For example, a search for R will find all matches to A or G or R, but a search for "R" will find only bases that are actually represented as R.

## Review Recent Searches



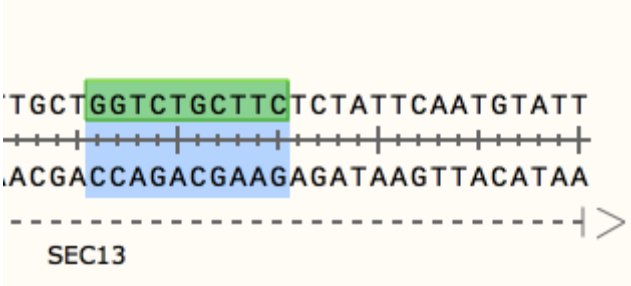

To review or reuse recent searches, click the arrow button on the search box.

## See the Matches in Map View



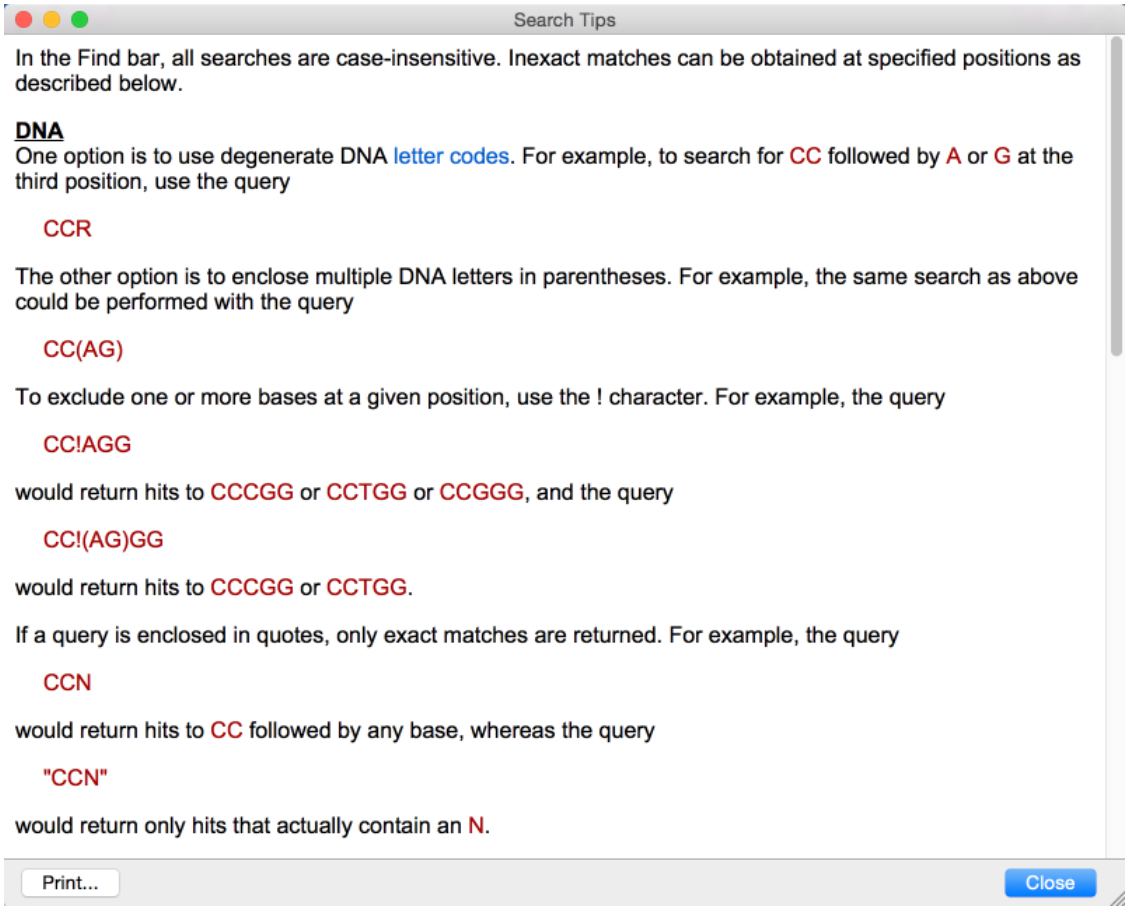

In Map view, a matching DNA sequence will be highlighted in blue.

## See the Matches in Sequence View

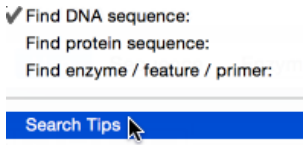



In Sequence view, a matching DNA sequence on either strand will be highlighted in green.

## Show the Search Tips




To view examples of DNA searches, click **Help** → **Search Tips**.

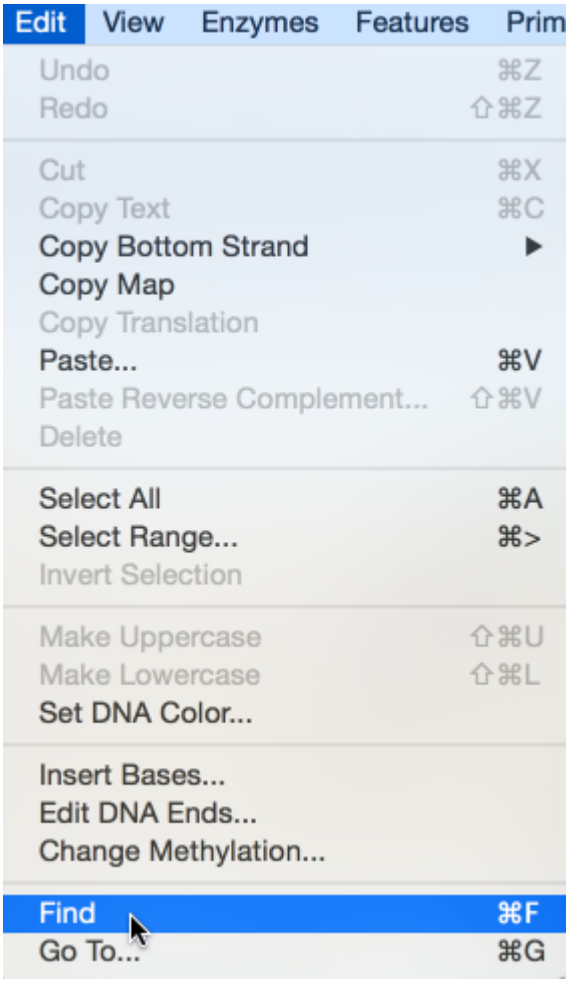



Alternatively, choose **Search Tips** from the Find controls.

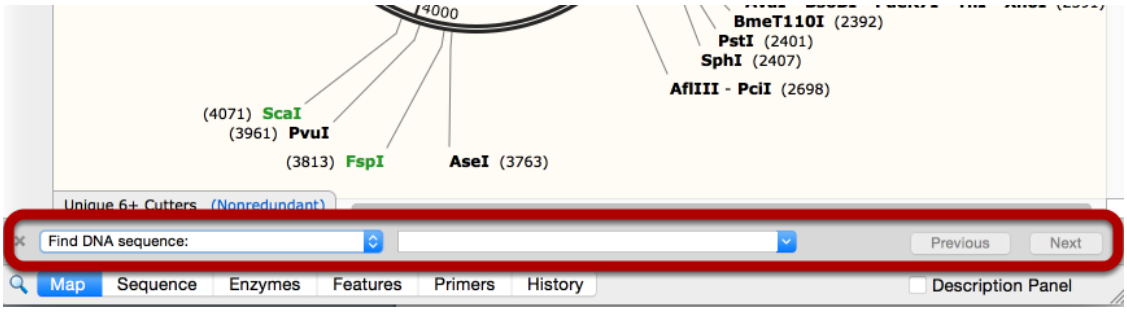
# Search for a Protein Sequence

How do I search for a protein sequence?

## Show the Find Controls




To show the Find controls, click **Edit** → **Find**.

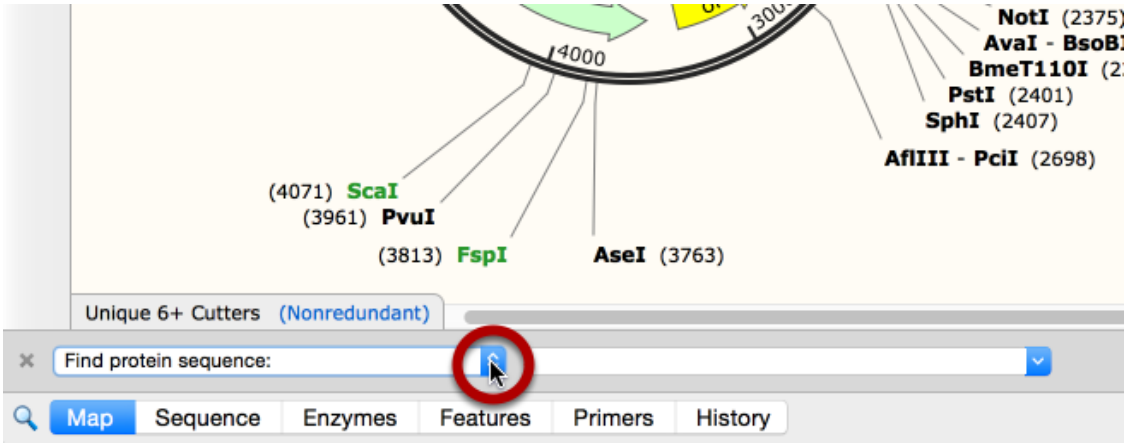



The Find controls will appear near the bottom of the window.

## Show Translations (Optional)

A protein sequence search is restricted to visible translations, which are typically present in translated features. If you also wish to show full-sequence translations, click the **Show Translations** button.

## Search for a Protein Sequence




Choose **Find protein sequence:**.

To search for a protein sequence, type the query in the search box, then click **Next**.

## Search with Ambiguous Amino Acids

To allow any amino acids at a specific position, use X.

## Search with Excluded Amino Acids

To exclude amino acids at a particular position, use an exclamation point in front of the forbidden amino acids. For example, !P means not P.

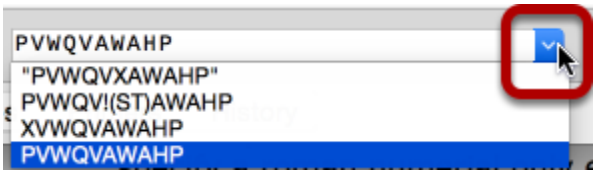
Alternatively, enclose the excluded amino acids in parentheses. For example, !(ST) means not S or T.

## Search for a Degenerate Amino Acid

To search for a degenerate amino acid, enclose the entire query in quotes. For example, a search for X will find all matches to any letter (A-Z), but a search for "X" will find only bases actually represented as X.

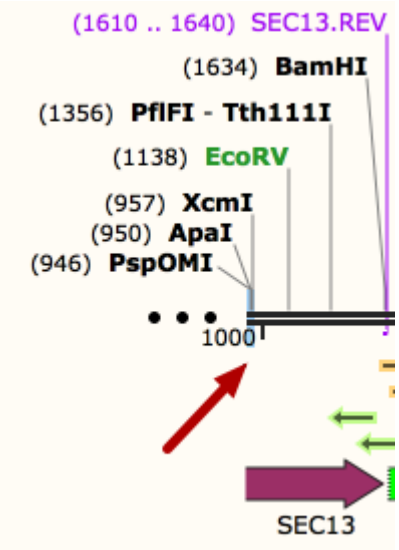
The other degenerate amino acids are B (matching either D or N) and Z (matching either E or Q).

## Review Recent Searches



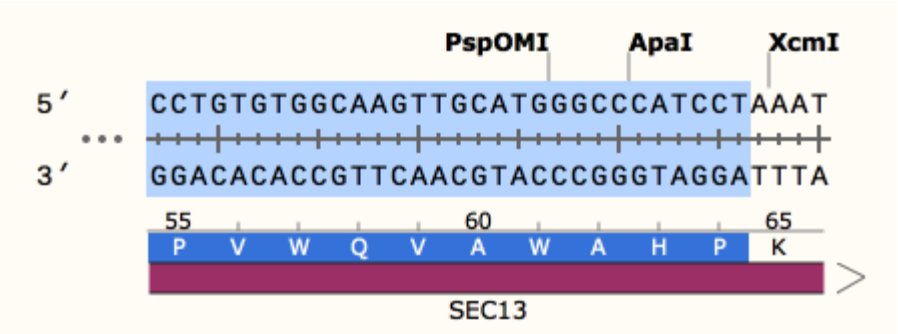

To review or reuse recent searches, click the arrow button on the search box.

## See the Matches in Map View



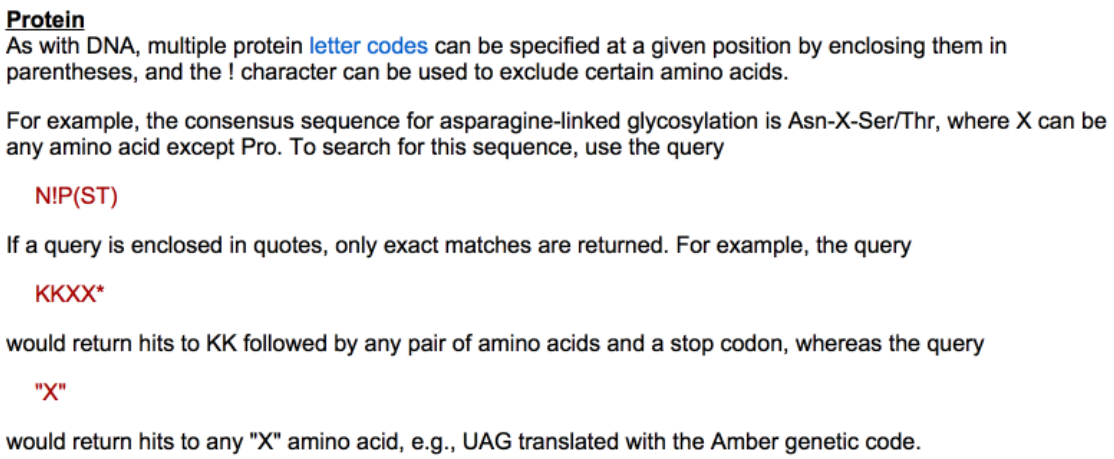

In Map view, a matching protein sequence will be highlighted in blue.

## See the Matches in Sequence View




In Sequence view, a matching protein sequence will be highlighted in blue.

## Show the Search Tips



**Protein**
As with DNA, multiple protein letter codes can be specified at a given position by enclosing them in parentheses, and the ! character can be used to exclude certain amino acids.

For example, the consensus sequence for asparagine-linked glycosylation is Asn-X-Ser/Thr, where X can be any amino acid except Pro. To search for this sequence, use the query

N!P(ST)

If a query is enclosed in quotes, only exact matches are returned. For example, the query

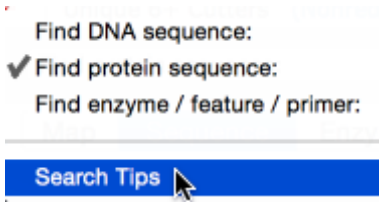
KKXX*

would return hits to KK followed by any pair of amino acids and a stop codon, whereas the query

"X"

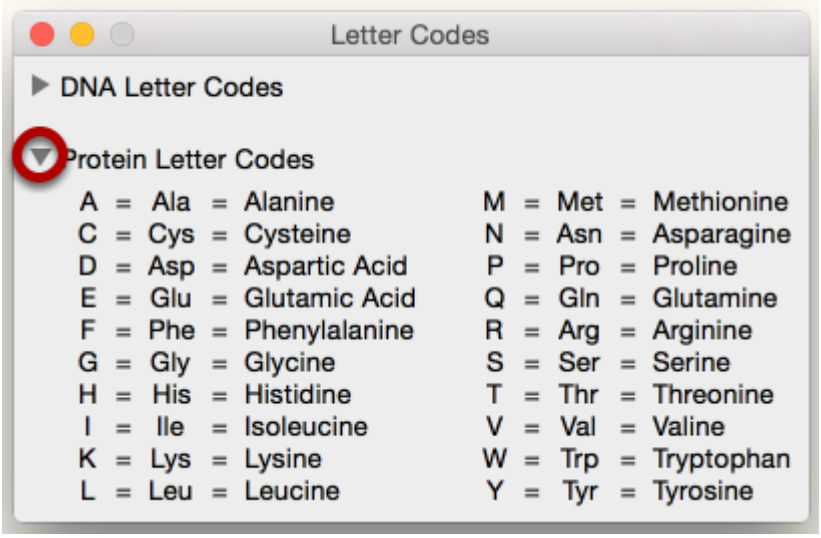
would return hits to any "X" amino acid, e.g., UAG translated with the Amber genetic code.

To view examples of protein searches, click **Help** → **Search Tips**, then scroll down.




Alternatively, choose **Search Tips** from the Find controls.

## View the Protein Letter Codes

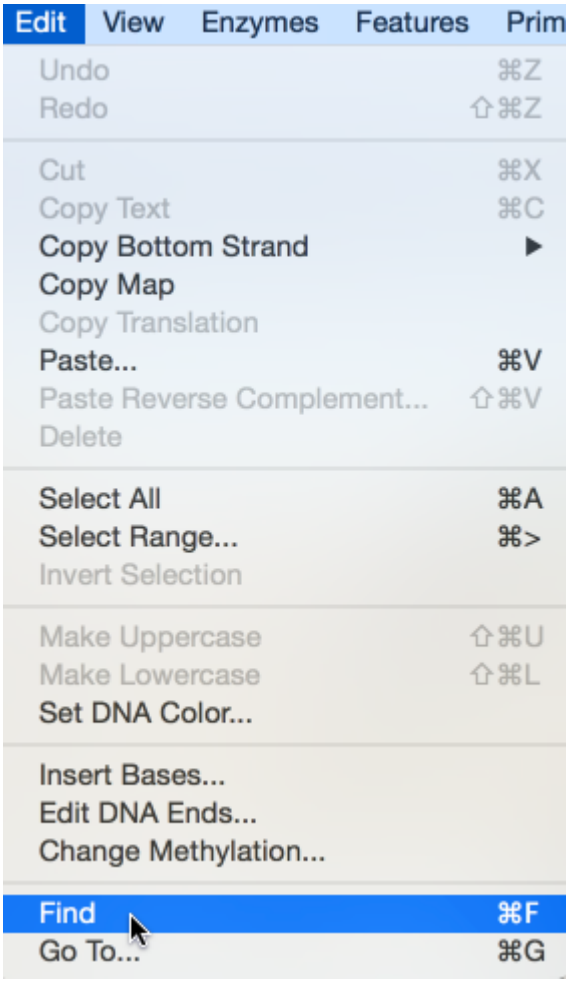



To view a list of protein letter codes, click **Tools** → **Letter Codes**, then click the triangle next to **Protein Letter Codes**.

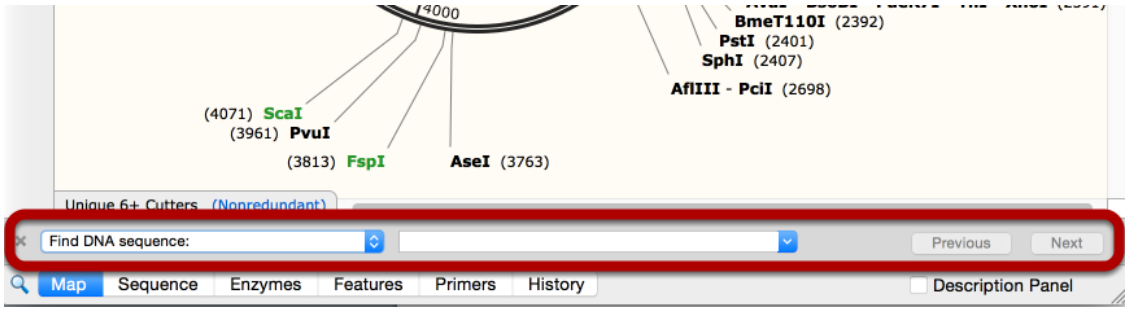
# Search for an Enzyme, Feature, or Primer

How do I search for an enzyme, feature, or primer?

## Show the Find Controls

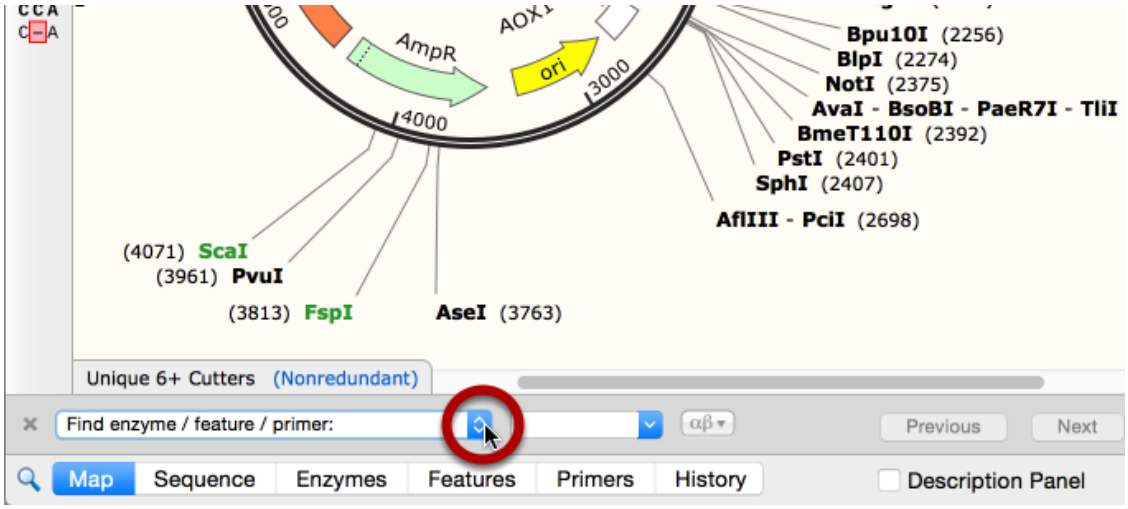



To show the Find controls, click **Edit** → **Find**.




The Find controls will appear near the bottom of the window.

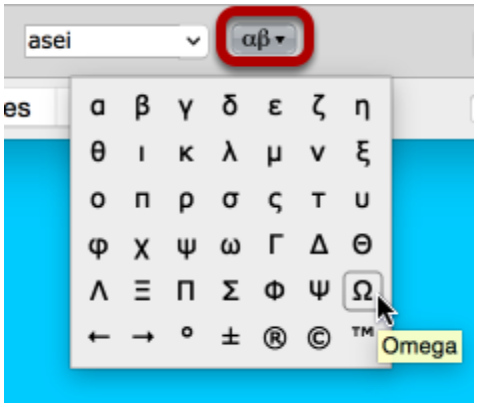
## Search for an Enzyme, Feature, or Primer




Choose **Find enzyme/feature/primer:**.

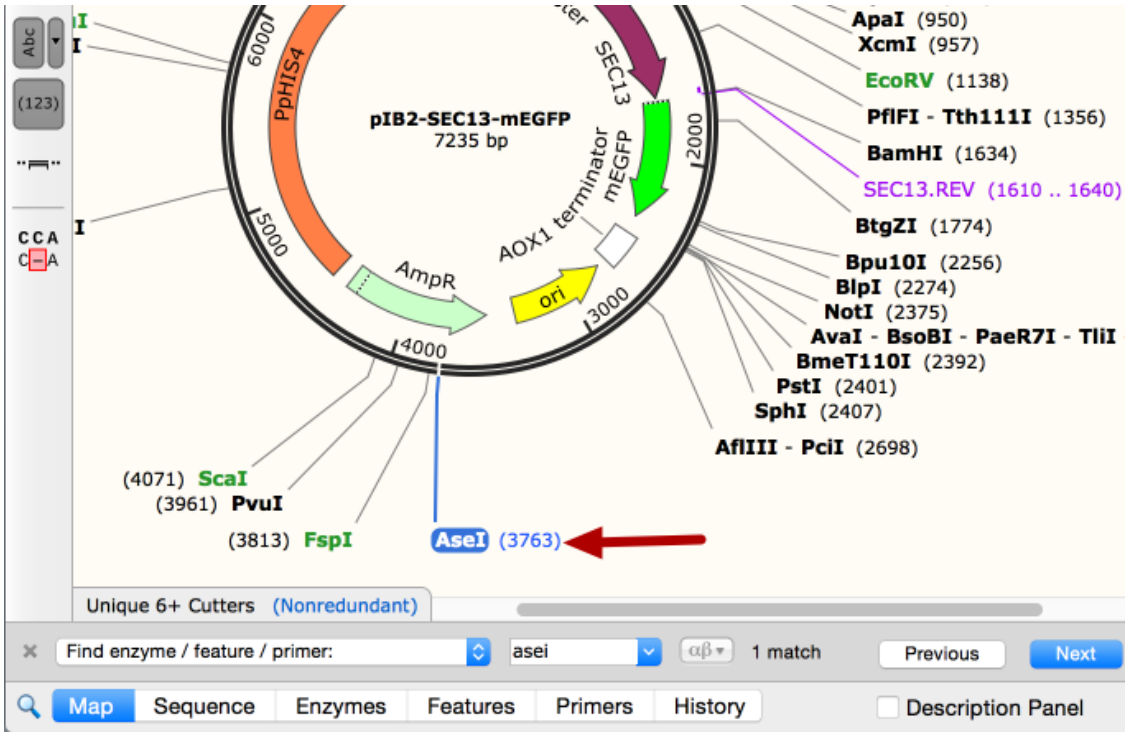
To search for an enzyme, feature, or primer, type the query in the search box, then click **Next**.

Note: Queries are not case sensitive.



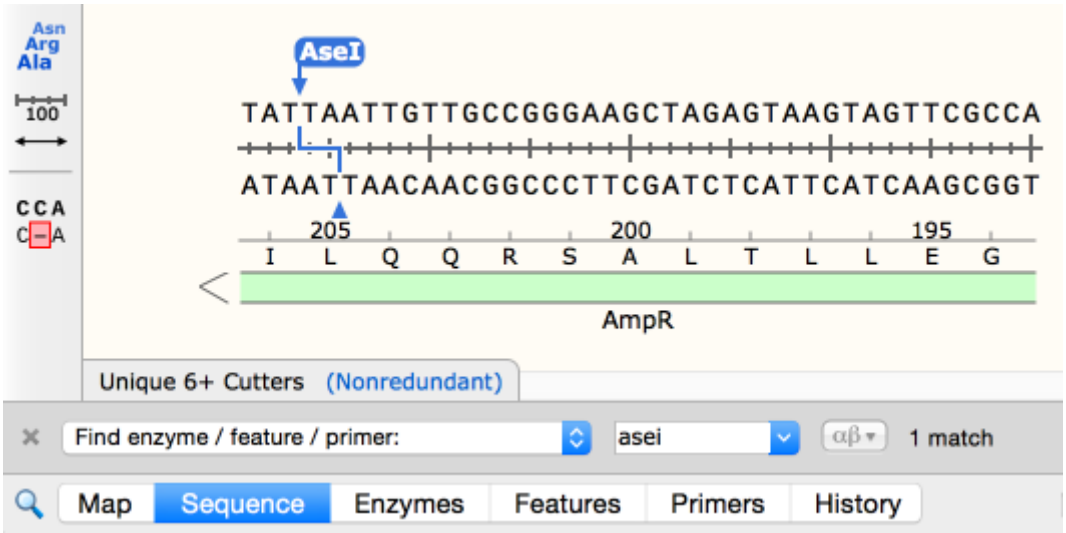

To add a symbol to the query, click the "Insert symbol" button, then select the symbol.

## See the Match in Map View



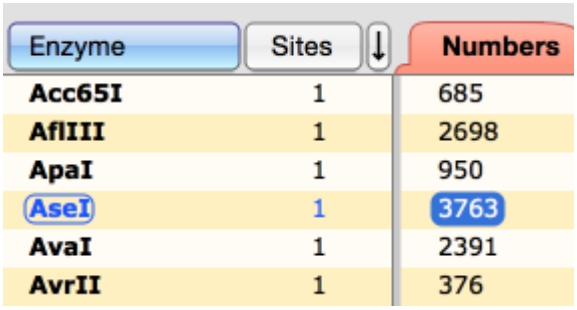

In Map view, a matching enzyme, feature, or primer will be highlighted in blue.

## See the Match in Sequence View




In Sequence view, a matching enzyme, feature, or primer will be highlighted in blue.

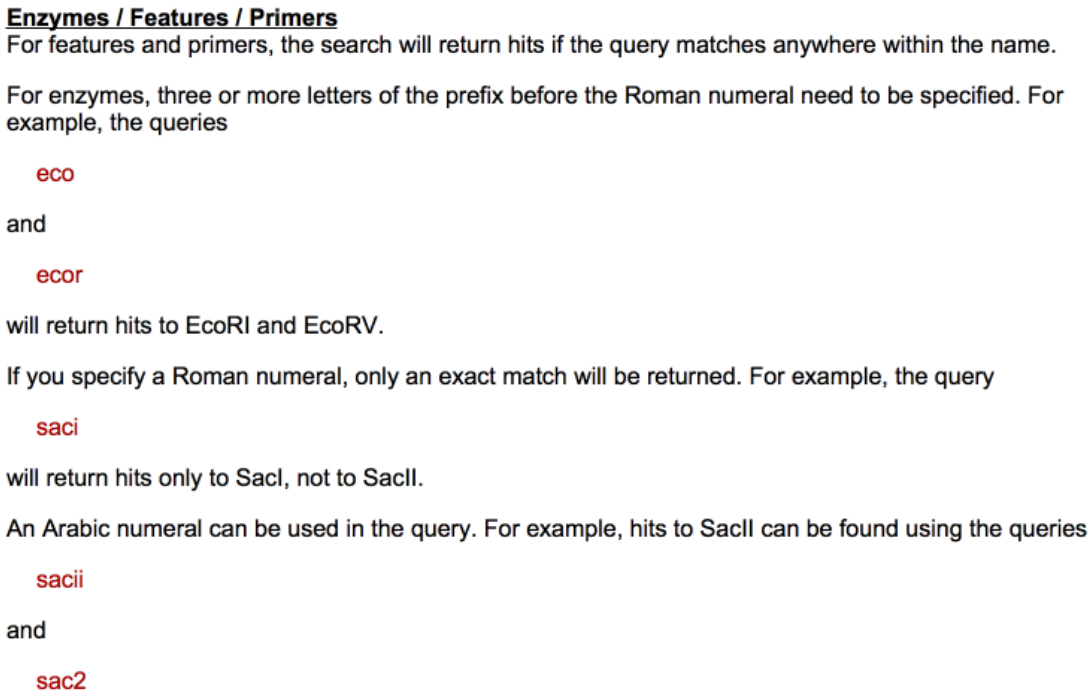
## See the Match in Enzymes View



| Enzyme | Sites | Numbers |
|---|---|---|
| Acc65I | 1 | 685 |
| AflIII | 1 | 2698 |
| ApaI | 1 | 950 |
| AseI | 1 | 3763 |
| AvaI | 1 | 2391 |
| AvrII | 1 | 376 |

In Enzymes view, a matching enzyme will be highlighted in blue.

## Show the Search Tips



**Enzymes / Features / Primers**
For features and primers, the search will return hits if the query matches anywhere within the name.

For enzymes, three or more letters of the prefix before the Roman numeral need to be specified. For example, the queries

eco

and

ecor

will return hits to EcoRI and EcoRV.

If you specify a Roman numeral, only an exact match will be returned. For example, the query

saci

will return hits only to SacI, not to SacII.

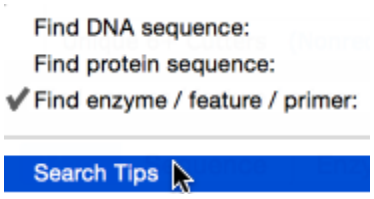
An Arabic numeral can be used in the query. For example, hits to SacII can be found using the queries

sacii

and

sac2

To view examples of enzyme, feature, and primer searches, click **Help** → **Search Tips**, then scroll down.



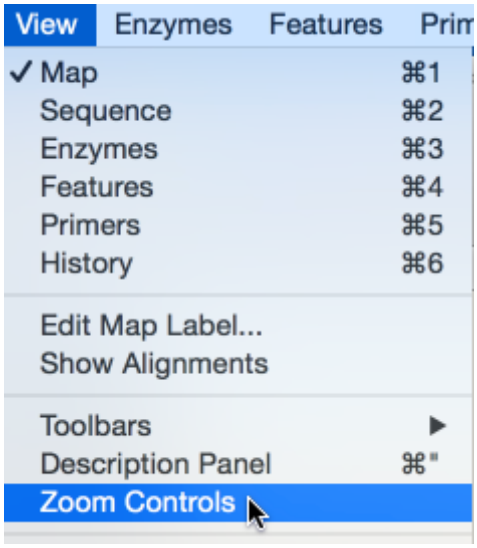

Alternatively, choose **Search Tips** from the Find controls.

# Zooming

## Show or Hide the Zoom Controls

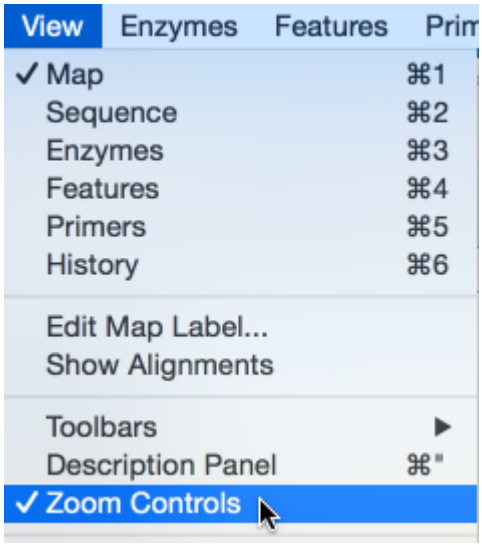
How do I show or hide the zoom controls?

### Show the Zoom Controls



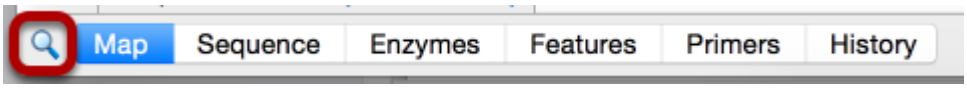

To show the zoom controls, click **View** → **Zoom Controls**.

### Hide the Zoom Controls




To hide the zoom controls, click **View** → **Zoom Controls** again.

## Toggle the Zoom Controls Using the Button

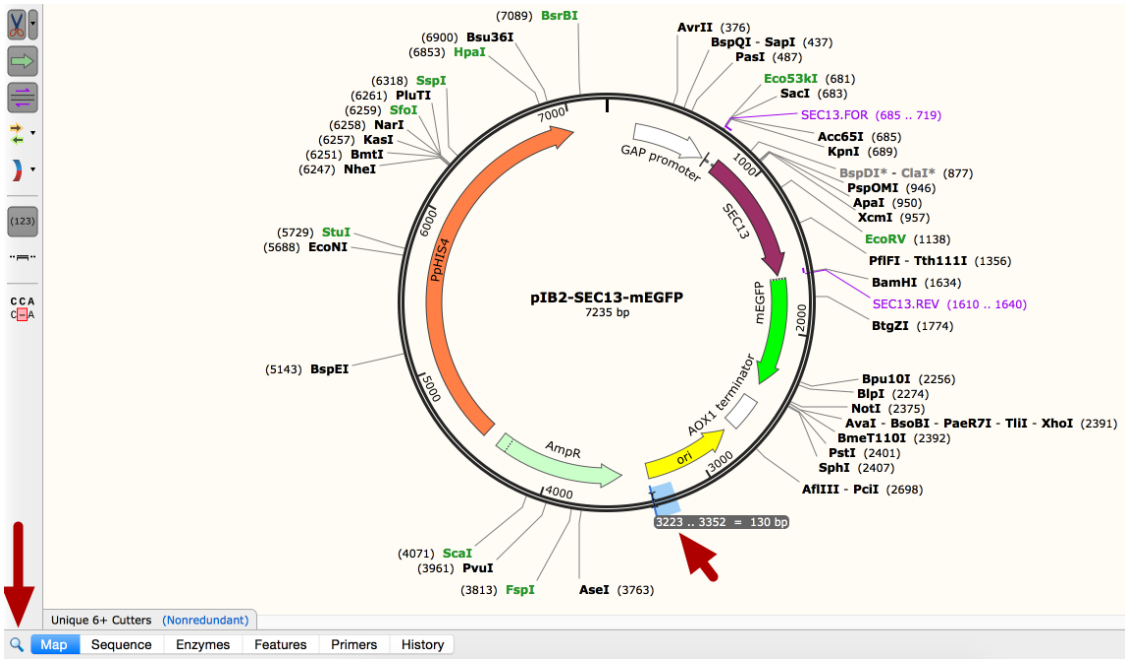



Clicking the **Zoom Controls** button (magnifying glass) can also be used to toggle between showing and hiding the zoom controls.

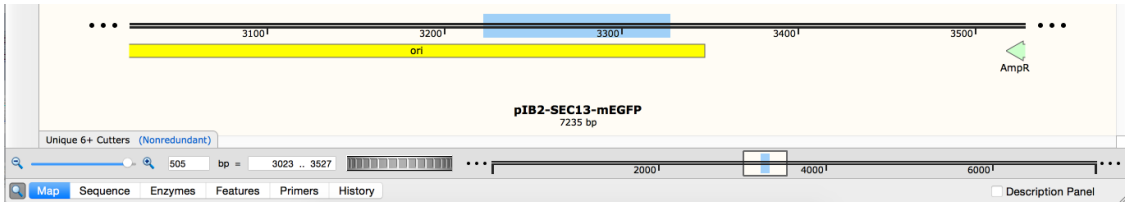
# Zoom In

How can I zoom in on a sequence, region, or feature?

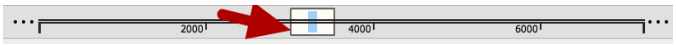
## Show the Zoom Controls




To show the zoom controls, click the **Zoom Controls** button.




If desired, first select the sequence region or feature that will be zoomed into, then click the **Zoom Controls** button. The blue selection will now be centered in a zoom window.

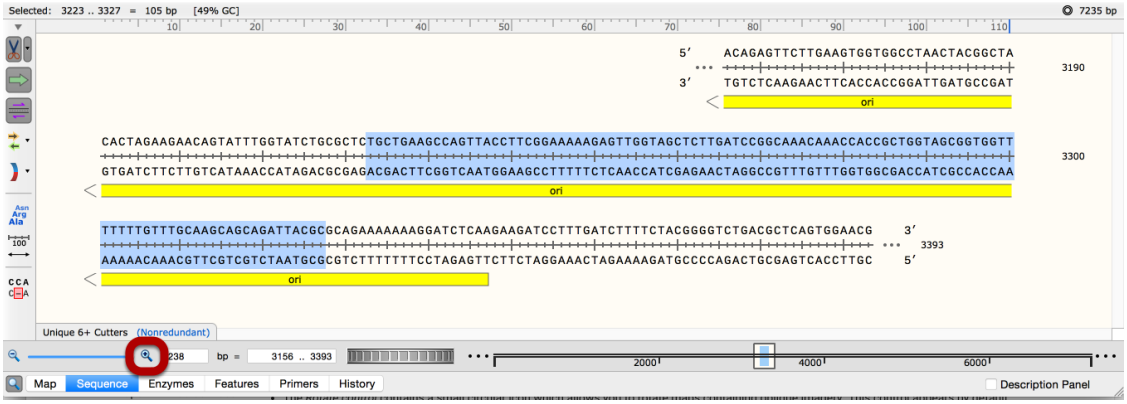



The white rectangle in the map overview shows the relative zoom window location within the entire sequence.

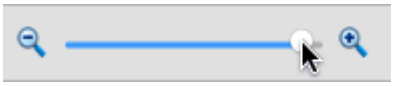
## View the Zoomed Region in Sequence View

To view the zoomed region in Sequence view, click the **Sequence** tab. Bases in the zoom window upstream and downstream of the selection are shown.

## Zoom in on a Region

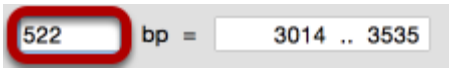



To reduce the number of bases shown on either side of the selection, click the **Zoom In** button (+) in the zoom controls.

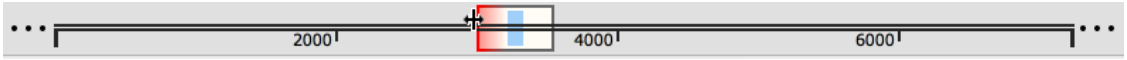


Alternatively, drag the zoom slider.

## Specify the Zoom Window Width




To specify the width of the zoom window, edit either the width value or the range bounds.

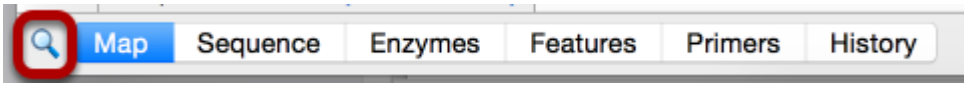



Alternatively, drag either side of the zoom window boundary.

# Pan Across a Sequence

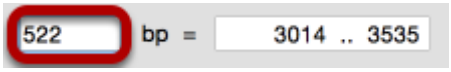
How can I pan across a sequence using a zoom window?

## Show the Zoom Controls

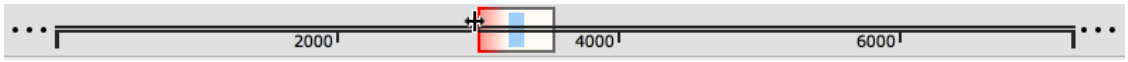



To show the zoom controls, click the **Zoom Controls** button.

## Specify the Zoom Window Width




To specify the width of the zoom window, edit either the width value or the range bounds.




Alternatively, drag either side of the zoom window boundary.

## Pan Across the Sequence

Drag the zoom window to pan across the sequence.

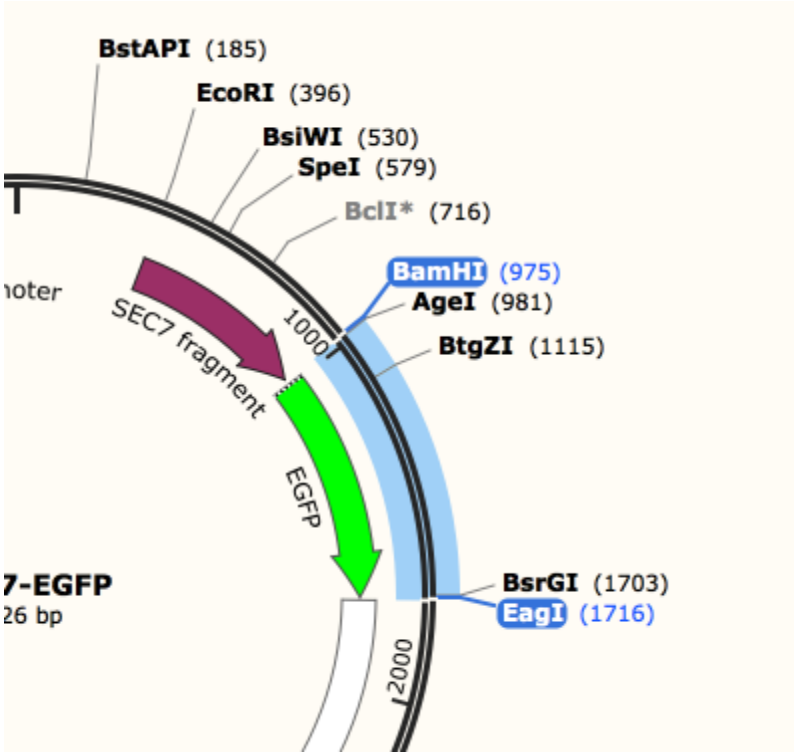
Alternatively, drag the wheel to pan across the sequence.

# Restriction Cloning and Linear Ligation

# Insert Fragment

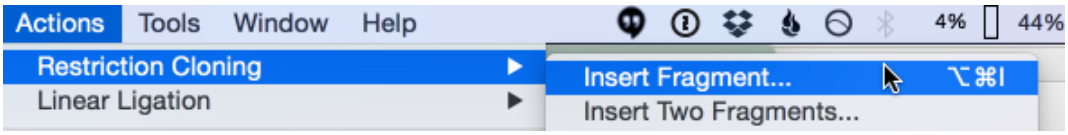
How can I insert a restriction fragment into a circular vector?

## Specify the Insertion Site



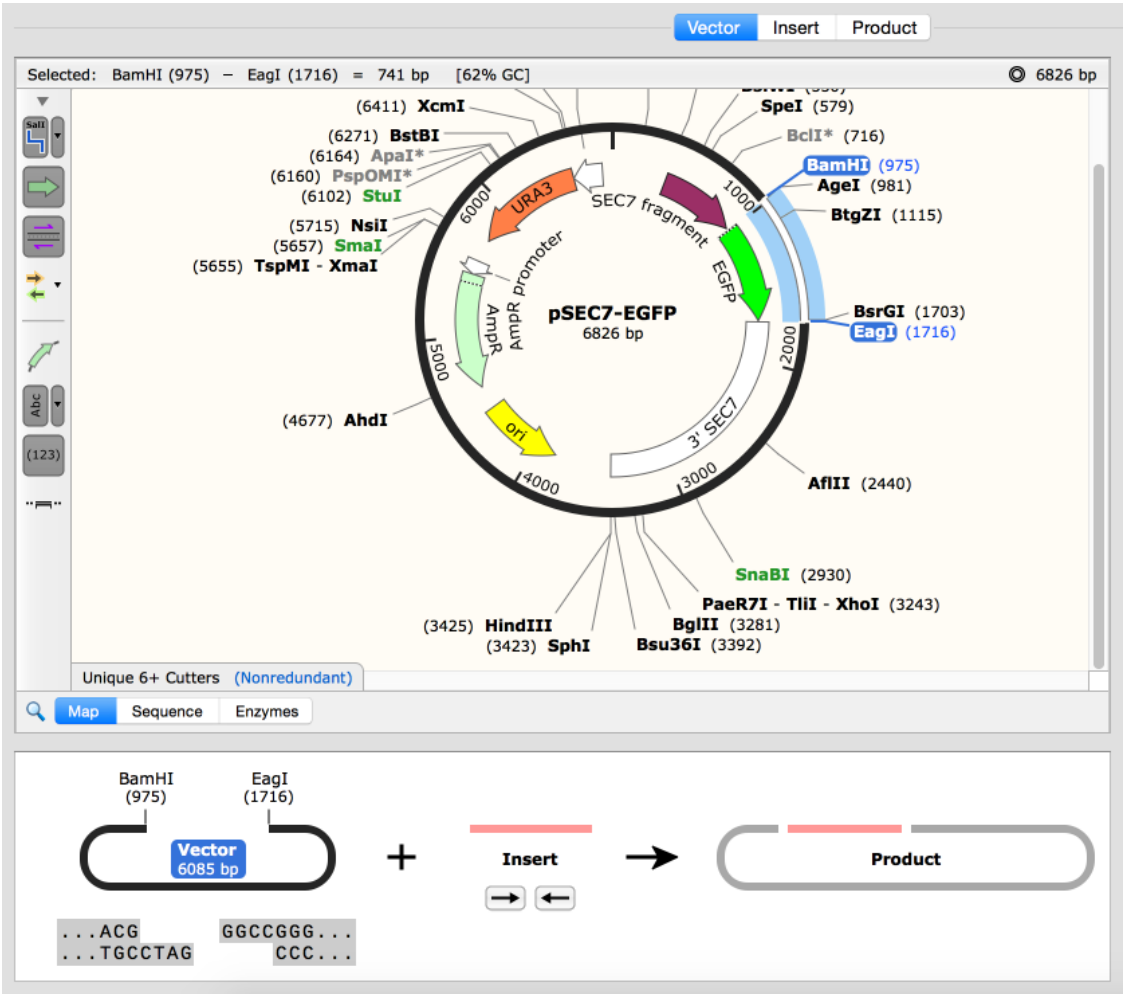

To insert a restriction fragment into a vector, begin by opening the vector file. The restriction fragment can be inserted at a restriction site in the vector or can replace an existing restriction fragment in the vector. To select a restriction site in the vector, click the enzyme site. To select an existing restriction fragment in the vector, click on the first enzyme site, then either Shift-click on the second enzyme site or drag to the second enzyme site.

## Open the Insert Fragment Dialog

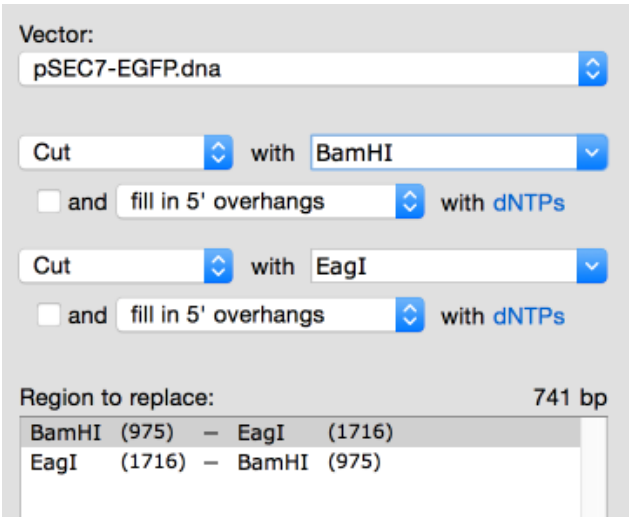



To open the Insert Fragment dialog, click **Actions** → **Restriction Cloning** → **Insert Fragment...** .

## Preview the Vector

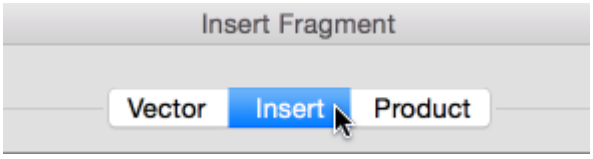



The Insert Fragment dialog will show the insertion site.

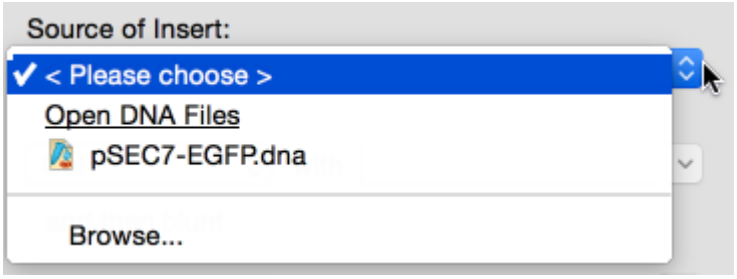



Review the vector blunting, enzyme selection, and fragment selection at the top right of the Insert Fragment dialog.

## Specify the Insert File

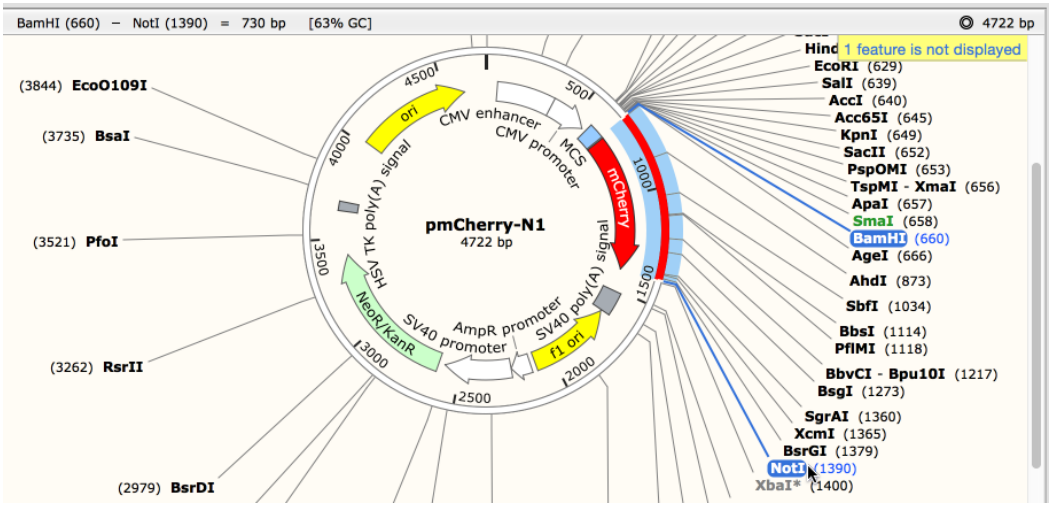



To specify the source of the insert, click the **Insert** tab at the top.

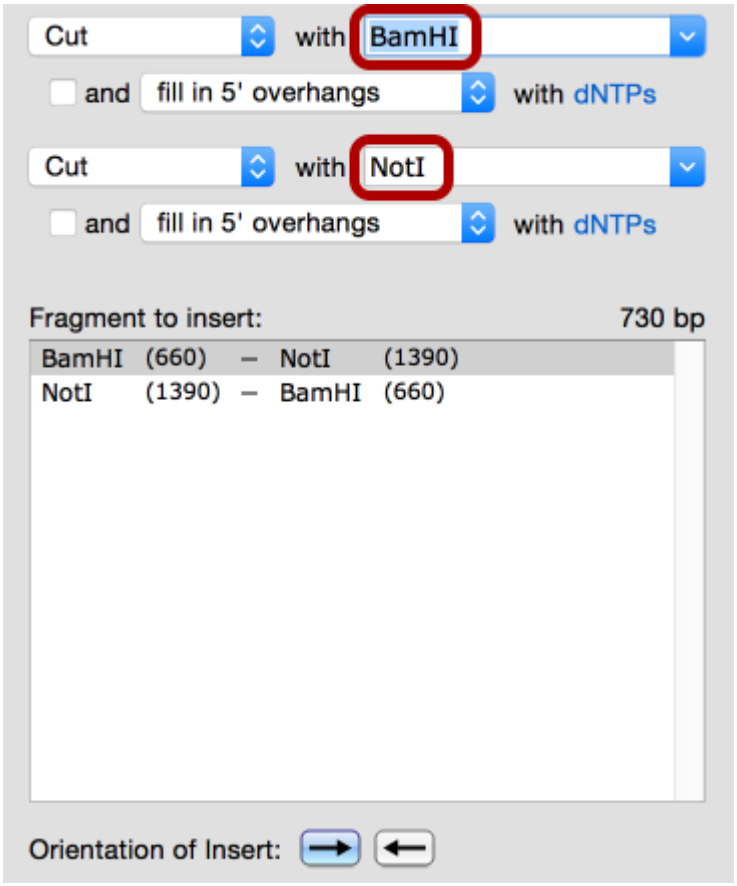



Choose the file that contains the insert from the **Source of Insert:** menu.

## Specify the Inserted Fragment

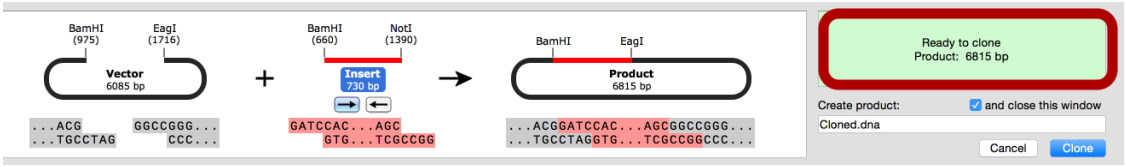



To specify the fragment to be inserted, click on the first enzyme site, then either Shift-click on the second enzyme site or drag to the second enzyme site.



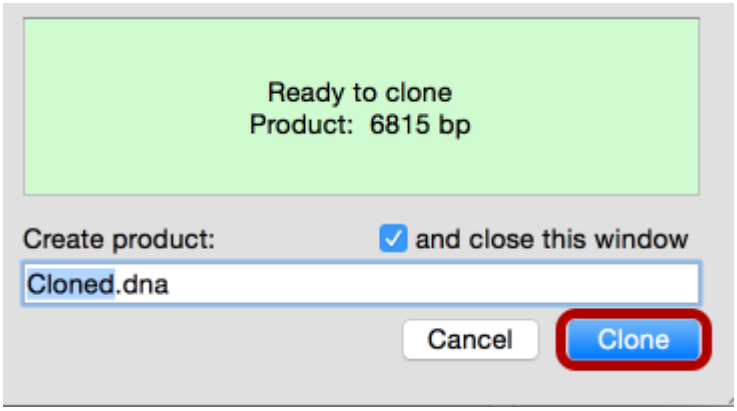

Alternatively, type the enzyme names in the boxes at the upper right. Review the blunting, enzyme selection, fragment selection, and orientation.

## Preview the Product



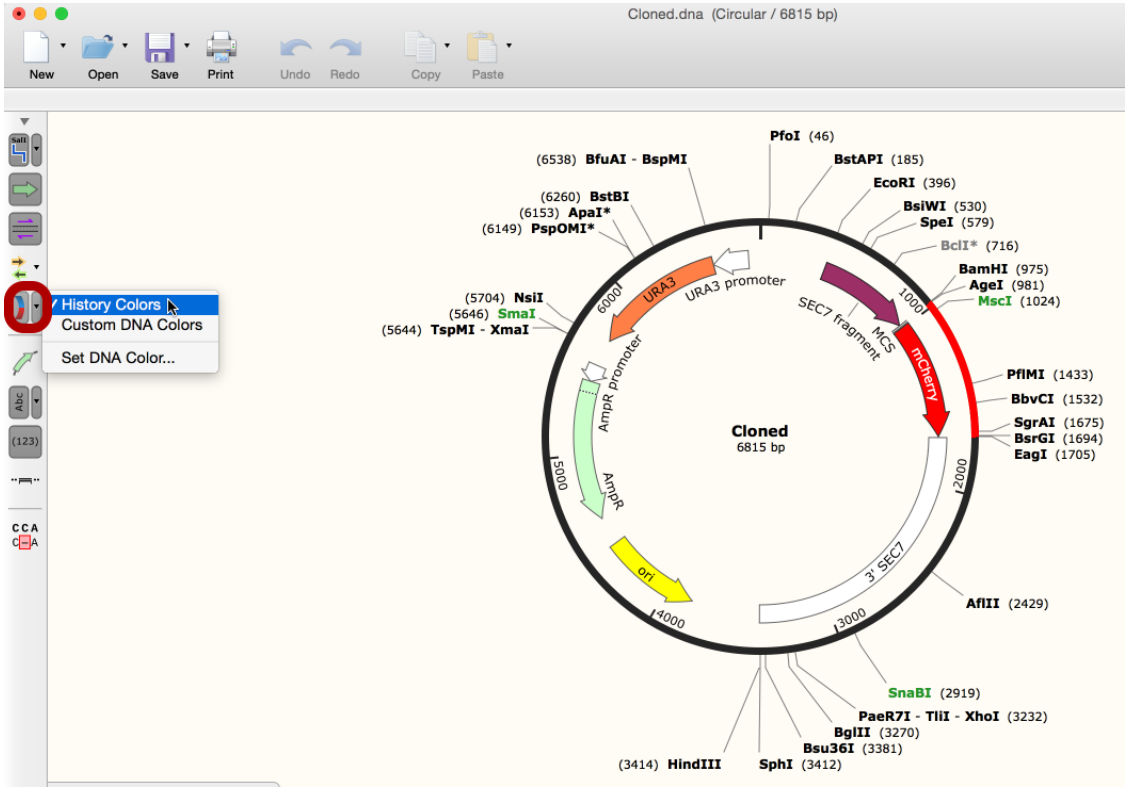

The overview at the bottom summarizes the reaction product. The green box at the bottom right indicates you are ready to clone. If cloning is not possible, this box will remain yellow and will explain what steps must be taken to proceed.

## Name the Product



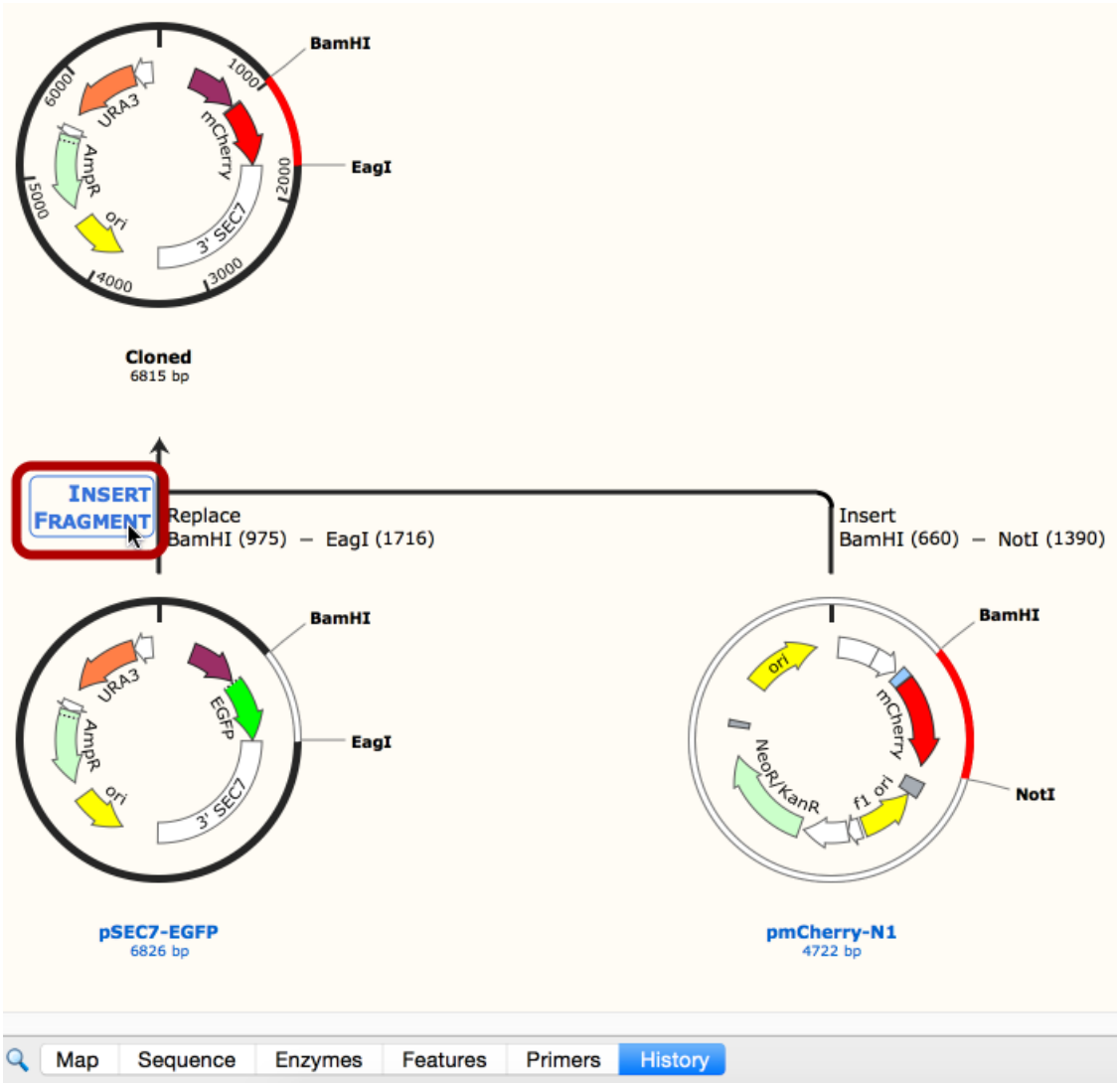

When you are ready to clone, type the name of the product, then click **Clone**.

## View the Product History Colors



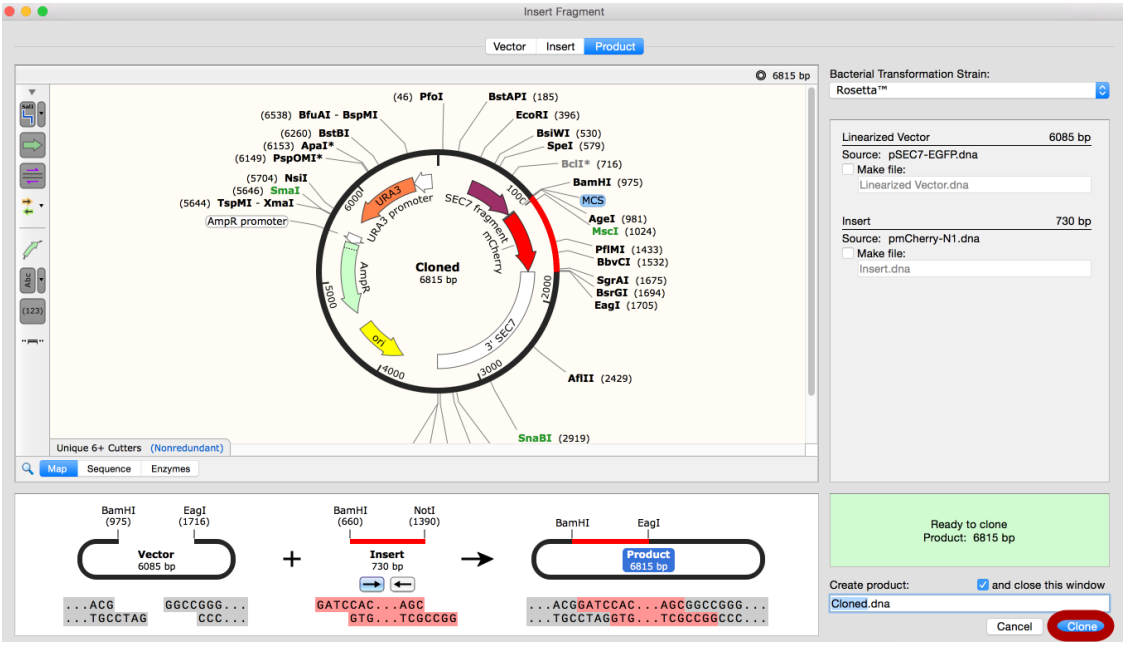

In the product file, to view the restriction fragment insertion site, first turn on the history colors in Map and Sequence views, then click the "Show colors" button in the side toolbar. The red color represents the inserted restriction fragment.

In Sequence view, the history colors illustrate more precisely where the restriction fragment was inserted.




To illustrate in History view where the insertion occurred, click the Insert Fragment operation name.

## Alternative Copy/Paste Approach

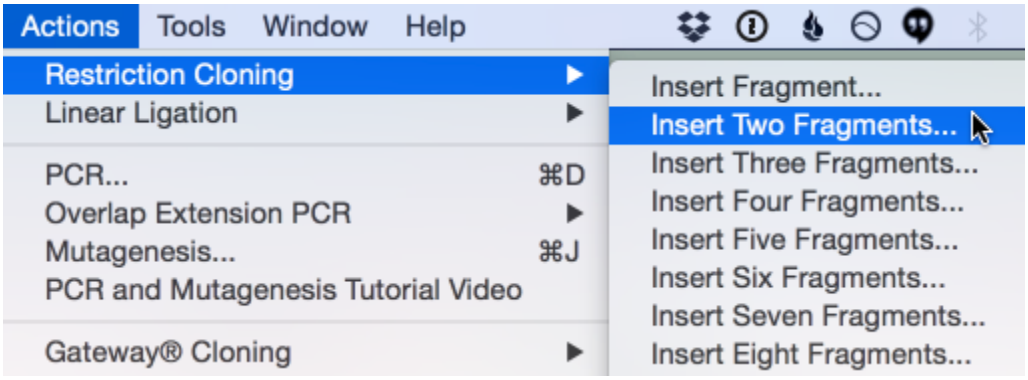



Alternatively, copy and paste can be used to populate the Insert Fragment dialog. To copy the fragment to be inserted, open the file for the insert, select the restriction fragment, and click **Edit** → **Copy Restriction Fragment**. Then switch to the vector file, select the insertion site or the restriction fragment to be replaced, and click **Edit** → **Paste**. The Insert Fragment dialog will open to the populated **Product** tab. Type the name of the product, then click **Clone**.

# Insert Multiple Fragments

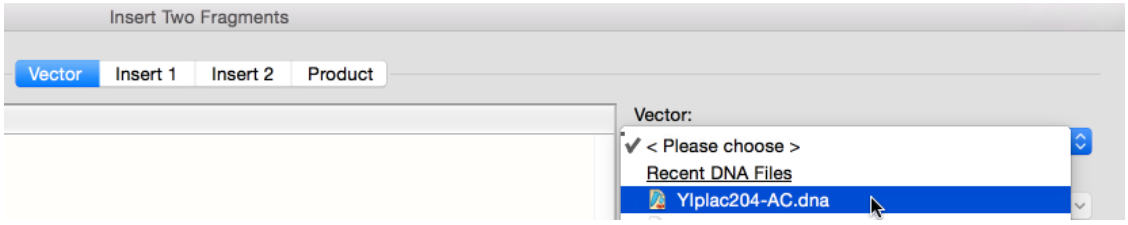
How can I insert multiple restriction fragments into a circular vector?

## Open the Insert [N] Fragments Dialog



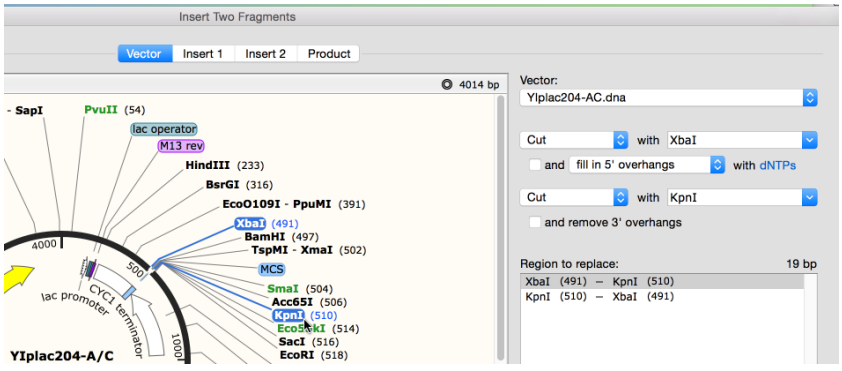

To open the Insert [N] Fragments dialog, click **Actions** → **Restriction Cloning** → **Insert [N] Fragments...** .

## Specify the Vector File




Choose the vector file from the **Vector:** menu.

## Specify the Vector Insertion Site




The restriction fragments can be inserted at a restriction site in the vector or can replace an existing restriction fragment in the vector. To select a restriction site in the vector, click the enzyme site. To select an existing restriction fragment in the vector, click on the first enzyme site, then

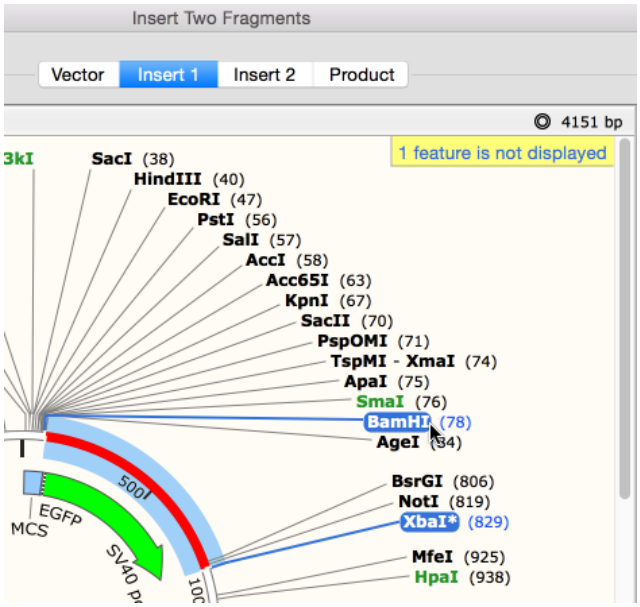
either Shift-click on the second enzyme site or drag to the second enzyme site. Alternatively, type the enzyme names in the boxes at the upper right. Review the blunting, enzyme selection, fragment selection, and orientation.

## Specify the First Insert File

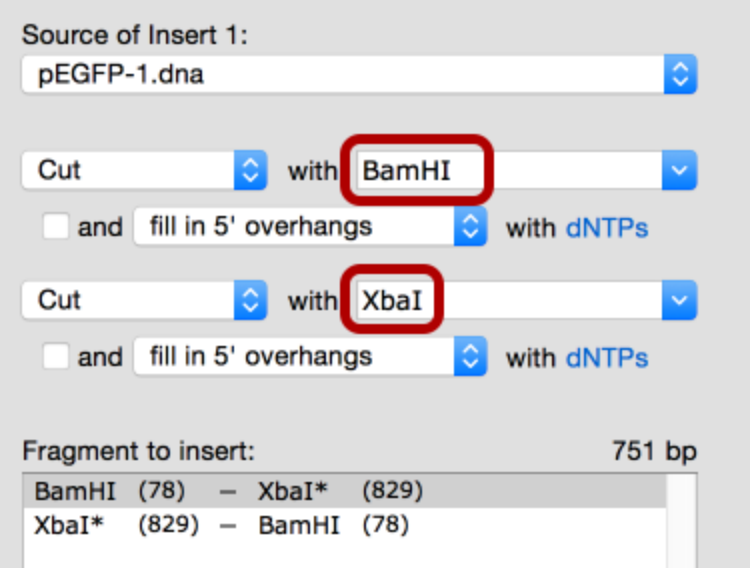
To specify the source of the first insert, click the **Insert 1** tab at the top of the dialog.

Choose the file that contains the first insert from the **Source of Insert 1:** menu.

## Specify the First Inserted Fragment




To specify the first inserted fragment, click on the first enzyme site, then either Shift-click on the second enzyme site or drag to the second enzyme site.



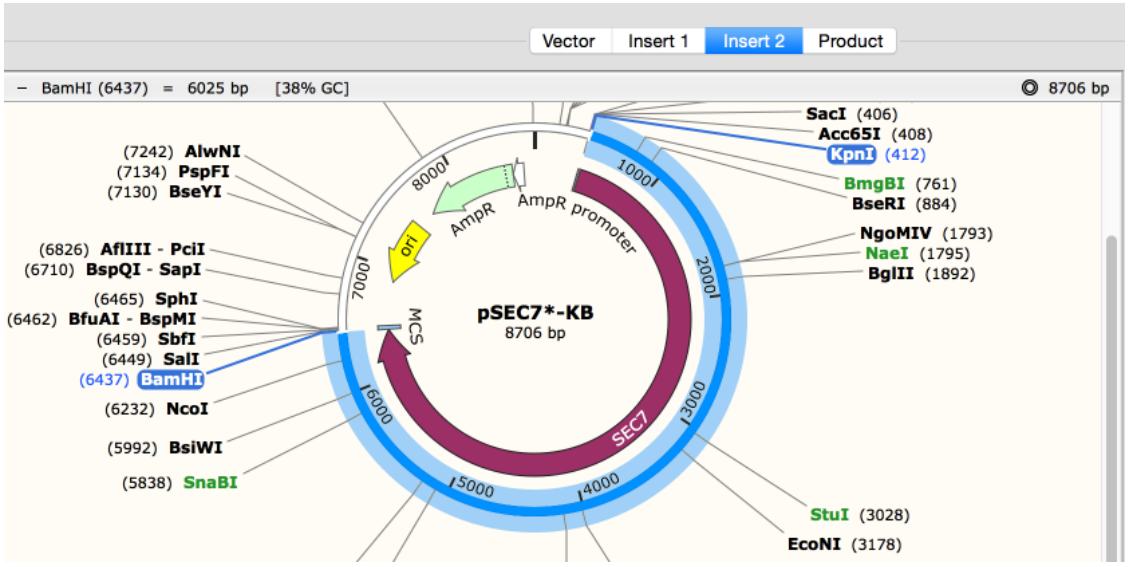

Alternatively, type the enzyme names into the boxes at the upper right. Review the insert blunting, enzyme selection, fragment selection, and orientation.

## Specify the Second Insert File

To specify the source of the second insert, click the **Insert 2** tab at the top of the dialog.

Choose the file that contains the second insert from the **Source of Insert 2:** menu.

## Specify the Second Inserted Fragment




To specify the second inserted fragment, click the first enzyme site, then either Shift-click on the second enzyme site or drag to the second enzyme site.

## Specify Any Additional Inserted Fragments

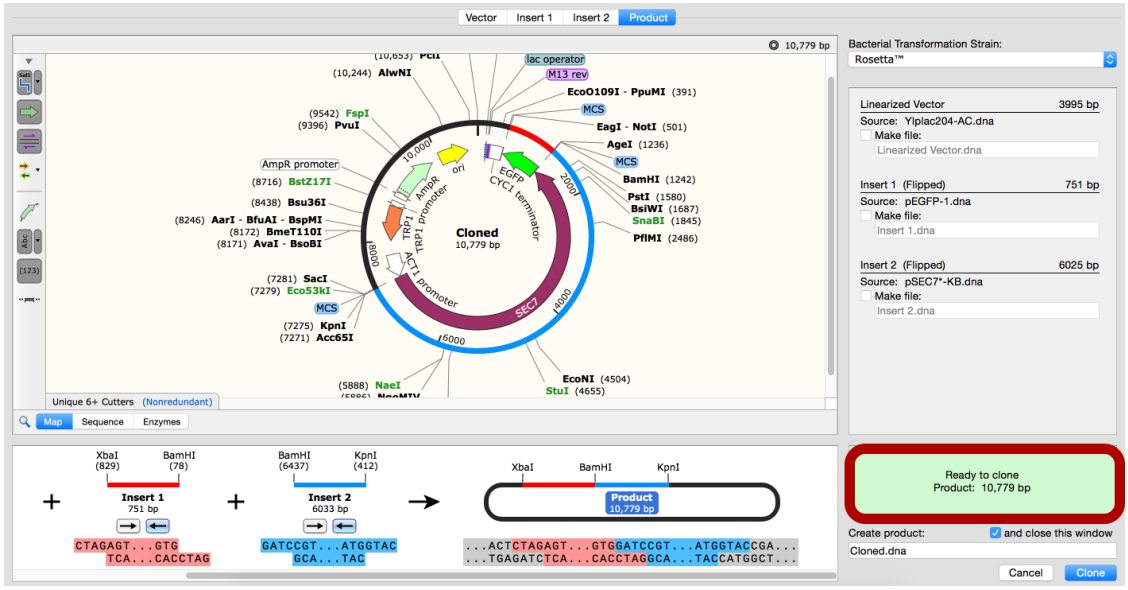
If necessary, repeat the steps described above to specify additional source files and inserted fragments.

## Make the Ends Compatible

If cloning is not possible, the box at the lower right will remain yellow and will explain what steps must be taken to proceed. Correct the problem by some combination of flipping inserts, or blunting insert or vector ends, to make the ends compatible for ligation.

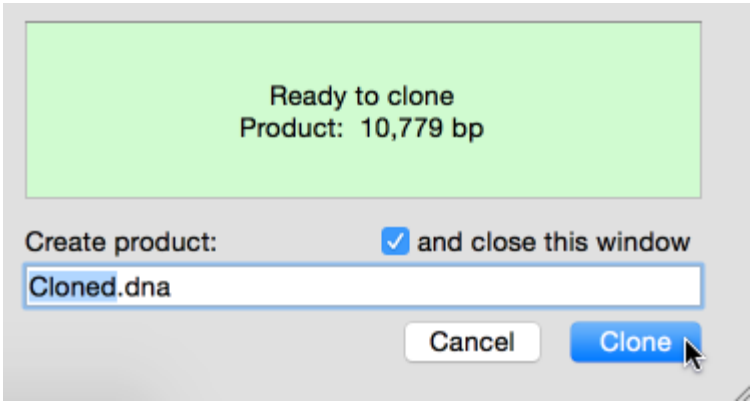
In the illustrated example, the yellow box indicates that the ends are not compatible for ligation. This problem can be corrected by clicking the reverse directional arrows to flip the orientations of both inserts.

## Preview the Product



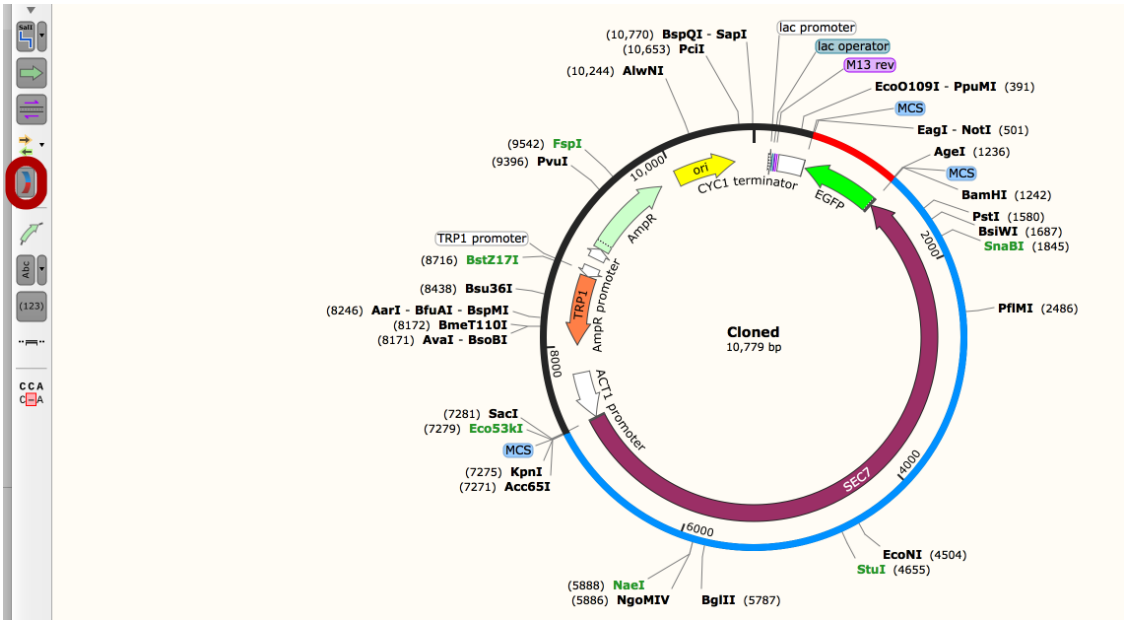

To preview the product, click the **Product** tab. The box will turn green when you are ready to clone.

## Name the Product

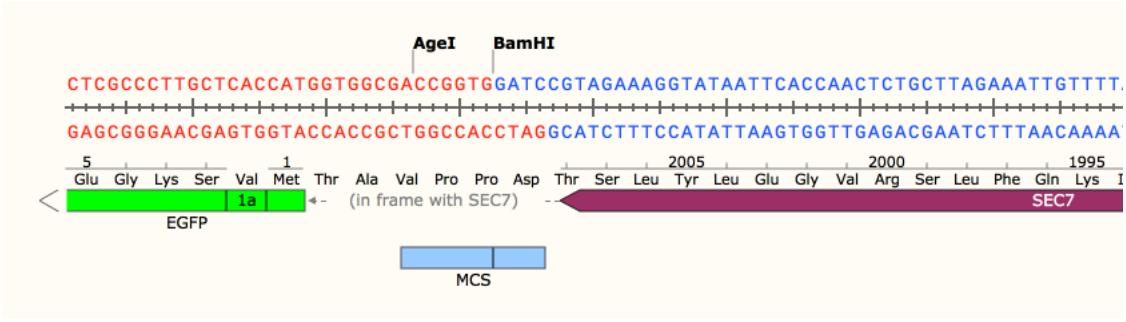



Type the name of the product, then click **Clone**.

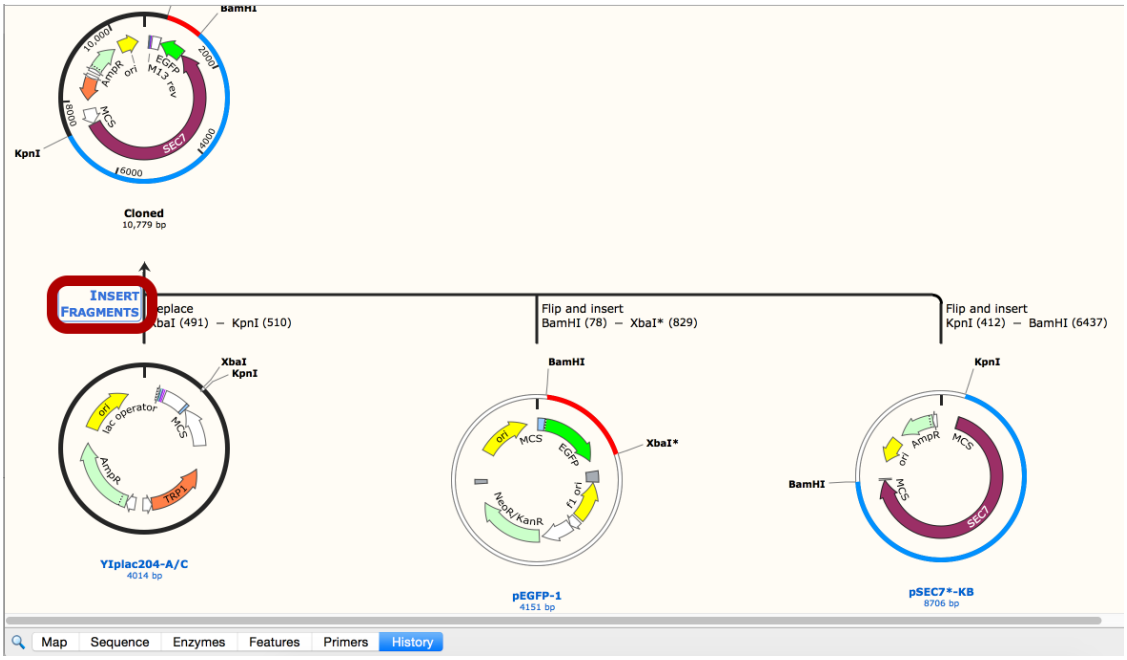
## View the Product History Colors




To turn on the product history colors in Map and Sequence views, click the "Show colors" button in the side toolbar. Alternating red and blue colors represent the multiple inserted restriction fragments.

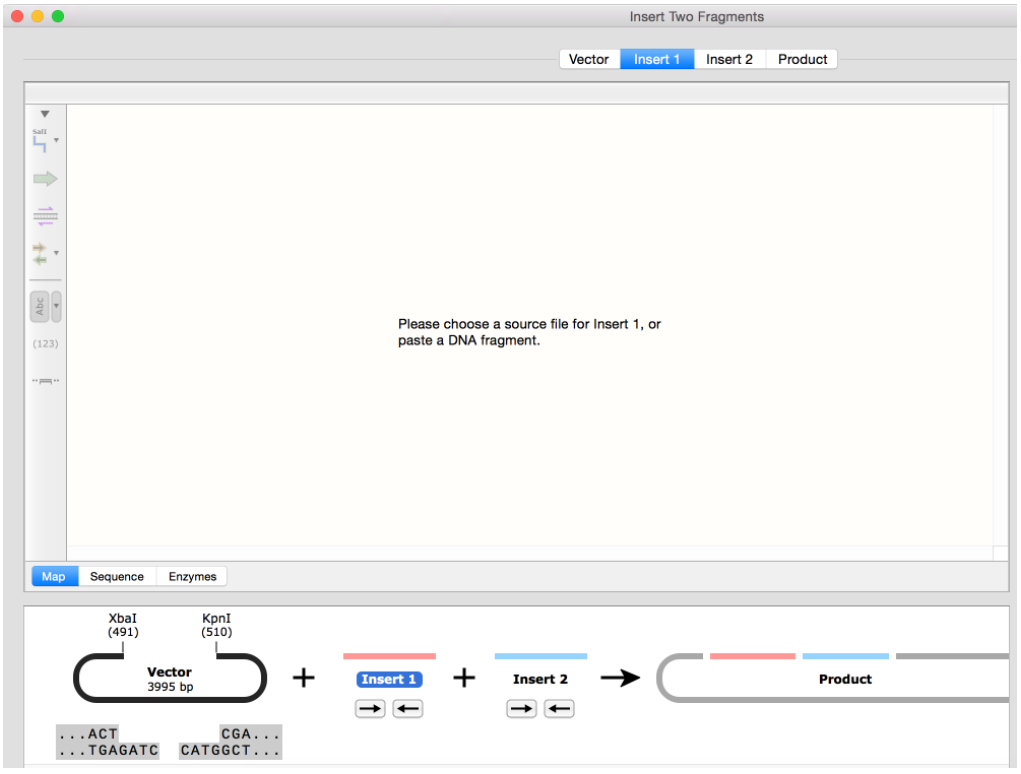



In Sequence view, the history colors illustrate more precisely where the restriction fragments were inserted.




To illustrate in History view where the insertions occurred, click the Insert Fragments operation name.

## Alternative Copy/Paste Approach




Alternatively, copy and paste can be used to populate the Insert [N] Fragments dialog.

To select the vector fragment to be replaced, click on the first enzyme site in Map or Sequence

view, then either Shift-click on the second enzyme site or drag to the second enzyme site. Click **Actions** → **Restriction Cloning** → **Insert [N] Fragments...** . The Insert [N] Fragments dialog will open to the populated **Vector** tab.

Switch to an open file containing the fragment to be inserted. Select that restriction fragment, and then click **Edit** → **Copy Restriction Fragment.**

Switch back to the Insert [N] Fragments dialog. Click the **Insert 1** tab at the top of the dialog. Click **Edit** → **Paste** to populate this part of the dialog.

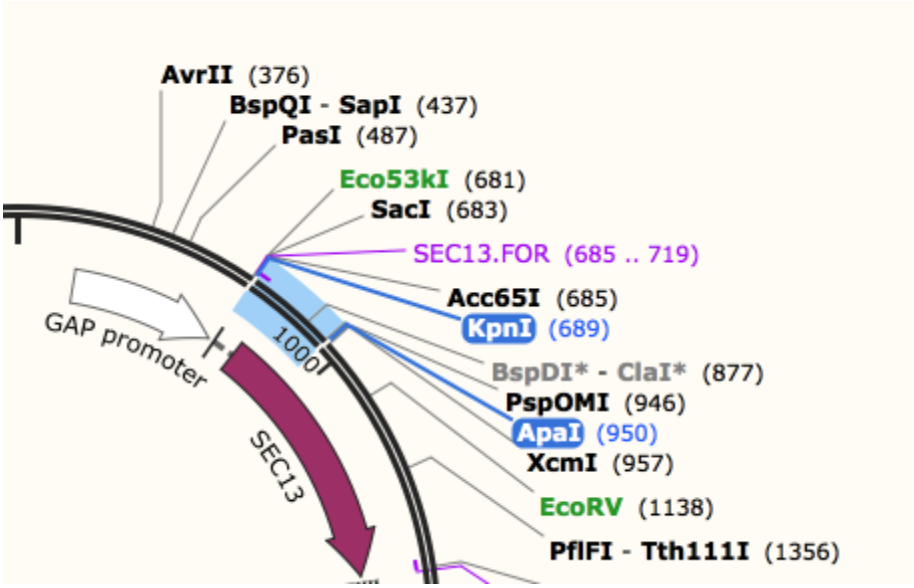
Repeat this process for the remaining **Insert** tabs in the dialog.

When you are ready to clone, type the name of the product, then click **Clone**.

# Delete Restriction Fragment

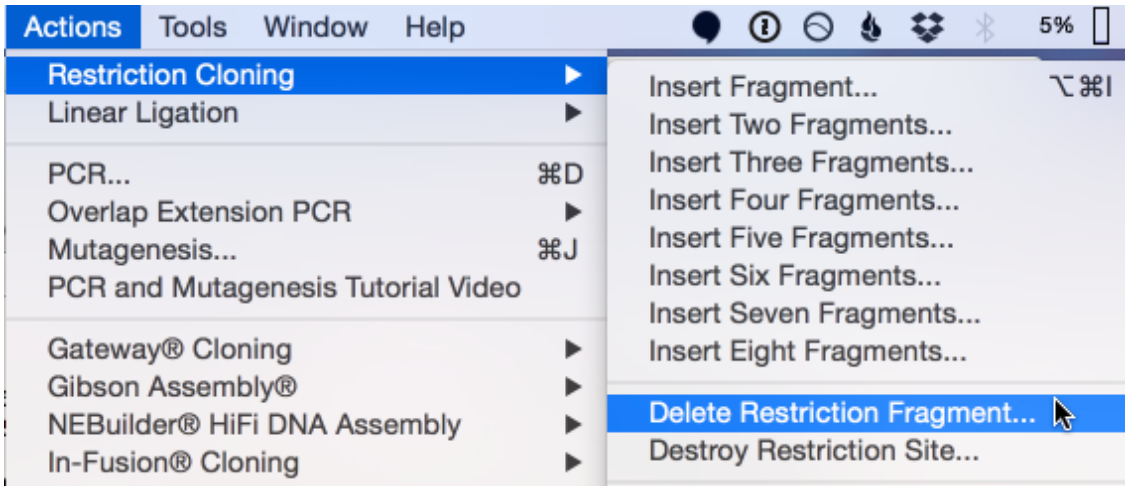
How can I delete an internal restriction fragment in a circular vector?

## Specify the Restriction Fragment



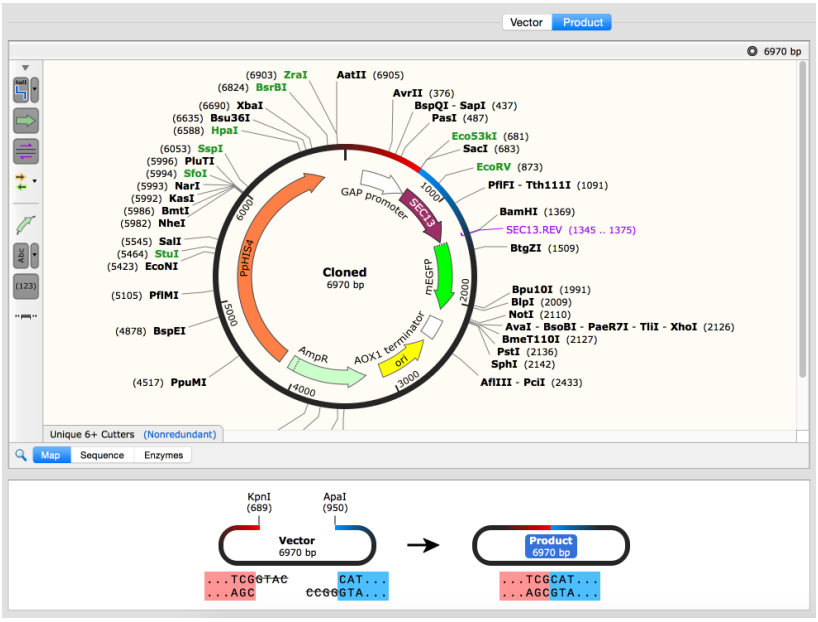

To select the restriction fragment, click on the first enzyme site, then either Shift-click on the second enzyme site or drag to the second enzyme site.

## Open the Delete Restriction Fragment Dialog

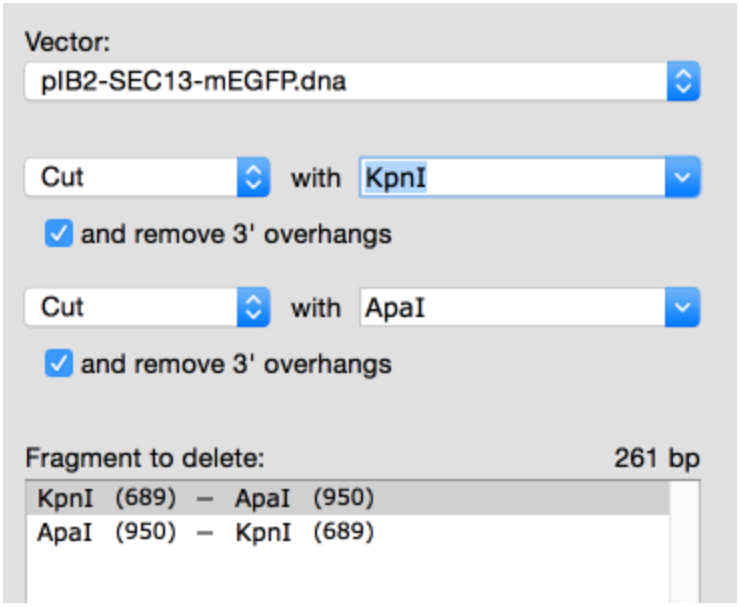



To open the Delete Restriction Fragment dialog, click **Actions** → **Restriction Cloning** → **Delete Restriction Fragment...**, or simply press the Delete key.

## Preview the Product

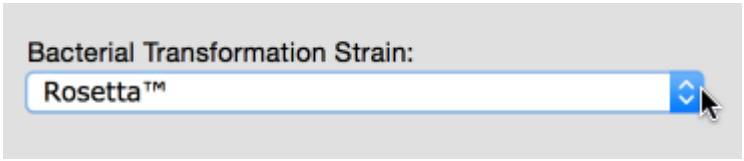



The Delete Restriction Fragment dialog will show the product preview in the **Product** tab. If the sticky ends were incompatible, they will be blunted automatically as shown in the overview at the bottom.



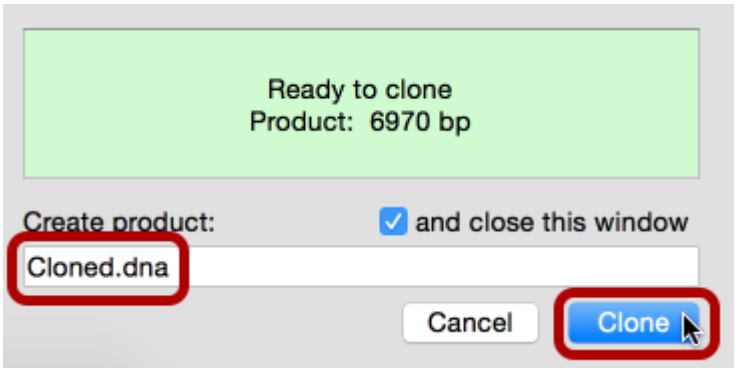

To view the settings, click the **Vector** tab at the top of the window, then review the blunting, enzyme selection, and fragment selection.

## Specify the Bacterial Transformation Strain



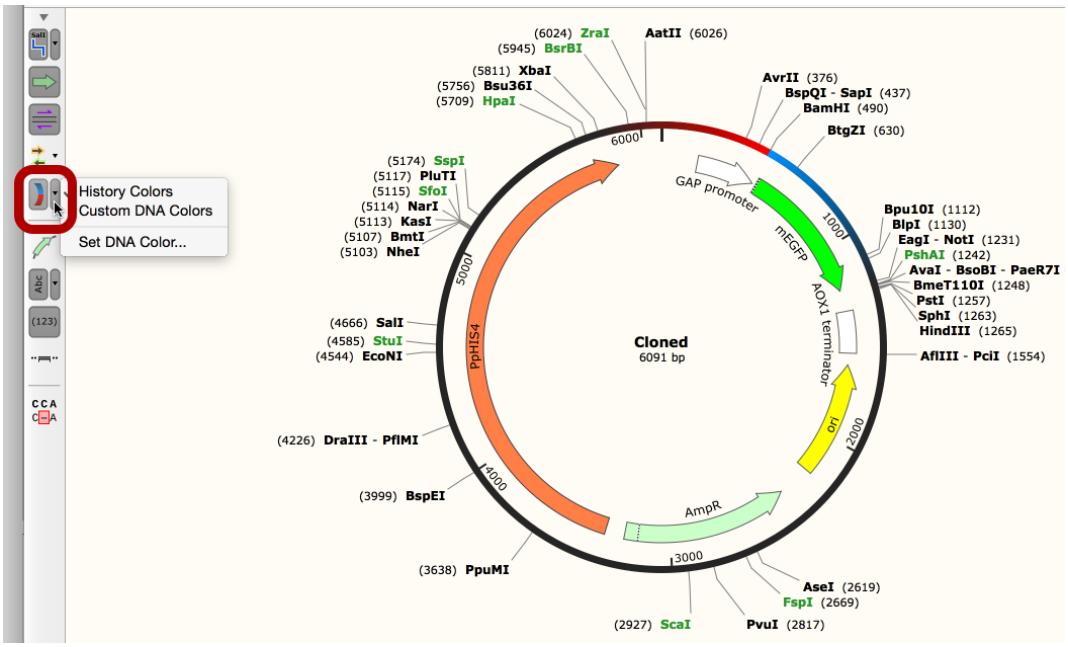

To specify the bacterial transformation strain, click the **Product** tab, then click the **Bacterial Transformation Strain** menu to change the chosen strain or to edit the strains list.

## Name the Product



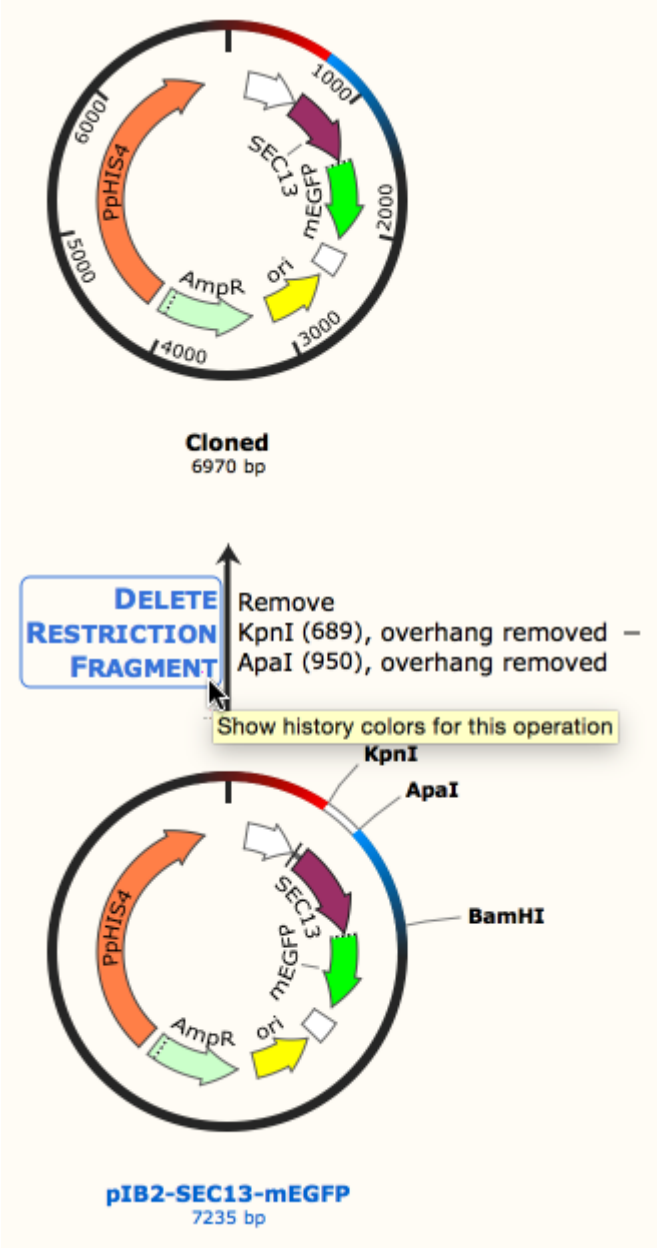

Type the name of the product, then click **Clone**.

## View the Product History Colors




To turn on the product history colors in Map and Sequence views, click the "Show colors" button in the side toolbar. The red and blue colors illustrate where the restriction fragment was deleted.

In Sequence view, the history colors illustrate more precisely where the restriction fragment was deleted.

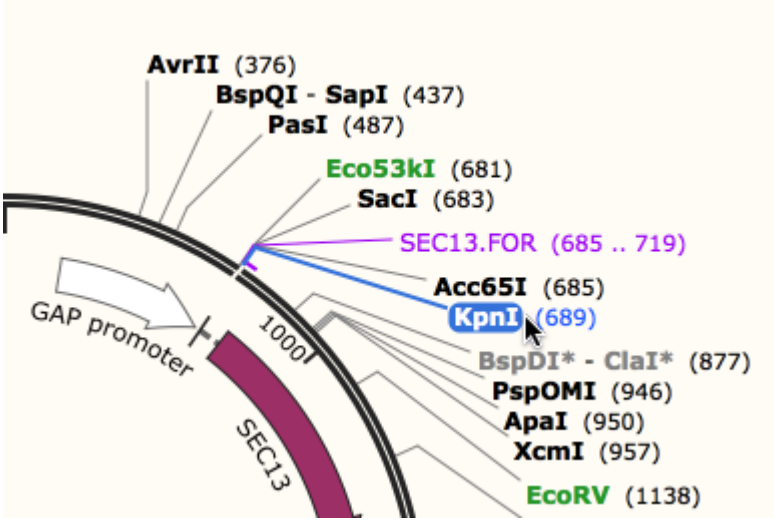



To illustrate in History view where the deletion occurred, click the Delete Restriction Fragment operation name.

# Destroy Restriction Site

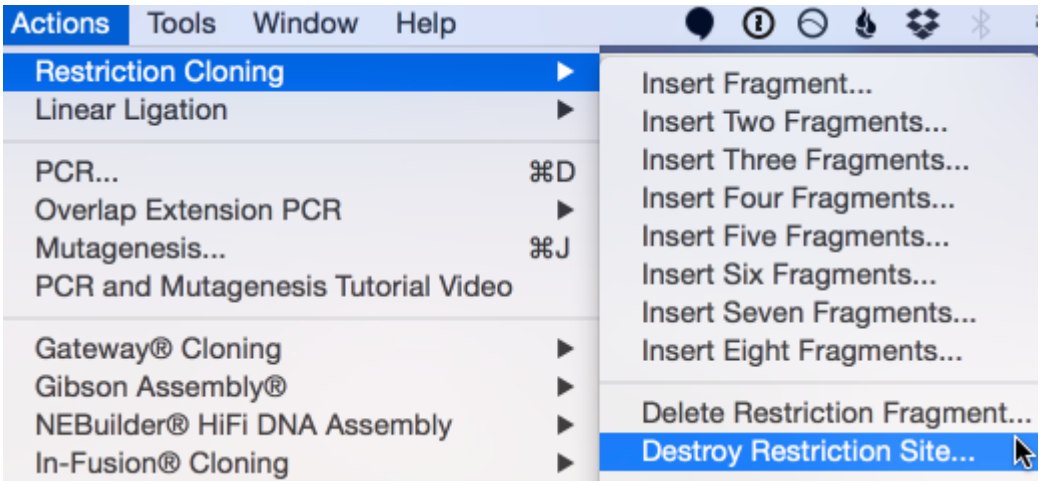
How can I destroy a restriction site in a circular plasmid?

## Specify the Restriction Site



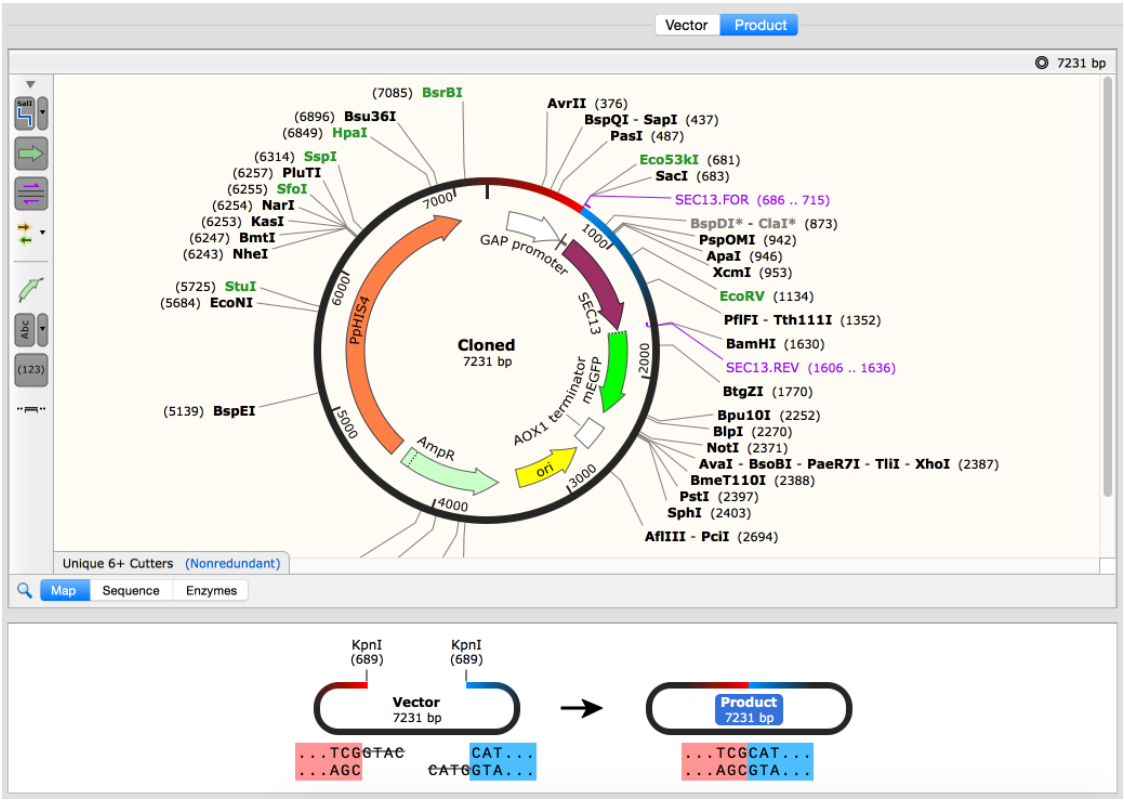

To select the restriction site, click the enzyme name.

## Open the Destroy Restriction Site Dialog

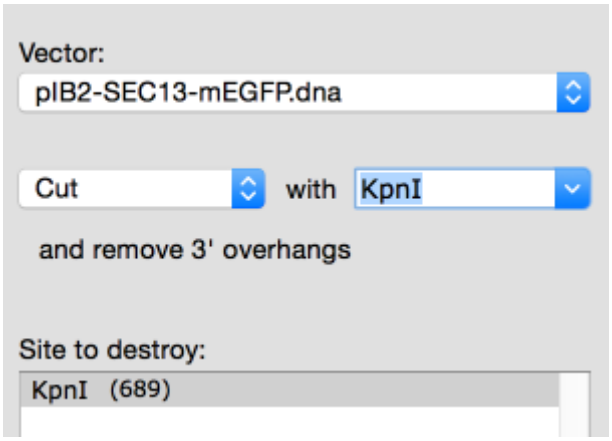



To open the Destroy Restriction Site dialog, click **Actions → Restriction Cloning → Destroy Restriction Site...**, or simply press the Delete key.

## Preview the Product

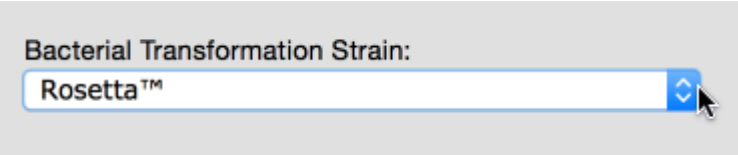



The Destroy Restriction Site dialog will show the product preview. If the sticky ends were incompatible, they will be blunted automatically as shown in the overview at the bottom.



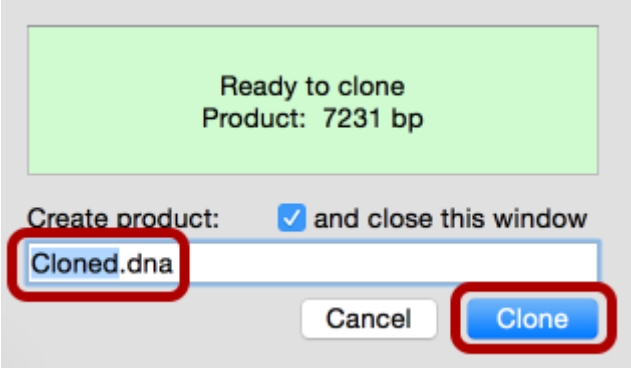

To view the settings, click the **Vector** tab at the top of the window, then review the blunting, enzyme selection, and selected site.

## Specify the Bacterial Transformation Strain



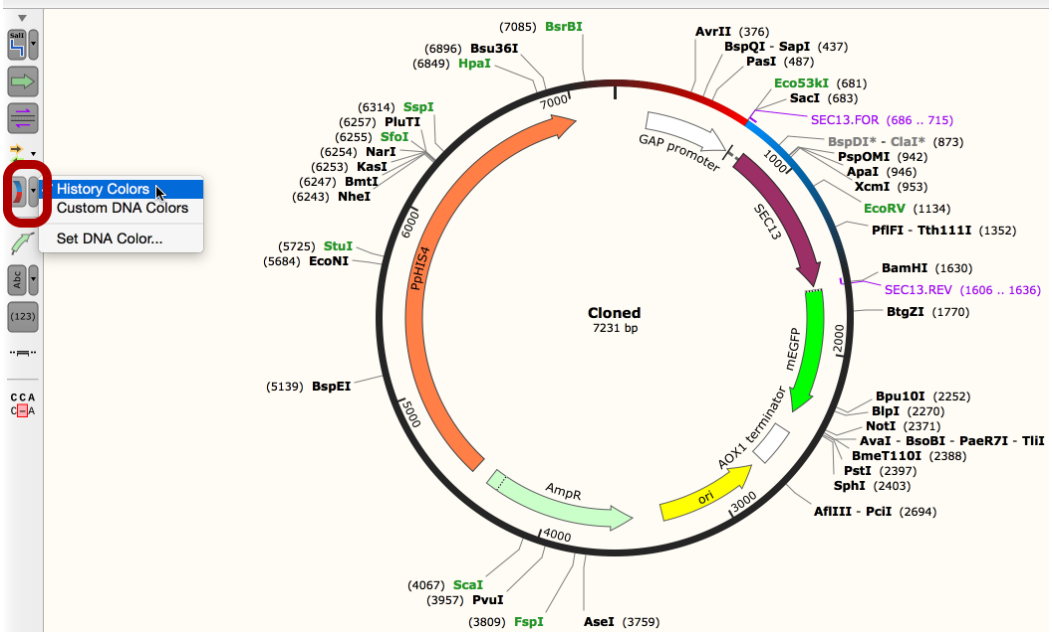

To specify the bacterial transformation strain, click the **Product** tab, then click the **Bacterial Transformation Strain** menu to change the chosen strain or to edit the strains list.

## Name the Product



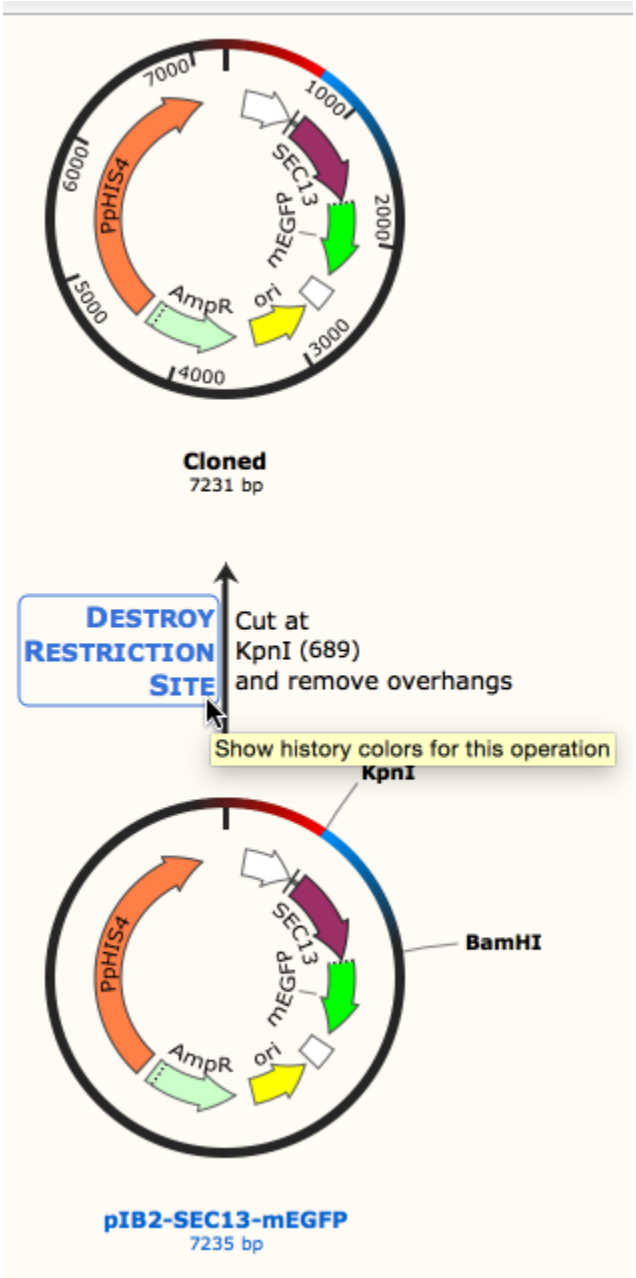

Type the name of the product, then click **Clone**.

## View the Product History Colors




To turn on the product history colors in Map and Sequence views, click the "Show colors" button in the side toolbar. The red and blue colors represent where the restriction site was destroyed.

In Sequence view, the history colors illustrate more precisely where the restriction site was destroyed.

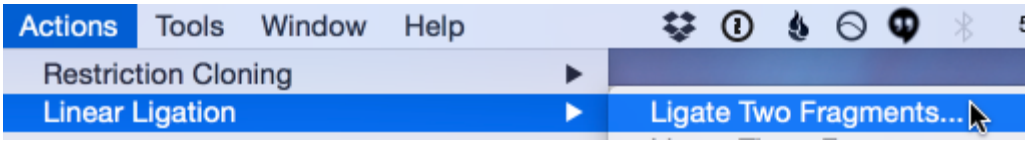



To illustrate in History view where the restriction site was destroyed, click the Destroy Restriction Site operation name.

# Linear Ligation

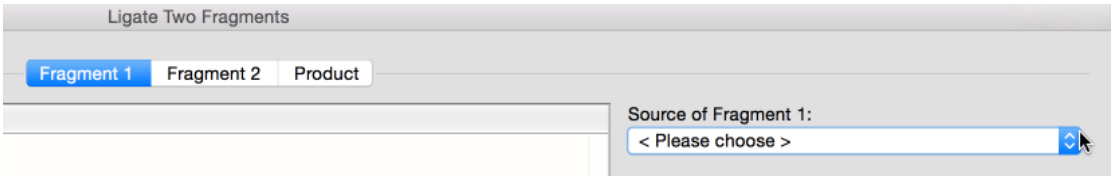
How can I ligate two or more linear fragments?

## Open the Ligate [N] Fragments Dialog



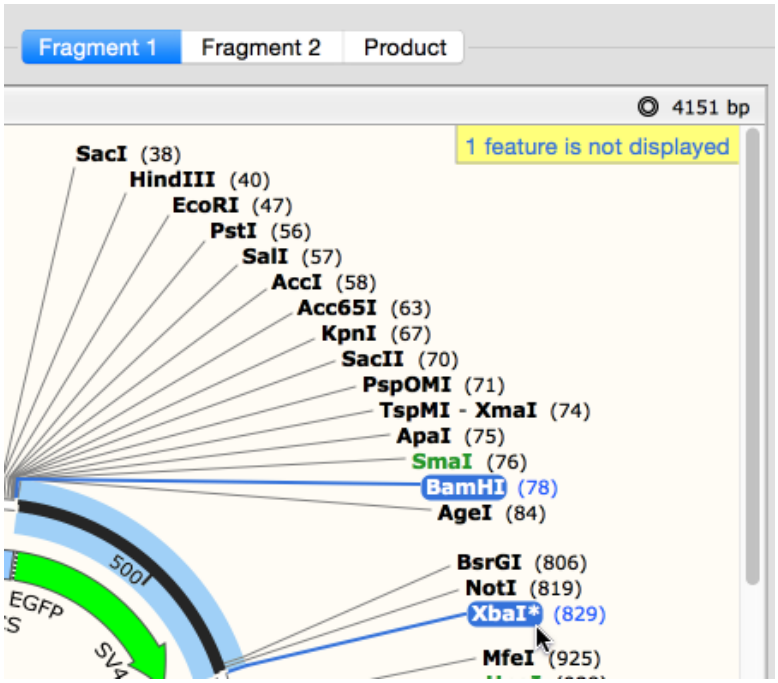

To open the Ligate [N] Fragments dialog, click **Actions** → **Linear Ligation** → **Ligate [N] Fragments...** .

## Specify the First Fragment File

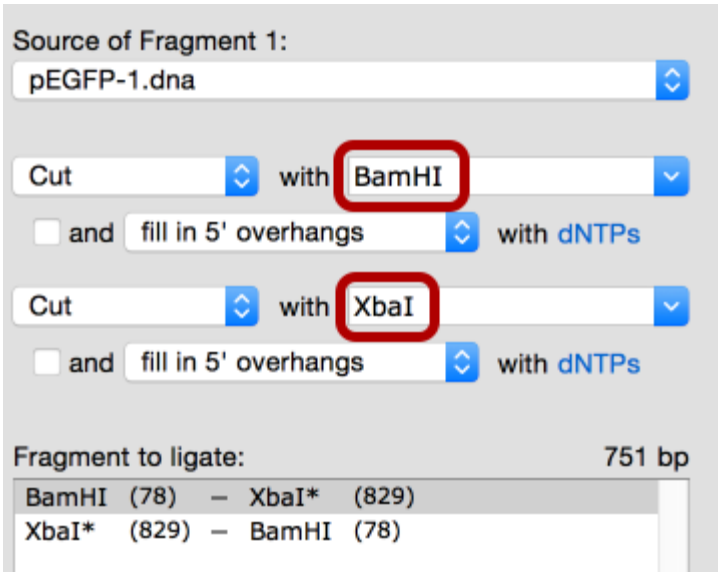



Choose the file that contains the first fragment from the **Source of Fragment 1:** menu.

## Specify the First Fragment




To specify the first fragment, click on the first enzyme site, then either Shift-click on the second enzyme site or drag to the second enzyme site.



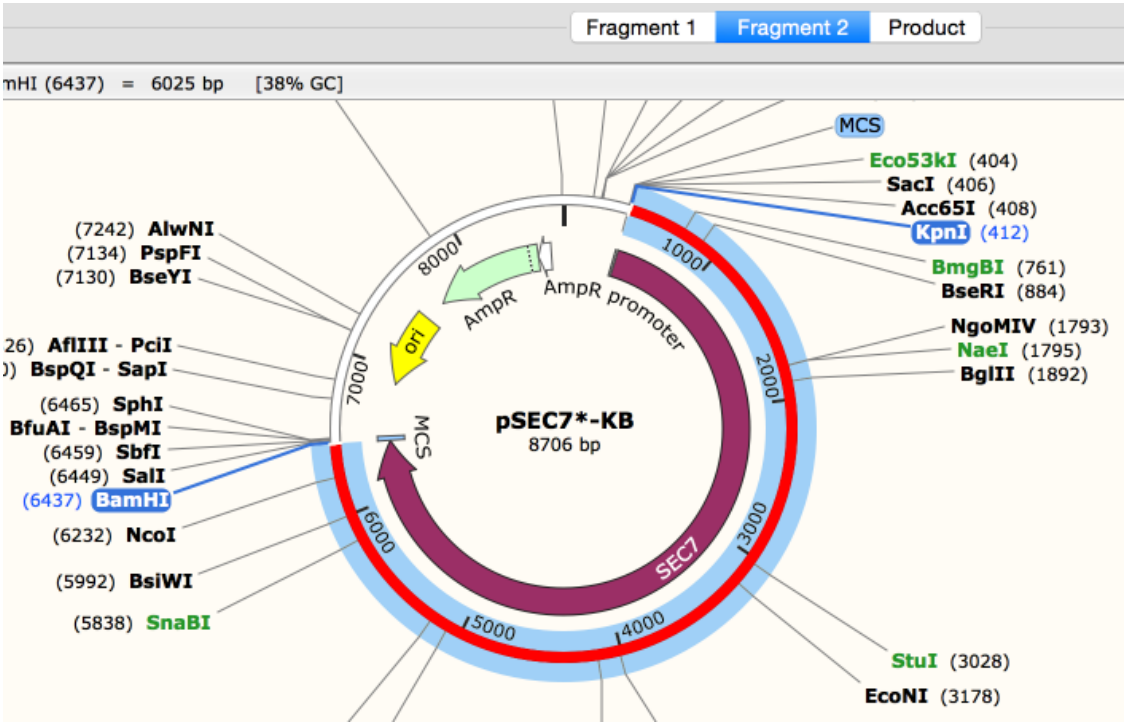

Alternatively, type the enzyme names into the boxes at the upper right. Review the blunting, enzyme selection, fragment selection, and orientation.

## Specify the Second Fragment File

To specify the source of the second fragment, click the **Fragment 2** tab at the top of the dialog.

Choose the file that contains the second fragment from the **Source of Fragment 2:** menu.

## Specify the Second Fragment




To specify the second fragment, click on the first enzyme site, then either Shift-click on the second enzyme site or drag to the second enzyme site.

## Specify Any Additional Fragments

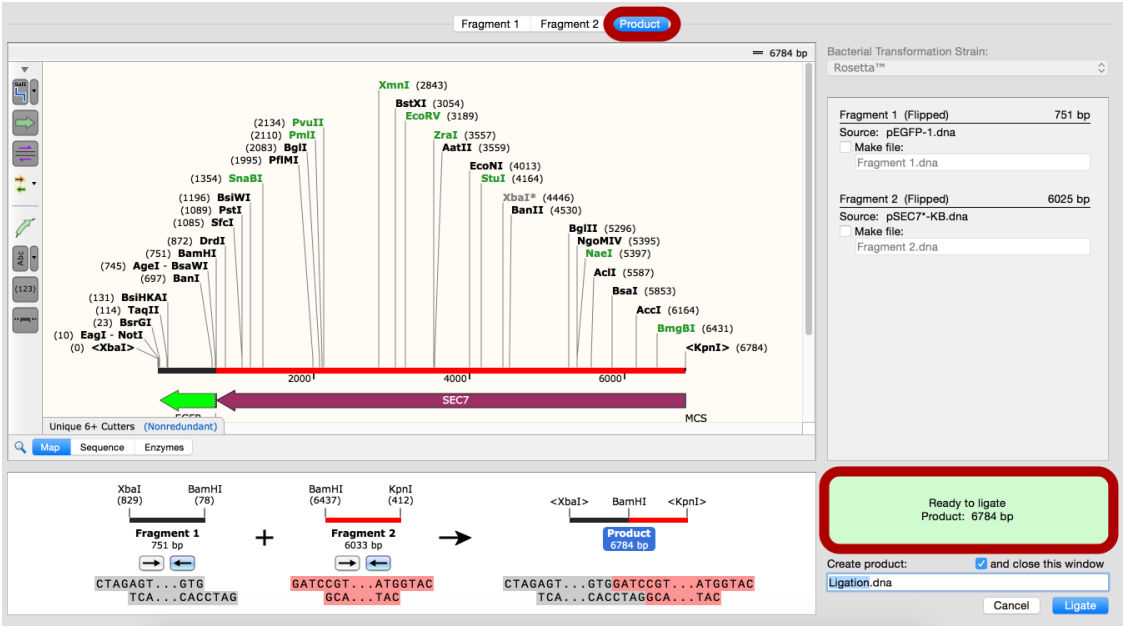
If necessary, repeat the steps described above to specify additional source files and restriction fragments.

## Make the Ends Compatible

If ligation is not possible, the box at the lower right will remain yellow and will explain what steps must be taken to proceed. Correct the problem by some combination of flipping fragments, or blunting ends, in order to make the ends compatible for ligation.

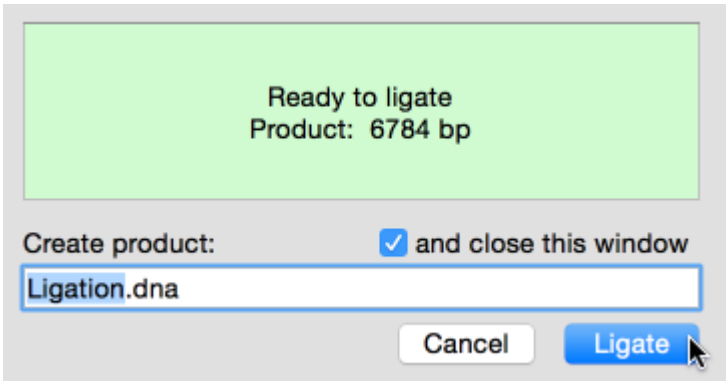
In the illustrated example, the yellow box indicates that the ends are not compatible for ligation. This problem can be corrected by clicking the reverse directional arrows to flip the orientations of both fragments.

## Preview the Product



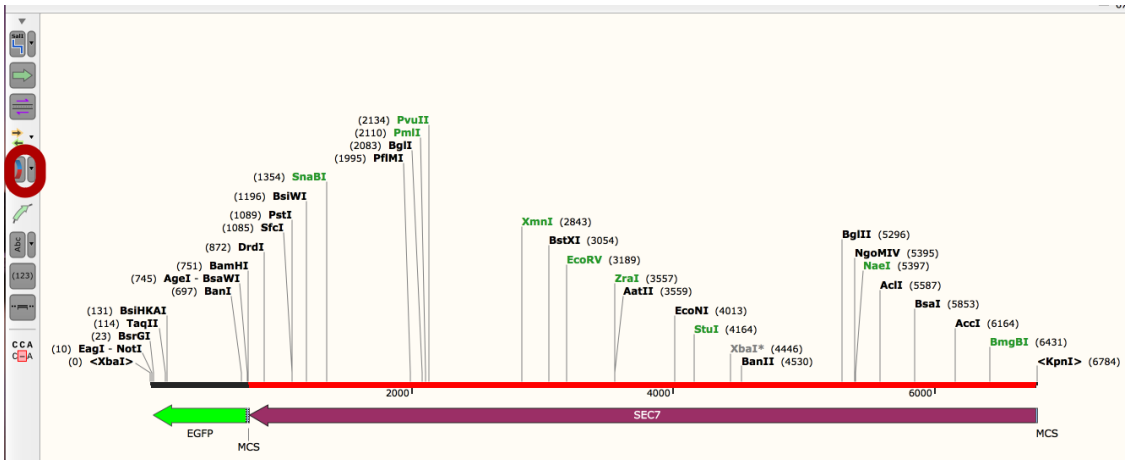

To preview the product, click the **Product** tab. The box will turn green when you are ready to ligate.

## Name the Product

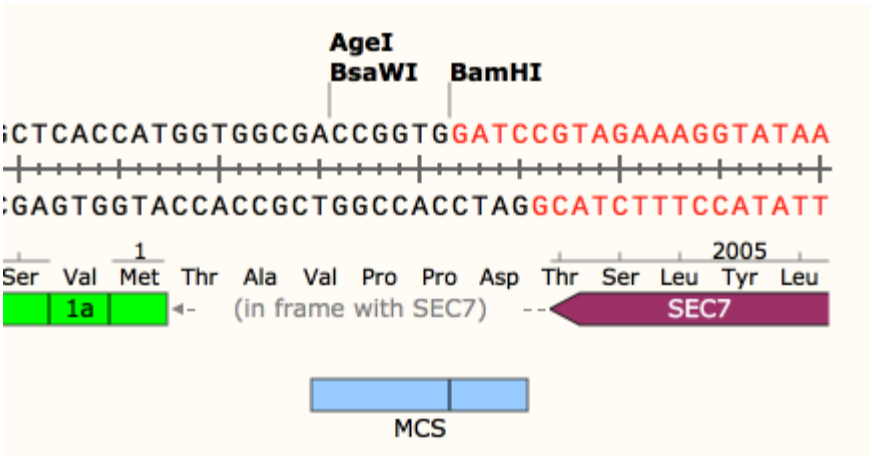



Type the name of the product, then click **Ligate**.

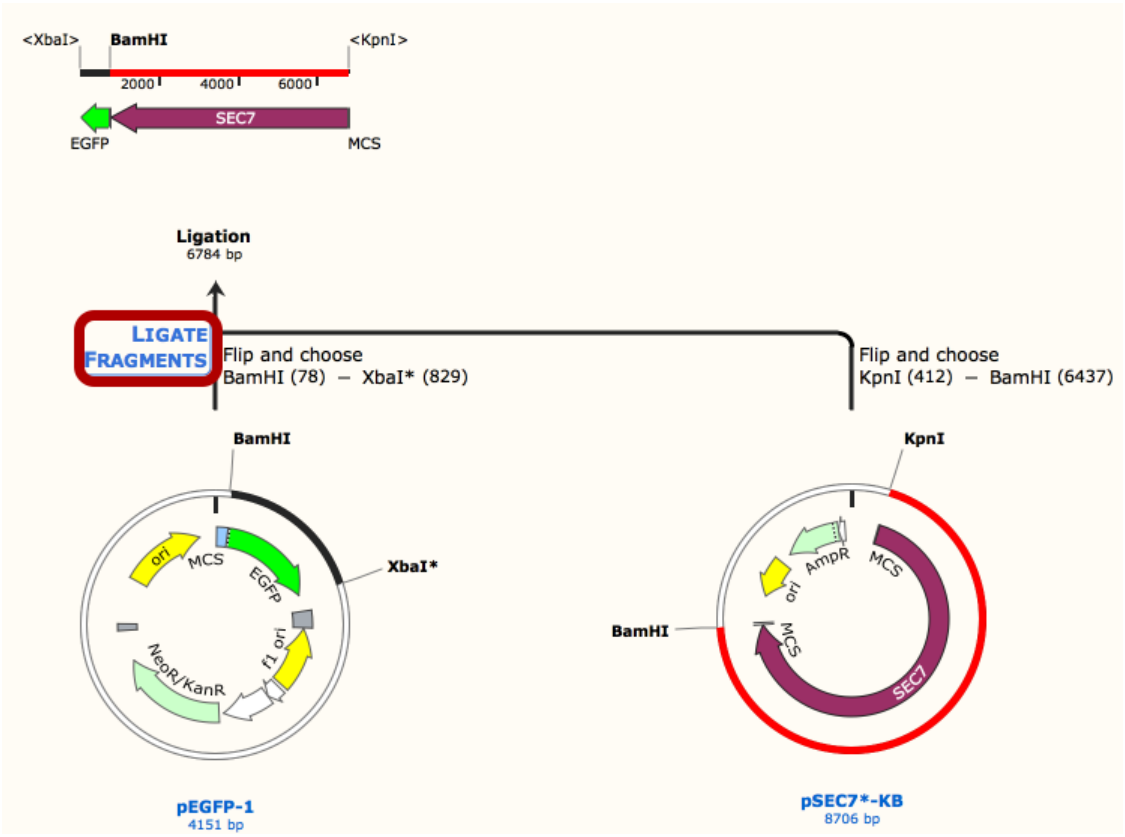
## View the Product History Colors




To turn on the product history colors in Map and Sequence views, click the "Show colors" button in the side toolbar. Alternating black, red and blue colors represent multiple ligated fragments.

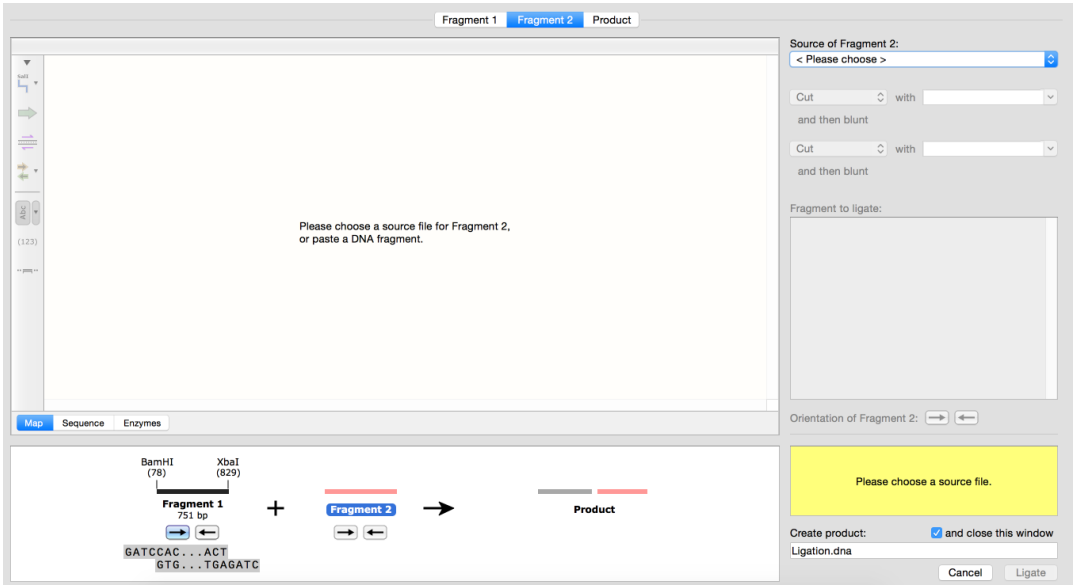



In Sequence view, the history colors illustrate more precisely where the fragments were ligated.




To illustrate in History view where the ligations occurred, click the Ligate Fragments operation name.

## Alternative Copy/Paste Approach




Alternatively, copy and paste can be used to populate the LIgate [N] Fragments dialog.

To select the first fragment to be ligated, click on the first enzyme site in Map or Sequence view,

then either Shift-click on the second enzyme site or drag to the second enzyme site. Click **Actions → Linear Ligation → Ligate [N] Fragments...** . The Ligate [N] Fragments dialog will open to the populated **Fragment 1** tab.

Switch to an open file containing the second fragment to be inserted. Select that restriction fragment, and then click **Edit → Copy Restriction Fragment**.

Switch back to the Ligate [N] fragments dialog. Click the **Fragment 2** tab at the top of the dialog. Click **Edit → Paste** to populate this part of the dialog.

Repeat this process for any remaining **Fragment** tabs in the dialog.

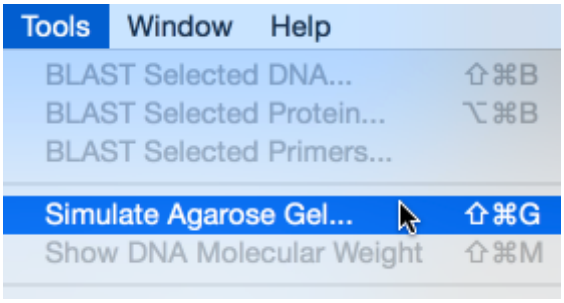
When you are ready to ligate, type the name of the product, then click **Ligate**.

# Agarose Gel Simulation

# Simulate an Agarose Gel

How do I open and configure the Simulate Agarose Gel dialog?

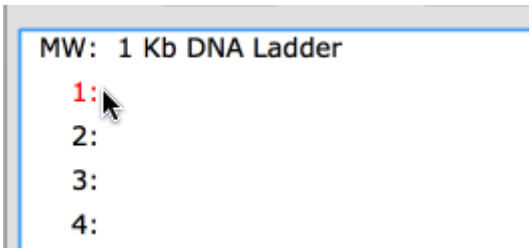
## Simulate an Empty Agarose Gel




To open the Simulate Agarose Gel dialog, click **Tools** → **Simulate Agarose Gel...** .

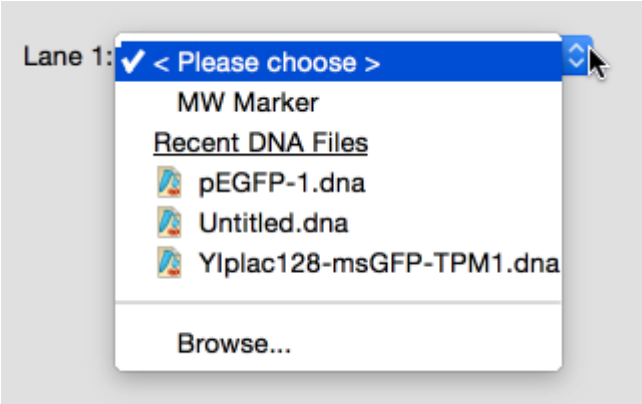
## Select a Gel Lane

Click a number above the gel to choose a lane.




Alternatively, click the lane number in the list below the gel.

## Specify the Sequence for Digestion or PCR

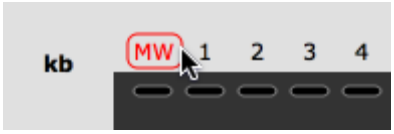



To specify the sequence for digestion or PCR, expand the **Lane [N]** menu, then choose the desired file.

## Choose a MW Marker

How do I choose a MW marker for an agarose gel simulation?

### Select the MW Marker Lane



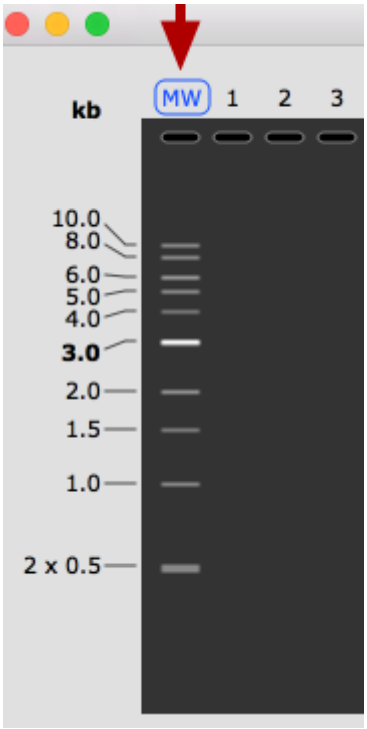

To select the MW Marker lane, click "MW" above the gel.

### Choose the Ladder

Expand the **MW Marker** menu, then choose the desired ladder.

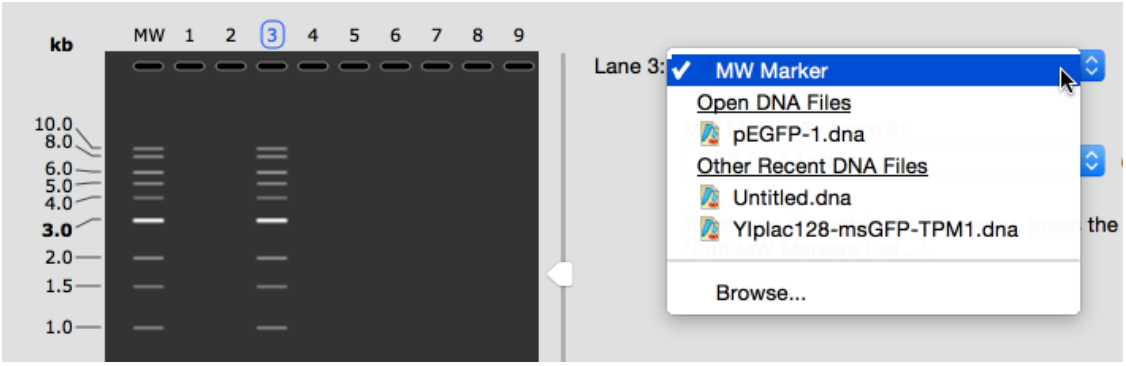
If we have overlooked a MW marker that you use, please contact us and we will add it to the list.

### View the Ladder




The selected marker will be shown in the MW lane of the agarose gel simulation.

## Add Another Ladder




To add another ladder, click the lane number, expand the **Lane [N]** menu, choose **MW Marker**, then specify the desired ladder.

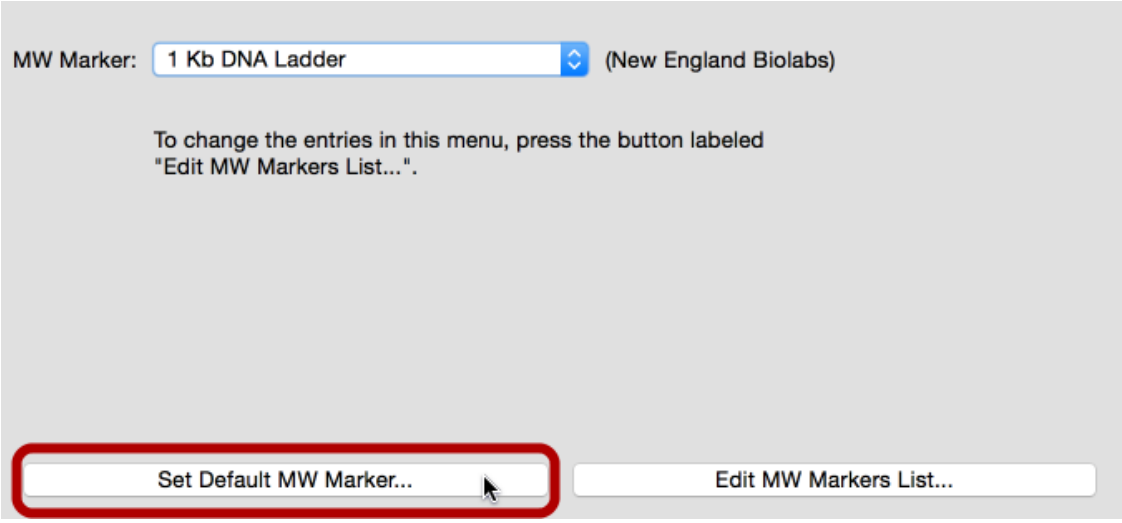
# Set the Default MW marker

How do I set the MW marker?

## Select the MW Marker Lane

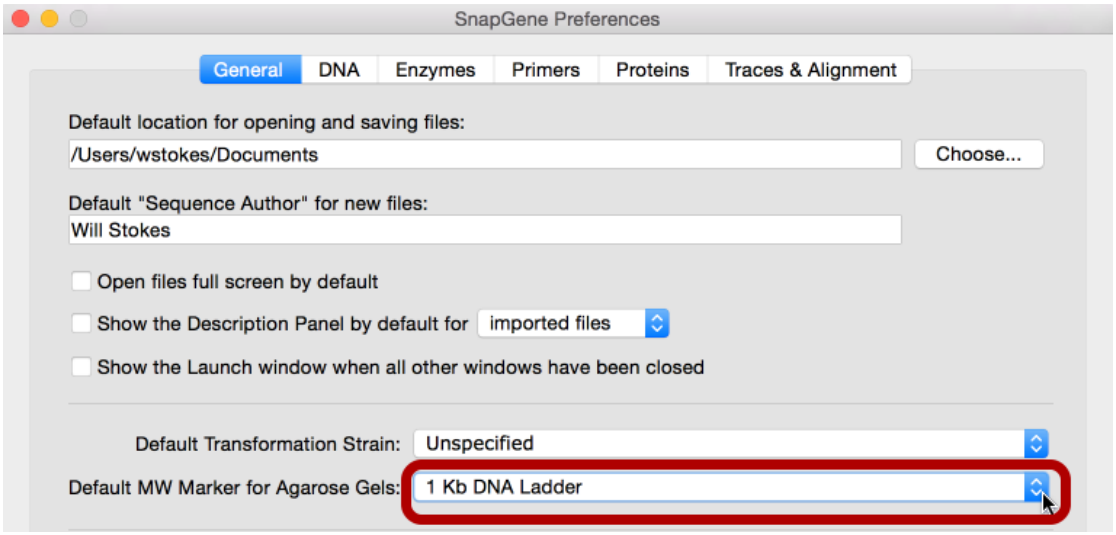
To select the MW Marker lane, click "MW" above the gel.

## Set the Default MW Marker

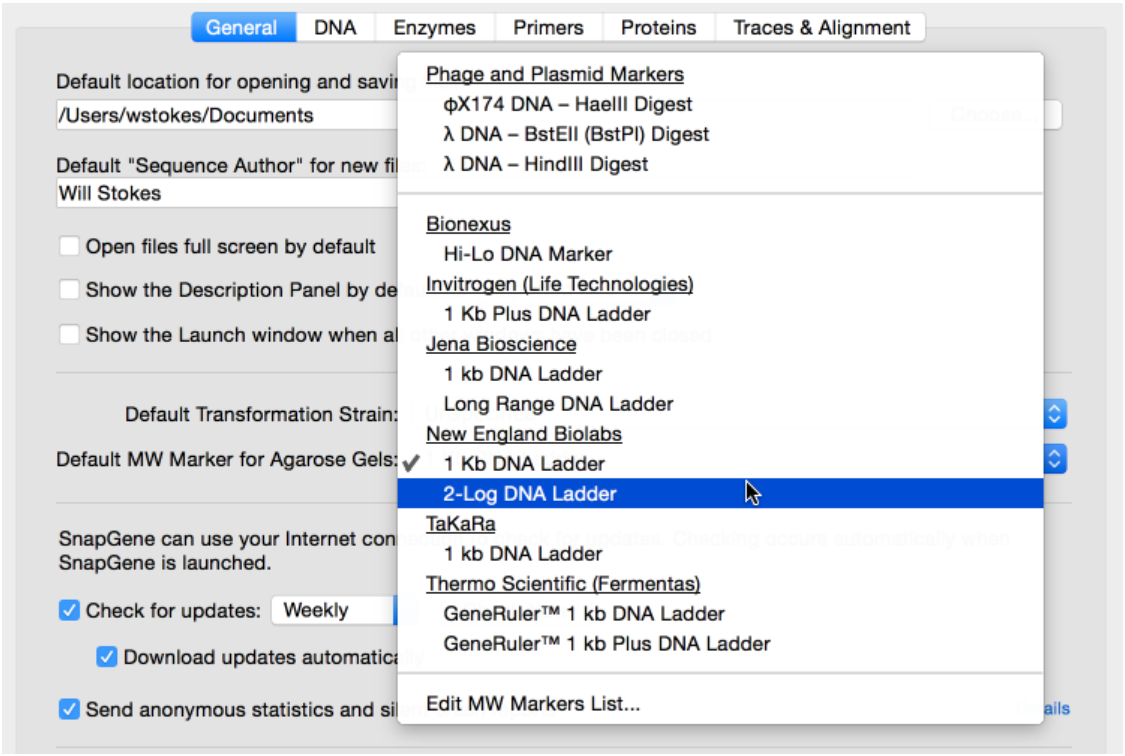



If you would like to change the default shown in the **MW Marker** menu, click the **Set Default MW Marker...** button.

Alternatively, on a Mac click **SnapGene** → **Preferences**, or on a Windows or Linux computer, click **Edit** → **Settings.**



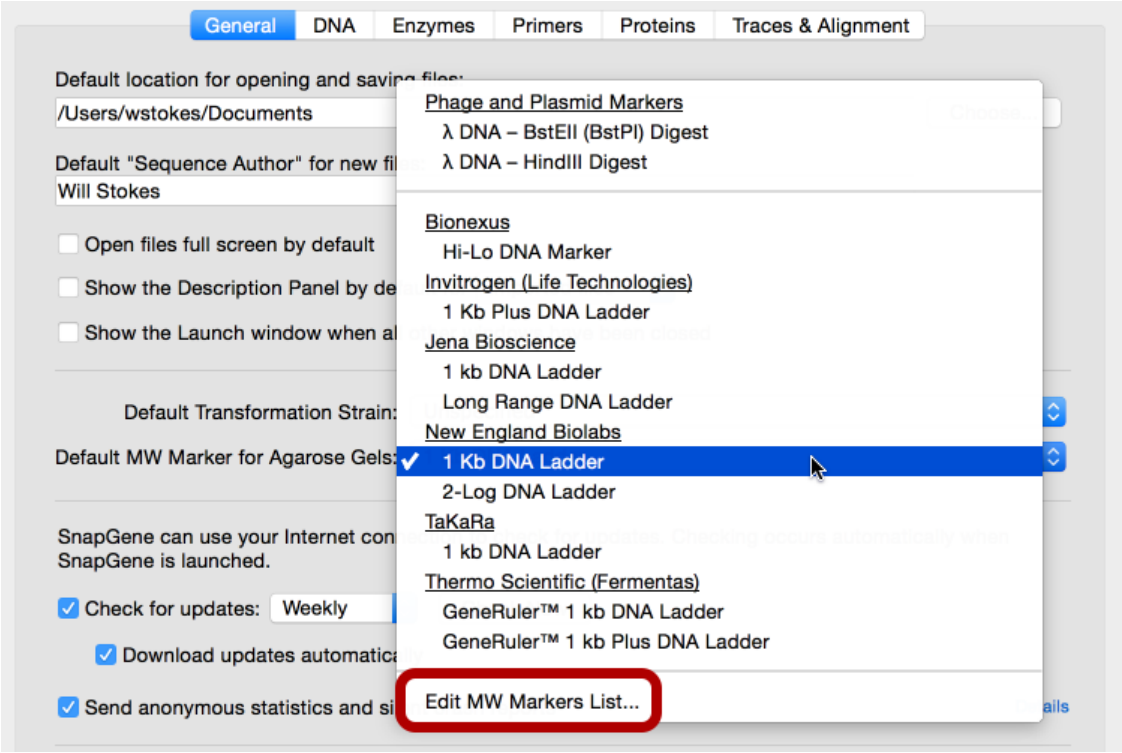

In the SnapGene Preferences dialog General tab, expand the **Default MW Marker for Agarose Gels** menu.




Choose the desired default ladder, then close the Preferences dialog.

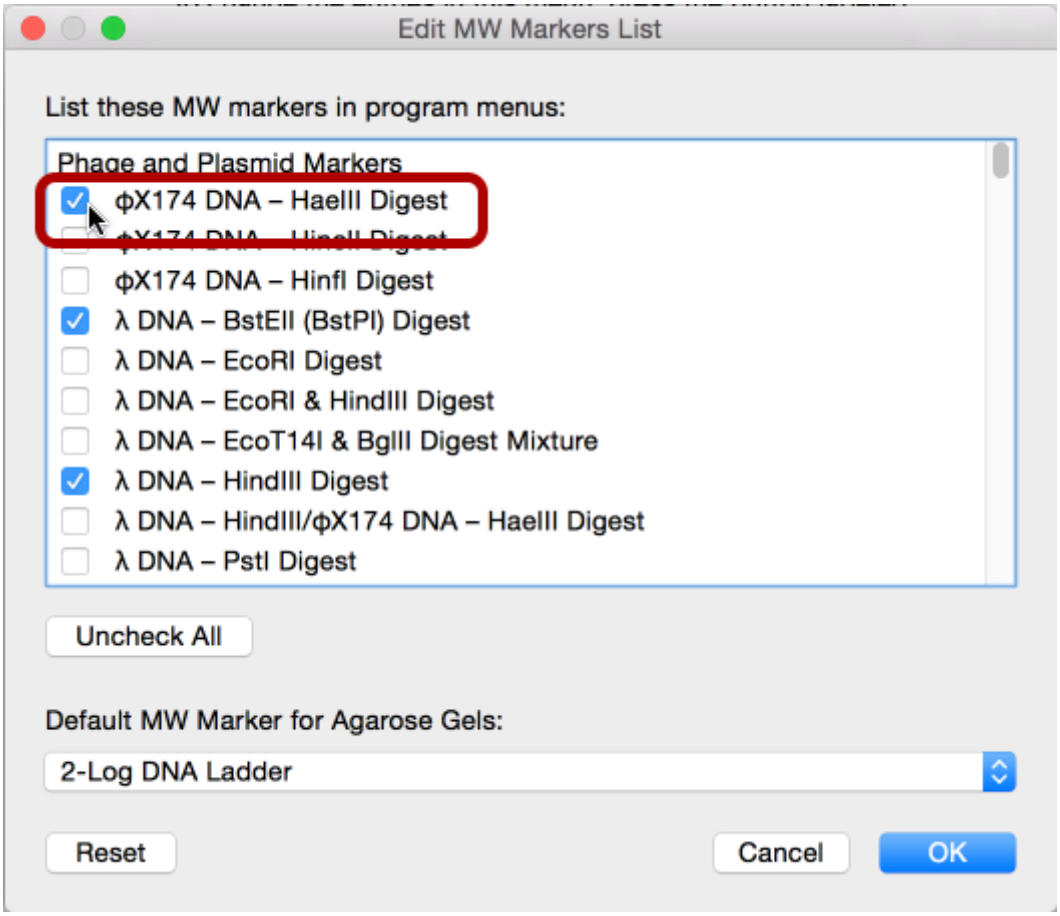
Note: To update the default **MW Marker** selection if the Simulate Agarose Gel dialog is already open, close the dialog then reopen it.

## Customize the MW Markers List




If the desired MW marker is not listed in the **Default MW Marker for Agarose Gels** menu, choose **Edit MW Markers List...** .

Alternatively, click the **Edit MW Markers List...** button in the Simulate Agarose Gel dialog.

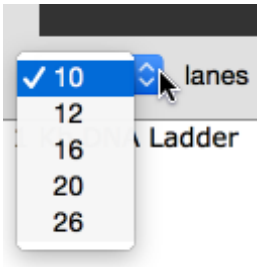



Click the check box next to each desired MW marker.

## Configure the Gel Properties

How do I configure the number of gel lanes, the % agarose, and the electrophoresis time?

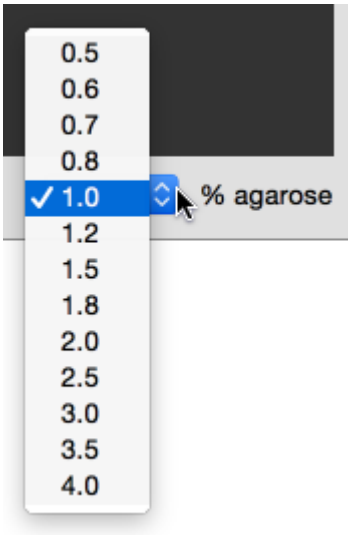
### Change the Number of Lanes




To adjust the number of lanes in the agarose gel, expand the **lanes** menu.

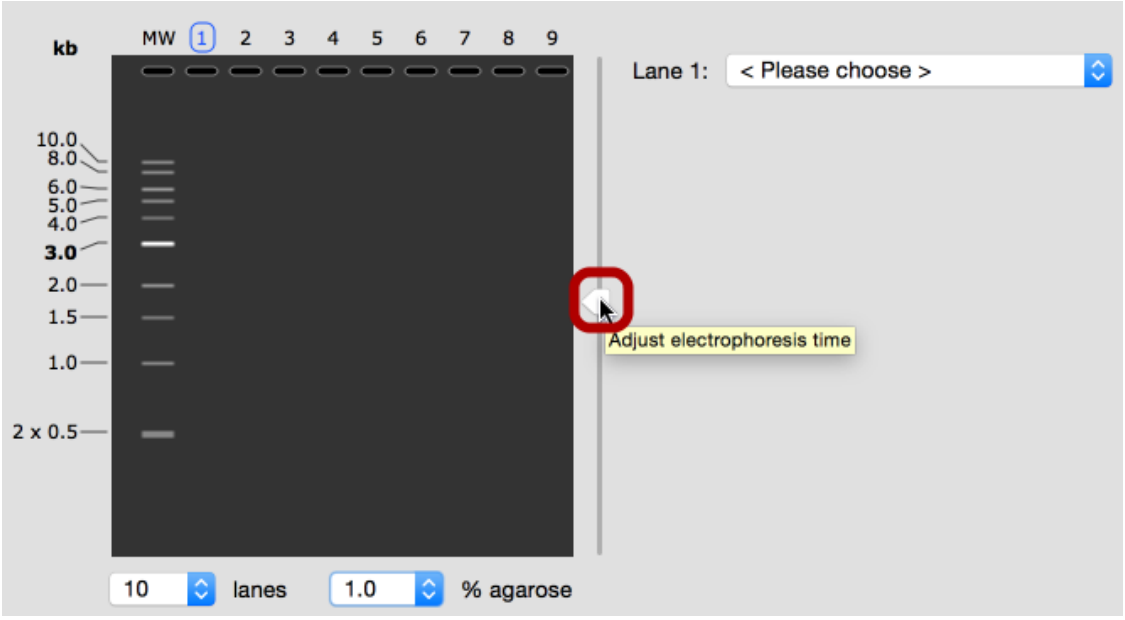
Note: A large screen is required to view a 26-lane simulation.

### Change the % Agarose




Expand the **% agarose** menu to adjust this gel property.

## Adjust the Electrophoresis Time




Drag the "Adjust electrophoresis time" slider up or down to adjust this gel property.

Alternatively, mouse-over the gel background, and use a two-finger scroll (or mouse wheel) to adjust the electrophoresis time.

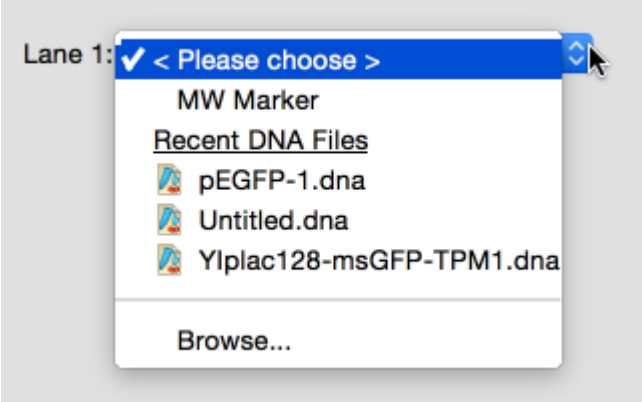
## Choose a Sequence for Digestion or PCR

How do I choose a sequence for restriction digestion or PCR?

### Select a Lane

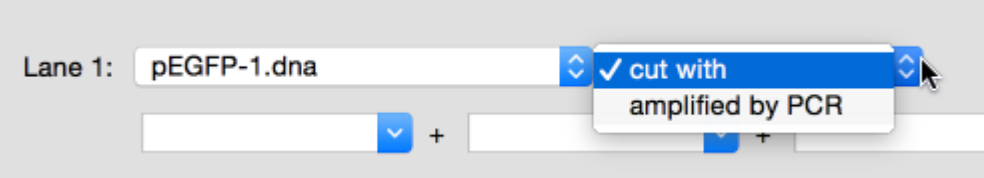
Click a number above the gel to choose a lane.

### Specify the Sequence for Digestion or PCR




To specify the sequence, expand the **Lane [N]** menu, then choose the desired file.

### Specify Digestion or PCR

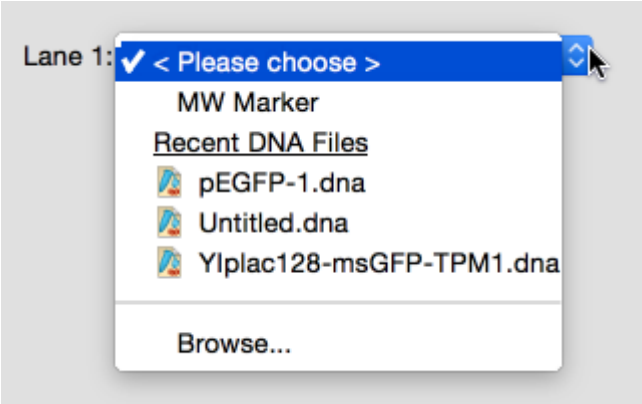



To specify how the gel fragments are produced, choose from the menu at the upper right.

## Simulate a Restriction Digest

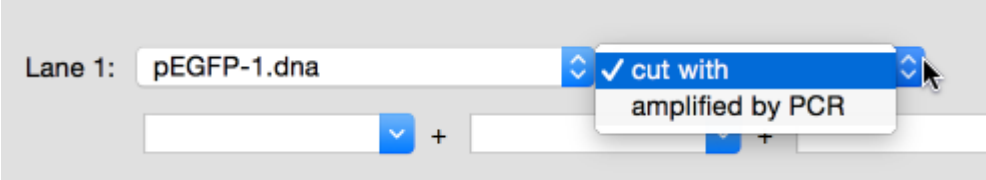
How do I simulate an agarose gel for a restriction digest?

### Specify the Sequence for Digestion



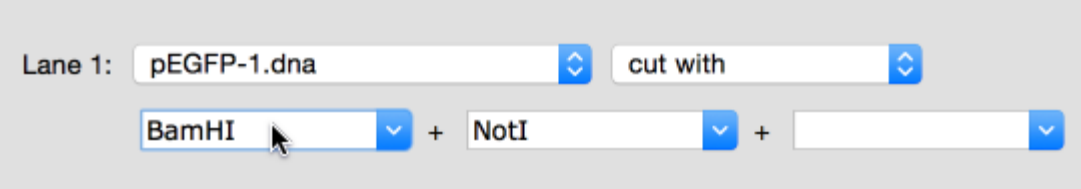

To specify the sequence, expand the **Lane [N]** menu, then choose the desired file.

### Choose Restriction Digestion




To specify a restriction digest, choose **cut with** (default) from the expanded menu.

### Specify the Digestion Enzymes



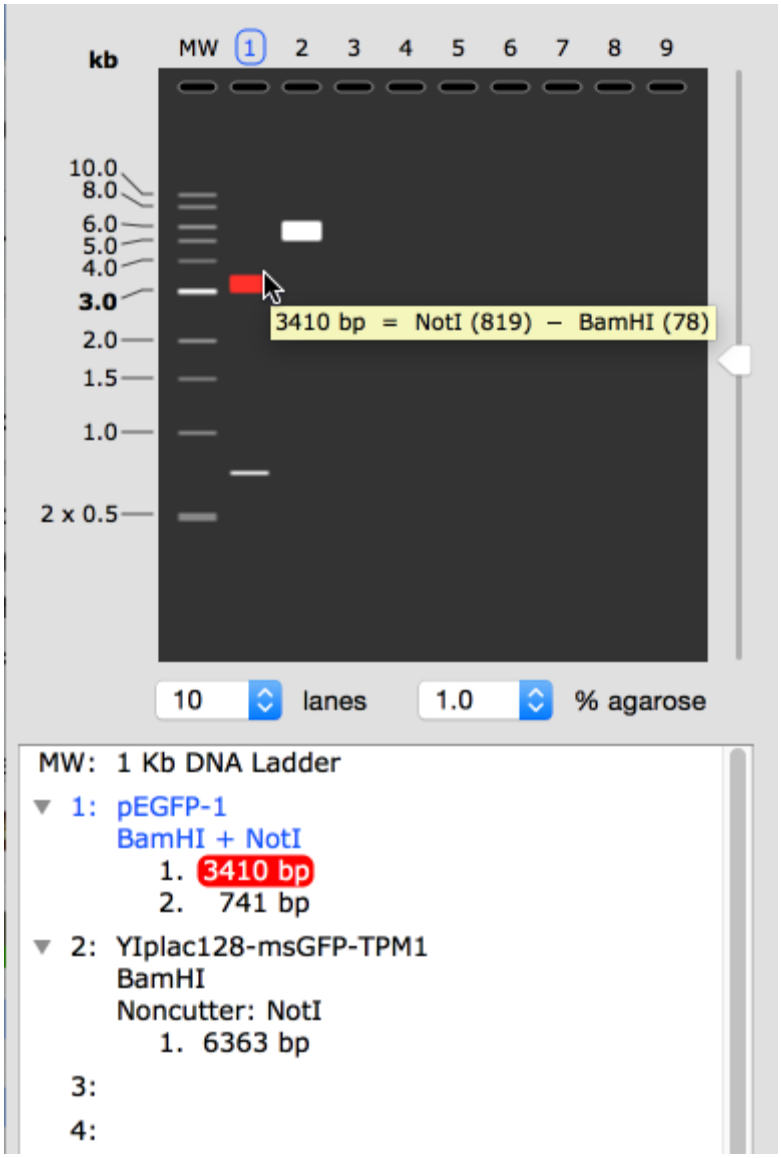

To specify the digestion enzymes, type the enzyme name in the menu box to auto-search the menu. **Tab** to specify other digestion enzymes. Alternatively, choose the enzyme name from the expanded menu (type the name to auto-search within the expanded menu).

The resulting digest simulation will appear.

## Apply to All Lanes

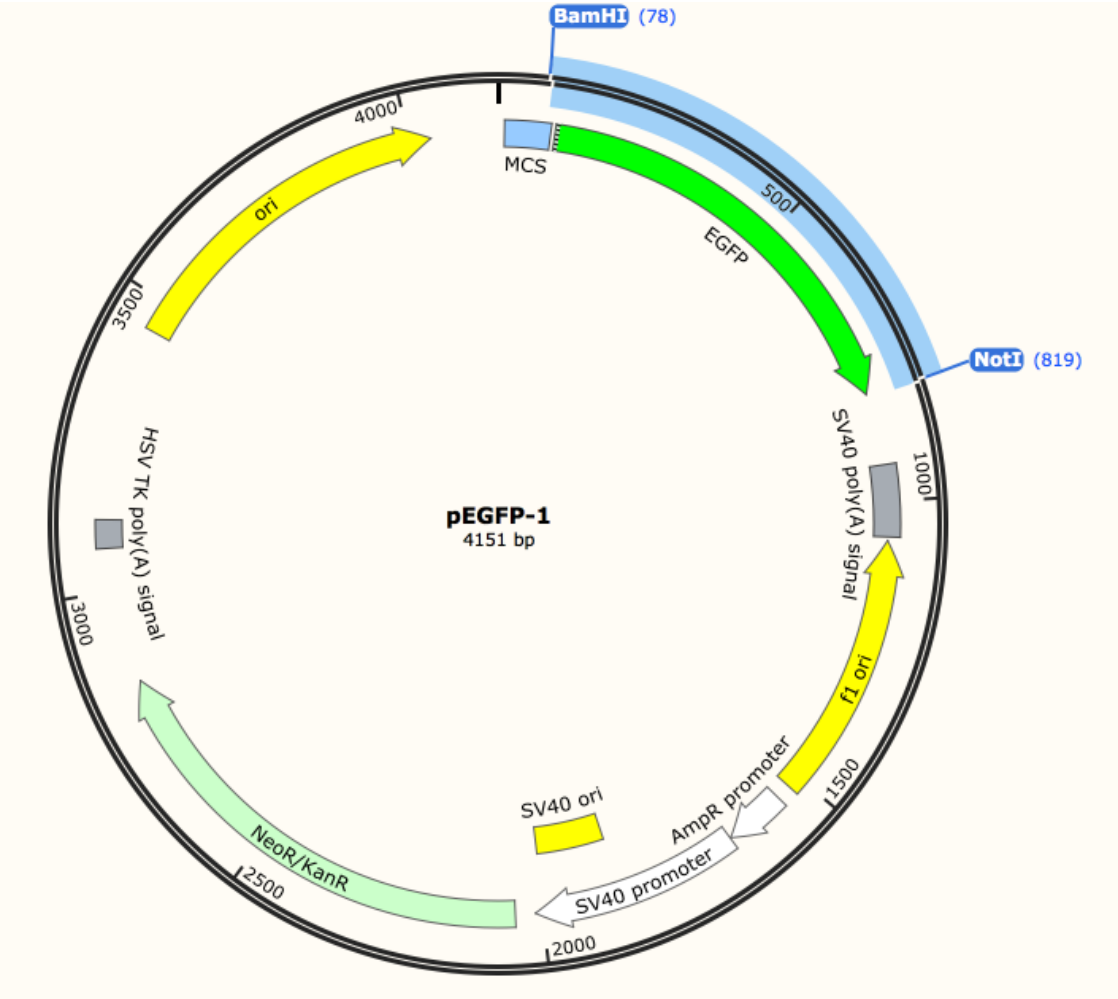
If you wish to simulate the digestion of multiple sequences with the same enzyme(s), click the **Apply to All Lanes** button.

## View the Simulated Agarose Gel

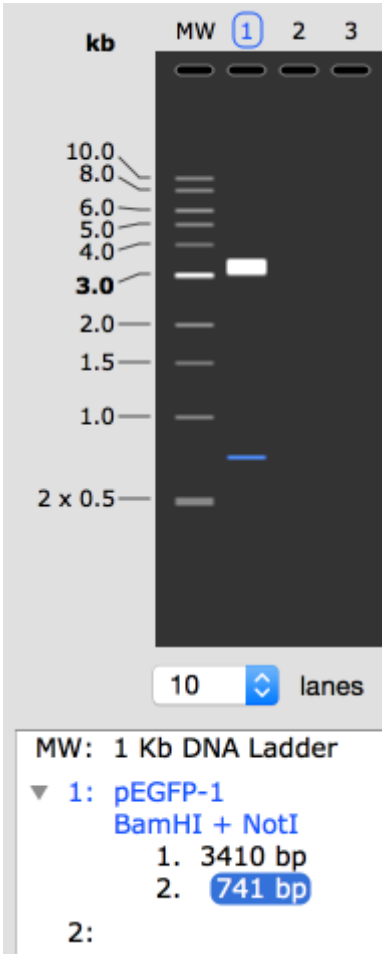



The simulated agarose gel for the restriction digest will be shown.

## Rapidly Simulate a Restriction Digest




To simulate a restriction digest from a sequence window, click one or more enzymes, then click **Tools → Simulate Agarose Gel**.



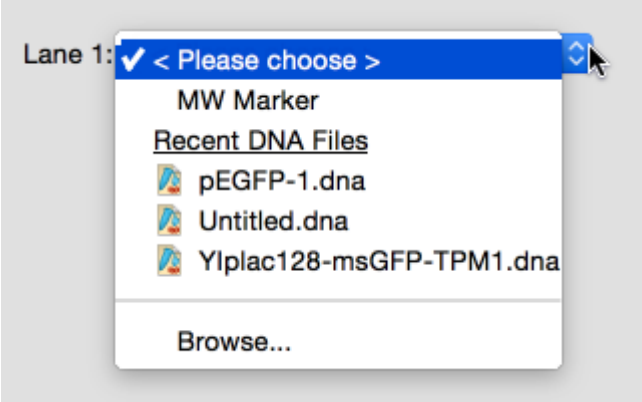

The resulting digest will appear in the next available lane in the Agarose Gel Simulation dialog.

Repeat this step to populate other lanes.

# Simulate a PCR Amplification

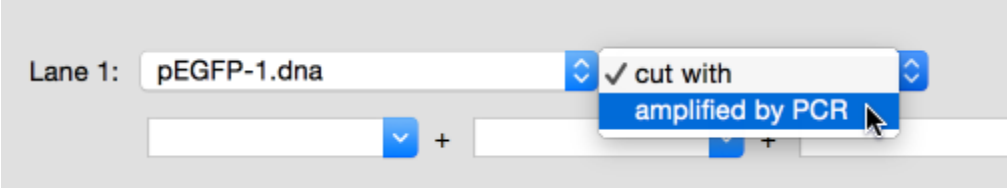
How do I simulate an agarose gel for a PCR amplification?

## Specify the Template for PCR Amplification



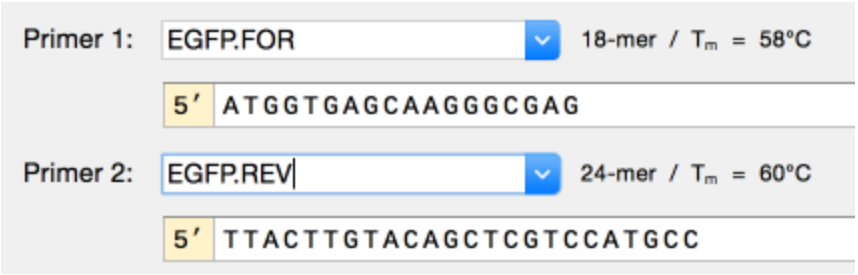

To specify the template, expand the **Lane [N]** menu, then choose the desired file.

## Choose PCR Amplification




To specify a PCR amplification, choose **amplified by PCR** from the expanded menu.

## Specify the Primers

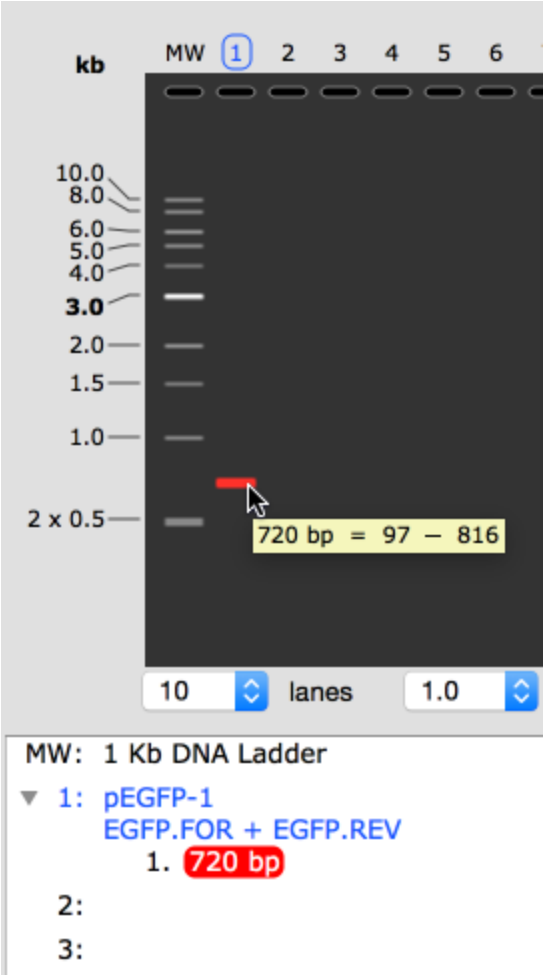



To specify the primers, choose the primer name from the expanded **Primer [N]** menu. **Tab** to specify the other primer.

Alternatively, paste the primer sequence into the **5' ... 3'** box.

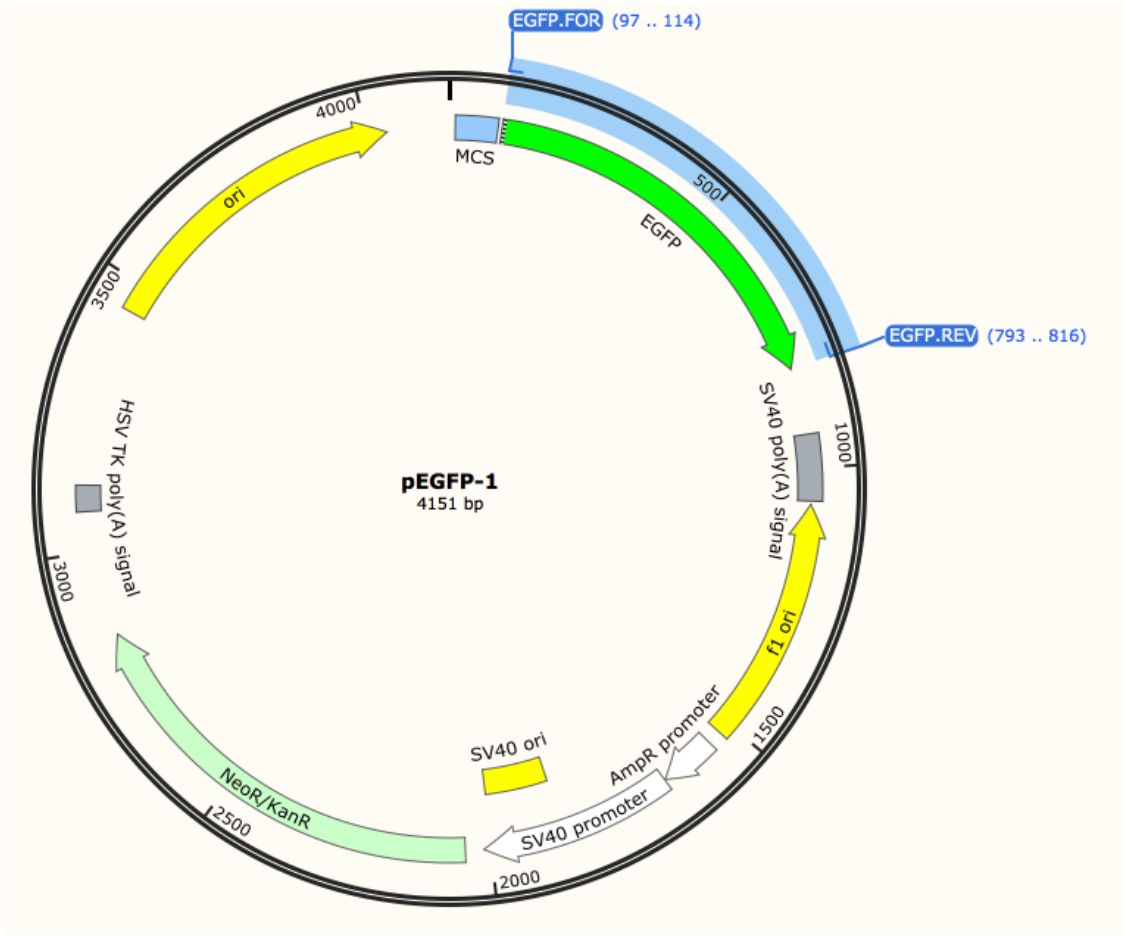
Alternatively, to specify the PCR primers, click to select two primers in Map, Sequence, or Primers view.

## View the Simulated Agarose Gel

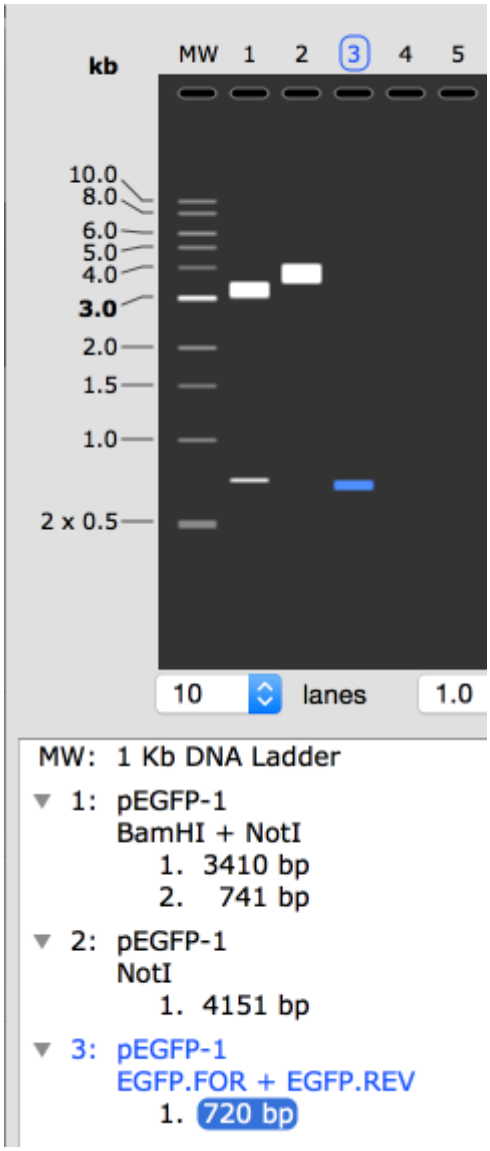



The simulated agarose gel for the PCR amplification simulation will be shown.

## Rapidly Simulate a PCR Amplification




To simulate a PCR amplification from a sequence window, select a pair of primers, then click **Tools → Simulate Agarose Gel**.




The resulting PCR amplification will appear in the next available lane in the Agarose Gel Simulation dialog.

Repeat this step to populate other lanes.

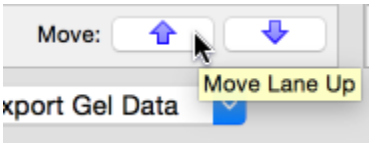
# Rearrange the Gel Lanes

How do I rearrange the gel lanes?

## Select a Lane

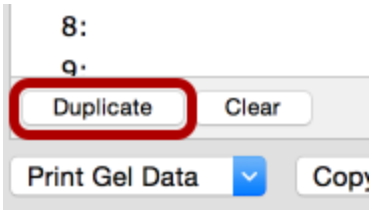
Click a number above the gel to choose a lane.

## Reorder the Lanes



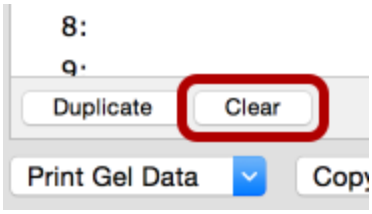

To change the lane position, click the "Move Lane Up" or "Move Lane Down" button.

## Duplicate a Lane




To duplicate lane data into an adjacent lane, select a lane, then click **Duplicate**. Subsequent lanes will shift if necessary to make room for the additional lane.

## Clear a Lane




To clear the chosen lane, click **Clear**.

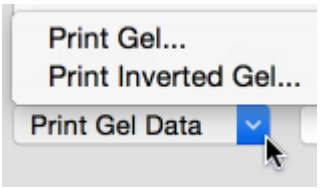
## Clear All Lanes

To clear all lanes, click **Reset**.

# Export the Gel Data

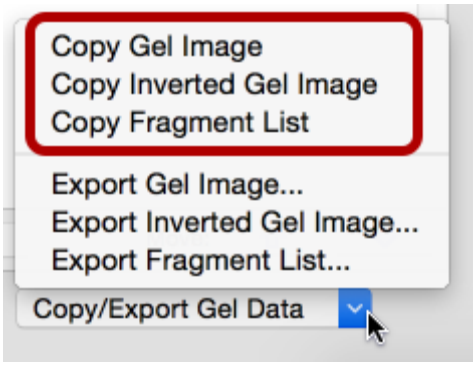
How do I print, copy or save the gel and/or fragment list?

## Print the Gel Data



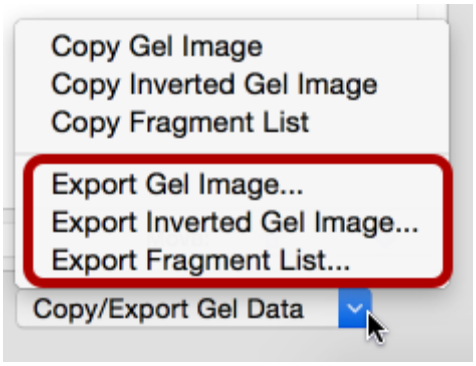

To print the simulated gel and the fragment list, expand the menu, then choose the format: standard or inverted.

## Copy the Gel Data

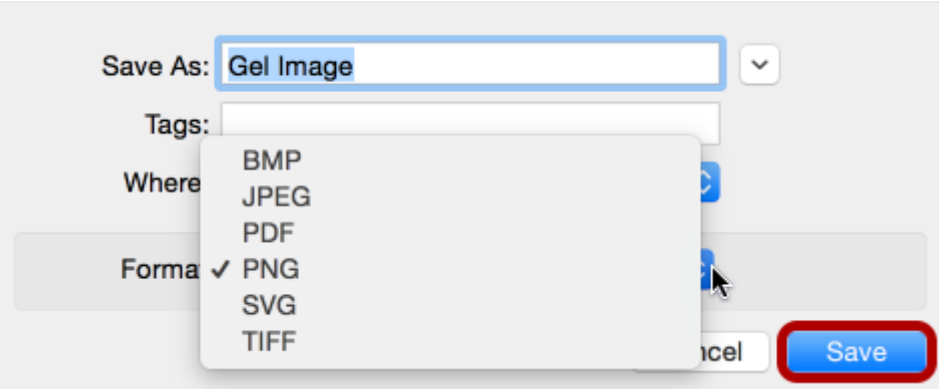



To copy the gel image or fragment list to the clipboard, expand the menu, then click the desired option.

## Export the Gel Data




To export the gel image or fragment list to a file, expand the menu, then click the desired option.




Specify the file format, then click **Save**.